FHIR-STU3 — Workflow Module (sections 12 – 12.25.10)

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12 Workflow Module

URL: https://hl7.org/fhir/STU3/workflow-module.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

Work Group FHIR Infrastructure | Ballot Status: Informative

12.0 Workflow Module

The workflow module focuses on the coordination of activities within and across systems. This includes three primary aspects:

* How do we ask for another person, device or system to do something?

* How do we track the linkages and dependencies between activities - actions to their authorizations, complex activities to individual steps, protocols to plans to orders, etc.?

* How do we define what activities are possible and the expected order and dependencies of the steps within those activities? I.e. process/orchestration definition

12.0 Index

Infrastructure | Resource: Task Patterns: Definition, Request, Event Documentation: Overview, Overview, Communication Patterns, Ad-Hoc PatternsManagement Patterns & Examples
Scheduling | Appointments: Appointment / AppointmentResponse Availability: Schedule / Slot
Clinical Process | Referrals: ReferralRequest, ProcedureRequest Orders: NutritionOrder, VisionPrescription Definitions: ActivityDefinition, PlanDefinition Miscellaineous: ProcessRequest & ProcessResponse, DeviceRequest & DeviceUseStatement, SupplyRequest & SupplyDelivery

12.0.1 Introduction

Workflows can be performed through direct posting of resources to a target server (combined with a specific tag), by using the Task resource, through the use of messaging or via FHIR services. This specification includes a workflow page that describes the concepts underlying the discussion of workflows, and points to a number of different communication and architectural workflow patterns.

In addition to the Task resource, this specification defines three logical models - Definition, Request and Event that define the patterns for resources that are typically involved in workflow. These patterns include elements defining common attributes of each type of resource as well as relationships between them. These relationships are summarized on the workflow page, along with a complete list of resources that follow (or are hoped to soon follow) the request and event patterns.

Finally the PlanDefinition and ActivityDefinition resources combine to support the creation of protocols, orders sets, guidelines and other workflow definitions by describing the types of activities that can occur and setting rules about their composition, sequencing, interdependencies and flow.

12.0.2 Common use Cases

Workflow manifests in many places in the healthcare environment:

* Creating a lab order, drug prescription, referral or other clinical order or an insurance claim, enrollment request, Appointment or similar administrative request and asking for it to be actioned by a specific organization or practitioner

* Negotiating a fulfillment process, such as requesting further information before accepting a claim or referral or proposing an alternative therapy when processing an order

* Letting an ordering physician know what the current progress is in fulfilling an order (e.g. blood has been drawn, sample is being processed, preliminary results are in, etc.)

* Defining a plan or recommendation for a set of clinical and/or administrative activities to manage a patient's care and then tracking how those plans and recommendations are (or are not) acted upon over time.

* Communicating a state change to a request or order (e.g. suspension, update, cancellation, etc.) to a fulfilling system so that they can take appropriate action

* Asking for a state change, requesting the merge of a couple of patients or the invoking of some operation or decision supported in an asynchronous manner - for example, one where human intervention is required

* Designing or adhering to a study protocol, chemotherapy protocol, instantiating an order set or other plan definition

FHIR provides multiple ways to enable these scenarios (and many others). Common mechanisms, along with their pros and cons can be found in the workflow sections on patterns.

12.0.3 Security and Privacy

Resources related to workflow need to adhere to the same security and privacy guidelines that apply to all FHIR resource, including specific considerations for those that may contain personally-identifying information. There are a couple of

additional security and privacy considerations specific to workflow:

1. Some workflows are ad-hoc without pre-defined participants or flows. These can be challenging for security and privacy processes to manage appropriately

2. Workflow can drive automated behavior. I.e. The mere existence of an electronic record can cause information to flow, procedures to be performed, records to be changed and money to be transferred, potentially without any intervention, oversight or sanity checking by a human being. As such, even greater care must be taken to ensure that:

* constraints are placed on what systems (and users) can initiate workflow processes

* requests for action are appropriately authenticated before action is taken

* patient consents and other relevant policies are enforced either by the system storing the request or the system acting upon it (and that if enforcement is not performed by the actor, that they are confident that relevant policies have been enforced on the request prior to action)

For more general considerations, see the Security and Privacy module.

12.0.4 Developmental Roadmap

Initial work has taken place on aligning most (though not yet all) resources with the Definition, Request and Event patterns. In the lead-up to R4, we'll be looking to align the remaining resources, reviewing and potentially increasing the consistency of existing alignments (where beneficial to implementers) and examining the potential for exposing alignment with the patterns in a computably useful manner (e.g. as interfaces). We will look at the possibility of collapsing the ProcessRequest and ProcessResponse

Work will continue on the workflow patterns, including vetting the patterns against various clinical scenarios and enhancing pattern documentation. We also hope to examine both messaging and services in more detail with further guidance about when and how such mechanisms should be used for workflow and how they relate to the Task resource. As well, we'll examine the possibility for developing "standardized" workflows for certain domains and how such patterns might be documented. We will look for implementer feedback to guide this work.

The PlanDefinition and ActivityDefinition resources will continue to evolve based on feedback from the implementer community. We'll explore using them in a variety of ways, including clinical order sets, medication protocols, workflow protocols, clinical pathways, administrative protocols, etc.

Additional topics for future work include:

* The initial effort to align with workflow patterns has been a bit over-zealous for some resources, resulting in the loss of domain-specific context or occasionally the introduction of elements that might be more properly represented as extensions. In R4, we'll endeavor to improve the balance, ensuring that consistency with patterns does not overshadow the essential requirements for implementer intuitiveness and simplicity

* Resolving the overlap between the SupplyRequest, DeviceRequest and VisionPrescription resources

* The pattern of capturing Practitioner plus "onBehalfOf" Organization and possibly their role is somewhat ugly. We will explore whether instead we can add PractitionerRole as an alternative choice with Practitioner to simplify the models

* Improving mapping and alignment of the elements and status codes of the Task resource with the WS-HumanTask specification

* Creating "best practice" guides for how to implement workflow for different business patterns

* Examining how workflow is used for "compensating actions", for example account transactions and reversals

12.1 Resource Task - Content

URL: https://hl7.org/fhir/STU3/task.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

Orders and Observations Work Group | Maturity Level: 2 | Trial Use | Compartments: Not linked to any defined compartments

A task to be performed.

12.1.1 Scope and Usage

A task resource describes an activity that can be performed and tracks the state of completion of that activity. It is a representation that an activity should be or has been initiated, and eventually, represents the successful or unsuccessful completion of that activity.

Note that there are a variety of processes associated with making and processing orders. Some orders may be handled immediately by automated systems but most require real world actions by one or more humans. Some orders can only be processed when other real world actions happen, such as a patient presenting themselves so that the action to be performed can actually be performed. Often these real world dependencies are only implicit in the order details.

12.1.2 Background and Context

12.1.2.1 Using Tasks in a RESTful context

In a RESTful context, a server functions as a repository of tasks. The server itself, or other agents are expected to monitor task activity and initiate appropriate actions to ensure task completion, updating the status of the task as it proceeds through its various stages of completion. These agents can be coordinated, following well choreographed business logic to ensure that tasks are completed. Task management may also be centrally controlled using some form of workflow engine, in which case, the workflow engine itself may update and maintain the task resources in the server, and through its orchestration activities ensure that tasks are completed. The task resource enables either model of task management, yet provides a consistent view of the status of tasks being executed in support of healthcare workflows.

The assignment of tasks into categories by type of task, type of performer and task status enable the server to function as a queue of work items. This queue can be polled or subscribed to by various agents, enabling automation of workflows in FHIR using existing search and subscription mechanisms. Owners, requesters, other agents (e.g. workflow managers) can thus be ready to initiate the next steps in a complex workflow.

12.1.2.2 Tasks State Machine

Tasks start in a Created state. Once they have been assigned to an owner they transition to the Ready state, indicating that they are ready to be performed. Once the owner initiates activity on the task, the task transitions to the In Progress state, indicating that work is being performed. Upon normal completion, the task enters the Completed state. If there is a failure during the task execution that prevents the task from being completed, it can also enter a Failed state, indicating an abnormal termination of the task. A task in any non-terminal state may also be Cancelled, representing an abnormal termination of the task due to external forces, rather than an error condition.

Tasks in any non-terminal state (Created, Ready, In Progress) can be suspended and resumed. When a task is suspended, it is typically resumed in the state it was in when it was originally suspended. Suspending a task suspends all of its children as well. Resuming a task resumes all of its children.

An In-progress task can also be stopped, returning it to a Ready state. This may be in preparation for delegation or reassignment (e.g., because it cannot be completed by the current owner), to restart a task due to a temporary failure (e.g., to reattempt completion of the activity), or in preparation to allow others to claim the task.

12.1.2.3 Using Tasks with operations

The Task resource defines several operations to enable rich control over the task execution environment. A server managing task resources may allow unfettered read access to all tasks that it maintains, but restrict write access to the task resources to itself in order to exert control over the task state machine. Instead, it might offer operations on tasks that enable strictly controlled write operations following commonly accepted business rules for task management. For example, only tasks that are in the "In Process" state might be allowed to be "Completed" or "Failed". Use or support of these operations is not essential in simple workflow environments where task activity needs little management.

There is no distinct operation for creating a new task because the FHIR RESTful API already distinguishes between the create and update operations.

The task history allows applications monitoring the state of a workflow to identify tasks that are long running, perhaps stuck in some queue, to enable management activities that could ensure completion. It also enables tracking of task statistics such as wait time, or time in progress, or time to completion, enabling capture of important task metrics in support of optimization and quality improvement.

12.1.3 Boundaries and Relationships

The task resource tracks the state of a task, enabling systems to ensure that tasks are completed. This information is kept separate from the operational details necessary to complete the task, as those details vary across and even within workflows. That detail is expected to be carried as the subject of the task.

Tasks may have named inputs and outputs. Inputs represent information that may or must be present in order for a task to complete. Outputs represent intermediate or final results produced by a task. For example, in a task supporting reading of radiology image, the inputs might include both the imaging study to be read, as well as relevant prior images. Outputs could represent radiology measurements as well as the Radiologist's diagnostic report.

Inputs and outputs are tracked by the task because workflow management activity may automate the transer of outputs from one task to inputs to a subsequent task.

To facilitate the integration of off the shelf workflow applications with FHIR, the task resource may reference a definition. This definition can represent a description of the workflow activity to be performed, using a standard workflow description language such as BPEL, BPMN, or XPDL, a workflow definition such as those defined in IHE profiles, or even simple written instructions explaining a process to be performed.

This resource is referenced by careplan

12.1.4 Resource Content

* Structure

* UML

* XML

* JSON

* Turtle

* R2 Diff

* All

Structure

Name | Flags | Card. | Type | Description & Constraints
Task | I | | DomainResource | A task to be performed + Last modified date must be greater than or equal to authored-on date. Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | | 0..* | Identifier | Task Instance Identifier
definition[x] | ? | 0..1 | | Formal definition of task
definitionUri | | | uri
definitionReference | | | Reference(ActivityDefinition)
basedOn | ? | 0..* | Reference(Any) | Request fulfilled by this task
groupIdentifier | ? | 0..1 | Identifier | Requisition or grouper id
partOf | ? | 0..* | Reference(Task) | Composite task
status | ? | 1..1 | code | draft | requested | received | accepted | + TaskStatus (Required)
statusReason | ? | 0..1 | CodeableConcept | Reason for current status
businessStatus | ? | 0..1 | CodeableConcept | E.g. "Specimen collected", "IV prepped"
intent | ? | 1..1 | code | proposal | plan | order + RequestIntent (Required)
priority | | 0..1 | code | normal | urgent | asap | stat RequestPriority (Required)
code | ? | 0..1 | CodeableConcept | Task Type
description | ? | 0..1 | string | Human-readable explanation of task
focus | ? | 0..1 | Reference(Any) | What task is acting on
for | ? | 0..1 | Reference(Any) | Beneficiary of the Task
context | ? | 0..1 | Reference(Encounter | EpisodeOfCare) | Healthcare event during which this task originated
executionPeriod | ? | 0..1 | Period | Start and end time of execution
authoredOn | I | 0..1 | dateTime | Task Creation Date
lastModified | ?I | 0..1 | dateTime | Task Last Modified Date
requester | ? | 0..1 | BackboneElement | Who is asking for task to be done
agent | ? | 1..1 | Reference(Device | Organization | Patient | Practitioner | RelatedPerson) | Individual asking for task
onBehalfOf | | 0..1 | Reference(Organization) | Organization individual is acting for
performerType | | 0..* | CodeableConcept | requester | dispatcher | scheduler | performer | monitor | manager | acquirer | reviewer TaskPerformerType (Preferred)
owner | ? | 0..1 | Reference(Device | Organization | Patient | Practitioner | RelatedPerson) | Responsible individual
reason | | 0..1 | CodeableConcept | Why task is needed
note | | 0..* | Annotation | Comments made about the task
relevantHistory | | 0..* | Reference(Provenance) | Key events in history of the Task
restriction | | 0..1 | BackboneElement | Constraints on fulfillment tasks
repetitions | | 0..1 | positiveInt | How many times to repeat
period | | 0..1 | Period | When fulfillment sought
recipient | | 0..* | Reference(Patient | Practitioner | RelatedPerson | Group | Organization) | For whom is fulfillment sought?
input | | 0..* | BackboneElement | Information used to perform task
type | | 1..1 | CodeableConcept | Label for the input
value[x] | | 1..1 | * | Content to use in performing the task
output | | 0..* | BackboneElement | Information produced as part of task
type | | 1..1 | CodeableConcept | Label for output
value[x] | | 1..1 | * | Result of output
Documentation for this format

UML Diagram (Legend)

XML Template

```
<Task xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Task Instance Identifier --></identifier>
 <definition[x]><!-- 0..1 uri|Reference(ActivityDefinition) Formal definition of task --></definition[x]>
 <basedOn><!-- 0..* Reference(Any) Request fulfilled by this task --></basedOn>
 <groupIdentifier><!-- 0..1 Identifier Requisition or grouper id --></groupIdentifier>
 <partOf><!-- 0..* Reference(Task) Composite task --></partOf>
 <status value="[code]"/><!-- 1..1 draft | requested | received | accepted | + -->
 <statusReason><!-- 0..1 CodeableConcept Reason for current status --></statusReason>
 <businessStatus><!-- 0..1 CodeableConcept E.g. "Specimen collected", "IV prepped" --></businessStatus>
 <intent value="[code]"/><!-- 1..1 proposal | plan | order + -->
 <priority value="[code]"/><!-- 0..1 normal | urgent | asap | stat -->
 <code><!-- 0..1 CodeableConcept Task Type --></code>
 <description value="[string]"/><!-- 0..1 Human-readable explanation of task -->
 <focus><!-- 0..1 Reference(Any) What task is acting on --></focus>
 <for><!-- 0..1 Reference(Any) Beneficiary of the Task --></for>
 <context><!-- 0..1 Reference(Encounter|EpisodeOfCare) Healthcare event during which this task originated --></context>
 <executionPeriod><!-- 0..1 Period Start and end time of execution --></executionPeriod>
 <authoredOn value="[dateTime]"/><!--  0..1 Task Creation Date -->
 <lastModified value="[dateTime]"/><!--  0..1 Task Last Modified Date -->
 <requester>  <!-- 0..1 Who is asking for task to be done -->
  <agent><!-- 1..1 Reference(Device|Organization|Patient|Practitioner|
    RelatedPerson) Individual asking for task --></agent>
  <onBehalfOf><!-- 0..1 Reference(Organization) Organization individual is acting for --></onBehalfOf>
 </requester>
 <performerType><!-- 0..* CodeableConcept requester | dispatcher | scheduler | performer | monitor | manager | acquirer
```

```
    | reviewer --></performerType>
 <owner><!-- 0..1 Reference(Device|Organization|Patient|Practitioner|
   RelatedPerson) Responsible individual --></owner>
 <reason><!-- 0..1 CodeableConcept Why task is needed --></reason>
 <note><!-- 0..* Annotation Comments made about the task --></note>
 <relevantHistory><!-- 0..* Reference(Provenance) Key events in history of the Task --></relevantHistory>
 <restriction>  <!-- 0..1 Constraints on fulfillment tasks -->
  <repetitions value="[positiveInt]"/><!-- 0..1 How many times to repeat -->
  <period><!-- 0..1 Period When fulfillment sought --></period>
  <recipient><!-- 0..* Reference(Patient|Practitioner|RelatedPerson|Group|
    Organization) For whom is fulfillment sought? --></recipient>
 </restriction>
 <input>  <!-- 0..* Information used to perform task -->
  <type><!-- 1..1 CodeableConcept Label for the input --></type>
  <value[x]><!-- 1..1 * Content to use in performing the task --></value[x]>
 </input>
 <output>  <!-- 0..* Information produced as part of task -->
  <type><!-- 1..1 CodeableConcept Label for output --></type>
  <value[x]><!-- 1..1 * Result of output --></value[x]>
 </output>
</Task>
```

JSON Template

```
{
  "resourceType" : "Task",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Task Instance Identifier
  // definition[x]: Formal definition of task. One of these 2:
  "definitionUri" : "<uri>",
  "definitionReference" : { Reference(ActivityDefinition) },
  "basedOn" : [{ Reference(Any) }], // Request fulfilled by this task
  "groupIdentifier" : { Identifier }, // Requisition or grouper id
  "partOf" : [{ Reference(Task) }], // Composite task
  "status" : "<code>", // R!  draft | requested | received | accepted | +
  "statusReason" : { CodeableConcept }, // Reason for current status
  "businessStatus" : { CodeableConcept }, // E.g. "Specimen collected", "IV prepped"
  "intent" : "<code>", // R!  proposal | plan | order +
  "priority" : "<code>", // normal | urgent | asap | stat
  "code" : { CodeableConcept }, // Task Type
  "description" : "<string>", // Human-readable explanation of task
  "focus" : { Reference(Any) }, // What task is acting on
  "for" : { Reference(Any) }, // Beneficiary of the Task
  "context" : { Reference(Encounter|EpisodeOfCare) }, // Healthcare event during which this task originated
  "executionPeriod" : { Period }, // Start and end time of execution
  "authoredOn" : "<dateTime>", // C? Task Creation Date
  "lastModified" : "<dateTime>", // C? Task Last Modified Date
  "requester" : { // Who is asking for task to be done
    "agent" : { Reference(Device|Organization|Patient|Practitioner|
    RelatedPerson) }, // R!  Individual asking for task
    "onBehalfOf" : { Reference(Organization) } // Organization individual is acting for
  },
  "performerType" : [{ CodeableConcept }], // requester | dispatcher | scheduler | performer | monitor | manager |
    acquirer | reviewer
  "owner" : { Reference(Device|Organization|Patient|Practitioner|
   RelatedPerson) }, // Responsible individual
  "reason" : { CodeableConcept }, // Why task is needed
  "note" : [{ Annotation }], // Comments made about the task
  "relevantHistory" : [{ Reference(Provenance) }], // Key events in history of the Task
  "restriction" : { // Constraints on fulfillment tasks
    "repetitions" : "<positiveInt>", // How many times to repeat
    "period" : { Period }, // When fulfillment sought
    "recipient" : [{ Reference(Patient|Practitioner|RelatedPerson|Group|
    Organization) }] // For whom is fulfillment sought?
  },
  "input" : [{ // Information used to perform task
    "type" : { CodeableConcept }, // R!  Label for the input
    // value[x]: Content to use in performing the task. One of these 23:
    "valueInteger" : <integer>
    "valueDecimal" : <decimal>
    "valueDateTime" : "<dateTime>"
    "valueDate" : "<date>"
    "valueInstant" : "<instant>"
    "valueString" : "<string>"
    "valueUri" : "<uri>"
    "valueBoolean" : <boolean>
    "valueCode" : "<code>"
    "valueBase64Binary" : "<base64Binary>"
    "valueCoding" : { Coding }
    "valueCodeableConcept" : { CodeableConcept }
    "valueAttachment" : { Attachment }
    "valueIdentifier" : { Identifier }
    "valueQuantity" : { Quantity }
    "valueRange" : { Range }
    "valuePeriod" : { Period }
    "valueRatio" : { Ratio }
    "valueHumanName" : { HumanName }
    "valueAddress" : { Address }
    "valueContactPoint" : { ContactPoint }
```

```
    "valueSchedule" : { Schedule }
    "valueReference" : { Reference }
  }],
  "output" : [{ // Information produced as part of task
    "type" : { CodeableConcept }, // R!  Label for output
    // value[x]: Result of output. One of these 23:
    "valueInteger" : <integer>
    "valueDecimal" : <decimal>
    "valueDateTime" : "<dateTime>"
    "valueDate" : "<date>"
    "valueInstant" : "<instant>"
    "valueString" : "<string>"
    "valueUri" : "<uri>"
    "valueBoolean" : <boolean>
    "valueCode" : "<code>"
    "valueBase64Binary" : "<base64Binary>"
    "valueCoding" : { Coding }
    "valueCodeableConcept" : { CodeableConcept }
    "valueAttachment" : { Attachment }
    "valueIdentifier" : { Identifier }
    "valueQuantity" : { Quantity }
    "valueRange" : { Range }
    "valuePeriod" : { Period }
    "valueRatio" : { Ratio }
    "valueHumanName" : { HumanName }
    "valueAddress" : { Address }
    "valueContactPoint" : { ContactPoint }
    "valueSchedule" : { Schedule }
    "valueReference" : { Reference }
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:Task;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:Task.identifier [ Identifier ], ... ; # 0..* Task Instance Identifier
  # Task.definition[x] : 0..1 Formal definition of task. One of these 2
    fhir:Task.definitionUri [ uri ]
    fhir:Task.definitionReference [ Reference(ActivityDefinition) ]
  fhir:Task.basedOn [ Reference(Any) ], ... ; # 0..* Request fulfilled by this task
  fhir:Task.groupIdentifier [ Identifier ]; # 0..1 Requisition or grouper id
  fhir:Task.partOf [ Reference(Task) ], ... ; # 0..* Composite task
  fhir:Task.status [ code ]; # 1..1 draft | requested | received | accepted | +
  fhir:Task.statusReason [ CodeableConcept ]; # 0..1 Reason for current status
  fhir:Task.businessStatus [ CodeableConcept ]; # 0..1 E.g. "Specimen collected", "IV prepped"
  fhir:Task.intent [ code ]; # 1..1 proposal | plan | order +
  fhir:Task.priority [ code ]; # 0..1 normal | urgent | asap | stat
  fhir:Task.code [ CodeableConcept ]; # 0..1 Task Type
  fhir:Task.description [ string ]; # 0..1 Human-readable explanation of task
  fhir:Task.focus [ Reference(Any) ]; # 0..1 What task is acting on
  fhir:Task.for [ Reference(Any) ]; # 0..1 Beneficiary of the Task
  fhir:Task.context [ Reference(Encounter|EpisodeOfCare) ]; # 0..1 Healthcare event during which this task originated
  fhir:Task.executionPeriod [ Period ]; # 0..1 Start and end time of execution
  fhir:Task.authoredOn [ dateTime ]; # 0..1 Task Creation Date
  fhir:Task.lastModified [ dateTime ]; # 0..1 Task Last Modified Date
  fhir:Task.requester [ # 0..1 Who is asking for task to be done
    fhir:Task.requester.agent [ Reference(Device|Organization|Patient|Practitioner|RelatedPerson) ]; # 1..1 Individual
    asking for task
    fhir:Task.requester.onBehalfOf [ Reference(Organization) ]; # 0..1 Organization individual is acting for
  ];
  fhir:Task.performerType [ CodeableConcept ], ... ; # 0..* requester | dispatcher | scheduler | performer | monitor |
    manager | acquirer | reviewer
  fhir:Task.owner [ Reference(Device|Organization|Patient|Practitioner|RelatedPerson) ]; # 0..1 Responsible individual
  fhir:Task.reason [ CodeableConcept ]; # 0..1 Why task is needed
  fhir:Task.note [ Annotation ], ... ; # 0..* Comments made about the task
  fhir:Task.relevantHistory [ Reference(Provenance) ], ... ; # 0..* Key events in history of the Task
  fhir:Task.restriction [ # 0..1 Constraints on fulfillment tasks
    fhir:Task.restriction.repetitions [ positiveInt ]; # 0..1 How many times to repeat
    fhir:Task.restriction.period [ Period ]; # 0..1 When fulfillment sought
    fhir:Task.restriction.recipient [ Reference(Patient|Practitioner|RelatedPerson|Group|Organization) ], ... ; # 0..*
    For whom is fulfillment sought?
  ];
  fhir:Task.input [ # 0..* Information used to perform task
    fhir:Task.input.type [ CodeableConcept ]; # 1..1 Label for the input
    # Task.input.value[x] : 1..1 Content to use in performing the task. One of these 38
      fhir:Task.input.valueBase64Binary [ base64Binary ]
      fhir:Task.input.valueBoolean [ boolean ]
      fhir:Task.input.valueCode [ code ]
      fhir:Task.input.valueDate [ date ]
      fhir:Task.input.valueDateTime [ dateTime ]
      fhir:Task.input.valueDecimal [ decimal ]
      fhir:Task.input.valueId [ id ]
      fhir:Task.input.valueInstant [ instant ]
```

```
      fhir:Task.input.valueInteger [ integer ]
      fhir:Task.input.valueMarkdown [ markdown ]
      fhir:Task.input.valueOid [ oid ]
      fhir:Task.input.valuePositiveInt [ positiveInt ]
      fhir:Task.input.valueString [ string ]
      fhir:Task.input.valueTime [ time ]
      fhir:Task.input.valueUnsignedInt [ unsignedInt ]
      fhir:Task.input.valueUri [ uri ]
      fhir:Task.input.valueAddress [ Address ]
      fhir:Task.input.valueAge [ Age ]
      fhir:Task.input.valueAnnotation [ Annotation ]
      fhir:Task.input.valueAttachment [ Attachment ]
      fhir:Task.input.valueCodeableConcept [ CodeableConcept ]
      fhir:Task.input.valueCoding [ Coding ]
      fhir:Task.input.valueContactPoint [ ContactPoint ]
      fhir:Task.input.valueCount [ Count ]
      fhir:Task.input.valueDistance [ Distance ]
      fhir:Task.input.valueDuration [ Duration ]
      fhir:Task.input.valueHumanName [ HumanName ]
      fhir:Task.input.valueIdentifier [ Identifier ]
      fhir:Task.input.valueMoney [ Money ]
      fhir:Task.input.valuePeriod [ Period ]
      fhir:Task.input.valueQuantity [ Quantity ]
      fhir:Task.input.valueRange [ Range ]
      fhir:Task.input.valueRatio [ Ratio ]
      fhir:Task.input.valueReference [ Reference ]
      fhir:Task.input.valueSampledData [ SampledData ]
      fhir:Task.input.valueSignature [ Signature ]
      fhir:Task.input.valueTiming [ Timing ]
      fhir:Task.input.valueMeta [ Meta ]
  ], ...;
  fhir:Task.output [ # 0..* Information produced as part of task
    fhir:Task.output.type [ CodeableConcept ]; # 1..1 Label for output
    # Task.output.value[x] : 1..1 Result of output. One of these 38
      fhir:Task.output.valueBase64Binary [ base64Binary ]
      fhir:Task.output.valueBoolean [ boolean ]
      fhir:Task.output.valueCode [ code ]
      fhir:Task.output.valueDate [ date ]
      fhir:Task.output.valueDateTime [ dateTime ]
      fhir:Task.output.valueDecimal [ decimal ]
      fhir:Task.output.valueId [ id ]
      fhir:Task.output.valueInstant [ instant ]
      fhir:Task.output.valueInteger [ integer ]
      fhir:Task.output.valueMarkdown [ markdown ]
      fhir:Task.output.valueOid [ oid ]
      fhir:Task.output.valuePositiveInt [ positiveInt ]
      fhir:Task.output.valueString [ string ]
      fhir:Task.output.valueTime [ time ]
      fhir:Task.output.valueUnsignedInt [ unsignedInt ]
      fhir:Task.output.valueUri [ uri ]
      fhir:Task.output.valueAddress [ Address ]
      fhir:Task.output.valueAge [ Age ]
      fhir:Task.output.valueAnnotation [ Annotation ]
      fhir:Task.output.valueAttachment [ Attachment ]
      fhir:Task.output.valueCodeableConcept [ CodeableConcept ]
      fhir:Task.output.valueCoding [ Coding ]
      fhir:Task.output.valueContactPoint [ ContactPoint ]
      fhir:Task.output.valueCount [ Count ]
      fhir:Task.output.valueDistance [ Distance ]
      fhir:Task.output.valueDuration [ Duration ]
      fhir:Task.output.valueHumanName [ HumanName ]
      fhir:Task.output.valueIdentifier [ Identifier ]
      fhir:Task.output.valueMoney [ Money ]
      fhir:Task.output.valuePeriod [ Period ]
      fhir:Task.output.valueQuantity [ Quantity ]
      fhir:Task.output.valueRange [ Range ]
      fhir:Task.output.valueRatio [ Ratio ]
      fhir:Task.output.valueReference [ Reference ]
      fhir:Task.output.valueSampledData [ SampledData ]
      fhir:Task.output.valueSignature [ Signature ]
      fhir:Task.output.valueTiming [ Timing ]
      fhir:Task.output.valueMeta [ Meta ]
  ], ...;
]
```

Changes since DSTU2

This resource did not exist in Release 2

This analysis is available as XML or JSON.

Structure

Name | Flags | Card. | Type | Description & Constraints

Task | I | | DomainResource | A task to be performed + Last modified date must be greater than or equal to authored-on date. Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension

identifier | | 0..* | Identifier | Task Instance Identifier

definition[x] | ? | 0..1 | | Formal definition of task

definitionUri | | | uri

definitionReference | | | Reference(ActivityDefinition)
basedOn | ? | 0..* | Reference(Any) | Request fulfilled by this task
groupIdentifier | ? | 0..1 | Identifier | Requisition or grouper id
partOf | ? | 0..* | Reference(Task) | Composite task
status | ? | 1..1 | code | draft | requested | received | accepted | + TaskStatus (Required)
statusReason | ? | 0..1 | CodeableConcept | Reason for current status
businessStatus | ? | 0..1 | CodeableConcept | E.g. "Specimen collected", "IV prepped"
intent | ? | 1..1 | code | proposal | plan | order + RequestIntent (Required)
priority | | 0..1 | code | normal | urgent | asap | stat RequestPriority (Required)
code | ? | 0..1 | CodeableConcept | Task Type
description | ? | 0..1 | string | Human-readable explanation of task
focus | ? | 0..1 | Reference(Any) | What task is acting on
for | ? | 0..1 | Reference(Any) | Beneficiary of the Task
context | ? | 0..1 | Reference(Encounter | EpisodeOfCare) | Healthcare event during which this task originated
executionPeriod | ? | 0..1 | Period | Start and end time of execution
authoredOn | I | 0..1 | dateTime | Task Creation Date
lastModified | ?I | 0..1 | dateTime | Task Last Modified Date
requester | ? | 0..1 | BackboneElement | Who is asking for task to be done
agent | ? | 1..1 | Reference(Device | Organization | Patient | Practitioner | RelatedPerson) | Individual asking for task
onBehalfOf | | 0..1 | Reference(Organization) | Organization individual is acting for
performerType | | 0..* | CodeableConcept | requester | dispatcher | scheduler | performer | monitor | manager | acquirer | reviewer TaskPerformerType (Preferred)
owner | ? | 0..1 | Reference(Device | Organization | Patient | Practitioner | RelatedPerson) | Responsible individual
reason | | 0..1 | CodeableConcept | Why task is needed
note | | 0..* | Annotation | Comments made about the task
relevantHistory | | 0..* | Reference(Provenance) | Key events in history of the Task
restriction | | 0..1 | BackboneElement | Constraints on fulfillment tasks
repetitions | | 0..1 | positiveInt | How many times to repeat
period | | 0..1 | Period | When fulfillment sought
recipient | | 0..* | Reference(Patient | Practitioner | RelatedPerson | Group | Organization) | For whom is fulfillment sought?
input | | 0..* | BackboneElement | Information used to perform task
type | | 1..1 | CodeableConcept | Label for the input
value[x] | | 1..1 | * | Content to use in performing the task
output | | 0..* | BackboneElement | Information produced as part of task
type | | 1..1 | CodeableConcept | Label for output
value[x] | | 1..1 | * | Result of output
Documentation for this format

UML Diagram (Legend)

XML Template

```
<Task xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Task Instance Identifier --></identifier>
 <definition[x]><!-- 0..1 uri|Reference(ActivityDefinition) Formal definition of task --></definition[x]>
 <basedOn><!-- 0..* Reference(Any) Request fulfilled by this task --></basedOn>
 <groupIdentifier><!-- 0..1 Identifier Requisition or grouper id --></groupIdentifier>
 <partOf><!-- 0..* Reference(Task) Composite task --></partOf>
 <status value="[code]"/><!-- 1..1 draft | requested | received | accepted | + -->
 <statusReason><!-- 0..1 CodeableConcept Reason for current status --></statusReason>
 <businessStatus><!-- 0..1 CodeableConcept E.g. "Specimen collected", "IV prepped" --></businessStatus>
 <intent value="[code]"/><!-- 1..1 proposal | plan | order + -->
 <priority value="[code]"/><!-- 0..1 normal | urgent | asap | stat -->
 <code><!-- 0..1 CodeableConcept Task Type --></code>
 <description value="[string]"/><!-- 0..1 Human-readable explanation of task -->
 <focus><!-- 0..1 Reference(Any) What task is acting on --></focus>
 <for><!-- 0..1 Reference(Any) Beneficiary of the Task --></for>
 <context><!-- 0..1 Reference(Encounter|EpisodeOfCare) Healthcare event during which this task originated --></context>
 <executionPeriod><!-- 0..1 Period Start and end time of execution --></executionPeriod>
 <authoredOn value="[dateTime]"/><!--  0..1 Task Creation Date -->
 <lastModified value="[dateTime]"/><!--  0..1 Task Last Modified Date -->
 <requester>  <!-- 0..1 Who is asking for task to be done -->
  <agent><!-- 1..1 Reference(Device|Organization|Patient|Practitioner|
    RelatedPerson) Individual asking for task --></agent>
  <onBehalfOf><!-- 0..1 Reference(Organization) Organization individual is acting for --></onBehalfOf>
 </requester>
 <performerType><!-- 0..* CodeableConcept requester | dispatcher | scheduler | performer | monitor | manager | acquirer
    | reviewer --></performerType>
 <owner><!-- 0..1 Reference(Device|Organization|Patient|Practitioner|
   RelatedPerson) Responsible individual --></owner>
 <reason><!-- 0..1 CodeableConcept Why task is needed --></reason>
 <note><!-- 0..* Annotation Comments made about the task --></note>
 <relevantHistory><!-- 0..* Reference(Provenance) Key events in history of the Task --></relevantHistory>
 <restriction>  <!-- 0..1 Constraints on fulfillment tasks -->
  <repetitions value="[positiveInt]"/><!-- 0..1 How many times to repeat -->
  <period><!-- 0..1 Period When fulfillment sought --></period>
  <recipient><!-- 0..* Reference(Patient|Practitioner|RelatedPerson|Group|
    Organization) For whom is fulfillment sought? --></recipient>
 </restriction>
 <input>  <!-- 0..* Information used to perform task -->
```

```
  <type><!-- 1..1 CodeableConcept Label for the input --></type>
  <value[x]><!-- 1..1 * Content to use in performing the task --></value[x]>
 </input>
 <output>  <!-- 0..* Information produced as part of task -->
  <type><!-- 1..1 CodeableConcept Label for output --></type>
  <value[x]><!-- 1..1 * Result of output --></value[x]>
 </output>
</Task>
```

JSON Template

```
{
  "resourceType" : "Task",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Task Instance Identifier
  // definition[x]: Formal definition of task. One of these 2:
  "definitionUri" : "<uri>",
  "definitionReference" : { Reference(ActivityDefinition) },
  "basedOn" : [{ Reference(Any) }], // Request fulfilled by this task
  "groupIdentifier" : { Identifier }, // Requisition or grouper id
  "partOf" : [{ Reference(Task) }], // Composite task
  "status" : "<code>", // R!  draft | requested | received | accepted | +
  "statusReason" : { CodeableConcept }, // Reason for current status
  "businessStatus" : { CodeableConcept }, // E.g. "Specimen collected", "IV prepped"
  "intent" : "<code>", // R!  proposal | plan | order +
  "priority" : "<code>", // normal | urgent | asap | stat
  "code" : { CodeableConcept }, // Task Type
  "description" : "<string>", // Human-readable explanation of task
  "focus" : { Reference(Any) }, // What task is acting on
  "for" : { Reference(Any) }, // Beneficiary of the Task
  "context" : { Reference(Encounter|EpisodeOfCare) }, // Healthcare event during which this task originated
  "executionPeriod" : { Period }, // Start and end time of execution
  "authoredOn" : "<dateTime>", // C? Task Creation Date
  "lastModified" : "<dateTime>", // C? Task Last Modified Date
  "requester" : { // Who is asking for task to be done
    "agent" : { Reference(Device|Organization|Patient|Practitioner|
    RelatedPerson) }, // R!  Individual asking for task
    "onBehalfOf" : { Reference(Organization) } // Organization individual is acting for
  },
  "performerType" : [{ CodeableConcept }], // requester | dispatcher | scheduler | performer | monitor | manager |
    acquirer | reviewer
  "owner" : { Reference(Device|Organization|Patient|Practitioner|
   RelatedPerson) }, // Responsible individual
  "reason" : { CodeableConcept }, // Why task is needed
  "note" : [{ Annotation }], // Comments made about the task
  "relevantHistory" : [{ Reference(Provenance) }], // Key events in history of the Task
  "restriction" : { // Constraints on fulfillment tasks
    "repetitions" : "<positiveInt>", // How many times to repeat
    "period" : { Period }, // When fulfillment sought
    "recipient" : [{ Reference(Patient|Practitioner|RelatedPerson|Group|
    Organization) }] // For whom is fulfillment sought?
  },
  "input" : [{ // Information used to perform task
    "type" : { CodeableConcept }, // R!  Label for the input
    // value[x]: Content to use in performing the task. One of these 23:
    "valueInteger" : <integer>
    "valueDecimal" : <decimal>
    "valueDateTime" : "<dateTime>"
    "valueDate" : "<date>"
    "valueInstant" : "<instant>"
    "valueString" : "<string>"
    "valueUri" : "<uri>"
    "valueBoolean" : <boolean>
    "valueCode" : "<code>"
    "valueBase64Binary" : "<base64Binary>"
    "valueCoding" : { Coding }
    "valueCodeableConcept" : { CodeableConcept }
    "valueAttachment" : { Attachment }
    "valueIdentifier" : { Identifier }
    "valueQuantity" : { Quantity }
    "valueRange" : { Range }
    "valuePeriod" : { Period }
    "valueRatio" : { Ratio }
    "valueHumanName" : { HumanName }
    "valueAddress" : { Address }
    "valueContactPoint" : { ContactPoint }
    "valueSchedule" : { Schedule }
    "valueReference" : { Reference }
  }],
  "output" : [{ // Information produced as part of task
    "type" : { CodeableConcept }, // R!  Label for output
    // value[x]: Result of output. One of these 23:
    "valueInteger" : <integer>
    "valueDecimal" : <decimal>
    "valueDateTime" : "<dateTime>"
    "valueDate" : "<date>"
    "valueInstant" : "<instant>"
    "valueString" : "<string>"
    "valueUri" : "<uri>"
```

```
    "valueBoolean" : <boolean>
    "valueCode" : "<code>"
    "valueBase64Binary" : "<base64Binary>"
    "valueCoding" : { Coding }
    "valueCodeableConcept" : { CodeableConcept }
    "valueAttachment" : { Attachment }
    "valueIdentifier" : { Identifier }
    "valueQuantity" : { Quantity }
    "valueRange" : { Range }
    "valuePeriod" : { Period }
    "valueRatio" : { Ratio }
    "valueHumanName" : { HumanName }
    "valueAddress" : { Address }
    "valueContactPoint" : { ContactPoint }
    "valueSchedule" : { Schedule }
    "valueReference" : { Reference }
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:Task;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:Task.identifier [ Identifier ], ... ; # 0..* Task Instance Identifier
  # Task.definition[x] : 0..1 Formal definition of task. One of these 2
    fhir:Task.definitionUri [ uri ]
    fhir:Task.definitionReference [ Reference(ActivityDefinition) ]
  fhir:Task.basedOn [ Reference(Any) ], ... ; # 0..* Request fulfilled by this task
  fhir:Task.groupIdentifier [ Identifier ]; # 0..1 Requisition or grouper id
  fhir:Task.partOf [ Reference(Task) ], ... ; # 0..* Composite task
  fhir:Task.status [ code ]; # 1..1 draft | requested | received | accepted | +
  fhir:Task.statusReason [ CodeableConcept ]; # 0..1 Reason for current status
  fhir:Task.businessStatus [ CodeableConcept ]; # 0..1 E.g. "Specimen collected", "IV prepped"
  fhir:Task.intent [ code ]; # 1..1 proposal | plan | order +
  fhir:Task.priority [ code ]; # 0..1 normal | urgent | asap | stat
  fhir:Task.code [ CodeableConcept ]; # 0..1 Task Type
  fhir:Task.description [ string ]; # 0..1 Human-readable explanation of task
  fhir:Task.focus [ Reference(Any) ]; # 0..1 What task is acting on
  fhir:Task.for [ Reference(Any) ]; # 0..1 Beneficiary of the Task
  fhir:Task.context [ Reference(Encounter|EpisodeOfCare) ]; # 0..1 Healthcare event during which this task originated
  fhir:Task.executionPeriod [ Period ]; # 0..1 Start and end time of execution
  fhir:Task.authoredOn [ dateTime ]; # 0..1 Task Creation Date
  fhir:Task.lastModified [ dateTime ]; # 0..1 Task Last Modified Date
  fhir:Task.requester [ # 0..1 Who is asking for task to be done
    fhir:Task.requester.agent [ Reference(Device|Organization|Patient|Practitioner|RelatedPerson) ]; # 1..1 Individual
    asking for task
    fhir:Task.requester.onBehalfOf [ Reference(Organization) ]; # 0..1 Organization individual is acting for
  ];
  fhir:Task.performerType [ CodeableConcept ], ... ; # 0..* requester | dispatcher | scheduler | performer | monitor |
    manager | acquirer | reviewer
  fhir:Task.owner [ Reference(Device|Organization|Patient|Practitioner|RelatedPerson) ]; # 0..1 Responsible individual
  fhir:Task.reason [ CodeableConcept ]; # 0..1 Why task is needed
  fhir:Task.note [ Annotation ], ... ; # 0..* Comments made about the task
  fhir:Task.relevantHistory [ Reference(Provenance) ], ... ; # 0..* Key events in history of the Task
  fhir:Task.restriction [ # 0..1 Constraints on fulfillment tasks
    fhir:Task.restriction.repetitions [ positiveInt ]; # 0..1 How many times to repeat
    fhir:Task.restriction.period [ Period ]; # 0..1 When fulfillment sought
    fhir:Task.restriction.recipient [ Reference(Patient|Practitioner|RelatedPerson|Group|Organization) ], ... ; # 0..*
    For whom is fulfillment sought?
  ];
  fhir:Task.input [ # 0..* Information used to perform task
    fhir:Task.input.type [ CodeableConcept ]; # 1..1 Label for the input
    # Task.input.value[x] : 1..1 Content to use in performing the task. One of these 38
      fhir:Task.input.valueBase64Binary [ base64Binary ]
      fhir:Task.input.valueBoolean [ boolean ]
      fhir:Task.input.valueCode [ code ]
      fhir:Task.input.valueDate [ date ]
      fhir:Task.input.valueDateTime [ dateTime ]
      fhir:Task.input.valueDecimal [ decimal ]
      fhir:Task.input.valueId [ id ]
      fhir:Task.input.valueInstant [ instant ]
      fhir:Task.input.valueInteger [ integer ]
      fhir:Task.input.valueMarkdown [ markdown ]
      fhir:Task.input.valueOid [ oid ]
      fhir:Task.input.valuePositiveInt [ positiveInt ]
      fhir:Task.input.valueString [ string ]
      fhir:Task.input.valueTime [ time ]
      fhir:Task.input.valueUnsignedInt [ unsignedInt ]
      fhir:Task.input.valueUri [ uri ]
      fhir:Task.input.valueAddress [ Address ]
      fhir:Task.input.valueAge [ Age ]
      fhir:Task.input.valueAnnotation [ Annotation ]
      fhir:Task.input.valueAttachment [ Attachment ]
      fhir:Task.input.valueCodeableConcept [ CodeableConcept ]
```

```
      fhir:Task.input.valueCoding [ Coding ]
      fhir:Task.input.valueContactPoint [ ContactPoint ]
      fhir:Task.input.valueCount [ Count ]
      fhir:Task.input.valueDistance [ Distance ]
      fhir:Task.input.valueDuration [ Duration ]
      fhir:Task.input.valueHumanName [ HumanName ]
      fhir:Task.input.valueIdentifier [ Identifier ]
      fhir:Task.input.valueMoney [ Money ]
      fhir:Task.input.valuePeriod [ Period ]
      fhir:Task.input.valueQuantity [ Quantity ]
      fhir:Task.input.valueRange [ Range ]
      fhir:Task.input.valueRatio [ Ratio ]
      fhir:Task.input.valueReference [ Reference ]
      fhir:Task.input.valueSampledData [ SampledData ]
      fhir:Task.input.valueSignature [ Signature ]
      fhir:Task.input.valueTiming [ Timing ]
      fhir:Task.input.valueMeta [ Meta ]
  ], ...;
  fhir:Task.output [ # 0..* Information produced as part of task
    fhir:Task.output.type [ CodeableConcept ]; # 1..1 Label for output
    # Task.output.value[x] : 1..1 Result of output. One of these 38
      fhir:Task.output.valueBase64Binary [ base64Binary ]
      fhir:Task.output.valueBoolean [ boolean ]
      fhir:Task.output.valueCode [ code ]
      fhir:Task.output.valueDate [ date ]
      fhir:Task.output.valueDateTime [ dateTime ]
      fhir:Task.output.valueDecimal [ decimal ]
      fhir:Task.output.valueId [ id ]
      fhir:Task.output.valueInstant [ instant ]
      fhir:Task.output.valueInteger [ integer ]
      fhir:Task.output.valueMarkdown [ markdown ]
      fhir:Task.output.valueOid [ oid ]
      fhir:Task.output.valuePositiveInt [ positiveInt ]
      fhir:Task.output.valueString [ string ]
      fhir:Task.output.valueTime [ time ]
      fhir:Task.output.valueUnsignedInt [ unsignedInt ]
      fhir:Task.output.valueUri [ uri ]
      fhir:Task.output.valueAddress [ Address ]
      fhir:Task.output.valueAge [ Age ]
      fhir:Task.output.valueAnnotation [ Annotation ]
      fhir:Task.output.valueAttachment [ Attachment ]
      fhir:Task.output.valueCodeableConcept [ CodeableConcept ]
      fhir:Task.output.valueCoding [ Coding ]
      fhir:Task.output.valueContactPoint [ ContactPoint ]
      fhir:Task.output.valueCount [ Count ]
      fhir:Task.output.valueDistance [ Distance ]
      fhir:Task.output.valueDuration [ Duration ]
      fhir:Task.output.valueHumanName [ HumanName ]
      fhir:Task.output.valueIdentifier [ Identifier ]
      fhir:Task.output.valueMoney [ Money ]
      fhir:Task.output.valuePeriod [ Period ]
      fhir:Task.output.valueQuantity [ Quantity ]
      fhir:Task.output.valueRange [ Range ]
      fhir:Task.output.valueRatio [ Ratio ]
      fhir:Task.output.valueReference [ Reference ]
      fhir:Task.output.valueSampledData [ SampledData ]
      fhir:Task.output.valueSignature [ Signature ]
      fhir:Task.output.valueTiming [ Timing ]
      fhir:Task.output.valueMeta [ Meta ]
  ], ...;
]
```

Changes since DSTU2

This resource did not exist in Release 2

This analysis is available as XML or JSON.

Alternate definitions: Master Definition (XML, JSON), XML Schema/Schematron (for) + JSON Schema, ShEx (for Turtle)

12.1.4.1 Terminology Bindings

Path | Definition | Type | Reference

Task.status | The current status of the task. | Required | TaskStatus

Task.statusReason | Codes to identify the reason for current status. These will typically be specific to a particular workflow. | Unknown | No details provided yet

Task.businessStatus | The domain-specific business-contextual sub-state of the task. For example: "Blood drawn", "IV inserted", "Awaiting physician signature", etc. | Unknown | No details provided yet

Task.intent | Distinguishes whether the task is a proposal, plan or full order | Required | RequestIntent

Task.priority | The task's priority | Required | RequestPriority

Task.code | Codes to identify what the task involves. These will typically be specific to a particular workflow. | Unknown | No details provided yet

Task.performerType | The type(s) of task performers allowed | Preferred | TaskPerformerType

Task.reason | Indicates why the task is needed. E.g. Suspended because patient admitted to hospital. | Unknown | No details provided yet

Task.input.type | Codes to identify types of input parameters. These will typically be specific to a particular workflow. E.g. "Comparison source", "Applicable consent", "Concommitent Medications", etc. | Unknown | No details provided yet

Task.output.type | Codes to identify types of input parameters. These will typically be specific to a particular workflow. E.g. "Identified issues", "Preliminary results", "Filler order", "Final results", etc. | Unknown | No details provided yet

12.1.4.2 Constraints

* inv-1: Last modified date must be greater than or equal to authored-on date. (expression : lastModified.exists().not() or authoredOn.exists().not() or lastModified >= authoredOn)

12.1.5 Task Titles

Tasks often have titles (eg "My Tasks", "Outstanding Tasks for Patient X") which can be presented in a list. The task title should go into the Task.code as a coded concept and/or text.

12.1.6 Task state machine

The following diagram reflects the "typical" state machine for Task. Note that not all states will be supported by all workflows and that some workflows may support additional transitions, including transitions from terminal states (e.g. back to "in-progress" from "failed" or "completed").

12.1.7 The Cancelled state

While the intention of a "cancelled" task is that all work authorized by the task should cease, this may not always be possible practice. It is possible that the originally requested action could still be completed and still attached to the Task but this would not change the status of the task. If the placer cancels a task, it signals they no longer care about the outcome of the task.

12.1.8 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
authored-on | date | Search by creation date | Task.authoredOn
based-on | reference | Search by requests this task is based on | Task.basedOn (Any)
business-status | token | Search by business status | Task.businessStatus
code | token | Search by task code | Task.code
context | reference | Search by encounter or episode | Task.context (EpisodeOfCare, Encounter)
focus | reference | Search by task focus | Task.focus (Any)
group-identifier | token | Search by group identifier | Task.groupIdentifier
identifier | token | Search for a task instance by its business identifier | Task.identifier
intent | token | Search by task intent | Task.intent
modified | date | Search by last modification date | Task.lastModified
organization | reference | Search by responsible organization | Task.requester.onBehalfOf (Organization)
owner | reference | Search by task owner | Task.owner (Practitioner, Organization, Device, Patient, RelatedPerson)
part-of | reference | Search by task this task is part of | Task.partOf (Task)
patient | reference | Search by patient | Task.for (Patient)
performer | token | Search by recommended type of performer (e.g., Requester, Performer, Scheduler). | Task.performerType
period | date | Search by period Task is/was underway | Task.executionPeriod
priority | token | Search by task priority | Task.priority
requester | reference | Search by task requester | Task.requester.agent (Practitioner, Organization, Device, Patient, RelatedPerson)
status | token | Search by task status | Task.status
subject | reference | Search by subject | Task.for (Any)

12.1.9 Resource Task - Examples

URL: https://hl7.org/fhir/STU3/task-examples.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

12.1.9 Resource Task - Examples

Orders and Observations Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Not linked to any defined compartments

Example List:

Example of in-progress/awaiting specimen task for a lipid panel order | example1 | XML | JSON | Turtle
Example of an accepted, filler generated specimen collection subtask | example2 | XML | JSON | Turtle
Example of a completed filler generated specimen collection subtask | example4 | XML | JSON | Turtle
Example of in-progress/test results pending task for a lipid panel order | example5 | XML | JSON | Turtle
Example of completed task for a lipid panel order | example6 | XML | JSON | Turtle
Example of simple prescription task | example3 | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

12.1.10 Resource Task - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/task-definitions.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

12.1.10 Resource Task - Detailed Descriptions

Orders and Observations Work Group | Maturity Level: 2 | Trial Use | Compartments: Not linked to any defined compartments

Detailed Descriptions for the elements in the Task resource.

Task
Definition | A task to be performed.
Control | 1..1
Invariants | Defined on this element inv-1: Last modified date must be greater than or equal to authored-on date. (expression : lastModified.exists().not() or authoredOn.exists().not() or lastModified >= authoredOn, xpath: not(exists(f:lastModified/@value)) or not(exists(f:authoredOn/@value)) or f:lastModified/@value >= f:authoredOn/@value)
Task.identifier
Definition | The business identifier for this task.
Note | This is a business identifer, not a resource identifier (see discussion)
Control | 0..*
Type | Identifier
Task.definition[x]
Definition | A reference to a formal or informal definition of the task. For example, a protocol, a step within a defined workflow definition, etc.
Control | 0..1
Type | uri|Reference(ActivityDefinition)
[x] Note | See Choice of Data Types for further information about how to use [x]
Requirements | Enables a formal definition of how he task is to be performed (e.g. using BPMN, BPEL, XPDL or other formal notation) to be associated with a task, enabling automation.
Summary | true
Task.basedOn
Definition | BasedOn refers to a higher-level authorization that triggered the creation of the task. It references a "request" resource such as a ProcedureRequest, MedicationRequest, ProcedureRequest, CarePlan, etc. which is distinct from the "request" resource the task is seeking to fulfil. This latter resource is referenced by FocusOn. For example, based on a ProcedureRequest (= BasedOn), a task is created to fulfil a procedureRequest (= FocusOn) to collect a specimen from a patient.
Control | 0..*
Type | Reference(Any)
Summary | true
Task.groupIdentifier
Definition | An identifier that links together multiple tasks and other requests that were created in the same context.
Control | 0..1
Type | Identifier
Requirements | Billing and/or reporting can be linked to whether multiple requests were created as a single unit.
Summary | true
Task.partOf
Definition | Task that this particular task is part of.
Control | 0..*
Type | Reference(Task)
Requirements | Allows tasks to be broken down into sub-steps (and this division can occur independent of the original task).
Summary | true
Comments | This should usually be 0..1.
Task.status
Definition | The current status of the task.
Control | 1..1
Terminology Binding | TaskStatus (Required)
Type | code
Requirements | These states enable coordination of task status with off-the-shelf workflow solutions that support automation of tasks.
Summary | true
Task.statusReason
Definition | An explanation as to why this task is held, failed, was refused, etc.
Control | 0..1
Terminology Binding | TaskStatusReason:
Type | CodeableConcept
Summary | true
Comments | This applies to the current status. Look at the history of the task to see reasons for past statuses.
Task.businessStatus
Definition | Contains business-specific nuances of the business state.
Control | 0..1
Terminology Binding | TaskBusinessStatus:
Type | CodeableConcept
Requirements | There's often a need to track substates of a task - this is often variable by specific workflow implementation.
Summary | true
Task.intent
Definition | Indicates the "level" of actionability associated with the Task. I.e. Is this a proposed task, a planned task, an actionable task, etc.
Control | 1..1
Terminology Binding | RequestIntent (Required)
Type | code
Summary | true

Comments | This element is immutable. Proposed tasks, planned tasks, etc. must be distinct instances. In most cases, Tasks will have an intent of "order".
Task.priority
Definition | Indicates how quickly the Task should be addressed with respect to other requests.
Control | 0..1
Terminology Binding | RequestPriority (Required)
Type | code
Meaning if Missing | If missing, this task should be performed with normal priority
Requirements | Used to identify the service level expected while performing a task.
Task.code
Definition | A name or code (or both) briefly describing what the task involves.
Control | 0..1
Terminology Binding | TaskCode:
Type | CodeableConcept
Summary | true
Comments | The title (eg "My Tasks", "Outstanding Tasks for Patient X") should go into the code.
Task.description
Definition | A free-text description of what is to be performed.
Control | 0..1
Type | string
Summary | true
Task.focus
Definition | The request being actioned or the resource being manipulated by this task.
Control | 0..1
Type | Reference(Any)
Requirements | Used to identify the thing to be done.
Summary | true
Comments | If multiple resources need to be manipulated, use sub-tasks. (This ensures that status can be tracked independently for each referenced resource.).
Task.for
Definition | The entity who benefits from the performance of the service specified in the task (e.g., the patient).
Control | 0..1
Type | Reference(Any)
Requirements | Used to track tasks outstanding for a beneficiary. Do not use to track the task owner or creator (see owner and creator respectively). This can also affect access control.
Alternate Names | Patient
Summary | true
Task.context
Definition | The healthcare event (e.g. a patient and healthcare provider interaction) during which this task was created.
Control | 0..1
Type | Reference(Encounter | EpisodeOfCare)
Requirements | For some tasks it may be important to know the link between the task or episode of care the task originated within.
Summary | true
Task.executionPeriod
Definition | Identifies the time action was first taken against the task (start) and/or the time final action was taken against the task prior to marking it as completed (end).
Control | 0..1
Type | Period
Summary | true
Task.authoredOn
Definition | The date and time this task was created.
Control | 0..1
Type | dateTime
Requirements | Most often used along with lastUpdated to track duration of task to supporting monitoring and management.
Alternate Names | Created Date
Invariants | Affect this element inv-1: Last modified date must be greater than or equal to authored-on date. (expression : lastModified.exists().not() or authoredOn.exists().not() or lastModified >= authoredOn, xpath: not(exists(f:lastModified/@value)) or not(exists(f:authoredOn/@value)) or f:lastModified/@value >= f:authoredOn/@value)
Task.lastModified
Definition | The date and time of last modification to this task.
Control | 0..1
Type | dateTime
Requirements | Used along with history to track task activity and time in a particular task state. This enables monitoring and management.
Alternate Names | Update Date
Summary | true
Invariants | Affect this element inv-1: Last modified date must be greater than or equal to authored-on date. (expression : lastModified.exists().not() or authoredOn.exists().not() or lastModified >= authoredOn, xpath: not(exists(f:lastModified/@value)) or not(exists(f:authoredOn/@value)) or f:lastModified/@value >= f:authoredOn/@value)
Task.requester
Definition | The creator of the task.
Control | 0..1
Requirements | Identifies who created this task. May be used by access control mechanisms (e.g., to ensure that only the creator can cancel a task).
Summary | true

Task.requester.agent
Definition | The device, practitioner, etc. who initiated the task.
Control | 1..1
Type | Reference(Device | Organization | Patient | Practitioner | RelatedPerson)
Alternate Names | Initiator; Author
Summary | true
Task.requester.onBehalfOf
Definition | The organization the device or practitioner was acting on behalf of when they initiated the task.
Control | 0..1
Type | Reference(Organization)
Requirements | Practitioners and Devices can be associated with multiple organizations. This element indicates which organization they were acting on behalf of when authoring the request.
Task.performerType
Definition | The type of participant that can execute the task.
Control | 0..*
Terminology Binding | TaskPerformerType (Preferred)
Type | CodeableConcept
Requirements | Use to distinguish tasks on different activity queues.
Task.owner
Definition | Individual organization or Device currently responsible for task execution.
Control | 0..1
Type | Reference(Device | Organization | Patient | Practitioner | RelatedPerson)
Requirements | Identifies who is expected to perform this task.
Alternate Names | Performer; Executer
Summary | true
Comments | Tasks may be created with an owner not yet identified.
Task.reason
Definition | A description or code indicating why this task needs to be performed.
Control | 0..1
Terminology Binding | TaskReason:
Type | CodeableConcept
Comments | This should only be included if there is no focus or if it differs from the reason indicated on the focus.
Task.note
Definition | Free-text information captured about the task as it progresses.
Control | 0..*
Type | Annotation
Task.relevantHistory
Definition | Links to Provenance records for past versions of this Task that identify key state transitions or updates that are likely to be relevant to a user looking at the current version of the task.
Control | 0..*
Type | Reference(Provenance)
Alternate Names | Status History
Comments | This element does not point to the Provenance associated with the current version of the resource - as it would be created after this version existed. The Provenance for the current version can be retrieved with a _revinclude.
Task.restriction
Definition | If the Task.focus is a request resource and the task is seeking fulfillment (i.e is asking for the request to be actioned), this element identifies any limitations on what parts of the referenced request should be actioned.
Control | 0..1
Requirements | Sometimes when fulfillment is sought, you don't want full fulfillment.
Task.restriction.repetitions
Definition | Indicates the number of times the requested action should occur.
Control | 0..1
Type | positiveInt
Requirements | E.g. order that requests monthly lab tests, fulfillment is sought for 1.
Task.restriction.period
Definition | Over what time-period is fulfillment sought.
Control | 0..1
Type | Period
Requirements | E.g. order that authorizes 1 year's services. Fulfillment is sought for next 3 months.
Comments | Note that period.high is the due date representing the time by which the task should be completed.
Task.restriction.recipient
Definition | For requests that are targeted to more than on potential recipient/target, for whom is fulfillment sought?
Control | 0..*
Type | Reference(Patient | Practitioner | RelatedPerson | Group | Organization)
Task.input
Definition | Additional information that may be needed in the execution of the task.
Control | 0..*
Requirements | Resources and data used to perform the task. This data is used in the business logic of task execution, and is stored separately because it varies between workflows.
Alternate Names | Supporting Information
Task.input.type
Definition | A code or description indicating how the input is intended to be used as part of the task execution.
Control | 1..1

Terminology Binding | TaskInputParameterType:
Type | CodeableConcept
Requirements | Inputs are named to enable task automation to bind data and pass it from one task to the next.
Alternate Names | Name
Comments | If referencing a BPMN workflow or Protocol, the "system" is the URL for the workflow definition and the code is the "name" of the required input.
Task.input.value[x]
Definition | The value of the input parameter as a basic type.
Control | 1..1
Type | *
[x] Note | See Choice of Data Types for further information about how to use [x]
Task.output
Definition | Outputs produced by the Task.
Control | 0..*
Requirements | Resources and data produced during the execution the task. This data is generated by the business logic of task execution, and is stored separately because it varies between workflows.
Task.output.type
Definition | The name of the Output parameter.
Control | 1..1
Terminology Binding | TaskOutputParameterType:
Type | CodeableConcept
Requirements | Outputs are named to enable task automation to bind data and pass it from one task to the next.
Alternate Names | Name
Task.output.value[x]
Definition | The value of the Output parameter as a basic type.
Control | 1..1
Type | *
[x] Note | See Choice of Data Types for further information about how to use [x]
Requirements | Task outputs can take any form.

12.1.11 Resource Task - Mappings

URL: https://hl7.org/fhir/STU3/task-mappings.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

12.1.11 Resource Task - Mappings

Orders and Observations Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Not linked to any defined compartments

Mappings:

RIM Mapping

Workflow Mapping

Mappings for the task resource.

12.1.11.1 Mappings for RIM Mapping (http://hl7.org/v3)

Task | ControlAct[moodCode=INT]
identifier | .id
definition[x] | .outboundRelationship[typeCode=DEFN].target
basedOn | .outboundRelationship[typeCode=FLFS].target[moodCode=INT]
groupIdentifier | .inboundRelationship[typeCode=COMP].source[moodCode=INT].id
partOf | .inboundRelationship[typeCode=COMP].source[moodCode=INT]
status | .statusCode
statusReason | .inboundRelationship[typeCode=SUBJ].source[classCode=CACT, moodCode=EVN, code="status change"].reasonCode
businessStatus | .inboundRelationship[typeCode=SUBJ].source[classCode=OBS, moodCode=EVN, code="business status"]
intent | .moodCode
priority | .priorityCode
code | .code
description | .text
focus | .outboundRelationship[typeCode=SUBJ].target
for | .participation[typeCode=RCT].role
context | .inboundRelationship[typeCode=COMP].source[classCode=PCPR, moodCode=EVN]
executionPeriod | .effectiveTime
authoredOn | .participation[typeCode=AUT].time
lastModified | .inboundRelationship[typeCode=SUBJ,].source[classCode=CACT, moodCode=EVN].effectiveTime
requester | .participation[typeCode=AUT].role
agent | .player
onBehalfOf | .scoper
performerType | .participation[typeCode=PRF].role.code
owner | .participation[typeCode=PRF].role
reason | .reasonCode
note | .inboundRelationship[typeCode=SUBJ,].source[classCode=OBS, moodCode=EVN, code="annotation"].value(string)

relevantHistory | .inboundRelationship(typeCode=SUBJ].source[classCode=CACT, moodCode=EVN]
restriction | Instead of pointing to request, would point to component of request, having these characteristics
repetitions | .repeatNumber
period | .effectiveTime(IVL<TS>)
recipient | .participation[typeCode=SBJ].role
input | ???
type | ???
value[x] | ???
output | ???
type | ???
value[x] | ???

12.1.11.2 Mappings for Workflow Mapping (http://hl7.org/fhir/workflow)

Task | Request, Event
identifier | Request.identifier, Event.identifier
definition[x] | Request.definition, Event.definition
basedOn | Request.basedOn, Definition.basedOn
groupIdentifier | Request.groupIdentifier
partOf | Event.partOf
status | Request.status, Event.status
statusReason
businessStatus
intent | Request.intent
priority | Request.priority
code | Request.code, Event.code
description
focus
for | Request.subject, Event.subject
context | Request.context, Event.context
executionPeriod | Event.occurrence[x]
authoredOn | Request.authoredOn
lastModified
requester | Request.requester
agent | Request.requester.agent
onBehalfOf | Request.requester.onBehalfOf
performerType | Event.performer.role, Request.performerType
owner | Event.performer.actor, Request.performer
reason | Request.reasonCode, Event.reasonCodeableConcept
note | Request.note, Event.note
relevantHistory | Request.relevantHistory
restriction
repetitions
period
recipient
input
type
value[x]
output
type
value[x]

12.1.12 Resource Task - Design Notes

URL: https://hl7.org/fhir/STU3/task-explanations.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

12.1.12 Resource Task - Design Notes

to do

12.1.13 Resource Task - Extensions & Profiles

URL: https://hl7.org/fhir/STU3/task-profiles.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

12.1.13 Resource Task - Extensions & Profiles

Orders and Observations Work Group | Maturity Level: N/A | Ballot Status: Informative

This table lists profiles and extensions for the Task resource. For background information, see Profiling Resources and Extensibility. Additional profiles and resources may be found in published Implementation Guides , or in the Conformance resource registry .

12.1.13.1 Profiles

No Profiles defined for this resource

12.1.13.2 Extensions

task-candidateList | candidateList | for Task HL7 Extensions
task-replaces | replaces | for Task HL7 Extensions

Extensions that reference this resource:

task-replaces | replaces | for Task HL7 Extensions

12.1.13.3 Search Extensions

No Search Extensions defined for this resource

12.1.14 Task HL7 Extensions

URL: https://hl7.org/fhir/STU3/task-extensions.html

Defines common extensions used with or related to the Task resource

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

12.1.14 Task HL7 Extensions

(No assigned work group) | Maturity Level: N/A | Ballot Status: Informative

12.1.14.1 Content

Extensions:
task-candidateList | candidateList : Identifies the individuals who are candidates for being the owner of the task.
task-replaces | replaces : Completed or terminated task(s) whose function is taken by this new task.

12.5 Workflow Description

URL: https://hl7.org/fhir/STU3/workflow.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

FHIR Infrastructure Work Group | Maturity Level: 2 | Ballot Status: Trial Use

12.5 Workflow Description

Workflow is an essential part of healthcare - orders, care protocols, referrals are the drivers of most activity within in-patient settings and a great deal of activity in community care as well. FHIR is concerned with workflow when there's a need to share information about workflow state or relationships, when there's a need to coordinate or drive the execution of workflow across systems and when there's a need to define allowed actions, dependencies and conditions on behavior.

Workflow state & relationships

FHIR does not need to be used for the execution of workflow. Orders, care plans, lab results, hospital admissions, claim payments and other records can all be shared using FHIR resources without the process to solicit fulfillment of those orders or requesting payment of those claims being driven by a FHIR transaction. Interoperable support for workflow execution is a more advanced FHIR activity because it requires a higher degree of standardization. Rather than merely standardizing the data to exchange, interoperable workflow execution requires standardization of the processes, roles and activities across the different systems. However, even without using FHIR for workflow execution, there's still a need to standardize the data elements related to workflow: how does an event or a result point to the order that authorized it? How do parent steps and child steps get linked together? How does a care plan identify what protocol it's adhering to?

FHIR defines three categories of resources that are involved in activities - requests, events and definitions. Each of these categories has a "pattern" associated with it. Resources that fall into that category are encouraged to adhere to their respective pattern. These patterns provide standard elements that are typical for most resources of each category. Strict adherence is not required as work groups are expected to align with typical domain behavior and requirements as more authoritative than "desired" architectural patterns. In some cases, capabilities might be supported with extensions rather than core elements where a pattern capability is deemed to be "not common, but still relevant" for a given resource.

A full description of the patterns and their interrelationships can be found in the Workflow Resource Patterns section of this page.

Workflow execution

In addition to defining patterns for resources used in workflow processes, FHIR supports the execution of those processes as well. However, FHIR does not define a "one size fits all" solution for workflow architecture. FHIR supports a variety of

interoperability paradigms and most of them (REST, Messaging and Services) provide support for driving workflow execution. (The Document paradigm does not directly support driving behavior, though it can be combined with one of the other patterns to do so.) In addition, several of these paradigms allow multiple approaches to supporting workflow, depending on the context and needs of the workflow process.

The Workflow Execution and Communication Patterns section describes many options for workflow execution, summarizes their respective pros and cons and makes recommendations for the circumstances in which they might best be used.

Workflow definition

The definition of protocols, order sets, guidelines and other structures that define what sorts of activities should occur, what order they should occur on, what dependencies they have, in what circumstances they should start or end, etc. is handled by a pair of resources:

* PlanDefinition defines the interrelationships of steps and the rules around their execution

* ActivityDefinition defines an activity to be performed as a single step

The use of these two artifacts is discussed here.

12.5.1 Workflow Resource Patterns

Not all resources in FHIR are related to workflow - many are used to describe entities and roles (patients, medications, etc.) or infrastructure (structure definitions, value sets, etc.). However, a large proportion of the FHIR resources are devoted to the description of activities in one fashion or another and almost all of these fall into the realm of workflow - they describe things that can be done (definitions), are desired to be done (requests) or that have been done (events). The table below summarizes the list of workflow-relevant resources:

12.5.1.1 Workflow resources

Requests | Resources that ask for or express a desire/intention for something to be done

Appointment* CarePlan Claim CommunicationRequest | DeviceRequest EligibilityRequest EnrollmentRequest ImmunizationRecommendation | MedicationRequest NutritionOrder ProcedureRequest ProcessRequest† | ReferralRequest Task‡ SupplyRequest VisionPrescription

Events | Resources that express that something has been done and which can potentially be done because of a request

AppointmentResponse* ClaimResponse ClinicalImpression Communication Composition Condition (aka Problem) Consent Contract Coverage | DeviceUseStatement DiagnosticReport DocumentManifest DocumentReference EligibilityResponse Encounter EnrollmentResponse EpisodeOfCare ExplanationOfBenefit | FamilyMemberHistory GuidanceResponse ImagingStudy Immunization MeasureReport Media MedicationAdministration MedicationDispense MedicationStatement | Observation PaymentNotice PaymentReconciliation Procedure ProcessResponse† QuestionnaireResponse RiskAssessment SupplyDelivery Task‡

Definitions | Resources that define something that can potentially happen in a patient and time-independent manner

ActivityDefinition DataElement | HealthcareService Measure | OperationDefinition PlanDefinition | Questionnaire

* | The Appointment and AppointmentResponse resources do not follow the same sort of request/response pattern as the other resources. Their design is based on iCal conventions, so their model won't reflect the same alignment as most other resources. They are included here for completeness.

† | ProcessRequest and ProcessResponse are candidates for retirement with their function subsumed by Task

‡ | The Task resource takes on characteristics of both "requests" and "events" and thus shares characteristics from both patterns

Note that requests, events and definitions don't exist in a 1:1:1 relationship. Some requests and events have obvious pairings. For example, a SupplyRequest will generally always pair with a SupplyDelivery. The same goes for EnrollmentRequest/EnrollmentResponse, etc. On the other hand, for other resources there isn't a strict pairing. A ReferralRequest might be responded to by an Encounter, DiagnosticReport, Procedure, RiskAssessment, etc. Similarly, a Procedure might be triggered by a ProcedureRequest or ReferralRequest. The set of common linkages should be asserted in their respective resources. The specific types of responses for a given request will be governed by the Request.code, any workflow definitions/protocols referenced and local convention.

12.5.1.2 Workflow Resource Relationships

These three patterns of resources have a standard set of relationships, both with themselves, as well as with each other.

Specifically:

* requests, events and definitions can point to their respective definitions

* events and requests can point to the proposals, plans or orders they are based on

* events and definitions can be organized into parent-child relationships of parents and components

* definitions and requests can both replace prior versions of the same type of artifact

This list of relationships is not exhaustive, but covers those that are "standardized" as part of the patterns. Further description and guidance on these relationships can be found in the Request, Event and Definition logical patterns.

12.5.1.3 Request Resource Pattern

Requests are resources that represent the proposal, plan or order for an activity to occur. A Request pattern defines the common elements typically present on all request resources.

The amount of information needed for a Request to be actionable can vary by circumstance. Some request instances may not be "fully specified" - additional information from protocol, patient preference and/or professional decision-making may be necessary before the authorized action can occur. For example, a MedicationRequest might be specified without indicating a strength or route in situations where the pharmacy (or even nursing station) has the authority to determine those parameters. A VisionPrescription might not be actionable until frames have been chosen and the necessary measurements of the patient's

face have been taken to allow the lenses to be positioned appropriately within the frames.

All requests with an intent of "order" authorize something. Whether what is authorized is sufficient to be immediately actionable depends on who is fulfilling the order and the context in which the fulfillment request is made. The determination of whether a given "request" is actionable may be made by the systems involved or the humans being asked to act.

As well, the existence of a "Request" instance doesn't necessarily imply that fulfillment will be requested immediately - or even ever. The decision to request fulfillment may be delegated to the patient or to down-stream practitioners. Such fulfilling practitioners may need to capture additional information prior to executing the fulfillment.

12.5.1.4 Event Resource Pattern

Events are resources that represent the ongoing or completed execution of some activity or observation. For example, a clinical procedure, a financial transaction, the recording of a diagnosis, etc. An Event pattern defines the common elements typically present on all event resources.

12.5.1.5 Definition Resource Pattern

Definitions are resources that represent activities that could be performed in a time and subject-independent manner such as a protocol, order set, clinical guideline, etc. A Definition pattern defines the common elements typically present on all event resources.

12.5.2 Open Issues

STU Notes:

* It is possible to replace some portions of the MessageHeader with a reference to the Task resource. Doing so would mean consistency in how asynchronous requests are represented using REST and messaging. However, it introduces an additional layer of complexity and formality into the messaging paradigm that may be unwelcome, particularly for those systems that do not currently foresee a need to support both RESTful and messaging invocations of workflow

* The OperationDefinition resource could be used to define types of tasks and the sets of parameters that can go with them. Is this an appropriate use of the OperationDefinition resource?

* The SupplyRequest, DeviceRequest and VisionPrescription resources have a significant degree of overlap. Should they remain distinct resources?

Feedback is welcome here .

12.6 Workflow Execution and Communication Patterns

URL: https://hl7.org/fhir/STU3/workflow-communications.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

FHIR Infrastructure Work Group | Maturity Level: 2 | Ballot Status: Trial Use

12.6 Workflow Execution and Communication Patterns

As described in previous sections, many FHIR resources can be used within workflows without using FHIR to manage the execution of the workflows. When we want to manage workflow execution with FHIR, there are several mechanisms available. In addition to managing workflow with FHIR, there are also cases where management of workflow execution is not necessary, and supporting workflow becomes simply the use of the resource patterns in an ad-hoc fashion. In deciding how best to interoperate around workflow with FHIR, there are several considerations:

* Is sharing of the state of the workflow necessary among the participants?

* Which paradigm do you want to use (REST, messaging, services, a mix)?

* Who owns/manages the various resources involved in the workflow (placer, filler, another participant)?

* Is there infrastructure in place to support polling, push notifications via subscriptions or both?

* Is there a need for confirmation that the desired performer agrees to act, or can that be presumed?

* Is there a need to negotiate whether/how the requested action will be performed?

* Can the requesting and performing system communicate directly? Are they able to post to each other's servers (if using REST)?

* Is there an ability/need to have a queue server to facilitate workflow execution?

* How many potential actors are involved?

* Will the workflow always be directed or is there a pool of potential performers who could choose to perform the requested action?

The answers to these (and other) questions will guide the selection of communication patterns to be used for a specific interoperability use case. It is recommended that domain workgroups analyze the support for workflow execution within their domains and provide recommendations for an appropriate subset of patterns, with a discussion on where and how to use them. Implementation guides, by their nature, should prescribe which patterns to use for workflow execution management.

12.6.1 Communication Patterns Conventions

This section highlights some of the more common patterns and identifies their characteristics and limitations and provides recommendations on when each approach may be most useful or relevant. Please note that this list of patterns is not exhaustive. Patterns can be combined in various ways and there are likely some possibilities we haven't thought about yet (feel free to submit additional patterns using the 'submit a change' link at the bottom of the page). As well, the recommendations given here are just that - recommendations. Implementers are free to choose which patterns they wish to support. Because of this, tight interoperability around workflow execution (as with any other tight interoperability using FHIR) will depend on communicating participants doing some up-front negotiation around how they plan to support workflow execution or all communicating partners will need to adhere to an implementation guide that sets out clear interoperability expectations.

Prior to reviewing this list of options, readers are encouraged to be familiar with the following pages and resources: REST, messaging, operations, services and the Subscription resource.

The scenarios used to illustrate the patterns below make use of a few conventions:

* The focus here is on a "request" and the actioning of that request. Almost all workflows can be broken down to a sequence of these steps, though the responsibilities of the different parties may shift for each interaction and there can be more than two parties involved in the overall workflow

* The request could be as simple as "please look at this information" and the response could be as simple as an implicit "it's been looked at" or the request could be for some more involved action that may include reporting back multiple interim and final steps

* The requester is referred to as the "placer" and the performer is referred to as the "filler", which are often seen as order-specific terms. However, in this context, the terms hold whether the request is expressed as a proposal, plan or full-blown order

* Each of the patterns defines the set of steps involved in processing the request, lists some of the benefits and limitations associated with the approach, provides a scenario to illustrate the use of the pattern, and then makes recommendations about when the pattern is most appropriate

* The descriptions of these patterns focus on the notion of requesting fulfillment of a request. However most of these patterns are also applicable to requests for status change, requests for information, etc. If a pattern is limited in the types of execution it can trigger, this will be noted in the "limitations" section.

12.6.2 Workflow State

One of the key distinguishing characteristics in the patterns below is whether the pattern supports the tracking of the workflow's state by both the placer and the filler. The workflow state is represented by the Task resource. The Task state machine, which is also presented here, shows the use of Task.status to represent the general infrastructure state of the Task resource.

The combination of Task.status, Task.statusReason and Task.businessStatus provide the mechanism for tracking the workflow state. The following two examples illustrate how that could be accomplished with regards of specific workflows. Note that these examples avoid any specificity on where the described resources exist, or any reference to a specific communication pattern. The goal of the examples is to demonstrate the use of the Task attributes to represent the workflow state.

12.6.2.1 Example: Workflow states for a lab order for a blood test

The following table shows the steps for a simple blood test order, and the corresponding values of the Task attributes. A lot of details on other attributes have been omitted in order to focus on the state changes.

Workflow step | Task.status | Task.statusReason | Task.businessStatus

Provider orders a blood test for a patient. A ProcedureRequest resource is created to represent the order, and a Task resource is created to represent the request to the lab. Task.basedOn points to the ProcedureRequest instance. | Requested | New order | Ordered | N/A

The lab accepts the order. | Accepted | Able to perform the test. | Accepted | N/A

The patient is at the lab, and blood is drawn. | In progress | Obtained the specimen | Specimen available

Work on the blood sample is in progress, preliminary results are available, a DiagnosticReport resource is created, and Task.output.value[1] references the DiagnosticReport resource. | In progress | Preliminary results available | Preliminary results

Final results are available | Complete | Final results are available | Final results

12.6.2.2 Example: Workflow states for a referral from a PCP to a specialist

The following table shows the steps for a referral workflow, and the corresponding values of the Task attributes. A lot of details on other attributes have been omitted in order to focus on the state changes.

Workflow step | Task.status | Task.statusReason | Task.businessStatus

Provider A is seeing a patient with a specific complaint. Based on the exam, Provider A refers the patient to a specialist at clinic B. It is an urgent referral, requesting to see the specialist within 4 days. A ReferralRequest resource is created to represent the referral order, and a Task resource is created with Task.priority set to Urgent, and Task.restriction.period.end set to 4 days in the future. | Requested | New referral | Referred | N/A

Clinic B receives the referral and puts it on the urgent queue to be reviewed. | Received | Received by organization | Pending | N/A

Upon reviewing the referral, the clinic determines they have no capacity to see the patient within the specified time period, and they decline the referral. | Rejected | Provider unavailable | Declined | N/A

Provider A changes the referral request to clinic C. This creates a new Task resource, with the appropriate information | Requested | Changed the referred to provider | Referred | N/A

Clinic C accepts the referral and it is placed on a scheduler's queue. | Accepted | Able to provide the service. | Accepted | N/A

An appointment is scheduled for the patient at Clinic C. | In progress | Appointment scheduled | Scheduled

Soon after the appointment is scheduled, the patient calls provider A to ask for a different location for the referral appointment, as he lost his transportation, and he cannot reach Clinic C. The provider requests a cancel for the referral. | Suspended | Patient unable to make appointment. | Cancel requested

Clinic C confirms the cancellation of the existing task. | In progress Failed | Referral cancelled | Cancelled as requested

Provider A changes the referral to Clinic D, which is located close to where the patient lives, and can be conveniently reached using public transportation. This creates a new Task resource, with the appropriate information. | Requested | Changed the referred to provider | Referred | N/A

Clinic D accepts the referral and it is placed on a scheduler's queue. | Accepted | Able to provide the service. | Accepted | N/A

An appointment is scheduled for the patient at Clinic D. | In progress | Appointment scheduled | Scheduled

Patient is seen at Clinic D. A diagnosis is made by the specialist, a prescription is given, and a follow-up appointment is made at clinic D. Task.output.value[1] references a Compositions resource, which contains the encounter summary. | In progress | Intermediary consultation note available | Preliminary notes

Patient comes for follow-up appointment at clinic D. The specialist is satisfied by the progress the patient has made and creates the final consultation note. Task.output.value[2] references a Compositions resource, which contains consultation note. | Completed | Final consultation note available | Final notes

12.6.3 List of patterns

The patterns that facilitate the execution of workflow using the Task resource are grouped in the Workflow Management Patterns section. The patterns where no Task resource is used are grouped in the Ad-hoc Workflow Patterns section. The list of patterns is as follows:

Ad-Hoc Workflow Patterns

Option A: Simple RESTful POST or PUT

Option B: Direct POST of request to fulfiller's system

Option C: POST of request to placer/queue server system, receiver uses polling or subscription

Option D: Messaging request from placer to filler & acknowledgment

Option E: Services request from placer to filler & acknowledgment

Workflow Management Patterns

Option F: Creation of Task on placer's system

Option G: POST of Task on fulfiller's system

Option H: POST of Task to a workflow broker

Option I: POST of Task on fulfiller's system, POST of sub-Task on placer's system

Option J: Messaging Task from placer to fulfiller

Option K: Service request referencing Task from placer to fullfiller

Additional Scenarios and Variations

TODO: Insert Jose's decision tree here?

12.7 Ad-Hoc Workflow Communication Patterns

URL: https://hl7.org/fhir/STU3/workflow-ad-hoc.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

FHIR Infrastructure Work Group | Maturity Level: 2 | Ballot Status: Trial Use

12.7 Ad-Hoc Workflow Communication Patterns

TODO: Discussion (or reference to one) on Polling and Subscription

12.7.1 Option A: Simple RESTful POST or PUT

12.7.1.1 Steps

* The placer makes a RESTful call to create or update a record or a POST to invoke an operation over HTTP

* The receiver responds with a 2xx HTTP response indicating whether the request was successfully processed or not and, if appropriate, provides the response to the request in the payload of the HTTP response

12.7.1.2 Benefits

* Simplest of all the possible workflow architectures

* Placer knows whether the request was accepted or not and knows when the task has been done

12.7.1.3 Limitations

* Only works for automated execution where the decision to perform the request and the execution of the request can be done synchronously within the HTTP timeout period (generally on the order of 10s of seconds).

* Requires that the placer have authority to post directly to the placer's system

* Requires that the "request" be expressible as a simple creation, update or operation invocation

* Only works for "fulfillment" requests for Request resources - can't handle request for state changes or information

12.7.1.4 Usage Recommendations

This is by far the most common pattern in FHIR for simple changes as it requires the least overhead. However, it covers only automated responses, and does not support any actual workflows of more than one step. If human processing is involved in the request execution, then this approach won't suffice. This approach is listed here to make sure that implementers consider

whether they can make this one work first before falling back to one of the more sophisticated patterns.

12.7.1.5 Usage Examples

???

12.7.2 Option B: Direct POST of request to fulfiller's system

12.7.2.1 Steps

* Placer system invokes a create by POSTing a 'request' resource (e.g. MedicationRequest, ProcedureRequest, ReferralRequest, etc.) to the appropriate RESTful resource endpoint (e.g. [base]/MedicationRequest) on the filler system and places an actionable tag on the resource that indicates the request is intended to be acted upon, not merely stored.

* The filler synchronously responds with a "201" indicating that that they have received and stored (created) the resource on their system

* At some later point, the filler POSTs an 'event' resource (e.g. MedicationDispense, DiagosticReport, Encounter, etc.) to the appropriate resource endpoint on the placer system, including a basedOn link to the 'request' resource that the action was performed in fulfillment of.

* The placer system synchronously responds with a "201" indicating they've received and store (created) the resource on their system

12.7.2.2 Benefits

* Lowest amount of overhead. No need for Task. No need for polling or subscriptions

* Explicit acknowledgement that filler has received the request

12.7.2.3 Limitations

* Can only use when requesting fulfillment (can't use to request status change or other updates)

* Placer and filler must be able to communicate directly (i.e. know each other's respective endpoints), must each have a FHIR server, and must have "write" permissions to each other's servers. This could become unmanageable if there are a large (or dynamic) number of placers and fillers that need to communicate

* No indication of agreement to act on the request

* There's no ability to negotiate fulfillment - no ability to say "no"

* This completely reverses the usual provenance of resources, as the request only exists on the fulfiller's system, and the event only exists on the placer's system. This greatly limits the use of these resources for any other purpose.

12.7.2.4 Usage Recommendations

Use this approach when there's no ability to have queue servers and no support/need for complexity of Task, polling or pub/sub (and no need for negotiation or the ability for the filler to say "no"). This is a pseudo-messaging architecture that doesn't actually use messaging architecture.

12.7.2.5 Usage Examples

12.7.3 Option C: POST of request to placer/queue server system, receiver uses polling or subscription

12.7.3.1 Steps

* Placer system creates a 'request' resource (e.g. MedicationRequest, DiagnosticOrder, ReferralRequest, etc.) by a system action or a POST on either its own system or a third party queue server system and places an "actionable" tag on the resource that indicates the request is intended to be acted upon. The request explicitly identifies the intended fullfiller

* The filler system uses polling (or a subscription) at the placer or queue server system to see if there are any "new" requests that: are tagged as "actionable", have the filler identified as the intended performer, and are a type of request "of interest" to the filler.

* At some later point, the filler creates an 'event' resource (e.g. MedicationDispense, DiagosticReport, Encounter, etc.) via a system action, or a POST on either its own system, the same queue server as the request was placed on, or some alternate queue server, including a link to the 'request' resource that the action was performed in fulfillment of

* The placer system uses polling (or a subscription) to the filler or queue server system to see if there are any "new" events that are tied to any outstanding requests the placer has initiated

12.7.3.2 Benefits

* Placer and fulfiller don't have to communicate directly (can act through queue server). This can reduce the number of point-to-point interfaces that need to be supported

* Preserves the conventional management and ownership of the request and event at the placer and fulfiller's systems respectively

* No need for Task

12.7.3.3 Limitations

* Can only use when requesting fulfillment (can't use to request status change or other updates)

* Additional complexity of using subscription or polling

* Polling by the placer for "anything related to these 500 open orders" could be onerous, especially if some orders never get closed.

* Placer and fulfiller must know where to poll or subscribe for content - this could be a large number of systems

* No indication of agreement to act on the request

* There's no ability to negotiate fulfillment - no ability to say "no"

* Placer may not know when (or if) filler system has retrieved the request

12.7.3.4 Usage Recommendations

This pattern could be used when there's no support/need for complexity of Task, usually for simple two step workflows. This is a more typically RESTful approach where data resides on the server "owned" by the data creator and is accessed by other systems.

12.7.3.5 Usage Examples

12.7.4 Option D: Messaging request from placer to filler & acknowledgment

TODO: needs review and update

12.7.4.1 Steps

* Placer sends message to filler system including Request resource (and other relevant resources) along with a MessageHeader with an "event" code saying "please fulfill" and "data" element pointing to the Request resource as the item to fulfill. Message could potentially use Task instead of MessageHeader.event to convey desired action (ongoing discussion)

* Filler system sends a response indicating receipt of the message and, optionally an indication of their intention to fulfill the request

* Filler system may send incremental messages to the placer showing progress (e.g. specimen collected, preliminary results, final results)

12.7.4.2 Benefits

* Reduced number of communications

* All relevant data sent in one package

* Responses can be asynchronous and content may be routed

* There's an ability to negotiate fulfillment - i.e. the ability to say "no"

* Can request things other than just fulfillment (e.g. please suspend)

* Explicit acknowledgement that filler has received and agreed to act on the request (though no need for the placer to check)

12.7.4.3 Limitations

* Messaging is "heavy"

* Need to negotiate what allowed responses are and what data can be present in request and response messages

* Additional complexity of setting up and maintaining a subscription or polling infrastructure

* Additional complexity of using Task

* Need message delivery infrastructure in place

12.7.4.4 Usage Recommendations

Existing messaging infrastructure (e.g. v2 LTP, MLTP, WSI Web Services, Direct, VISA, REST, etc.) and a need to stay consistent with that architecture

12.7.5 Option E: Services request from placer to filler & acknowledgment

TODO: Needs review, update and more work - there's not a lot of experience using FHIR services to manage the fulfillment process

12.7.5.1 Steps

* Placer may create and store a Request resource on their own system or a queue server.

* Placer invokes a service on the filler system saying "please fulfill this order", including the content or a reference to the request resource and any other relevant data

* Filler system responds (synchronously if using HTTP, but may be asynchronous if using SOAP or other transport mechanisms) with conformation of receipt and, optionally indication of intention to fulfill and/or results

12.7.5.2 Benefits

* ???

12.7.5.3 Limitations

* ???

12.7.5.4 Usage Recommendations

TBD

12.7.6 Additional Scenarios

TODO: needs review and update. Possibly add options about using messaging and/or services instead of polling/subscription in above scenarios

12.7.6.1 POST of "request" resource for filler system, response via Task

This is a variation ofOption H, where the Workflow broker is essentially merged with the fulfiller. It still allows the placer to only use a POST of the request, and be made aware of the changes to the other resources via subscpriotn or polling.

* Placer system invokes a "create" action by POSTing a 'request' resource (e.g. MedicationRequest, DiagnosticOrder, ReferralRequest, etc.) to the appropriate RESTful resource endpoint (e.g. [base]/MedicationRequest) on the filler, placer or queue server system and sets a "tag" on the resource that indicates the request is "actionable"

* Filler POSTs a Task resource to its own system or a queue server system, pointing to the request resource and indicating intent to fulfill or refusal to fulfill

* Placer system uses either polling or pub/sub to become aware of the existence of the task and fulfillment intent

* Fulfiller may update the Task to indicate interim progress notes

* Placer either polls the Task to note acceptance and changes or uses a subscription to determine the same

* Fulfiller POSTs an event resource to its own system or to a queue server system

* Fulfiller Updates the Task resource to change its status to completed and to point to the event resource

* Placer system becomes aware of the update via polling or subscription

* Placer system retrieves the event

* Placer system marks the request as "complete"

12.7.6.2 Querying the status of a workflow using REST

* Placer sends query for Task(s) that have a focus of the request of interest to a system (placer system, queue server or filler) that holds tasks related to their request.

* System returns a query response showing all related tasks (typically just one). Task shows current status.

12.7.6.3 Querying the status of a workflow using services

* Placer invokes a "what's the status of this order" service, passing the request business identifier or URL of the request

* Services responds with a Task showing the current state of the fulfillment of the request

12.7.6.4 Cancellation of a Task using REST - placer owns

* Placer sends an update to the Task setting the status to "cancelled" signalling they are no longer concerned about the outcome of the task

* Filler receives notification of the update (because the task is on their system, or because they poll it, or are subscribed to it) and ceases work if they are able

12.7.6.5 Cancellation of a Task using REST - filler owns

* Placer creates a new task requesting cancellation of the original fulfillment task Fulfillment of the "cancellation task" can be requested using any of the mechanisms above

* Filler decides whether they are able to cancel the task and update the "cancellation" task to indicate either cancellation is complete or has been refused

12.8 Workflow Management Communication Patterns

URL: https://hl7.org/fhir/STU3/workflow-management.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

FHIR Infrastructure Work Group | Maturity Level: 2 | Ballot Status: Trial Use

12.8 Workflow Management Communication Patterns

TODO: Discussion (or reference to one) on Polling and Subscription

12.8.1 Option F: Creation of Task on placer's system

12.8.1.1 Steps

* Placer creates the request in its own system via POST, an internal action, or POSTs it to a queue server system

* Placer creates a Task resource in its own system via POST, an internal action, or POSTs it to a queue server system, pointing to the request resource and seeking fulfillment. The Task may have a specified performer, in which case step 3 is expected to be done by that performer. If the Task does not have a specified "performer" (although may have performer type),

then this is a case of an "open" task, where any number of fulfillers may attempt to "claim" the task. Who succeeds is determined by local policies and procedures.

* Fulfiller system uses either polling or pub/sub to become aware of the existence of the task

* A common case may be the conveyance of the Task id to a fulfiller by other means. For example, a lab test is ordered, and the patient takes a requisition to the lab of his choice. The requisition contains the Task id (as a bar code, or stored in the patient's healthcare smart card), and the lab system can execute a direct GET for the Task, thus eliminating the need for subscription or polling.

* Fulfiller system queries to retrieve the referenced request and updates the Task to indicate "acceptance" and agreement to fulfill

* Fulfiller may update the Task to indicate interim progress notes

* Placer is madeaware of the acceptance of the Task and any changes to the Task either through its ownership of the resource or using polling or a subscription to a queue server system to determine the same

* Fulfiller creates an event resource in its own system via POST or internal action, or POSTs it to a queue server system

* Fulfiller PUTs an update to the Task resource to change its status to completed and to point to the event resource

* Placer is aware of the completion of the Task either through the ownership of the resource, or via polling or subscription to a queue server system, and retrieves the referenced event resource

* Placer updates the request resource to indicate completion via PUT or an internal action, or PUTs the update to a queue server system

12.8.1.2 Benefits

* Uses the FHIR REST API for managing the workflow

* Both placer and fulfiller can track the state of the workflow

* Can use this approach for request other than just fulfillment (e.g. to request status change or other updates)

* There's an ability to negotiate fulfillment - i.e. the ability to say "no"

* Explicit acknowledgement that filler has received and agreed to act on the request

12.8.1.3 Limitations

* Additional complexity of using Task

* Additional complexity of setting up and maintaining a subscription or polling infrastructure

* Placer and fulfiller must have a FHIR server, and placer's FHIR server must allow "write" updates to the Task resource

Placer and filler may need to be able to communicate directly (i.e. know each other's respective endpoints)

* This could become unmanageable if there are a large (or dynamic) number of placers and fillers that need to communicate

* Placer and fulfiller must know where to subscribe for content - this could be a large number of systems

* May not apply if there's a queue server

12.8.1.4 Usage Recommendations

* When following the state of the workflow is important

* When there is a need to know that a fulfiller has accepted to perform the request

* When there is a need to be able to change the performance of the request

* When there are multiple steps in fulfilling the request

* When requests are not (always) directed to a specific filler

* When the placer needs to manage the Task resource

12.8.1.5 Usage examples

Lab order to known performer

A provider orders a bilirubin test for a patient, to be done at the provider's facility lab....

"Open" lab order

A patient is scheduled for their physical, and the provider sends her a secure message to have a fasting blood test performed before the office visit. The message has a link to a requisition, and a list of possible lab locations where the test can be done. The patient chooses a lab location near her place of employment and a test is scheduled. The lab gets the requisition from the patient and, using the information on how to retrieve the Task from the provider's system, claims the test to be performed...

12.8.2 Option G: POST of Task on fulfiller's system

12.8.2.1 Steps

* Placer creates the request in its own system via POST or an internal action, or POSTs it to a queue server system

* Placer POSTs a Task resource to the filler system, pointing to the request resource and seeking fulfillment

* Fulfiller system GETs the referenced request

* Fulfiller updates the Task to indicate acceptance of the task

* Placer either polls the Task to note acceptance or uses a subscription to determine the same

* Fulfiller may further update the Task to reflect the progress made. Using the same method as in step 5, the placer becomes aware of these updates

* Fulfiller creates an event resource in its own system via POST or an internal action, or POSTs it to a queue server system

* Fulfiller Updates the Task resource to change its status to completed and to point to the event resource

* Placer either polls the Task to note completion and changes or uses a subscription to determine the same

* Placer system queries to retrieve the referenced event resource

* Placer updates the request resource to indicate completion via PUT or an internal action, or PUTs the update to a queue server system

12.8.2.2 Benefits

* Uses the FHIR REST API for managing the workflow

* Subscription/polling is targeted to a specific resource instance - much simpler than general subscription/polling infrastructure

* Both placer and fulfiller can track the state of the workflow

* Can use this approach for request other than just when requesting fulfillment (e.g. to request status change or other updates)

* There's an ability to negotiate fulfillment - i.e. the ability to say "no"

* Explicit acknowledgement that filler has received and agreed to act on the request

12.8.2.3 Limitations

* Additional complexity of using Task

* Additional complexity of using subscription or polling

* Placer and fulfiller must have a FHIR server and fulfiller's FHIR server must allow "write" creation of the Task resource by the placer

* Placer and filler may need to be able to communicate directly (i.e. know each other's respective endpoints)

* This could become unmanageable if there are a large (or dynamic) number of placers and fillers that need to communicate

* May not apply if there's a queue server

* Placer may not know immediately when filler system has retrieved the request

12.8.2.4 Usage Recommendations

* When following the state of the workflow is important

* When there is a need to know that a fulfiller has accepted to perform the request

* When there is a need to be able to change the status of the workflow by either the placer or the fulfiller

* When there are multiple steps in fulfilling the request

* When the fulfiller needs to manage the Task resource

12.8.2.5 Usage Examples

Outpatient prescription

A patient is suffering from a poison ivy rash and is prescribed a steroid medication to manage the reaction. The provider asks the patient for his preferred pharmacy and creates the Task at the specified pharmacy's system. The patient is running late and by the time he arrives at the pharmacy, it is closed. He calls the clinic where the provider practices and requests that the pharmacy where the order is to be fulfilled is changed to one that is open around the clock. The staff at the clinic cancels the Task at the original pharmacy and creates a new Task at the requested new one. By the time the patient arrives at the new pharmacy, the medication is ready for pickup. Once the request is fulfilled, the clinic receives the notification that the Task is complete and the prescription is also marked as complete.

12.8.3 Option H: POST of Task to a workflow broker

TODO: Still needs review and update

12.8.3.1 Steps

* Placer POSTs the request to its own system or to a queue server system

* Broker detects that new un-assigned request (without a Task yet created and falling within the scope of the Broker to ensure fulfillment) via polling or subscription

* Broker POSTs a Task resource to its own system or a queue server system, pointing to the request resource and seeking fulfillment from a specific filler Task does not have a specified "performer" (but may have performer type)

* If the Task is rejected by one potential recipient, the broker may create a new task to seek fulfillment from others

* Continue as per Option G

12.8.3.2 Benefits

* Offloads responsibility for seeking fulfillment from the placer system, but more actively solicits fulfillment than a simple "post the task and see who takes it". Also, allows prioritized assignment of tasks (i.e. some fillers may be preferred over others)

12.8.3.3 Limitations

* Requires a broker to exist

* Broker must know all available fillers and their capabilities to allow appropriate assignment

12.8.3.4 Usage Recommendations

Appropriate in environments that have a workflow engine that takes on responsibility for ensuring fulfillment

12.8.3.5 Usage Examples

12.8.4 Option I: Option I: POST of Task on fulfiller's system, POST of sub-Task on placer's system

12.8.4.1 Steps

* Placer creates the request in its own system via POST or an internal action

* Placer POSTs a Task resource to the filler system, pointing to the request resource and seeking fulfillment

* Fulfiller system GETs the referenced request

* Fulfiller updates the Task to indicate acceptance of the task

* Fulfiller POSTs a Task, which also points to the request resource and uses the Task.partOf attribute to point to the Task from step 2 (indicating it is a sub-Task)

* Fulfiller may further update both Tasks to reflect the progress made. Since the sub-Task is on the placer's system, the placer is aware of these updates

* Fulfiller creates an event resource in its own system via POST or an internal action

* Fulfiller Updates the Task resource to change its status to completed and to point to the event resource

* Fulfiller updates the sub-Task resource as completed and points to the event resource. Since the sub-Task is on the placer's system, the placer is aware of the completion and changes.

* Placer system queries to retrieve the referenced event resource

* Placer updates the request resource to indicate completion via PUT or an internal action

12.8.4.2 Benefits

* Uses the FHIR REST API for managing the workflow

* No need to use subscription or polling

* Both placer and fulfiller can track and change the state of the workflow

* Can use this approach for request other than just when requesting fulfillment (e.g. to request status change or other updates)

* There's an ability to negotiate fulfillment - i.e. the ability to say "no"

* Explicit acknowledgement that filler has received and agreed to act on the request

12.8.4.3 Limitations

* Additional complexity of using Task

* Need to keep the two Task states in sync

* Placer and fulfiller must each have a FHIR server and give the other side "write" access to create and update Task resources.

* Placer and filler may need to be able to communicate directly (i.e. know each other's respective endpoints)

* This could become unmanageable if there are a large (or dynamic) number of placers and fillers that need to communicate

* May not apply if there's a queue server

12.8.4.4 Usage Recommendations

* When following the state of the workflow is important

* When there is a need to know that a fulfiller has accepted to perform the request

* When there is a need to be able to change the status of the workflow by either the placer or the fulfiller

* When there are multiple steps in fulfilling the request

* When there is no need for polling or subscription infrastructure

* When us of the FHIR REST API is preferred to any other method of communications

* When the intended fulfiller is known in advance

* When messaging-like functionality is needed, but a messaging infrastructure is not available

12.8.4.5 Usage Examples

Outpatient referral

A patient is seen by her Primary Care Provider (PCP) for shortness of breath and the provider decides to refer her to a cardiologist. Because of the referral, the provider's system creates ReferralRequest resource and POSTs a Task resource to a cardiologist's practice system, requesting a referral. This places the referral request on a referral coordinator's work list to review. Upon accepting the referral, the Task is updated as accepted and a new sub-Task, referencing both the ReferralRequest and the Task created by the provider's system, is created on the provider's system. The patient gets an appointment with the cardiologist and the sub-Task is updated with that information, allowing the provider's system to inform the provider of the change.

The patient misplaces the information about the appointment and misses the visit with the cardiologist. The cardiologist's system updates the sub-Task to indicate the no-show and the PCP is alerted of the missed appointment. The provider's staff gets hold of the patient and schedules another appointment for her with the cardiologist. Because of the cardiologist's system scheduling the appointment, the sub-Task is updated with that information. The patient is seen and the sub-Task is updated accordingly. The cardiologist document the findings about the patient and updates the sub-Task as complete, adding references to the resources representing the findings. The PCP is alerted of the updates and marks the referral as complete, which updates the original Task and the ReferralRequest as complete as well.

12.8.5 Option J: Messaging Task from placer to fulfiller

TODO: needs more details

12.8.5.1 Steps

* Placer sends message to filler with a MessageHeader, where the "data" element points to the Task resource, also contained in the message. The message may or may not contain any other relevant resources (e.g. the actual request resource), or an "event" code saying "please fulfill"

* Filler system sends a response containing the same Task resource, indicating receipt of the message and, optionally, an indication of their intention to fulfill the request

* Filler system may send incremental messages to the placer showing progress (e.g. specimen collected, preliminary results, final results) by including an updated Task resource

* Placer system may also send messages to the fulfiller containing the Task resource and updating the state of the workflow

12.8.5.2 Benefits

* Reduced number of communications

* All relevant data can be sent in one package

* Responses can be asynchronous and content may be routed

* There's an ability to negotiate fulfillment - i.e. the ability to say "no"

* Can request things other than just fulfillment (e.g. please suspend)

* Explicit acknowledgement that filler has received and agreed to act on the request (though no need for the placer to check)

12.8.5.3 Limitations

* Messaging is "heavy"

* Need to negotiate what allowed responses are and what data can be present in request and response messages

* Additional complexity of using Task

* Need message delivery infrastructure in place

12.8.5.4 Usage Recommendations

Appropriate when existing messaging infrastructure can be used (e.g. HL7 over HTTP, v2 LTP, MLTP, WSI Web Services, Direct, VISA, REST, etc.), and a need to stay consistent with that architecture.

12.8.5.5 Usage examples

12.8.6 Option K: Service request referencing Task from placer to fulfiller

TODO: This scenario needs work - there's not a lot of experience using FHIR services to manage the fulfillment process

12.8.6.1 Steps

* Placer creates a request resource on their own system or a queue server

* Placer may create a Task resource on their own system or a queue server

* Placer invokes a service on the filler system saying "please fulfill this order", including the content or a reference to the request resource and any other relevant data

* Filler system responds (synchronously if using HTTP, but may be asynchronous if using SOAP or other transport mechanisms) with conformation of receipt and, optionally indication of intention to fulfill and/or results

12.8.6.2 Benefits

* ???

12.8.6.3 Limitations

* ???

12.8.6.4 Usage Recommendations

TBD

12.8.6.5 Usage examples

12.9 Usage Examples for Workflow Patterns

URL: https://hl7.org/fhir/STU3/workflow-examples.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

FHIR Infrastructure Work Group | Maturity Level: 2 | Ballot Status: Trial Use

12.9 Usage Examples for Workflow Patterns

Future home for examples

12.10 Resource Appointment - Content

URL: https://hl7.org/fhir/STU3/appointment.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

Patient Administration Work Group | Maturity Level: 3 | Trial Use | Compartments: Device, Patient, Practitioner, RelatedPerson

A booking of a healthcare event among patient(s), practitioner(s), related person(s) and/or device(s) for a specific date/time. This may result in one or more Encounter(s).

12.10.1 Scope and Usage

Appointment resources are used to provide information about a planned meeting that may be in the future or past. The resource only describes a single meeting, a series of repeating visits would require multiple appointment resources to be created for each instance. Examples include a scheduled surgery, a follow-up for a clinical visit, a scheduled conference call between clinicians to discuss a case, the reservation of a piece of diagnostic equipment for a particular use, etc. The visit scheduled by an appointment may be in person or remote (by phone, video conference, etc.) All that matters is that the time and usage of one or more individuals, locations and/or pieces of equipment is being fully or partially reserved for a designated period of time.

This definition takes the concepts of appointments in a clinical setting and also extends them to be relevant in the community healthcare space, and to ease exposure to other appointment / calendar standards widely used outside of healthcare.

12.10.1.1 The basic workflow to create an appointment

Discovery/Addressing

Before an appointment can be made, the address/endpoint details of the resource that we want to schedule an appointment with must be determined. This is often based on the healthcare Service Type and any formatting information which indicates how to make the request. This is typically handled via the Schedule resource.

Checking Availability on the Schedule (optional)

This optional step permits the checking of any existing available times (slot resources associated with a selected schedule) that can be booked against. Just because a time is indicated as available doesn't guarantee that an appointment can be made. The booking system that is going to process the request may make other qualifying decisions to determine if the appointment can be made, such as permissions, assessments, availability of other resources, etc.

This step is optional, as the creation of the appointment is never a guaranteed action. But by performing this availability check, you can increase the chances of making a successful booking.

Making the Appointment Request

When an appointment is required, a requester creates new Appointment resource with the Appointment.status="proposed". All included participants (optional or mandatory) should have the status="needs-action" to allow filtering and displaying appointments to user-participants for accepting or rejecting new and updated requests. Based on internal system business rules, certain statuses may be automatically updated, for example: "reject because the requested participant is on vacation" or "this type of user is not allowed to request those specific appointments".

Replying to the request

The reply process is simply performed by the person/system handing the requests, updating the participant statuses on the appointment as needed. If there are multiple systems involved, then these will create AppointmentResponse entries with the desired statuses.

Once all participants have their participation status created/updated (and the main system marking the appointment participant records with the AppointmentResponse statuses) then the overall status of the Appointment is updated.

Checking the overall status (Requester)

The requester (organizer) of the appointment checks for the overall status of the appointment (and appointment responses, where applicable) using FHIR pub-sub techniques.

Where the participant statuses indicate that a re-scheduling is required, then the process may start again, with other systems replying to a new set of times.

12.10.1.2 There are 2 typical workflows that occur with appointments

Outlook Style - Community

These types of requests are typically handled by selecting a specific time from a list of available slots, then making the request for that timeslot.

Hospital Scheduling - Clinical

Clinical scheduling is often far more complex in its requirements and processing. Often this involves checking multiple availabilities across multiple systems and timing with other internal systems, not just those exposed by the Slot resources.

Consideration should be given to situations where scheduling needs to be handled in more of a queue-like process.

Note: This type of clinical appointment scheduling has not been specifically covered with this definition of the Appointment resource (and other related resources), however if you would like to contribute to the modification of this resource to cover these use cases, please contact the HL7 Patient Administration work-group.

12.10.2 Boundaries and Relationships

12.10.2.1 Appointment Request/Response Pattern

When using a request-response style of appointment this is done using Appointment and AppointmentResponse resources. The request is made in the form of an Appointment with a proposed or pending status, and the list of actors with a participation status of "needs-action".

Participants in the appointment respond with their acceptance (or not) to the appointment by creating AppointmentResponse resources. Once all the participants have replied, then the Appointment resource is able to be updated with an overall status which collates the results of all the participants and presents the approved details of the appointment.

The participant type property can be used to represent a specific role that a practitioner is required to perform for the appointment. This could be specified without an actor when the actual practitioner is not known, and will be filled in closer to the scheduled time. This property must be the same between the Appointment-participant and the AppointmentResponse so that the appropriate values can be allocated. If you need multiple actors of a specific type, then multiple participants with that type value are included on the appointment.

12.10.2.2 Appointment Statuses and Encounters

Appointments can be considered as Administrative only, and the Encounter is expected to have clinical implications.

In general, it is expected that appointments will result in the creation of an Encounter. The encounter is typically created when the service starts, not when the patient arrives. When the patient arrives, an appointment can be marked with a status of Arrived.

In an Emergency Room context, the appointment Resource is probably not appropriate to be used. In these cases an Encounter should be created.

The Appointment request pattern used is different from the order-response pattern used elsewhere in FHIR. This is due to the close relationship to the iCAL standard. Many non-clinical systems use generic non-health appointment systems which implement this standard, and the desire to integrate with the consumer who has no access to health based software is highly desirable. The mappings to the iCAL standard have been provided to guide implementation of gateways between FHIR servers and iCAL systems.

12.10.2.3 Appointment Locations and Participation

The location of the appointment is to be defined by using a participant that references a Location or HealthcareService resource. This permits the location to also have its availability checked via a schedule and any conflicts more easily managed.

This resource is referenced by AppointmentResponse, CarePlan, ClinicalImpression and Encounter

12.10.3 Resource Content

* Structure
* UML
* XML
* JSON
* Turtle
* R2 Diff
* All

Structure

Name | Flags | Card. | Type | Description & Constraints
Appointment | I | | DomainResource | A booking of a healthcare event among patient(s), practitioner(s), related person(s) and/or device(s) for a specific date/time. This may result in one or more Encounter(s) + Only proposed or cancelled appointments can be missing start/end dates + Either start and end are specified, or neither Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | ? | 0..* | Identifier | External Ids for this item
status | ?!? | 1..1 | code | proposed | pending | booked | arrived | fulfilled | cancelled | noshow | entered-in-error AppointmentStatus (Required)
serviceCategory | ? | 0..1 | CodeableConcept | A broad categorisation of the service that is to be performed during this appointment ServiceCategory (Example)
serviceType | ? | 0..* | CodeableConcept | The specific service that is to be performed during this appointment ServiceType (Example)
specialty | ? | 0..* | CodeableConcept | The specialty of a practitioner that would be required to perform the service requested in this appointment Practice Setting Code Value Set (Preferred)
appointmentType | ? | 0..1 | CodeableConcept | The style of appointment or patient that has been booked in the slot (not service type) v2 Appointment reason codes (Preferred)
reason | ? | 0..* | CodeableConcept | Reason this appointment is scheduled Encounter Reason Codes (Preferred)
indication | | 0..* | Reference(Condition | Procedure) | Reason the appointment is to takes place (resource)
priority | | 0..1 | unsignedInt | Used to make informed decisions if needing to re-prioritize
description | | 0..1 | string | Shown on a subject line in a meeting request, or appointment list
supportingInformation | | 0..* | Reference(Any) | Additional information to support the appointment
start | ? | 0..1 | instant | When appointment is to take place
end | ? | 0..1 | instant | When appointment is to conclude
minutesDuration | | 0..1 | positiveInt | Can be less than start/end (e.g. estimate)
slot | | 0..* | Reference(Slot) | The slots that this appointment is filling
created | | 0..1 | dateTime | The date that this appointment was initially created
comment | | 0..1 | string | Additional comments
incomingReferral | | 0..* | Reference(ReferralRequest) | The ReferralRequest provided as information to allocate to the Encounter
participant | I | 1..* | BackboneElement | Participants involved in appointment + Either the type or actor on the participant SHALL be specified
type | ? | 0..* | CodeableConcept | Role of participant in the appointment ParticipantType (Extensible)
actor | ? | 0..1 | Reference(Patient | Practitioner | RelatedPerson | Device | HealthcareService | Location) | Person, Location/HealthcareService or Device
required | ? | 0..1 | code | required | optional | information-only ParticipantRequired (Required)
status | | 1..1 | code | accepted | declined | tentative | needs-action ParticipationStatus (Required)
requestedPeriod | | 0..* | Period | Potential date/time interval(s) requested to allocate the appointment within
Documentation for this format

UML Diagram (Legend)

XML Template

```
<Appointment xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier External Ids for this item --></identifier>
 <status value="[code]"/><!-- 1..1 proposed | pending | booked | arrived | fulfilled | cancelled | noshow | entered-in-
    error -->
 <serviceCategory><!-- 0..1 CodeableConcept A broad categorisation of the service that is to be performed during this
    appointment --></serviceCategory>
 <serviceType><!-- 0..* CodeableConcept The specific service that is to be performed during this appointment
    --></serviceType>
 <specialty><!-- 0..* CodeableConcept The specialty of a practitioner that would be required to perform the service
    requested in this appointment --></specialty>
 <appointmentType><!-- 0..1 CodeableConcept The style of appointment or patient that has been booked in the slot (not
    service type) --></appointmentType>
 <reason><!-- 0..* CodeableConcept Reason this appointment is scheduled --></reason>
 <indication><!-- 0..* Reference(Condition|Procedure) Reason the appointment is to takes place (resource)
    --></indication>
 <priority value="[unsignedInt]"/><!-- 0..1 Used to make informed decisions if needing to re-prioritize -->
 <description value="[string]"/><!-- 0..1 Shown on a subject line in a meeting request, or appointment list -->
 <supportingInformation><!-- 0..* Reference(Any) Additional information to support the appointment
    --></supportingInformation>
 <start value="[instant]"/><!-- 0..1 When appointment is to take place -->
 <end value="[instant]"/><!-- 0..1 When appointment is to conclude -->
 <minutesDuration value="[positiveInt]"/><!-- 0..1 Can be less than start/end (e.g. estimate) -->
 <slot><!-- 0..* Reference(Slot) The slots that this appointment is filling --></slot>
 <created value="[dateTime]"/><!-- 0..1 The date that this appointment was initially created -->
 <comment value="[string]"/><!-- 0..1 Additional comments -->
 <incomingReferral><!-- 0..* Reference(ReferralRequest) The ReferralRequest provided as information to allocate to the
    Encounter --></incomingReferral>
 <participant>  <!-- 1..* Participants involved in appointment -->
  <type><!-- 0..* CodeableConcept Role of participant in the appointment --></type>
  <actor><!-- 0..1 Reference(Patient|Practitioner|RelatedPerson|Device|
    HealthcareService|Location) Person, Location/HealthcareService or Device --></actor>
  <required value="[code]"/><!-- 0..1 required | optional | information-only -->
  <status value="[code]"/><!-- 1..1 accepted | declined | tentative | needs-action -->
 </participant>
 <requestedPeriod><!-- 0..* Period Potential date/time interval(s) requested to allocate the appointment within
    --></requestedPeriod>
</Appointment>
```

JSON Template

```
{
  "resourceType" : "Appointment",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
```

```
  "identifier" : [{ Identifier }], // External Ids for this item
  "status" : "<code>", // R!  proposed | pending | booked | arrived | fulfilled | cancelled | noshow | entered-in-error
  "serviceCategory" : { CodeableConcept }, // A broad categorisation of the service that is to be performed during this
    appointment
  "serviceType" : [{ CodeableConcept }], // The specific service that is to be performed during this appointment
  "specialty" : [{ CodeableConcept }], // The specialty of a practitioner that would be required to perform the service
    requested in this appointment
  "appointmentType" : { CodeableConcept }, // The style of appointment or patient that has been booked in the slot (not
    service type)
  "reason" : [{ CodeableConcept }], // Reason this appointment is scheduled
  "indication" : [{ Reference(Condition|Procedure) }], // Reason the appointment is to takes place (resource)
  "priority" : "<unsignedInt>", // Used to make informed decisions if needing to re-prioritize
  "description" : "<string>", // Shown on a subject line in a meeting request, or appointment list
  "supportingInformation" : [{ Reference(Any) }], // Additional information to support the appointment
  "start" : "<instant>", // When appointment is to take place
  "end" : "<instant>", // When appointment is to conclude
  "minutesDuration" : "<positiveInt>", // Can be less than start/end (e.g. estimate)
  "slot" : [{ Reference(Slot) }], // The slots that this appointment is filling
  "created" : "<dateTime>", // The date that this appointment was initially created
  "comment" : "<string>", // Additional comments
  "incomingReferral" : [{ Reference(ReferralRequest) }], // The ReferralRequest provided as information to allocate to
    the Encounter
  "participant" : [{ // R!  Participants involved in appointment
    "type" : [{ CodeableConcept }], // Role of participant in the appointment
    "actor" : { Reference(Patient|Practitioner|RelatedPerson|Device|
    HealthcareService|Location) }, // Person, Location/HealthcareService or Device
    "required" : "<code>", // required | optional | information-only
    "status" : "<code>" // R!  accepted | declined | tentative | needs-action
  }],
  "requestedPeriod" : [{ Period }] // Potential date/time interval(s) requested to allocate the appointment within
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:Appointment;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:Appointment.identifier [ Identifier ], ... ; # 0..* External Ids for this item
  fhir:Appointment.status [ code ]; # 1..1 proposed | pending | booked | arrived | fulfilled | cancelled | noshow |
    entered-in-error
  fhir:Appointment.serviceCategory [ CodeableConcept ]; # 0..1 A broad categorisation of the service that is to be
    performed during this appointment
  fhir:Appointment.serviceType [ CodeableConcept ], ... ; # 0..* The specific service that is to be performed during
    this appointment
  fhir:Appointment.specialty [ CodeableConcept ], ... ; # 0..* The specialty of a practitioner that would be required to
     perform the service requested in this appointment
  fhir:Appointment.appointmentType [ CodeableConcept ]; # 0..1 The style of appointment or patient that has been booked
    in the slot (not service type)
  fhir:Appointment.reason [ CodeableConcept ], ... ; # 0..* Reason this appointment is scheduled
  fhir:Appointment.indication [ Reference(Condition|Procedure) ], ... ; # 0..* Reason the appointment is to takes place
    (resource)
  fhir:Appointment.priority [ unsignedInt ]; # 0..1 Used to make informed decisions if needing to re-prioritize
  fhir:Appointment.description [ string ]; # 0..1 Shown on a subject line in a meeting request, or appointment list
  fhir:Appointment.supportingInformation [ Reference(Any) ], ... ; # 0..* Additional information to support the
    appointment
  fhir:Appointment.start [ instant ]; # 0..1 When appointment is to take place
  fhir:Appointment.end [ instant ]; # 0..1 When appointment is to conclude
  fhir:Appointment.minutesDuration [ positiveInt ]; # 0..1 Can be less than start/end (e.g. estimate)
  fhir:Appointment.slot [ Reference(Slot) ], ... ; # 0..* The slots that this appointment is filling
  fhir:Appointment.created [ dateTime ]; # 0..1 The date that this appointment was initially created
  fhir:Appointment.comment [ string ]; # 0..1 Additional comments
  fhir:Appointment.incomingReferral [ Reference(ReferralRequest) ], ... ; # 0..* The ReferralRequest provided as
    information to allocate to the Encounter
  fhir:Appointment.participant [ # 1..* Participants involved in appointment
    fhir:Appointment.participant.type [ CodeableConcept ], ... ; # 0..* Role of participant in the appointment
    fhir:Appointment.participant.actor [ Reference(Patient|Practitioner|RelatedPerson|Device|HealthcareService|Location)
     ]; # 0..1 Person, Location/HealthcareService or Device
    fhir:Appointment.participant.required [ code ]; # 0..1 required | optional | information-only
    fhir:Appointment.participant.status [ code ]; # 1..1 accepted | declined | tentative | needs-action
  ], ...;
  fhir:Appointment.requestedPeriod [ Period ], ... ; # 0..* Potential date/time interval(s) requested to allocate the
    appointment within
]
```

Changes since DSTU2

Appointment

Appointment.serviceCategory | Added Element

Appointment.serviceType | Renamed from type to serviceType Max Cardinality changed from 1 to *

Appointment.specialty | Added Element

Appointment.appointmentType | Added Element

Appointment.reason | Max Cardinality changed from 1 to * Remove Binding http://hl7.org/fhir/ValueSet/encounter-reason (required)

Appointment.indication | Added Element

Appointment.supportingInformation | Added Element

Appointment.created | Added Element
Appointment.incomingReferral | Added Element
Appointment.participant.type | Change binding strength from required to extensible
Appointment.requestedPeriod | Added Element

See the Full Difference for further information

This analysis is available as XML or JSON.

See R2 <--> R3 Conversion Maps (status = 3 tests that all execute ok. All tests pass round-trip testing and all r3 resources are valid.).

Structure

Name | Flags | Card. | Type | Description & Constraints
Appointment | I | | DomainResource | A booking of a healthcare event among patient(s), practitioner(s), related person(s) and/or device(s) for a specific date/time. This may result in one or more Encounter(s) + Only proposed or cancelled appointments can be missing start/end dates + Either start and end are specified, or neither Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | ? | 0..* | Identifier | External Ids for this item
status | ?!? | 1..1 | code | proposed | pending | booked | arrived | fulfilled | cancelled | noshow | entered-in-error AppointmentStatus (Required)
serviceCategory | ? | 0..1 | CodeableConcept | A broad categorisation of the service that is to be performed during this appointment ServiceCategory (Example)
serviceType | ? | 0..* | CodeableConcept | The specific service that is to be performed during this appointment ServiceType (Example)
specialty | ? | 0..* | CodeableConcept | The specialty of a practitioner that would be required to perform the service requested in this appointment Practice Setting Code Value Set (Preferred)
appointmentType | ? | 0..1 | CodeableConcept | The style of appointment or patient that has been booked in the slot (not service type) v2 Appointment reason codes (Preferred)
reason | ? | 0..* | CodeableConcept | Reason this appointment is scheduled Encounter Reason Codes (Preferred)
indication | | 0..* | Reference(Condition | Procedure) | Reason the appointment is to takes place (resource)
priority | | 0..1 | unsignedInt | Used to make informed decisions if needing to re-prioritize
description | | 0..1 | string | Shown on a subject line in a meeting request, or appointment list
supportingInformation | | 0..* | Reference(Any) | Additional information to support the appointment
start | ? | 0..1 | instant | When appointment is to take place
end | ? | 0..1 | instant | When appointment is to conclude
minutesDuration | | 0..1 | positiveInt | Can be less than start/end (e.g. estimate)
slot | | 0..* | Reference(Slot) | The slots that this appointment is filling
created | | 0..1 | dateTime | The date that this appointment was initially created
comment | | 0..1 | string | Additional comments
incomingReferral | | 0..* | Reference(ReferralRequest) | The ReferralRequest provided as information to allocate to the Encounter
participant | I | 1..* | BackboneElement | Participants involved in appointment + Either the type or actor on the participant SHALL be specified
type | ? | 0..* | CodeableConcept | Role of participant in the appointment ParticipantType (Extensible)
actor | ? | 0..1 | Reference(Patient | Practitioner | RelatedPerson | Device | HealthcareService | Location) | Person, Location/HealthcareService or Device
required | ? | 0..1 | code | required | optional | information-only ParticipantRequired (Required)
status | | 1..1 | code | accepted | declined | tentative | needs-action ParticipationStatus (Required)
requestedPeriod | | 0..* | Period | Potential date/time interval(s) requested to allocate the appointment within
Documentation for this format

UML Diagram (Legend)

XML Template

```
<Appointment xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier External Ids for this item --></identifier>
 <status value="[code]"/><!-- 1..1 proposed | pending | booked | arrived | fulfilled | cancelled | noshow | entered-in-
    error -->
 <serviceCategory><!-- 0..1 CodeableConcept A broad categorisation of the service that is to be performed during this
    appointment --></serviceCategory>
 <serviceType><!-- 0..* CodeableConcept The specific service that is to be performed during this appointment
    --></serviceType>
 <specialty><!-- 0..* CodeableConcept The specialty of a practitioner that would be required to perform the service
    requested in this appointment --></specialty>
 <appointmentType><!-- 0..1 CodeableConcept The style of appointment or patient that has been booked in the slot (not
    service type) --></appointmentType>
 <reason><!-- 0..* CodeableConcept Reason this appointment is scheduled --></reason>
 <indication><!-- 0..* Reference(Condition|Procedure) Reason the appointment is to takes place (resource)
    --></indication>
 <priority value="[unsignedInt]"/><!-- 0..1 Used to make informed decisions if needing to re-prioritize -->
 <description value="[string]"/><!-- 0..1 Shown on a subject line in a meeting request, or appointment list -->
 <supportingInformation><!-- 0..* Reference(Any) Additional information to support the appointment
    --></supportingInformation>
 <start value="[instant]"/><!-- 0..1 When appointment is to take place -->
 <end value="[instant]"/><!-- 0..1 When appointment is to conclude -->
 <minutesDuration value="[positiveInt]"/><!-- 0..1 Can be less than start/end (e.g. estimate) -->
 <slot><!-- 0..* Reference(Slot) The slots that this appointment is filling --></slot>
 <created value="[dateTime]"/><!-- 0..1 The date that this appointment was initially created -->
 <comment value="[string]"/><!-- 0..1 Additional comments -->
 <incomingReferral><!-- 0..* Reference(ReferralRequest) The ReferralRequest provided as information to allocate to the
    Encounter --></incomingReferral>
 <participant>  <!-- 1..* Participants involved in appointment -->
  <type><!-- 0..* CodeableConcept Role of participant in the appointment --></type>
  <actor><!-- 0..1 Reference(Patient|Practitioner|RelatedPerson|Device|
```

```
    HealthcareService|Location) Person, Location/HealthcareService or Device --></actor>
  <required value="[code]"/><!-- 0..1 required | optional | information-only -->
  <status value="[code]"/><!-- 1..1 accepted | declined | tentative | needs-action -->
 </participant>
 <requestedPeriod><!-- 0..* Period Potential date/time interval(s) requested to allocate the appointment within
    --></requestedPeriod>
</Appointment>
```

JSON Template

```
{
  "resourceType" : "Appointment",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // External Ids for this item
  "status" : "<code>", // R!  proposed | pending | booked | arrived | fulfilled | cancelled | noshow | entered-in-error
  "serviceCategory" : { CodeableConcept }, // A broad categorisation of the service that is to be performed during this
    appointment
  "serviceType" : [{ CodeableConcept }], // The specific service that is to be performed during this appointment
  "specialty" : [{ CodeableConcept }], // The specialty of a practitioner that would be required to perform the service
    requested in this appointment
  "appointmentType" : { CodeableConcept }, // The style of appointment or patient that has been booked in the slot (not
    service type)
  "reason" : [{ CodeableConcept }], // Reason this appointment is scheduled
  "indication" : [{ Reference(Condition|Procedure) }], // Reason the appointment is to takes place (resource)
  "priority" : "<unsignedInt>", // Used to make informed decisions if needing to re-prioritize
  "description" : "<string>", // Shown on a subject line in a meeting request, or appointment list
  "supportingInformation" : [{ Reference(Any) }], // Additional information to support the appointment
  "start" : "<instant>", // When appointment is to take place
  "end" : "<instant>", // When appointment is to conclude
  "minutesDuration" : "<positiveInt>", // Can be less than start/end (e.g. estimate)
  "slot" : [{ Reference(Slot) }], // The slots that this appointment is filling
  "created" : "<dateTime>", // The date that this appointment was initially created
  "comment" : "<string>", // Additional comments
  "incomingReferral" : [{ Reference(ReferralRequest) }], // The ReferralRequest provided as information to allocate to
    the Encounter
  "participant" : [{ // R!  Participants involved in appointment
    "type" : [{ CodeableConcept }], // Role of participant in the appointment
    "actor" : { Reference(Patient|Practitioner|RelatedPerson|Device|
    HealthcareService|Location) }, // Person, Location/HealthcareService or Device
    "required" : "<code>", // required | optional | information-only
    "status" : "<code>" // R!  accepted | declined | tentative | needs-action
  }],
  "requestedPeriod" : [{ Period }] // Potential date/time interval(s) requested to allocate the appointment within
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:Appointment;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:Appointment.identifier [ Identifier ], ... ; # 0..* External Ids for this item
  fhir:Appointment.status [ code ]; # 1..1 proposed | pending | booked | arrived | fulfilled | cancelled | noshow |
    entered-in-error
  fhir:Appointment.serviceCategory [ CodeableConcept ]; # 0..1 A broad categorisation of the service that is to be
    performed during this appointment
  fhir:Appointment.serviceType [ CodeableConcept ], ... ; # 0..* The specific service that is to be performed during
    this appointment
  fhir:Appointment.specialty [ CodeableConcept ], ... ; # 0..* The specialty of a practitioner that would be required to
     perform the service requested in this appointment
  fhir:Appointment.appointmentType [ CodeableConcept ]; # 0..1 The style of appointment or patient that has been booked
    in the slot (not service type)
  fhir:Appointment.reason [ CodeableConcept ], ... ; # 0..* Reason this appointment is scheduled
  fhir:Appointment.indication [ Reference(Condition|Procedure) ], ... ; # 0..* Reason the appointment is to takes place
    (resource)
  fhir:Appointment.priority [ unsignedInt ]; # 0..1 Used to make informed decisions if needing to re-prioritize
  fhir:Appointment.description [ string ]; # 0..1 Shown on a subject line in a meeting request, or appointment list
  fhir:Appointment.supportingInformation [ Reference(Any) ], ... ; # 0..* Additional information to support the
    appointment
  fhir:Appointment.start [ instant ]; # 0..1 When appointment is to take place
  fhir:Appointment.end [ instant ]; # 0..1 When appointment is to conclude
  fhir:Appointment.minutesDuration [ positiveInt ]; # 0..1 Can be less than start/end (e.g. estimate)
  fhir:Appointment.slot [ Reference(Slot) ], ... ; # 0..* The slots that this appointment is filling
  fhir:Appointment.created [ dateTime ]; # 0..1 The date that this appointment was initially created
  fhir:Appointment.comment [ string ]; # 0..1 Additional comments
  fhir:Appointment.incomingReferral [ Reference(ReferralRequest) ], ... ; # 0..* The ReferralRequest provided as
    information to allocate to the Encounter
  fhir:Appointment.participant [ # 1..* Participants involved in appointment
    fhir:Appointment.participant.type [ CodeableConcept ], ... ; # 0..* Role of participant in the appointment
    fhir:Appointment.participant.actor [ Reference(Patient|Practitioner|RelatedPerson|Device|HealthcareService|Location)
     ]; # 0..1 Person, Location/HealthcareService or Device
    fhir:Appointment.participant.required [ code ]; # 0..1 required | optional | information-only
    fhir:Appointment.participant.status [ code ]; # 1..1 accepted | declined | tentative | needs-action
  ], ...;
  fhir:Appointment.requestedPeriod [ Period ], ... ; # 0..* Potential date/time interval(s) requested to allocate the
    appointment within
```

]

Changes since DSTU2

Appointment
Appointment.serviceCategory | Added Element
Appointment.serviceType | Renamed from type to serviceType Max Cardinality changed from 1 to *
Appointment.specialty | Added Element
Appointment.appointmentType | Added Element
Appointment.reason | Max Cardinality changed from 1 to * Remove Binding http://hl7.org/fhir/ValueSet/encounter-reason (required)
Appointment.indication | Added Element
Appointment.supportingInformation | Added Element
Appointment.created | Added Element
Appointment.incomingReferral | Added Element
Appointment.participant.type | Change binding strength from required to extensible
Appointment.requestedPeriod | Added Element

See the Full Difference for further information

This analysis is available as XML or JSON.

See R2 <--> R3 Conversion Maps (status = 3 tests that all execute ok. All tests pass round-trip testing and all r3 resources are valid.).

Alternate definitions: Master Definition (XML, JSON), XML Schema/Schematron (for) + JSON Schema, ShEx (for Turtle)

12.10.3.1 Terminology Bindings

Path | Definition | Type | Reference
Appointment.status | The free/busy status of an appointment. | Required | AppointmentStatus
Appointment.serviceCategory | | Example | ServiceCategory
Appointment.serviceType | | Example | ServiceType
Appointment.specialty | | Preferred | Practice Setting Code Value Set
Appointment.appointmentType | | Preferred | v2 Appointment reason codes
Appointment.reason | The Reason for the appointment to take place. | Preferred | Encounter Reason Codes
Appointment.participant.type | Role of participant in encounter. | Extensible | ParticipantType
Appointment.participant.required | Is the Participant required to attend the appointment. | Required | ParticipantRequired
Appointment.participant.status | The Participation status of an appointment. | Required | ParticipationStatus

12.10.3.2 Constraints

* app-1: On Appointment.participant: Either the type or actor on the participant SHALL be specified (expression on Appointment.participant: type.exists() or actor.exists())

* app-2: Either start and end are specified, or neither (expression : start.empty() xor end.exists())

* app-3: Only proposed or cancelled appointments can be missing start/end dates (expression : (start.exists() and end.exists()) or (status in ('proposed' | 'cancelled')))

12.10.4 Typical Status Transition Examples:

12.10.4.1 Typical flow of statuses for an appointment:

Activity Description | Slot | Appointment | Appointment Response | Encounter
The schedule is created/published (Role: Scheduler) | freeBusyType = FREE
An appointment request is created after locating an available slot (Role: Requester) | | status = pending participant.status = needs-action
The appointment request is processed and the slot status updated (Role: Scheduler) | freeBusyType = BUSY-TENTATIVE
The appointment is accepted as described – by all participants (Role: Participant(s)) | | | participantStatus = accepted
The appointment is confirmed as accepted by all participants (Role: Scheduler) | freeBusyType = BUSY | status = booked participant.status = accepted
Optional: Preparation for the appointment begins – could be preparing a room for the appointment, etc. (Role: Participants/Admin) | | | status = planned (optional) location.status = planned
The patient arrives for the appointment, often sitting in a waiting room (Role: Admin) | | status = arrived | | status = arrived location.status = present
The practitioner and the patient meet and the provision of the service begins (Role: Scheduler/Participant(s)/Admin) | | status = fulfilled | | status = in-progress
The encounter concludes (Role: Scheduler/Participant(s)/Admin) | | | | status = finished

12.10.4.2 Flow for the rejection/cancellation of an appointment

Activity Description | Slot | Appointment | Appointment Response
The schedule is created/published (Role: Scheduler) | freeBusyType = FREE
An appointment request is created (Role: Requester) | | status = pending participant.status = needs-action
The appointment request is processed and the slot status updated (Role: Scheduler) | freeBusyType = BUSY-TENTATIVE
Participant declines the Appointment (Role: Participant) | | | participantStatus = declined
The appointment is cancelled (Role: Scheduler) | freeBusyType = FREE | status = cancelled participant.status = declined

12.10.4.3 Flow for re-negotiation:

Activity Description | Slot | Appointment | Appointment Response
The schedule is created/published (Role: Scheduler) | freeBusyType = FREE

An appointment is requested (e.g.,with Brian and Peter) (Role: Requester) | | status = proposed participant(Brian).status = needs-action participant(Peter).status = needs-action
The schedule is updated to inform others of interest in the slot (Role: Scheduler) | freeBusyType = BUSY-TENTATIVE
Brian accepts the appointment (Role: Participant-Brian) | | | (Brian).participantStatus = accepted
Appointment is updated with Brian's status (Role: Scheduler) | | status = pending participant(Brian).status = accepted
Peter suggests a new time (Role: Participant-Peter) | | | (Peter).participantStatus = tentative (with new time)
Appointment is updated with new time, and indicates that action is needed by both participants (Role: Scheduler) | | (new time details updated) participant(Brian).status = needs-action participant(Peter).status = needs-action
Brian accepts the appointment (Role: Participant-Brian) | | | (Brian).participantStatus = accepted
Appointment updated (Role: Scheduler) | | participant(Brian).status = accepted
(Role: Participant-Peter) | | | (Peter).participantStatus = accepted
Appointment updated (Role: Scheduler) | freeBusyType = BUSY | status = booked participant(Peter).status = accepted

12.10.4.4 Flow for a patient no-show:

Activity Description | Slot | Appointment | Appointment Response | Encounter
(from typical status flow) | freeBusyType = BUSY | status = booked participant.status = accepted
Appointment is updated as a noshow (Role: Scheduler/Admin) | | status = noshow | | (no encounter created)

12.10.5 Notes:

* Placer/Filler (HL7 v2)

The appointment information is effectively the same between the filler and placer, and given the nature of the FHIR resource, there is only a single resource for both purposes. The placer is the actor that performs the PUT or POST operation on the resource, and the filler is the actor that receives these resource messages and processes the information and makes a decision if the appointment can be used.

* Interaction with other Standards

The strong desire is that implementers of this resource should consider providing this resource in the iCalendar format as an alternative representation. Many 3rd party applications and component providers have parsers and user interface controls to display this information. This may lower the entry point to integrate outside the health-care specific applications, and into the consumer space. This would permit the easier creation of a mobile application that creates appointments in the devices native calendar. The iCalendar specification can be found at http://www.ietf.org/rfc/rfc2445.txt .

STU Note: Implementer feedback is sought on the values for Appointment.priority and how interoperable they are. Using an extension to record a CodeableConcept for named values may be tested at a future Connectathon.

Feedback here .

12.10.6 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
actor | reference | Any one of the individuals participating in the appointment | Appointment.participant.actor (Practitioner, Device, Patient, HealthcareService, RelatedPerson, Location)
appointment-type | token | The style of appointment or patient that has been booked in the slot (not service type) | Appointment.appointmentType
date | date | Appointment date/time. | Appointment.start
identifier | token | An Identifier of the Appointment | Appointment.identifier
incomingreferral | reference | The ReferralRequest provided as information to allocate to the Encounter | Appointment.incomingReferral (ReferralRequest)
location | reference | This location is listed in the participants of the appointment | Appointment.participant.actor (Location)
part-status | token | The Participation status of the subject, or other participant on the appointment. Can be used to locate participants that have not responded to meeting requests. | Appointment.participant.status
patient | reference | One of the individuals of the appointment is this patient | Appointment.participant.actor (Patient)
practitioner | reference | One of the individuals of the appointment is this practitioner | Appointment.participant.actor (Practitioner)
service-type | token | The specific service that is to be performed during this appointment | Appointment.serviceType
status | token | The overall status of the appointment | Appointment.status

12.10.7 Resource Appointment - Examples

URL: https://hl7.org/fhir/STU3/appointment-examples.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

12.10.7 Resource Appointment - Examples

Patient Administration Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Device, Patient, Practitioner, RelatedPerson

Example List:

General Person Example | example | XML | JSON | Turtle
Appointment between 2 doctors to discuss MRI results | 2docs | XML | JSON | Turtle
Appointment request | examplereq | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

12.10.8 Resource Appointment - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/appointment-definitions.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

12.10.8 Resource Appointment - Detailed Descriptions

Patient Administration Work Group | Maturity Level: 3 | Trial Use | Compartments: Device, Patient, Practitioner, RelatedPerson

Detailed Descriptions for the elements in the Appointment resource.

Appointment
Definition | A booking of a healthcare event among patient(s), practitioner(s), related person(s) and/or device(s) for a specific date/time. This may result in one or more Encounter(s).
Control | 1..1
Invariants | Defined on this element app-2: Either start and end are specified, or neither (expression : start.empty() xor end.exists(), xpath: ((exists(f:start) and exists(f:end)) or (not(exists(f:start)) and not(exists(f:end))))) app-3: Only proposed or cancelled appointments can be missing start/end dates (expression : (start.exists() and end.exists()) or (status in ('proposed' | 'cancelled')), xpath: ((exists(f:start) and exists(f:end)) or (f:status/@value='proposed') or (f:status/@value='cancelled')))
Appointment.identifier
Definition | This records identifiers associated with this appointment concern that are defined by business processes and/or used to refer to it when a direct URL reference to the resource itself is not appropriate (e.g. in CDA documents, or in written / printed documentation).
Note | This is a business identifer, not a resource identifier (see discussion)
Control | 0..*
Type | Identifier
Summary | true
Appointment.status
Definition | The overall status of the Appointment. Each of the participants has their own participation status which indicates their involvement in the process, however this status indicates the shared status.
Control | 1..1
Terminology Binding | AppointmentStatus (Required)
Type | code
Is Modifier | true
Summary | true
Comments | If the Appointment's status is "cancelled" then all participants are expected to have their calendars released for the appointment period, and as such any Slots that were marked as BUSY can be re-set to FREE. This element is labeled as a modifier because the status contains the code entered-in-error that mark the Appointment as not currently valid.
Appointment.serviceCategory
Definition | A broad categorisation of the service that is to be performed during this appointment.
Control | 0..1
Terminology Binding | ServiceCategory (Example)
Type | CodeableConcept
Summary | true
Appointment.serviceType
Definition | The specific service that is to be performed during this appointment.
Control | 0..*
Terminology Binding | ServiceType (Example)
Type | CodeableConcept
Summary | true
Appointment.specialty
Definition | The specialty of a practitioner that would be required to perform the service requested in this appointment.
Control | 0..*
Terminology Binding | Practice Setting Code Value Set (Preferred)
Type | CodeableConcept
Summary | true
Appointment.appointmentType
Definition | The style of appointment or patient that has been booked in the slot (not service type).
Control | 0..1
Terminology Binding | v2 Appointment reason codes (Preferred)
Type | CodeableConcept
Summary | true
Appointment.reason
Definition | The reason that this appointment is being scheduled. This is more clinical than administrative.
Control | 0..*
Terminology Binding | Encounter Reason Codes (Preferred)
Type | CodeableConcept
Summary | true
Appointment.indication
Definition | Reason the appointment has been scheduled to take place, as specified using information from another resource. When the patient arrives and the encounter begins it may be used as the admission diagnosis. The indication will typically be a Condition (with other resources

referenced in the evidence.detail), or a Procedure.
Control | 0..*
Type | Reference(Condition | Procedure)
Appointment.priority
Definition | The priority of the appointment. Can be used to make informed decisions if needing to re-prioritize appointments. (The iCal Standard specifies 0 as undefined, 1 as highest, 9 as lowest priority).
Control | 0..1
Type | unsignedInt
Comments | Seeking implementer feedback on this property and how interoperable it is. Using an extension to record a codeableconcept for named values may be tested at a future connectathon.
Appointment.description
Definition | The brief description of the appointment as would be shown on a subject line in a meeting request, or appointment list. Detailed or expanded information should be put in the comment field.
Control | 0..1
Type | string
Appointment.supportingInformation
Definition | Additional information to support the appointment provided when making the appointment.
Control | 0..*
Type | Reference(Any)
Appointment.start
Definition | Date/Time that the appointment is to take place.
Control | 0..1
Type | instant
Summary | true
Appointment.end
Definition | Date/Time that the appointment is to conclude.
Control | 0..1
Type | instant
Summary | true
Appointment.minutesDuration
Definition | Number of minutes that the appointment is to take. This can be less than the duration between the start and end times (where actual time of appointment is only an estimate or is a planned appointment request).
Control | 0..1
Type | positiveInt
Appointment.slot
Definition | The slots from the participants' schedules that will be filled by the appointment.
Control | 0..*
Type | Reference(Slot)
Appointment.created
Definition | The date that this appointment was initially created. This could be different to the meta.lastModified value on the initial entry, as this could have been before the resource was created on the FHIR server, and should remain unchanged over the lifespan of the appointment.
Control | 0..1
Type | dateTime
Comments | This property is required for many use cases where the age of an appointment is considered in processing workflows for scheduling and billing of appointments.
Appointment.comment
Definition | Additional comments about the appointment.
Control | 0..1
Type | string
Comments | Additional text to aid in facilitating the appointment. For instance, a comment might be, "patient should proceed immediately to infusion room upon arrival" Where this is a planned appointment and the start/end dates are not set then this field can be used to provide additional guidance on the details of the appointment request, including any restrictions on when to book it.
Appointment.incomingReferral
Definition | The referral request this appointment is allocated to assess (incoming referral).
Control | 0..*
Type | Reference(ReferralRequest)
Appointment.participant
Definition | List of participants involved in the appointment.
Control | 1..*
Invariants | Defined on this element app-1: Either the type or actor on the participant SHALL be specified (expression : type.exists() or actor.exists(), xpath: (exists(f:type) or exists(f:actor)))
Appointment.participant.type
Definition | Role of participant in the appointment.
Control | 0..*
Terminology Binding | ParticipantType (Extensible)
Type | CodeableConcept
Summary | true
Comments | The role of the participant can be used to declare what the actor will be doing in the scope of this appointment. If the actor is not specified, then it is expected that the actor will be filled in at a later stage of planning. This value SHALL be the same when creating an AppointmentResponse so that they can be matched, and subsequently update the Appointment.
Appointment.participant.actor
Definition | A Person, Location/HealthcareService or Device that is participating in the appointment.
Control | 0..1

Type | Reference(Patient | Practitioner | RelatedPerson | Device | HealthcareService | Location)
Summary | true
Appointment.participant.required
Definition | Is this participant required to be present at the meeting. This covers a use-case where 2 doctors need to meet to discuss the results for a specific patient, and the patient is not required to be present.
Control | 0..1
Terminology Binding | ParticipantRequired (Required)
Type | code
Summary | true
Appointment.participant.status
Definition | Participation status of the actor.
Control | 1..1
Terminology Binding | ParticipationStatus (Required)
Type | code
Appointment.requestedPeriod
Definition | A set of date ranges (potentially including times) that the appointment is preferred to be scheduled within. When using these values, the minutes duration should be provided to indicate the length of the appointment to fill and populate the start/end times for the actual allocated time.
Control | 0..*
Type | Period
Comments | This does not introduce a capacity for recurring appointments.

12.10.9 Resource Appointment - Mappings

URL: https://hl7.org/fhir/STU3/appointment-mappings.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

12.10.9 Resource Appointment - Mappings

Patient Administration Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Device, Patient, Practitioner, RelatedPerson

Mappings:

RIM Mapping

HL7 v2 Mapping

iCalendar

Mappings for the appointment resource.

12.10.9.1 Mappings for RIM Mapping (http://hl7.org/v3)

Appointment | Act[@moodCode <= 'INT']
identifier | .id
status | .statusCode
serviceCategory | n/a, might be inferred from the ServiceDeliveryLocation
serviceType | n/a, might be inferred from the ServiceDeliveryLocation
specialty | .performer.AssignedPerson.code
appointmentType | .code
reason | .reasonCode
indication | .inboundRelationship[@typeCode = 'RSON'].observation
priority | .priorityCode
description | .text
supportingInformation | .inboundRelationship[@typeCode = 'SPRT'].observation
start | .effectiveTime.low
end | .effectiveTime.high
minutesDuration | .activityTime[@xsi:type = ('SXPR_TS', 'PIVL_TS')].width
slot | .outboundRelationship[@typeCode = 'REFR'].act[@classCode = 'ACT'][@moodCode = 'SLOT']
created | .author.time
comment | .subjectOf.observation[@moodCode = 'EVN'][code = 'annotation'].value
incomingReferral | .outboundRelationship[@typeCode = 'FLFS'].act[@classCode < 'ActCareProvisionRequestType'][@moodCode = 'RQO']
participant | performer | reusableDevice | subject | location
type | (performer | reusableDevice | subject | location).@typeCode
actor | performer.person | reusableDevice.manufacturedDevice | subject.patient | location.serviceDeliveryLocation
required | (performer | reusableDevice | subject | location).@performInd
status | (performer | reusableDevice | subject | location).statusCode
requestedPeriod | .activityTime[@xsi:type = ('SXPR_TS', 'PIVL_TS')]

12.10.9.2 Mappings for HL7 v2 Mapping (http://hl7.org/v2)

Appointment
identifier | SCH-1, SCH-2, ARQ-1, ARQ-2
status | SCH-25
serviceCategory
serviceType
specialty

appointmentType | ARQ-7
reason | AIS-3, SCH-7
indication
priority | ARQ-12, SCH-11.6 / TQ1-9
description | NTE-3
supportingInformation
start | ARQ-11.1, SCH-11.4 / TQ1-7
end | SCH-11.5 / TQ1-8/ calculated based on the start and the duration which is in ARQ-9
minutesDuration
slot
created
comment | NTE-3
incomingReferral
participant | PID, AIP, AIG, AIL
type | AIP-4, AIG-4
actor | PID-3-Patient ID List | AIL-3 | AIG-3 | AIP-3
required
status | AIP-12, AIG-14
requestedPeriod | ARQ-11

12.10.9.3 Mappings for iCalendar (http://www.ietf.org/rfc/rfc2445.txt)

Appointment | VEVENT
identifier | UID
status
serviceCategory
serviceType
specialty
appointmentType
reason
indication
priority | PRIORITY
description | DESCRIPTION
supportingInformation
start
end
minutesDuration
slot
created
comment | COMMENT
incomingReferral
participant | ATTENDEE component to be created
type | n/a
actor | ATTENDEE: (Need to provide the common name CN and also the MAILTO properties from the practitioner resource) where the participant type is not "information-only"
required | ROLE=REQ-PARTICIPANT (roleparam)
status | ATTENDEE;CN="Brian Postlethwaite";RSVP=TRUE:mailto:BPostlethwaite@data.com (rsvpparam | partstatparam)
requestedPeriod

12.10.10 Resource Appointment - Design Notes

URL: https://hl7.org/fhir/STU3/appointment-explanations.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

12.10.10 Resource Appointment - Design Notes

to do

12.10.11 Resource Appointment - Extensions & Profiles

URL: https://hl7.org/fhir/STU3/appointment-profiles.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

12.10.11 Resource Appointment - Extensions & Profiles

Patient Administration Work Group | Maturity Level: N/A | Ballot Status: Informative

This table lists profiles and extensions for the Appointment resource. For background information, see Profiling Resources and Extensibility. Additional profiles and resources may be found in published Implementation Guides , or in the Conformance resource registry .

12.10.11.1 Profiles

No Profiles defined for this resource

12.10.11.2 Extensions

No Extensions defined for this resource

Extensions that reference this resource:

No Extensions refer to this resource

12.10.11.3 Search Extensions

No Search Extensions defined for this resource

12.10.12 Resource Appointment - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/appointment-version-maps.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

12.10.12 Resource Appointment - Detailed Descriptions

Patient Administration Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Device, Patient, Practitioner, RelatedPerson

R2 : R3 Converson maps for Appointment.

Functional status for this map: 3 tests that all execute ok. All tests pass round-trip testing and all r3 resources are valid. (see documentation)

12.10.12.1 R2 to R3

```
map "http://hl7.org/fhir/StructureMap/Appointment2to3" = "R2 to R3 Conversions for Appointment"

uses "http://hl7.org/fhir/DSTU2/StructureDefinition/Appointment" alias AppointmentR2 as source
uses "http://hl7.org/fhir/StructureDefinition/Appointment" alias Appointment as target

imports "http://hl7.org/fhir/StructureMap/*2to3"

group for type+types  Appointment extends DomainResource
  input src : AppointmentR2 as source
  input tgt : Appointment as target

  "Appointment-identifier" : for src.identifier make tgt.identifier
  "Appointment-status" : for src.status make tgt.status
  "Appointment-serviceType" : for src.type make tgt.serviceType
  "Appointment-reason" : for src.reason make tgt.reason
  "Appointment-priority" : for src.priority make tgt.priority
  "Appointment-description" : for src.description make tgt.description
  "Appointment-start" : for src.start make tgt.start
  "Appointment-end" : for src.end make tgt.end
  "Appointment-minutesDuration" : for src.minutesDuration make tgt.minutesDuration
  "Appointment-slot" : for src.slot make tgt.slot
  "Appointment-comment" : for src.comment make tgt.comment
  "Appointment-participant" : for src.participant as vs0 make tgt.participant as vt0 then appParticipant(vs0, vt0)
endgroup

group appParticipant extends BackboneElement
  input src as source
  input tgt as target

  "Appointment.participant-type" : for src.type make tgt.type
  "Appointment.participant-actor" : for src.actor make tgt.actor
  "Appointment.participant-required" : for src.required make tgt.required
  "Appointment.participant-status" : for src.status make tgt.status
endgroup
```

12.10.12.2 R3 to R2

```
map "http://hl7.org/fhir/StructureMap/Appointment3to2" = "R3 to R2 Conversion for Appointment"

uses "http://hl7.org/fhir/StructureDefinition/Appointment" alias Appointment as source
uses "http://hl7.org/fhir/DSTU2/StructureDefinition/Appointment" alias AppointmentR2 as target

imports "http://hl7.org/fhir/StructureMap/*3to2"

group for type+types  Appointment extends DomainResource
  input src : Appointment as source
  input tgt : AppointmentR2 as target

  "Appointment-identifier" : for src.identifier make tgt.identifier
  "Appointment-status" : for src.status make tgt.status
  "Appointment-type" : for src.serviceType make tgt.type
  "Appointment-reason" : for src.reason make tgt.reason
  "Appointment-priority" : for src.priority make tgt.priority
  "Appointment-description" : for src.description make tgt.description
  "Appointment-start" : for src.start make tgt.start
```

```
  "Appointment-end" : for src.end make tgt.end
  "Appointment-minutesDuration" : for src.minutesDuration make tgt.minutesDuration
  "Appointment-slot" : for src.slot make tgt.slot
  "Appointment-comment" : for src.comment make tgt.comment
  "Appointment-participant" : for src.participant as vs0 make tgt.participant as vt0 then appParticipant(vs0, vt0)
endgroup

group appParticipant extends BackboneElement
  input src as source
  input tgt as target

  "Appointment.participant-type" : for src.type make tgt.type
  "Appointment.participant-actor" : for src.actor make tgt.actor
  "Appointment.participant-required" : for src.required make tgt.required
  "Appointment.participant-status" : for src.status make tgt.status
endgroup
```

12.10.12.3 R3 Validation Errors

12.11 Resource AppointmentResponse - Content

URL: https://hl7.org/fhir/STU3/appointmentresponse.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

Patient Administration Work Group | Maturity Level: 3 | Trial Use | Compartments: Device, Patient, Practitioner, RelatedPerson

A reply to an appointment request for a patient and/or practitioner(s), such as a confirmation or rejection.

12.11.1 Scope and Usage

Appointment resources are used to provide information about a planned meeting that may be in the future or past. They may be for a single meeting or for a series of repeating visits. Examples include a scheduled surgery, a follow-up for a clinical visit, a scheduled conference call between clinicians to discuss a case, the reservation of a piece of diagnostic equipment for a particular use, etc. The visit scheduled by an appointment may be in person or remote (by phone, video conference, etc.) All that matters is that the time and usage of one or more individuals, locations and/or pieces of equipment is being fully or partially reserved for a designated period of time.

This definition takes the concepts of appointments in a clinical setting and also extends them to be relevant in the community healthcare space, and also ease exposure to other appointment / calendar standards widely used outside of Healthcare.

12.11.1.1 The basic workflow to create an appointment

Making the Appointment Request

When an appointment is required, a requester creates new Appointment resource with the Appointment.status="proposed". All included participants (optional or mandatory) should have the status="needs-action" to allow filtering and displaying appointments to user-participants for accepting or rejecting new and updated requests. Based on internal system business rules, certain statuses may be automatically updated, for example: "reject because the requested participant is on vacation" or "this type of user is not allowed to request those specific appointments".

Replying to the request

The reply process is simply performed by the person/system handing the requests updating the participant statuses as needed. If there are multiple systems involved, then these will create AppointmentResponse entries with the desired statuses.

Once all participants have their participation status created/updated (and the main system marking the appointment participant records with the AppointmentResponse statuses) then the overall status of the Appointment is updated.

12.11.2 Resource Content

* Structure

* UML

* XML

* JSON

* Turtle

* R2 Diff

* All

Structure

Name | Flags | Card. | Type | Description & Constraints

AppointmentResponse | I | | DomainResource | A reply to an appointment request for a patient and/or practitioner(s), such as a confirmation or rejection + Either the participantType or actor must be specified Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension

identifier | ? | 0..* | Identifier | External Ids for this item

appointment | ? | 1..1 | Reference(Appointment) | Appointment this response relates to

start | | 0..1 | instant | Time from appointment, or requested new start time

end | | 0..1 | instant | Time from appointment, or requested new end time

participantType | ? | 0..* | CodeableConcept | Role of participant in the appointment ParticipantType (Extensible)
actor | ? | 0..1 | Reference(Patient | Practitioner | RelatedPerson | Device | HealthcareService | Location) | Person, Location/HealthcareService or Device
participantStatus | ?!? | 1..1 | code | accepted | declined | tentative | in-process | completed | needs-action | entered-in-error ParticipationStatus (Required)
comment | | 0..1 | string | Additional comments
Documentation for this format

UML Diagram (Legend)

XML Template

```
<AppointmentResponse xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier External Ids for this item --></identifier>
 <appointment><!-- 1..1 Reference(Appointment) Appointment this response relates to --></appointment>
 <start value="[instant]"/><!-- 0..1 Time from appointment, or requested new start time -->
 <end value="[instant]"/><!-- 0..1 Time from appointment, or requested new end time -->
 <participantType><!-- 0..* CodeableConcept Role of participant in the appointment --></participantType>
 <actor><!-- 0..1 Reference(Patient|Practitioner|RelatedPerson|Device|
   HealthcareService|Location) Person, Location/HealthcareService or Device --></actor>
 <participantStatus value="[code]"/><!-- 1..1 accepted | declined | tentative | in-process | completed | needs-action |
    entered-in-error -->
 <comment value="[string]"/><!-- 0..1 Additional comments -->
</AppointmentResponse>
```

JSON Template

```
{
  "resourceType" : "AppointmentResponse",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // External Ids for this item
  "appointment" : { Reference(Appointment) }, // R!  Appointment this response relates to
  "start" : "<instant>", // Time from appointment, or requested new start time
  "end" : "<instant>", // Time from appointment, or requested new end time
  "participantType" : [{ CodeableConcept }], // Role of participant in the appointment
  "actor" : { Reference(Patient|Practitioner|RelatedPerson|Device|
   HealthcareService|Location) }, // Person, Location/HealthcareService or Device
  "participantStatus" : "<code>", // R!  accepted | declined | tentative | in-process | completed | needs-action |
    entered-in-error
  "comment" : "<string>" // Additional comments
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:AppointmentResponse;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:AppointmentResponse.identifier [ Identifier ], ... ; # 0..* External Ids for this item
  fhir:AppointmentResponse.appointment [ Reference(Appointment) ]; # 1..1 Appointment this response relates to
  fhir:AppointmentResponse.start [ instant ]; # 0..1 Time from appointment, or requested new start time
  fhir:AppointmentResponse.end [ instant ]; # 0..1 Time from appointment, or requested new end time
  fhir:AppointmentResponse.participantType [ CodeableConcept ], ... ; # 0..* Role of participant in the appointment
  fhir:AppointmentResponse.actor [ Reference(Patient|Practitioner|RelatedPerson|Device|HealthcareService|Location) ]; #
    0..1 Person, Location/HealthcareService or Device
  fhir:AppointmentResponse.participantStatus [ code ]; # 1..1 accepted | declined | tentative | in-process | completed |
     needs-action | entered-in-error
  fhir:AppointmentResponse.comment [ string ]; # 0..1 Additional comments
]
```

Changes since DSTU2

AppointmentResponse
AppointmentResponse.participantType | Change binding strength from required to extensible
AppointmentResponse.participantStatus | Change value set from http://hl7.org/fhir/ValueSet/participantstatus to http://hl7.org/fhir/ValueSet/participationstatus

See the Full Difference for further information

This analysis is available as XML or JSON.

See R2 <--> R3 Conversion Maps (status = 2 tests that all execute ok. All tests pass round-trip testing and all r3 resources are valid.).

Structure

Name | Flags | Card. | Type | Description & Constraints
AppointmentResponse | I | | DomainResource | A reply to an appointment request for a patient and/or practitioner(s), such as a confirmation or rejection + Either the participantType or actor must be specified Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | ? | 0..* | Identifier | External Ids for this item
appointment | ? | 1..1 | Reference(Appointment) | Appointment this response relates to
start | | 0..1 | instant | Time from appointment, or requested new start time
end | | 0..1 | instant | Time from appointment, or requested new end time

participantType | ? | 0..* | CodeableConcept | Role of participant in the appointment ParticipantType (Extensible)
actor | ? | 0..1 | Reference(Patient | Practitioner | RelatedPerson | Device | HealthcareService | Location) | Person, Location/HealthcareService or Device
participantStatus | ?!? | 1..1 | code | accepted | declined | tentative | in-process | completed | needs-action | entered-in-error ParticipationStatus (Required)
comment | | 0..1 | string | Additional comments
Documentation for this format

UML Diagram (Legend)

XML Template

```
<AppointmentResponse xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier External Ids for this item --></identifier>
 <appointment><!-- 1..1 Reference(Appointment) Appointment this response relates to --></appointment>
 <start value="[instant]"/><!-- 0..1 Time from appointment, or requested new start time -->
 <end value="[instant]"/><!-- 0..1 Time from appointment, or requested new end time -->
 <participantType><!-- 0..* CodeableConcept Role of participant in the appointment --></participantType>
 <actor><!-- 0..1 Reference(Patient|Practitioner|RelatedPerson|Device|
   HealthcareService|Location) Person, Location/HealthcareService or Device --></actor>
 <participantStatus value="[code]"/><!-- 1..1 accepted | declined | tentative | in-process | completed | needs-action |
    entered-in-error -->
 <comment value="[string]"/><!-- 0..1 Additional comments -->
</AppointmentResponse>
```

JSON Template

```
{
  "resourceType" : "AppointmentResponse",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // External Ids for this item
  "appointment" : { Reference(Appointment) }, // R!  Appointment this response relates to
  "start" : "<instant>", // Time from appointment, or requested new start time
  "end" : "<instant>", // Time from appointment, or requested new end time
  "participantType" : [{ CodeableConcept }], // Role of participant in the appointment
  "actor" : { Reference(Patient|Practitioner|RelatedPerson|Device|
   HealthcareService|Location) }, // Person, Location/HealthcareService or Device
  "participantStatus" : "<code>", // R!  accepted | declined | tentative | in-process | completed | needs-action |
    entered-in-error
  "comment" : "<string>" // Additional comments
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:AppointmentResponse;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:AppointmentResponse.identifier [ Identifier ], ... ; # 0..* External Ids for this item
  fhir:AppointmentResponse.appointment [ Reference(Appointment) ]; # 1..1 Appointment this response relates to
  fhir:AppointmentResponse.start [ instant ]; # 0..1 Time from appointment, or requested new start time
  fhir:AppointmentResponse.end [ instant ]; # 0..1 Time from appointment, or requested new end time
  fhir:AppointmentResponse.participantType [ CodeableConcept ], ... ; # 0..* Role of participant in the appointment
  fhir:AppointmentResponse.actor [ Reference(Patient|Practitioner|RelatedPerson|Device|HealthcareService|Location) ]; #
    0..1 Person, Location/HealthcareService or Device
  fhir:AppointmentResponse.participantStatus [ code ]; # 1..1 accepted | declined | tentative | in-process | completed |
     needs-action | entered-in-error
  fhir:AppointmentResponse.comment [ string ]; # 0..1 Additional comments
]
```

Changes since DSTU2

AppointmentResponse
AppointmentResponse.participantType | Change binding strength from required to extensible
AppointmentResponse.participantStatus | Change value set from http://hl7.org/fhir/ValueSet/participantstatus to http://hl7.org/fhir/ValueSet/participationstatus

See the Full Difference for further information

This analysis is available as XML or JSON.

See R2 <--> R3 Conversion Maps (status = 2 tests that all execute ok. All tests pass round-trip testing and all r3 resources are valid.).

Alternate definitions: Master Definition (XML, JSON), XML Schema/Schematron (for) + JSON Schema, ShEx (for Turtle)

12.11.2.1 Terminology Bindings

Path | Definition | Type | Reference
AppointmentResponse.participantType | Role of participant in encounter. | Extensible | ParticipantType
AppointmentResponse.participantStatus | The Participation status of an appointment. | Required | ParticipationStatus

12.11.2.2 Constraints

* apr-1: Either the participantType or actor must be specified (expression : participantType.exists() or actor.exists())

12.11.3 Notes:

* Time zones and recurring appointments

Recurring appointments need to have the time zone defined in which the values were entered. Knowing that the start time was at 9:00:00Z+10 does not mean that the same time in 2 weeks is actually the same.

As if this was a time in Brisbane Australia, this time will be the same (in respect to its offset from UTC), however if this was for Melbourne Australia, during the daylight savings period of time Melbourne becomes +11. So without the additional information as to which time zone it was created in, scheduling a 9am appointment every Wednesday would not be possible.

12.11.4 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
actor | reference | The Person, Location/HealthcareService or Device that this appointment response replies for | AppointmentResponse.actor (Practitioner, Device, Patient, HealthcareService, RelatedPerson, Location)
appointment | reference | The appointment that the response is attached to | AppointmentResponse.appointment (Appointment)
identifier | token | An Identifier in this appointment response | AppointmentResponse.identifier
location | reference | This Response is for this Location | AppointmentResponse.actor (Location)
part-status | token | The participants acceptance status for this appointment | AppointmentResponse.participantStatus
patient | reference | This Response is for this Patient | AppointmentResponse.actor (Patient)
practitioner | reference | This Response is for this Practitioner | AppointmentResponse.actor (Practitioner)

12.11.5 Resource AppointmentResponse - Examples

URL: https://hl7.org/fhir/STU3/appointmentresponse-examples.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

12.11.5 Resource AppointmentResponse - Examples

Patient Administration Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Device, Patient, Practitioner, RelatedPerson

Example List:

General Person Example | example | XML | JSON | Turtle
Response to renegotiate an appointmet request | exampleresp | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

12.11.6 Resource AppointmentResponse - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/appointmentresponse-definitions.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

12.11.6 Resource AppointmentResponse - Detailed Descriptions

Patient Administration Work Group | Maturity Level: 3 | Trial Use | Compartments: Device, Patient, Practitioner, RelatedPerson

Detailed Descriptions for the elements in the AppointmentResponse resource.

AppointmentResponse
Definition | A reply to an appointment request for a patient and/or practitioner(s), such as a confirmation or rejection.
Control | 1..1
Invariants | Defined on this element apr-1: Either the participantType or actor must be specified (expression : participantType.exists() or actor.exists(), xpath: (exists(f:participantType) or exists(f:actor)))
AppointmentResponse.identifier
Definition | This records identifiers associated with this appointment response concern that are defined by business processes and/ or used to refer to it when a direct URL reference to the resource itself is not appropriate.
Note | This is a business identifer, not a resource identifier (see discussion)
Control | 0..*
Type | Identifier
Summary | true
AppointmentResponse.appointment
Definition | Appointment that this response is replying to.
Control | 1..1
Type | Reference(Appointment)
Summary | true
AppointmentResponse.start
Definition | Date/Time that the appointment is to take place, or requested new start time.
Control | 0..1

Type | instant
Comments | This may be either the same as the appointment request to confirm the details of the appointment, or alternately a new time to request a re-negotiation of the start time.
AppointmentResponse.end
Definition | This may be either the same as the appointment request to confirm the details of the appointment, or alternately a new time to request a re-negotiation of the end time.
Control | 0..1
Type | instant
AppointmentResponse.participantType
Definition | Role of participant in the appointment.
Control | 0..*
Terminology Binding | ParticipantType (Extensible)
Type | CodeableConcept
Summary | true
Comments | The role of the participant can be used to declare what the actor will be doing in the scope of the referenced appointment. If the actor is not specified, then it is expected that the actor will be filled in at a later stage of planning. This value SHALL be the same as specified on the referenced Appointment so that they can be matched, and subsequently updated.
AppointmentResponse.actor
Definition | A Person, Location/HealthcareService or Device that is participating in the appointment.
Control | 0..1
Type | Reference(Patient | Practitioner | RelatedPerson | Device | HealthcareService | Location)
Summary | true
AppointmentResponse.participantStatus
Definition | Participation status of the participant. When the status is declined or tentative if the start/end times are different to the appointment, then these times should be interpreted as a requested time change. When the status is accepted, the times can either be the time of the appointment (as a confirmation of the time) or can be empty.
Control | 1..1
Terminology Binding | ParticipationStatus (Required)
Type | code
Is Modifier | true
Summary | true
Comments | This element is labeled as a modifier because the status contains the code entered-in-error that marks the participant as not currently valid.
AppointmentResponse.comment
Definition | Additional comments about the appointment.
Control | 0..1
Type | string
Comments | This comment is particularly important when the responder is declining, tentatively accepting or requesting another time to indicate the reasons why.

12.11.7 Resource AppointmentResponse - Mappings

URL: https://hl7.org/fhir/STU3/appointmentresponse-mappings.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

12.11.7 Resource AppointmentResponse - Mappings

Patient Administration Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Device, Patient, Practitioner, RelatedPerson

Mappings:

RIM Mapping

HL7 v2 Mapping

iCalendar

Mappings for the appointmentresponse resource.

12.11.7.1 Mappings for RIM Mapping (http://hl7.org/v3)

AppointmentResponse | Appointment[@moodCode <= 'PRMS']
identifier | .id
appointment | scheduleRequest.ActAppointmentRequest[@moodCode <= 'ARQ']
start | .effectiveTime.low
end | .effectiveTime.high
participantType | (performer | reusableDevice | subject | location).@typeCode
actor | performer.person | reusableDevice.manufacturedDevice | subject.patient | location.serviceDeliveryLocation
participantStatus | (performer | reusableDevice | subject | location).statusCode
comment | .subjectOf.observation[@moodCode = 'EVN'][code = 'annotation'].value

12.11.7.2 Mappings for HL7 v2 Mapping (http://hl7.org/v2)

AppointmentResponse
identifier
appointment
start

end
participantType
actor | PID-3-Patient ID List
participantStatus
comment

12.11.7.3 Mappings for iCalendar (http://www.ietf.org/rfc/rfc2445.txt)

AppointmentResponse | VEVENT
identifier | UID
appointment
start
end
participantType | n/a
actor | ATTENDEE: (Need to provide the common name CN and also the MAILTO properties from the practitioner resource)
participantStatus | ATTENDEE;CN="Brian Postlethwaite";RSVP=TRUE:mailto:BPostlethwaite@data.com (rsvpparam | partstatparam)
comment | COMMENT

12.11.8 Resource AppointmentResponse - Design Notes

URL: https://hl7.org/fhir/STU3/appointmentresponse-explanations.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

12.11.8 Resource AppointmentResponse - Design Notes

to do

12.11.9 Resource AppointmentResponse - Extensions & Profiles

URL: https://hl7.org/fhir/STU3/appointmentresponse-profiles.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

12.11.9 Resource AppointmentResponse - Extensions & Profiles

Patient Administration Work Group | Maturity Level: N/A | Ballot Status: Informative

This table lists profiles and extensions for the AppointmentResponse resource. For background information, see Profiling Resources and Extensibility. Additional profiles and resources may be found in published Implementation Guides , or in the Conformance resource registry .

12.11.9.1 Profiles

No Profiles defined for this resource

12.11.9.2 Extensions

No Extensions defined for this resource

Extensions that reference this resource:

No Extensions refer to this resource

12.11.9.3 Search Extensions

No Search Extensions defined for this resource

12.11.10 Resource AppointmentResponse - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/appointmentresponse-version-maps.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

12.11.10 Resource AppointmentResponse - Detailed Descriptions

Patient Administration Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Device, Patient, Practitioner, RelatedPerson

R2 : R3 Converson maps for AppointmentResponse.

Functional status for this map: 2 tests that all execute ok. All tests pass round-trip testing and all r3 resources are valid. (see documentation)

12.11.10.1 R2 to R3

```
map "http://hl7.org/fhir/StructureMap/AppointmentResponse2to3" = "R2 to R3 Conversions for AppointmentResponse"

uses "http://hl7.org/fhir/DSTU2/StructureDefinition/AppointmentResponse" alias AppointmentResponseR2 as source
uses "http://hl7.org/fhir/StructureDefinition/AppointmentResponse" alias AppointmentResponse as target
```

```
imports "http://hl7.org/fhir/StructureMap/*2to3"

group for type+types  AppointmentResponse extends DomainResource
  input src : AppointmentResponseR2 as source
  input tgt : AppointmentResponse as target

  "AppointmentResponse-identifier" : for src.identifier make tgt.identifier
  "AppointmentResponse-appointment" : for src.appointment make tgt.appointment
  "AppointmentResponse-start" : for src.start make tgt.start
  "AppointmentResponse-end" : for src.end make tgt.end
  "AppointmentResponse-participantType" : for src.participantType make tgt.participantType
  "AppointmentResponse-actor" : for src.actor make tgt.actor
  "AppointmentResponse-participantStatus" : for src.participantStatus make tgt.participantStatus
  "AppointmentResponse-comment" : for src.comment make tgt.comment

endgroup
```

12.11.10.2 R3 to R2

```
map "http://hl7.org/fhir/StructureMap/AppointmentResponse3to2" = "R3 to R2 Conversion for AppointmentResponse"

uses "http://hl7.org/fhir/StructureDefinition/AppointmentResponse" alias AppointmentResponse as source
uses "http://hl7.org/fhir/DSTU2/StructureDefinition/AppointmentResponse" alias AppointmentResponseR2 as target

imports "http://hl7.org/fhir/StructureMap/*3to2"

group for type+types  AppointmentResponse extends DomainResource
  input src : AppointmentResponse as source
  input tgt : AppointmentResponseR2 as target

  "AppointmentResponse-identifier" : for src.identifier make tgt.identifier
  "AppointmentResponse-appointment" : for src.appointment make tgt.appointment
  "AppointmentResponse-start" : for src.start make tgt.start
  "AppointmentResponse-end" : for src.end make tgt.end
  "AppointmentResponse-participantType" : for src.participantType make tgt.participantType
  "AppointmentResponse-actor" : for src.actor make tgt.actor
  "AppointmentResponse-participantStatus" : for src.participantStatus make tgt.participantStatus
  "AppointmentResponse-comment" : for src.comment make tgt.comment

endgroup
```

12.11.10.3 R3 Validation Errors

12.12 Resource Schedule - Content

URL: https://hl7.org/fhir/STU3/schedule.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

Patient Administration Work Group | Maturity Level: 3 | Trial Use | Compartments: Device, Patient, Practitioner, RelatedPerson

A container for slots of time that may be available for booking appointments.

12.12.1 Scope and Usage

Schedule resources provide a container for time-slots that can be booked using an appointment. It provides the window of time (period) that slots are defined for and what type of appointments can be booked. The schedule does not provide any information about actual appointments. This separation also greatly assists where access to the appointments would not be permitted for security or privacy reasons, while still being able to determine if an appointment might be available.

Note: A schedule is not used for the delivery of medication, the Timing data type should be used for that purpose.

12.12.1.1 Context

A schedule controls the dates and times available for the performance of a service and/or the use of a resource. One schedule applies to one service or resource, since each service or resource can be reserved independently of the others. If two or more services, people, locations, or things cannot be reserved independently of one another, they are considered to be one activity or resource.

A schedule consists of slots of time during which the controlled service or resource is potentially available for provision or use. Slots are categorized as open, booked, or blocked. An open slot on a schedule indicates that the service or resource is available for provision or use during that period of time. A booked slot indicates that the service or resource is not available during the time period, because an appointment has been scheduled. A blocked slot indicates that a service or resource is unavailable for reasons other than a scheduled appointment.

The real-world, non-automated analogue of the schedule described above is a standard appointment book. These books are generally organized with rows of time slots, during which a service or resource is available.

A slot is one unit on a schedule. A slot represents the smallest unit of time or quantity that a service or resource may be booked. Depending on the nature of the service or resource, there may be more than one defined slot at a given instant of time. For example, if a service is an open group therapy session with twelve available seats, then there are twelve slots for the given block of time.

12.12.1.2 Actor - What the schedule applies to

The schedule belongs to a single instance of a service or resource. This is normally a HealthcareService, Practitioner, Location or Device. In the case where a single resource can provide different services, potentially at different location, then the schedulable resource is considered the composite of the actors. For example, if a practitioner can provide services at multiple locations, they might have one schedule per location, where each schedule includes both the practitioner and location actors. When booking an appointment with multiple schedulable resources, multiple schedules may need to be checked depending on the configuration of the system.

If an appointment has two practitioners, a specific medical device and a room then there could be a schedule for each of these resources that may need to be consulted to ensure that no collisions occur. If the schedule needed to be consulted, then there would be one created covering the planning horizon for the time of the appointment.

12.12.1.3 Checking availability - Searching

When checking availability for an appointment, the creator of the appointment should determine which schedules are applicable, then check for available slots within each schedule.

Determining which schedules should be consulted often will involve searching via the properties of the referenced actors, such as the ServiceCategory on the HealthcareService, or the Address on a Location.

The type parameter can be used to filter the type of services that can be booked within the associated slots.

If all slots are busy, the caller may attempt to book an appointment into a already booked slot, but the server business rules will dictate whether overbooking is allowed, or whether the appointment may be given a higher precedence and allocated the overbooked slot.

This resource is referenced by slot

12.12.2 Resource Content

* Structure

* UML

* XML

* JSON

* Turtle

* R2 Diff

* All

Structure

Name | Flags | Card. | Type | Description & Constraints

Schedule | | | DomainResource | A container for slots of time that may be available for booking appointments Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension

identifier | ? | 0..* | Identifier | External Ids for this item

active | ?!? | 0..1 | boolean | Whether this schedule is in active use

serviceCategory | ? | 0..1 | CodeableConcept | A broad categorisation of the service that is to be performed during this appointment ServiceCategory (Example)

serviceType | ? | 0..* | CodeableConcept | The specific service that is to be performed during this appointment ServiceType (Example)

specialty | ? | 0..* | CodeableConcept | The specialty of a practitioner that would be required to perform the service requested in this appointment Practice Setting Code Value Set (Preferred)

actor | ? | 1..* | Reference(Patient | Practitioner | PractitionerRole | RelatedPerson | Device | HealthcareService | Location) | The resource this Schedule resource is providing availability information for. These are expected to usually be one of HealthcareService, Location, Practitioner, PractitionerRole, Device, Patient or RelatedPerson

planningHorizon | ? | 0..1 | Period | The period of time that the slots that are attached to this Schedule resource cover (even if none exist). These cover the amount of time that an organization's planning horizon; the interval for which they are currently accepting appointments. This does not define a "template" for planning outside these dates

comment | | 0..1 | string | Comments on the availability to describe any extended information. Such as custom constraints on the slots that may be associated

Documentation for this format

UML Diagram (Legend)

XML Template

```
<Schedule xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier External Ids for this item --></identifier>
 <active value="[boolean]"/><!-- 0..1 Whether this schedule is in active use -->
 <serviceCategory><!-- 0..1 CodeableConcept A broad categorisation of the service that is to be performed during this
    appointment --></serviceCategory>
 <serviceType><!-- 0..* CodeableConcept The specific service that is to be performed during this appointment
    --></serviceType>
 <specialty><!-- 0..* CodeableConcept The specialty of a practitioner that would be required to perform the service
    requested in this appointment --></specialty>
 <actor><!-- 1..* Reference(Patient|Practitioner|PractitionerRole|RelatedPerson|
   Device|HealthcareService|Location) The resource this Schedule resource is providing availability information for.
    These are expected to usually be one of HealthcareService, Location, Practitioner, PractitionerRole, Device, Patient
     or RelatedPerson --></actor>
 <planningHorizon><!-- 0..1 Period The period of time that the slots that are attached to this Schedule resource cover
    (even if none exist). These  cover the amount of time that an organization's planning horizon; the interval for
```

```
    which they are currently accepting appointments. This does not define a "template" for planning outside these dates
    --></planningHorizon>
 <comment value="[string]"/><!-- 0..1 Comments on the availability to describe any extended information. Such as custom
    constraints on the slots that may be associated -->
</Schedule>
```

JSON Template

```
{
  "resourceType" : "Schedule",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // External Ids for this item
  "active" : <boolean>, // Whether this schedule is in active use
  "serviceCategory" : { CodeableConcept }, // A broad categorisation of the service that is to be performed during this
    appointment
  "serviceType" : [{ CodeableConcept }], // The specific service that is to be performed during this appointment
  "specialty" : [{ CodeableConcept }], // The specialty of a practitioner that would be required to perform the service
    requested in this appointment
  "actor" : [{ Reference(Patient|Practitioner|PractitionerRole|RelatedPerson|
   Device|HealthcareService|Location) }], // R!  The resource this Schedule resource is providing availability
    information for. These are expected to usually be one of HealthcareService, Location, Practitioner,
    PractitionerRole, Device, Patient or RelatedPerson
  "planningHorizon" : { Period }, // The period of time that the slots that are attached to this Schedule resource cover
     (even if none exist). These  cover the amount of time that an organization's planning horizon; the interval for
    which they are currently accepting appointments. This does not define a "template" for planning outside these dates
  "comment" : "<string>" // Comments on the availability to describe any extended information. Such as custom
    constraints on the slots that may be associated
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:Schedule;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:Schedule.identifier [ Identifier ], ... ; # 0..* External Ids for this item
  fhir:Schedule.active [ boolean ]; # 0..1 Whether this schedule is in active use
  fhir:Schedule.serviceCategory [ CodeableConcept ]; # 0..1 A broad categorisation of the service that is to be
    performed during this appointment
  fhir:Schedule.serviceType [ CodeableConcept ], ... ; # 0..* The specific service that is to be performed during this
    appointment
  fhir:Schedule.specialty [ CodeableConcept ], ... ; # 0..* The specialty of a practitioner that would be required to
    perform the service requested in this appointment
  fhir:Schedule.actor [ Reference(Patient|Practitioner|PractitionerRole|RelatedPerson|Device|HealthcareService|
  Location) ], ... ; # 1..* The resource this Schedule resource is providing availability information for. These are
    expected to usually be one of HealthcareService, Location, Practitioner, PractitionerRole, Device, Patient or
    RelatedPerson
  fhir:Schedule.planningHorizon [ Period ]; # 0..1 The period of time that the slots that are attached to this Schedule
    resource cover (even if none exist). These  cover the amount of time that an organization's planning horizon; the
    interval for which they are currently accepting appointments. This does not define a "template" for planning outside
     these dates
  fhir:Schedule.comment [ string ]; # 0..1 Comments on the availability to describe any extended information. Such as
    custom constraints on the slots that may be associated
]
```

Changes since DSTU2

Schedule

Schedule.active | Added Element

Schedule.serviceCategory | Added Element

Schedule.serviceType | Renamed from type to serviceType

Schedule.specialty | Added Element

Schedule.actor | Max Cardinality changed from 1 to * Add Reference(PractitionerRole)

See the Full Difference for further information

This analysis is available as XML or JSON.

See R2 <--> R3 Conversion Maps (status = 1 test that all execute ok. All tests pass round-trip testing and all r3 resources are valid.).

Structure

Name | Flags | Card. | Type | Description & Constraints

Schedule | | | DomainResource | A container for slots of time that may be available for booking appointments Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension

identifier | ? | 0..* | Identifier | External Ids for this item

active | ?!? | 0..1 | boolean | Whether this schedule is in active use

serviceCategory | ? | 0..1 | CodeableConcept | A broad categorisation of the service that is to be performed during this appointment ServiceCategory (Example)

serviceType | ? | 0..* | CodeableConcept | The specific service that is to be performed during this appointment ServiceType (Example)

specialty | ? | 0..* | CodeableConcept | The specialty of a practitioner that would be required to perform the service requested in this appointment Practice Setting Code Value Set (Preferred)

actor | ? | 1..* | Reference(Patient | Practitioner | PractitionerRole | RelatedPerson | Device | HealthcareService | Location) | The resource this Schedule resource is providing availability information for. These are expected to usually be one of HealthcareService, Location, Practitioner, PractitionerRole, Device, Patient or RelatedPerson

planningHorizon | ? | 0..1 | Period | The period of time that the slots that are attached to this Schedule resource cover (even if none exist). These cover the amount of time that an organization's planning horizon; the interval for which they are currently accepting appointments. This does not define a "template" for planning outside these dates

comment | | 0..1 | string | Comments on the availability to describe any extended information. Such as custom constraints on the slots that may be associated

Documentation for this format

UML Diagram (Legend)

XML Template

```
<Schedule xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier External Ids for this item --></identifier>
 <active value="[boolean]"/><!-- 0..1 Whether this schedule is in active use -->
 <serviceCategory><!-- 0..1 CodeableConcept A broad categorisation of the service that is to be performed during this
    appointment --></serviceCategory>
 <serviceType><!-- 0..* CodeableConcept The specific service that is to be performed during this appointment
    --></serviceType>
 <specialty><!-- 0..* CodeableConcept The specialty of a practitioner that would be required to perform the service
    requested in this appointment --></specialty>
 <actor><!-- 1..* Reference(Patient|Practitioner|PractitionerRole|RelatedPerson|
   Device|HealthcareService|Location) The resource this Schedule resource is providing availability information for.
    These are expected to usually be one of HealthcareService, Location, Practitioner, PractitionerRole, Device, Patient
     or RelatedPerson --></actor>
 <planningHorizon><!-- 0..1 Period The period of time that the slots that are attached to this Schedule resource cover
    (even if none exist). These  cover the amount of time that an organization's planning horizon; the interval for
    which they are currently accepting appointments. This does not define a "template" for planning outside these dates
    --></planningHorizon>
 <comment value="[string]"/><!-- 0..1 Comments on the availability to describe any extended information. Such as custom
    constraints on the slots that may be associated -->
</Schedule>
```

JSON Template

```
{
  "resourceType" : "Schedule",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // External Ids for this item
  "active" : <boolean>, // Whether this schedule is in active use
  "serviceCategory" : { CodeableConcept }, // A broad categorisation of the service that is to be performed during this
    appointment
  "serviceType" : [{ CodeableConcept }], // The specific service that is to be performed during this appointment
  "specialty" : [{ CodeableConcept }], // The specialty of a practitioner that would be required to perform the service
    requested in this appointment
  "actor" : [{ Reference(Patient|Practitioner|PractitionerRole|RelatedPerson|
   Device|HealthcareService|Location) }], // R!  The resource this Schedule resource is providing availability
    information for. These are expected to usually be one of HealthcareService, Location, Practitioner,
    PractitionerRole, Device, Patient or RelatedPerson
  "planningHorizon" : { Period }, // The period of time that the slots that are attached to this Schedule resource cover
     (even if none exist). These  cover the amount of time that an organization's planning horizon; the interval for
    which they are currently accepting appointments. This does not define a "template" for planning outside these dates
  "comment" : "<string>" // Comments on the availability to describe any extended information. Such as custom
    constraints on the slots that may be associated
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:Schedule;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:Schedule.identifier [ Identifier ], ... ; # 0..* External Ids for this item
  fhir:Schedule.active [ boolean ]; # 0..1 Whether this schedule is in active use
  fhir:Schedule.serviceCategory [ CodeableConcept ]; # 0..1 A broad categorisation of the service that is to be
    performed during this appointment
  fhir:Schedule.serviceType [ CodeableConcept ], ... ; # 0..* The specific service that is to be performed during this
    appointment
  fhir:Schedule.specialty [ CodeableConcept ], ... ; # 0..* The specialty of a practitioner that would be required to
    perform the service requested in this appointment
  fhir:Schedule.actor [ Reference(Patient|Practitioner|PractitionerRole|RelatedPerson|Device|HealthcareService|
  Location) ], ... ; # 1..* The resource this Schedule resource is providing availability information for. These are
    expected to usually be one of HealthcareService, Location, Practitioner, PractitionerRole, Device, Patient or
    RelatedPerson
  fhir:Schedule.planningHorizon [ Period ]; # 0..1 The period of time that the slots that are attached to this Schedule
    resource cover (even if none exist). These  cover the amount of time that an organization's planning horizon; the
    interval for which they are currently accepting appointments. This does not define a "template" for planning outside
     these dates
  fhir:Schedule.comment [ string ]; # 0..1 Comments on the availability to describe any extended information. Such as
    custom constraints on the slots that may be associated
]
```

Changes since DSTU2

Schedule
Schedule.active | Added Element
Schedule.serviceCategory | Added Element
Schedule.serviceType | Renamed from type to serviceType
Schedule.specialty | Added Element
Schedule.actor | Max Cardinality changed from 1 to * Add Reference(PractitionerRole)

See the Full Difference for further information

This analysis is available as XML or JSON.

See R2 <--> R3 Conversion Maps (status = 1 test that all execute ok. All tests pass round-trip testing and all r3 resources are valid.).

Alternate definitions: Master Definition (XML, JSON), XML Schema/Schematron (for) + JSON Schema, ShEx (for Turtle)

12.12.2.1 Terminology Bindings

Path | Definition | Type | Reference
Schedule.serviceCategory | | Example | ServiceCategory
Schedule.serviceType | | Example | ServiceType
Schedule.specialty | Additional details about where the content was created (e.g. clinical specialty) | Preferred | Practice Setting Code Value Set

12.12.3 Notes:

12.12.3.1 Interaction with other Standards

There is a strong desire that implementers of this resource should consider providing the resource in the iCalendar format as an alternative representation. Many 3rd party applications and component providers have parsers and user interface controls to display this information. This may lower the entry point to integrate outside the health-care specific applications, and into the consumer space. This would permit the easier creation of a mobile application that creates appointments in the devices native calendar. The iCalendar specification can be found at http://www.ietf.org/rfc/rfc2445.txt .

12.12.3.2 Slots - not a contained resource

Due to the dynamic nature of slots they are not included as a part of this resource.

It is anticipated that this resource is likely to be updated intermittently when the scope of slots is changed, i.e. to change the period of slots in the planning horizon. This could be performed each night to move the start and end date forward to keep the planning horizon as exactly 4 weeks.

The slot resource however is anticipated to be updated very regularly as the appointments that reference them are created/updated/cancelled.

12.12.4 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
active | token | Is the schedule in active use | Schedule.active
actor | reference | The individual(HealthcareService, Practitioner, Location, ...) to find a Schedule for | Schedule.actor (Practitioner, Device, Patient, HealthcareService, PractitionerRole, RelatedPerson, Location)
date | date | Search for Schedule resources that have a period that contains this date specified | Schedule.planningHorizon
identifier | token | A Schedule Identifier | Schedule.identifier
type | token | The type of appointments that can be booked into associated slot(s) | Schedule.serviceType

12.12.5 Resource Schedule - Examples

URL: https://hl7.org/fhir/STU3/schedule-examples.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

12.12.5 Resource Schedule - Examples

Patient Administration Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Device, Patient, Practitioner, RelatedPerson

Example List:

Example of an Schedule resource | example | XML | JSON | Turtle
Example 1 of a Schedule resource with a composite schedulable resource | exampleloc1 | XML | JSON | Turtle
Example 2 of a Schedule resource with a composite schedulable resource | exampleloc2 | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

12.12.6 Resource Schedule - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/schedule-definitions.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

12.12.6 Resource Schedule - Detailed Descriptions

Patient Administration Work Group | Maturity Level: 3 | Trial Use | Compartments: Device, Patient, Practitioner, RelatedPerson

Detailed Descriptions for the elements in the Schedule resource.

Schedule
Definition | A container for slots of time that may be available for booking appointments.
Control | 1..1
Alternate Names | Availability
Schedule.identifier
Definition | External Ids for this item.
Note | This is a business identifer, not a resource identifier (see discussion)
Control | 0..*
Type | Identifier
Summary | true
Schedule.active
Definition | Whether this schedule record is in active use, or should not be used (such as was entered in error).
Control | 0..1
Type | boolean
Is Modifier | true
Default Value | true
Summary | true
Comments | This element is labeled as a modifier because it may be used to mark that the resource was created in error.
Schedule.serviceCategory
Definition | A broad categorisation of the service that is to be performed during this appointment.
Control | 0..1
Terminology Binding | ServiceCategory (Example)
Type | CodeableConcept
Summary | true
Schedule.serviceType
Definition | The specific service that is to be performed during this appointment.
Control | 0..*
Terminology Binding | ServiceType (Example)
Type | CodeableConcept
Summary | true
Schedule.specialty
Definition | The specialty of a practitioner that would be required to perform the service requested in this appointment.
Control | 0..*
Terminology Binding | Practice Setting Code Value Set (Preferred)
Type | CodeableConcept
Summary | true
Schedule.actor
Definition | The resource this Schedule resource is providing availability information for. These are expected to usually be one of HealthcareService, Location, Practitioner, PractitionerRole, Device, Patient or RelatedPerson.
Control | 1..*
Type | Reference(Patient | Practitioner | PractitionerRole | RelatedPerson | Device | HealthcareService | Location)
Summary | true
Comments | The capacity to support multiple referenced resource types should be used in cases where the specific resources themselves cannot be scheduled without the other, and thus only make sense to the system exposing them as a group. Common examples of this are where the combination of a pracitioner and a room(location) are always required by a system.
Schedule.planningHorizon
Definition | The period of time that the slots that are attached to this Schedule resource cover (even if none exist). These cover the amount of time that an organization's planning horizon; the interval for which they are currently accepting appointments. This does not define a "template" for planning outside these dates.
Control | 0..1
Type | Period
Summary | true
Schedule.comment
Definition | Comments on the availability to describe any extended information. Such as custom constraints on the slots that may be associated.
Control | 0..1
Type | string

12.12.7 Resource Schedule - Mappings

URL: https://hl7.org/fhir/STU3/schedule-mappings.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

12.12.7 Resource Schedule - Mappings

Patient Administration Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Device, Patient, Practitioner, RelatedPerson

Mappings:

RIM Mapping

iCalendar

Mappings for the schedule resource.

12.12.7.1 Mappings for RIM Mapping (http://hl7.org/v3)

Schedule | n/a

12.12.7.2 Mappings for iCalendar (http://www.ietf.org/rfc/rfc2445.txt)

Schedule | BEGIN:VFREEBUSY ??? END:VFREEBUSY (If your fhir server provides this information via calDAV, then you may want to include addressing information in a URL section, and with the search, a DTSTART and DTEND component can be included to indicate the range of time that has been covered in the collection of slots, where these overlap the resources period values. For all slots that reference this resource, include a FREEBUSY row for each slot)
identifier | UID
active
serviceCategory
serviceType | n/a
specialty
actor | ATTENDEE:MAILTO:john_public@host2.com This can be populated with a value from the referenced resource
planningHorizon | DTSTART:20131201T003000Z DTEND:2014030101T003000Z
comment | COMMENT:(comment) If there is no comment, then the line can be excluded from the iCalendar entry

12.12.8 Resource Schedule - Design Notes

URL: https://hl7.org/fhir/STU3/schedule-explanations.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

12.12.8 Resource Schedule - Design Notes

to do

12.12.9 Resource Schedule - Extensions & Profiles

URL: https://hl7.org/fhir/STU3/schedule-profiles.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

12.12.9 Resource Schedule - Extensions & Profiles

Patient Administration Work Group | Maturity Level: N/A | Ballot Status: Informative

This table lists profiles and extensions for the Schedule resource. For background information, see Profiling Resources and Extensibility. Additional profiles and resources may be found in published Implementation Guides , or in the Conformance resource registry .

12.12.9.1 Profiles

No Profiles defined for this resource

12.12.9.2 Extensions

No Extensions defined for this resource

Extensions that reference this resource:

No Extensions refer to this resource

12.12.9.3 Search Extensions

No Search Extensions defined for this resource

12.12.10 Resource Schedule - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/schedule-version-maps.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

12.12.10 Resource Schedule - Detailed Descriptions

Patient Administration Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Device, Patient, Practitioner, RelatedPerson

R2 : R3 Converson maps for Schedule.

Functional status for this map: 1 test that all execute ok. All tests pass round-trip testing and all r3 resources are valid. (see documentation)

12.12.10.1 R2 to R3

```
map "http://hl7.org/fhir/StructureMap/Schedule2to3" = "R2 to R3 Conversions for Schedule"

uses "http://hl7.org/fhir/DSTU2/StructureDefinition/Schedule" alias ScheduleR2 as source
uses "http://hl7.org/fhir/StructureDefinition/Schedule" alias Schedule as target

imports "http://hl7.org/fhir/StructureMap/*2to3"

group for type+types  Schedule extends DomainResource
  input src : ScheduleR2 as source
  input tgt : Schedule as target

  "Schedule-identifier" : for src.identifier make tgt.identifier
  "Schedule-serviceType" : for src.type make tgt.serviceType
  "Schedule-actor" : for src.actor make tgt.actor
  "Schedule-planningHorizon" : for src.planningHorizon make tgt.planningHorizon
  "Schedule-comment" : for src.comment make tgt.comment

endgroup
```

12.12.10.2 R3 to R2

```
map "http://hl7.org/fhir/StructureMap/Schedule3to2" = "R3 to R2 Conversion for Schedule"

uses "http://hl7.org/fhir/StructureDefinition/Schedule" alias Schedule as source
uses "http://hl7.org/fhir/DSTU2/StructureDefinition/Schedule" alias ScheduleR2 as target

imports "http://hl7.org/fhir/StructureMap/*3to2"

group for type+types  Schedule extends DomainResource
  input src : Schedule as source
  input tgt : ScheduleR2 as target

  "Schedule-identifier" : for src.identifier make tgt.identifier
  "Schedule-type" : for src.serviceType make tgt.type
  "Schedule-actor" : for src.actor make tgt.actor
  "Schedule-planningHorizon" : for src.planningHorizon make tgt.planningHorizon
  "Schedule-comment" : for src.comment make tgt.comment

endgroup
```

12.12.10.3 R3 Validation Errors

12.13 Resource Slot - Content

URL: https://hl7.org/fhir/STU3/slot.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

Patient Administration Work Group | Maturity Level: 3 | Trial Use | Compartments: Not linked to any defined compartments

A slot of time on a schedule that may be available for booking appointments.

12.13.1 Scope and Usage

Slot resources are used to provide time-slots that can be booked using an appointment. They do not provide any information about appointments that are available, just the time, and optionally what the time can be used for. These are effectively spaces of free/busy time. Slots can also be marked as busy without having appointments associated.

A slot can have more than one appointment allocated to it. A scheduling system may permit multiple allocations up to a specific number of places. An example of this type of usage could be where the slot is being used for a group service that permits 5 participants at the same time.

A slot can be marked as over-booked indicating that there are too many appointments allocated to it.

In some situations a service may have a specific set of slots reserved for specific uses, such as "walk-ins" or a specific organization has a "standing booking" for Thursday mornings. These should be represented using the appointmentType field with a specified and agreed value. Security Permissions or specific business rules on the system could enforce that only eligible appointments are allocated to them.

If a service had a weekly schedule created that permitted eight 1 hour appointments each day of a working week (Monday - Friday), this would be constructed by a single Schedule resource with the dates for the start and end of the week set, and then 40 (5x8) Slot resources associated with it. As appointments fill up the schedule, these slots would individually be marked as busy as the appointments are filled into the slots. The slots in a schedule do not need to be the same size, and can be different for different days of the week.

Slot instances do not have any recurrence information included. If recurring information is desired, this will be managed outside these resources, or included as extensions.

Note that booking an appointment does not necessarily require that slot resources be identified. When attempting to book an appointment, if the requestor knows ahead of time which schedulable resources are required, then identifying individual slots

from the resources' schedules prior to creating the appointment is appropriate. However, in some medical scheduling scenarios, determining which resources are required for an appointment is very complex, and options other than using schedule+slot may be a better solution.

This resource is referenced by appointment

12.13.2 Resource Content

* Structure

* UML

* XML

* JSON

* Turtle

* R2 Diff

* All

Structure

Name | Flags | Card. | Type | Description & Constraints

Slot | | | DomainResource | A slot of time on a schedule that may be available for booking appointments Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension

identifier | ? | 0..* | Identifier | External Ids for this item

serviceCategory | ? | 0..1 | CodeableConcept | A broad categorisation of the service that is to be performed during this appointment ServiceCategory (Example)

serviceType | ? | 0..* | CodeableConcept | The type of appointments that can be booked into this slot (ideally this would be an identifiable service - which is at a location, rather than the location itself). If provided then this overrides the value provided on the availability resource ServiceType (Example)

specialty | ? | 0..* | CodeableConcept | The specialty of a practitioner that would be required to perform the service requested in this appointment Practice Setting Code Value Set (Preferred)

appointmentType | ? | 0..1 | CodeableConcept | The style of appointment or patient that may be booked in the slot (not service type) v2 Appointment reason codes (Preferred)

schedule | ? | 1..1 | Reference(Schedule) | The schedule resource that this slot defines an interval of status information

status | ? | 1..1 | code | busy | free | busy-unavailable | busy-tentative | entered-in-error SlotStatus (Required)

start | ? | 1..1 | instant | Date/Time that the slot is to begin

end | ? | 1..1 | instant | Date/Time that the slot is to conclude

overbooked | | 0..1 | boolean | This slot has already been overbooked, appointments are unlikely to be accepted for this time

comment | | 0..1 | string | Comments on the slot to describe any extended information. Such as custom constraints on the slot

Documentation for this format

UML Diagram (Legend)

XML Template

```
<Slot xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier External Ids for this item --></identifier>
 <serviceCategory><!-- 0..1 CodeableConcept A broad categorisation of the service that is to be performed during this
    appointment --></serviceCategory>
 <serviceType><!-- 0..* CodeableConcept The type of appointments that can be booked into this slot (ideally this would
    be an identifiable service - which is at a location, rather than the location itself). If provided then this
    overrides the value provided on the availability resource --></serviceType>
 <specialty><!-- 0..* CodeableConcept The specialty of a practitioner that would be required to perform the service
    requested in this appointment --></specialty>
 <appointmentType><!-- 0..1 CodeableConcept The style of appointment or patient that may be booked in the slot (not
    service type) --></appointmentType>
 <schedule><!-- 1..1 Reference(Schedule) The schedule resource that this slot defines an interval of status information
    --></schedule>
 <status value="[code]"/><!-- 1..1 busy | free | busy-unavailable | busy-tentative | entered-in-error -->
 <start value="[instant]"/><!-- 1..1 Date/Time that the slot is to begin -->
 <end value="[instant]"/><!-- 1..1 Date/Time that the slot is to conclude -->
 <overbooked value="[boolean]"/><!-- 0..1 This slot has already been overbooked, appointments are unlikely to be
    accepted for this time -->
 <comment value="[string]"/><!-- 0..1 Comments on the slot to describe any extended information. Such as custom
    constraints on the slot -->
</Slot>
```

JSON Template

```
{
  "resourceType" : "Slot",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // External Ids for this item
  "serviceCategory" : { CodeableConcept }, // A broad categorisation of the service that is to be performed during this
    appointment
  "serviceType" : [{ CodeableConcept }], // The type of appointments that can be booked into this slot (ideally this
    would be an identifiable service - which is at a location, rather than the location itself). If provided then this
    overrides the value provided on the availability resource
  "specialty" : [{ CodeableConcept }], // The specialty of a practitioner that would be required to perform the service
    requested in this appointment
  "appointmentType" : { CodeableConcept }, // The style of appointment or patient that may be booked in the slot (not
    service type)
```

```
  "schedule" : { Reference(Schedule) }, // R!  The schedule resource that this slot defines an interval of status
    information
  "status" : "<code>", // R!  busy | free | busy-unavailable | busy-tentative | entered-in-error
  "start" : "<instant>", // R!  Date/Time that the slot is to begin
  "end" : "<instant>", // R!  Date/Time that the slot is to conclude
  "overbooked" : <boolean>, // This slot has already been overbooked, appointments are unlikely to be accepted for this
    time
  "comment" : "<string>" // Comments on the slot to describe any extended information. Such as custom constraints on the
     slot
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:Slot;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:Slot.identifier [ Identifier ], ... ; # 0..* External Ids for this item
  fhir:Slot.serviceCategory [ CodeableConcept ]; # 0..1 A broad categorisation of the service that is to be performed
    during this appointment
  fhir:Slot.serviceType [ CodeableConcept ], ... ; # 0..* The type of appointments that can be booked into this slot
    (ideally this would be an identifiable service - which is at a location, rather than the location itself). If
    provided then this overrides the value provided on the availability resource
  fhir:Slot.specialty [ CodeableConcept ], ... ; # 0..* The specialty of a practitioner that would be required to
    perform the service requested in this appointment
  fhir:Slot.appointmentType [ CodeableConcept ]; # 0..1 The style of appointment or patient that may be booked in the
    slot (not service type)
  fhir:Slot.schedule [ Reference(Schedule) ]; # 1..1 The schedule resource that this slot defines an interval of status
    information
  fhir:Slot.status [ code ]; # 1..1 busy | free | busy-unavailable | busy-tentative | entered-in-error
  fhir:Slot.start [ instant ]; # 1..1 Date/Time that the slot is to begin
  fhir:Slot.end [ instant ]; # 1..1 Date/Time that the slot is to conclude
  fhir:Slot.overbooked [ boolean ]; # 0..1 This slot has already been overbooked, appointments are unlikely to be
    accepted for this time
  fhir:Slot.comment [ string ]; # 0..1 Comments on the slot to describe any extended information. Such as custom
    constraints on the slot
]
```

Changes since DSTU2

Slot

Slot.serviceCategory | Added Element

Slot.serviceType | Renamed from type to serviceType Max Cardinality changed from 1 to *

Slot.specialty | Added Element

Slot.appointmentType | Added Element

Slot.status | Renamed from freeBusyType to status

See the Full Difference for further information

This analysis is available as XML or JSON.

See R2 <--> R3 Conversion Maps (status = 4 tests that all execute ok. All tests pass round-trip testing and all r3 resources are valid.).

Structure

Name | Flags | Card. | Type | Description & Constraints

Slot | | | DomainResource | A slot of time on a schedule that may be available for booking appointments Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension

identifier | ? | 0..* | Identifier | External Ids for this item

serviceCategory | ? | 0..1 | CodeableConcept | A broad categorisation of the service that is to be performed during this appointment ServiceCategory (Example)

serviceType | ? | 0..* | CodeableConcept | The type of appointments that can be booked into this slot (ideally this would be an identifiable service - which is at a location, rather than the location itself). If provided then this overrides the value provided on the availability resource ServiceType (Example)

specialty | ? | 0..* | CodeableConcept | The specialty of a practitioner that would be required to perform the service requested in this appointment Practice Setting Code Value Set (Preferred)

appointmentType | ? | 0..1 | CodeableConcept | The style of appointment or patient that may be booked in the slot (not service type) v2 Appointment reason codes (Preferred)

schedule | ? | 1..1 | Reference(Schedule) | The schedule resource that this slot defines an interval of status information

status | ? | 1..1 | code | busy | free | busy-unavailable | busy-tentative | entered-in-error SlotStatus (Required)

start | ? | 1..1 | instant | Date/Time that the slot is to begin

end | ? | 1..1 | instant | Date/Time that the slot is to conclude

overbooked | | 0..1 | boolean | This slot has already been overbooked, appointments are unlikely to be accepted for this time

comment | | 0..1 | string | Comments on the slot to describe any extended information. Such as custom constraints on the slot

Documentation for this format

UML Diagram (Legend)

XML Template

```
<Slot xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
```

```
 <identifier><!-- 0..* Identifier External Ids for this item --></identifier>
 <serviceCategory><!-- 0..1 CodeableConcept A broad categorisation of the service that is to be performed during this
    appointment --></serviceCategory>
 <serviceType><!-- 0..* CodeableConcept The type of appointments that can be booked into this slot (ideally this would
    be an identifiable service - which is at a location, rather than the location itself). If provided then this
    overrides the value provided on the availability resource --></serviceType>
 <specialty><!-- 0..* CodeableConcept The specialty of a practitioner that would be required to perform the service
    requested in this appointment --></specialty>
 <appointmentType><!-- 0..1 CodeableConcept The style of appointment or patient that may be booked in the slot (not
    service type) --></appointmentType>
 <schedule><!-- 1..1 Reference(Schedule) The schedule resource that this slot defines an interval of status information
    --></schedule>
 <status value="[code]"/><!-- 1..1 busy | free | busy-unavailable | busy-tentative | entered-in-error -->
 <start value="[instant]"/><!-- 1..1 Date/Time that the slot is to begin -->
 <end value="[instant]"/><!-- 1..1 Date/Time that the slot is to conclude -->
 <overbooked value="[boolean]"/><!-- 0..1 This slot has already been overbooked, appointments are unlikely to be
    accepted for this time -->
 <comment value="[string]"/><!-- 0..1 Comments on the slot to describe any extended information. Such as custom
    constraints on the slot -->
</Slot>
```

JSON Template

```
{
  "resourceType" : "Slot",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // External Ids for this item
  "serviceCategory" : { CodeableConcept }, // A broad categorisation of the service that is to be performed during this
    appointment
  "serviceType" : [{ CodeableConcept }], // The type of appointments that can be booked into this slot (ideally this
    would be an identifiable service - which is at a location, rather than the location itself). If provided then this
    overrides the value provided on the availability resource
  "specialty" : [{ CodeableConcept }], // The specialty of a practitioner that would be required to perform the service
    requested in this appointment
  "appointmentType" : { CodeableConcept }, // The style of appointment or patient that may be booked in the slot (not
    service type)
  "schedule" : { Reference(Schedule) }, // R!  The schedule resource that this slot defines an interval of status
    information
  "status" : "<code>", // R!  busy | free | busy-unavailable | busy-tentative | entered-in-error
  "start" : "<instant>", // R!  Date/Time that the slot is to begin
  "end" : "<instant>", // R!  Date/Time that the slot is to conclude
  "overbooked" : <boolean>, // This slot has already been overbooked, appointments are unlikely to be accepted for this
    time
  "comment" : "<string>" // Comments on the slot to describe any extended information. Such as custom constraints on the
     slot
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:Slot;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:Slot.identifier [ Identifier ], ... ; # 0..* External Ids for this item
  fhir:Slot.serviceCategory [ CodeableConcept ]; # 0..1 A broad categorisation of the service that is to be performed
    during this appointment
  fhir:Slot.serviceType [ CodeableConcept ], ... ; # 0..* The type of appointments that can be booked into this slot
    (ideally this would be an identifiable service - which is at a location, rather than the location itself). If
    provided then this overrides the value provided on the availability resource
  fhir:Slot.specialty [ CodeableConcept ], ... ; # 0..* The specialty of a practitioner that would be required to
    perform the service requested in this appointment
  fhir:Slot.appointmentType [ CodeableConcept ]; # 0..1 The style of appointment or patient that may be booked in the
    slot (not service type)
  fhir:Slot.schedule [ Reference(Schedule) ]; # 1..1 The schedule resource that this slot defines an interval of status
    information
  fhir:Slot.status [ code ]; # 1..1 busy | free | busy-unavailable | busy-tentative | entered-in-error
  fhir:Slot.start [ instant ]; # 1..1 Date/Time that the slot is to begin
  fhir:Slot.end [ instant ]; # 1..1 Date/Time that the slot is to conclude
  fhir:Slot.overbooked [ boolean ]; # 0..1 This slot has already been overbooked, appointments are unlikely to be
    accepted for this time
  fhir:Slot.comment [ string ]; # 0..1 Comments on the slot to describe any extended information. Such as custom
    constraints on the slot
]
```

Changes since DSTU2

Slot

Slot.serviceCategory | Added Element

Slot.serviceType | Renamed from type to serviceType Max Cardinality changed from 1 to *

Slot.specialty | Added Element

Slot.appointmentType | Added Element

Slot.status | Renamed from freeBusyType to status

See the Full Difference for further information

This analysis is available as XML or JSON.

See R2 <--> R3 Conversion Maps (status = 4 tests that all execute ok. All tests pass round-trip testing and all r3 resources are valid.).

Alternate definitions: Master Definition (XML, JSON), XML Schema/Schematron (for) + JSON Schema, ShEx (for Turtle)

12.13.2.1 Terminology Bindings

Path | Definition | Type | Reference
Slot.serviceCategory | | Example | ServiceCategory
Slot.serviceType | | Example | ServiceType
Slot.specialty | Additional details about where the content was created (e.g. clinical specialty). | Preferred | Practice Setting Code Value Set
Slot.appointmentType | | Preferred | v2 Appointment reason codes
Slot.status | The free/busy status of the slot. | Required | SlotStatus

12.13.3 Notes:

* Interaction with other Standards

There is a strong desire that implementers of this resource should consider providing the resource in the iCalendar format as an alternative representation. Many 3rd party applications and component providers have parsers and user interface controls to display this information. This may lower the entry point to integrate outside the health-care specific applications, and into the consumer space. This would permit the easier creation of a mobile application that creates appointments in the devices native calendar. The iCalendar specification can be found at http://www.ietf.org/rfc/rfc2445.txt .

12.13.4 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
identifier | token | A Slot Identifier | Slot.identifier
schedule | reference | The Schedule Resource that we are seeking a slot within | Slot.schedule (Schedule)
slot-type | token | The type of appointments that can be booked into the slot | Slot.serviceType
start | date | Appointment date/time. | Slot.start
status | token | The free/busy status of the appointment | Slot.status

12.13.5 Resource Slot - Examples

URL: https://hl7.org/fhir/STU3/slot-examples.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

12.13.5 Resource Slot - Examples

Patient Administration Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Not linked to any defined compartments

Example List:

Example of an available slot (no appointments allocated) | example | XML | JSON | Turtle
Example for a slot that has an appointment allocated to it | 1 | XML | JSON | Turtle
Example for tentatively occupied slot | 2 | XML | JSON | Turtle
Example of a slot that has been marked as unavailable | 3 | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

12.13.6 Resource Slot - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/slot-definitions.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

12.13.6 Resource Slot - Detailed Descriptions

Patient Administration Work Group | Maturity Level: 3 | Trial Use | Compartments: Not linked to any defined compartments

Detailed Descriptions for the elements in the Slot resource.

Slot
Definition | A slot of time on a schedule that may be available for booking appointments.
Control | 1..1
Slot.identifier
Definition | External Ids for this item.
Note | This is a business identifer, not a resource identifier (see discussion)
Control | 0..*
Type | Identifier
Summary | true
Slot.serviceCategory

Definition | A broad categorisation of the service that is to be performed during this appointment.
Control | 0..1
Terminology Binding | ServiceCategory (Example)
Type | CodeableConcept
Summary | true
Slot.serviceType
Definition | The type of appointments that can be booked into this slot (ideally this would be an identifiable service - which is at a location, rather than the location itself). If provided then this overrides the value provided on the availability resource.
Control | 0..*
Terminology Binding | ServiceType (Example)
Type | CodeableConcept
Summary | true
Slot.specialty
Definition | The specialty of a practitioner that would be required to perform the service requested in this appointment.
Control | 0..*
Terminology Binding | Practice Setting Code Value Set (Preferred)
Type | CodeableConcept
Summary | true
Slot.appointmentType
Definition | The style of appointment or patient that may be booked in the slot (not service type).
Control | 0..1
Terminology Binding | v2 Appointment reason codes (Preferred)
Type | CodeableConcept
Summary | true
Slot.schedule
Definition | The schedule resource that this slot defines an interval of status information.
Control | 1..1
Type | Reference(Schedule)
Summary | true
Slot.status
Definition | busy | free | busy-unavailable | busy-tentative | entered-in-error.
Control | 1..1
Terminology Binding | SlotStatus (Required)
Type | code
Summary | true
Slot.start
Definition | Date/Time that the slot is to begin.
Control | 1..1
Type | instant
Summary | true
Slot.end
Definition | Date/Time that the slot is to conclude.
Control | 1..1
Type | instant
Summary | true
Slot.overbooked
Definition | This slot has already been overbooked, appointments are unlikely to be accepted for this time.
Control | 0..1
Type | boolean
Meaning if Missing | If overbooked is missing, systems may assume that there are still appointments available
Slot.comment
Definition | Comments on the slot to describe any extended information. Such as custom constraints on the slot.
Control | 0..1
Type | string

12.13.7 Resource Slot - Mappings

URL: https://hl7.org/fhir/STU3/slot-mappings.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

12.13.7 Resource Slot - Mappings

Patient Administration Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Not linked to any defined compartments

Mappings:

RIM Mapping

iCalendar

Mappings for the slot resource.

12.13.7.1 Mappings for RIM Mapping (http://hl7.org/v3)

Slot | n/a

12.13.7.2 Mappings for iCalendar (http://www.ietf.org/rfc/rfc2445.txt)

Slot | (refer to the Availability resource for the header mapping details)
identifier
serviceCategory
serviceType
specialty
appointmentType
schedule
status | FREEBUSY;FBTYPE=(freeBusyType):19980314T233000Z/19980315T003000Z If the freeBusyType is BUSY, then this value can be excluded
start | FREEBUSY;FBTYPE=FREE:(start)/19980315T003000Z If the freeBusyType is BUSY, then this value can be excluded
end | FREEBUSY:19980314T233000Z/(end) If the freeBusyType is BUSY, then this value can be excluded (as shown)
overbooked
comment

12.13.8 Resource Slot - Design Notes

URL: https://hl7.org/fhir/STU3/slot-explanations.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

12.13.8 Resource Slot - Design Notes

to do

12.13.9 Resource Slot - Extensions & Profiles

URL: https://hl7.org/fhir/STU3/slot-profiles.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

12.13.9 Resource Slot - Extensions & Profiles

Patient Administration Work Group | Maturity Level: N/A | Ballot Status: Informative

This table lists profiles and extensions for the Slot resource. For background information, see Profiling Resources and Extensibility. Additional profiles and resources may be found in published Implementation Guides , or in the Conformance resource registry .

12.13.9.1 Profiles

No Profiles defined for this resource

12.13.9.2 Extensions

No Extensions defined for this resource

Extensions that reference this resource:

No Extensions refer to this resource

12.13.9.3 Search Extensions

No Search Extensions defined for this resource

12.13.10 Resource Slot - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/slot-version-maps.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

12.13.10 Resource Slot - Detailed Descriptions

Patient Administration Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Not linked to any defined compartments

R2 : R3 Converson maps for Slot.

Functional status for this map: 4 tests that all execute ok. All tests pass round-trip testing and all r3 resources are valid. (see documentation)

12.13.10.1 R2 to R3

```
map "http://hl7.org/fhir/StructureMap/Slot2to3" = "R2 to R3 Conversions for Slot"

uses "http://hl7.org/fhir/DSTU2/StructureDefinition/Slot" alias SlotR2 as source
uses "http://hl7.org/fhir/StructureDefinition/Slot" alias Slot as target

imports "http://hl7.org/fhir/StructureMap/*2to3"
```

```
group for type+types  Slot extends DomainResource
  input src : SlotR2 as source
  input tgt : Slot as target

  "Slot-identifier" : for src.identifier make tgt.identifier
  "Slot-serviceType" : for src.type make tgt.serviceType
  "Slot-schedule" : for src.schedule make tgt.schedule
  "Slot-freeBusyType" : for src.freeBusyType make tgt.status
  "Slot-start" : for src.start make tgt.start
  "Slot-end" : for src.end make tgt.end
  "Slot-overbooked" : for src.overbooked make tgt.overbooked
  "Slot-comment" : for src.comment make tgt.comment

endgroup
```

12.13.10.2 R3 to R2

```
map "http://hl7.org/fhir/StructureMap/Slot3to2" = "R3 to R2 Conversion for Slot"

uses "http://hl7.org/fhir/StructureDefinition/Slot" alias Slot as source
uses "http://hl7.org/fhir/DSTU2/StructureDefinition/Slot" alias SlotR2 as target

imports "http://hl7.org/fhir/StructureMap/*3to2"

group for type+types  Slot extends DomainResource
  input src : Slot as source
  input tgt : SlotR2 as target

  "Slot-identifier" : for src.identifier make tgt.identifier
  "Slot-type" : for src.serviceType make tgt.type
  "Slot-schedule" : for src.schedule make tgt.schedule
  "Slot-freeBusyType" : for src.status check src.status != 'entered-in-error' make tgt.freeBusyType
  "Slot-start" : for src.start make tgt.start
  "Slot-end" : for src.end make tgt.end
  "Slot-overbooked" : for src.overbooked make tgt.overbooked
  "Slot-comment" : for src.comment make tgt.comment

endgroup
```

12.13.10.3 R3 Validation Errors

12.14 Resource ReferralRequest - Content

URL: https://hl7.org/fhir/STU3/referralrequest.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

Patient Care Work Group | Maturity Level: 1 | Trial Use | Compartments: Patient, Practitioner

Used to record and send details about a request for referral service or transfer of a patient to the care of another provider or provider organization.

12.14.1 Scope and Usage

ReferralRequest is one of the request resources in the FHIR workflow specification.

This resource is used to share relevant information required to support a referral request or a transfer of care request from one practitioner or organization to another. It is intended for use when a patient is required to be referred to another provider for a consultation/second opinion and/or for short term or longer term management of one or more health issues or problems.

Examples include:

* Request for a consultation from a specialist

* Referral for support from community services

* District nursing services referral

* Request for aged care placement assessment

* Request for a pharmacist medication review

* Referral for physiotherapy or occupational therapy

ReferralRequest is also intended for use when there is a complete and more permanent transfer of care responsibility from one practitioner/organization to another (for example, as in the case of requesting the transfer of care for a patient from an acute care setting to rehabilitation, aged care, or a skilled nursing facility).

12.14.2 Boundaries and Relationships

ReferralRequest is closely related to other types of "request" resources, particularly ProcedureRequest. In fact, for some services, it may be appropriate to use any one of these resources to request that the service be performed. Which one is used may be driven by organizational practice and by context. When it is unclear which to use, the following principles may be helpful:

* ProcedureRequest is typically used when the requesting clinician has and wishes to exercise the authority (and expertise) to decide exactly what action will be done.

* A ReferralRequest is used when the requesting practitioner is seeking another practitioner or organization to use their own expertise and/or authority to determine the specific action to take.

* Whether an activity is deemed to be a procedure or only a diagnostic request is typically driven by how invasive the diagnostic process is. More invasive processes are typically represented as procedures, though the dividing line may vary by organization.

Irrespective of the guidance above, systems should be prepared for some degree of overlap between these resources and should be prepared to execute searches against multiple resources in cases where differentiation cannot be guaranteed. Additionally, in some workflows more than one type of resource or even both might exist, such as upon receiving a ReferralRequest a practitioner might initiate a ProcedureRequest.

A "referral" is often thought of as a document that contains a great deal of information about the patient being referred. This resource does not actually contain the clinical background information for the patient. Instead, it supports references to the numerous other resources that define this information. For example, Condition, Family History, Allergy/Intolerance, Alerts, Medication, Diagnostic Reports, etc. Alternatively, some systems may choose to bundle up a ReferralRequest and this referenced information into a Document for delivery to the recipient. However, REST, Messaging and Services are also valid architectures for managing referrals and may be more appropriate where active workflow management is needed.

The details of the type of care desired as part of a referral may be conveyed using any of the "request" or "order" resources, likely with a status of "proposed". The CarePlan resource can be used to describe more sophisticated requests for combinations of services. Likewise, ReferralRequest may be referenced as part of a CarePlan.

A ReferralRequest may be targeted (identifying a specific Practitioner or Organization to perform the requested care) or untargeted (merely identifying the type of care desired). The Task resource may be used to help manage such workflows.

A ReferralRequest might be fulfilled by a DiagnosticReport, Encounter, Procedure, or other assessment-related resource.

A ReferralRequest should not be confused with an Appointment, as appointments are intended for booking/scheduling purposes.

This resource is referenced by Appointment, CarePlan, Claim, ClinicalImpression, DiagnosticReport, Encounter, EpisodeOfCare, ExplanationOfBenefit, ImagingStudy, MedicationRequest, MedicationStatement, Observation, Procedure and QuestionnaireResponse

12.14.3 Resource Content

* Structure

* UML

* XML

* JSON

* Turtle

* R2 Diff

* All

Structure

Name | Flags | Card. | Type | Description & Constraints
ReferralRequest | | | DomainResource | A request for referral or transfer of care Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | ? | 0..* | Identifier | Business identifier
definition | ? | 0..* | Reference(ActivityDefinition | PlanDefinition) | Instantiates protocol or definition
basedOn | ? | 0..* | Reference(ReferralRequest | CarePlan | ProcedureRequest) | Request fulfilled by this request
replaces | ? | 0..* | Reference(ReferralRequest) | Request(s) replaced by this request
groupIdentifier | ? | 0..1 | Identifier | Composite request this is part of
status | ?!? | 1..1 | code | draft | active | suspended | cancelled | completed | entered-in-error | unknown RequestStatus (Required)
intent | ?!? | 1..1 | code | proposal | plan | order RequestIntent (Required)
type | ? | 0..1 | CodeableConcept | Referral/Transition of care request type SNOMED CT Patient Referral (Example)
priority | ? | 0..1 | code | Urgency of referral / transfer of care request RequestPriority (Required)
serviceRequested | ? | 0..* | CodeableConcept | Actions requested as part of the referral Practice Setting Code Value Set (Example)
subject | ? | 1..1 | Reference(Patient | Group) | Patient referred to care or transfer
context | ? | 0..1 | Reference(Encounter | EpisodeOfCare) | Originating encounter
occurrence[x] | ? | 0..1 | | When the service(s) requested in the referral should occur
occurrenceDateTime | | | dateTime
occurrencePeriod | | | Period
authoredOn | ? | 0..1 | dateTime | Date of creation/activation
requester | ?I | 0..1 | BackboneElement | Who/what is requesting service + onBehalfOf can only be specified if agent is practitioner or device
agent | ? | 1..1 | Reference(Practitioner | Organization | Patient | RelatedPerson | Device) | Individual making the request
onBehalfOf | ?I | 0..1 | Reference(Organization) | Organization agent is acting for
specialty | | 0..1 | CodeableConcept | The clinical specialty (discipline) that the referral is requested for PractitionerSpecialty (Example)
recipient | ? | 0..* | Reference(Practitioner | Organization | HealthcareService) | Receiver of referral / transfer of care request
reasonCode | ? | 0..* | CodeableConcept | Reason for referral / transfer of care request SNOMED CT Clinical Findings (Example)
reasonReference | ? | 0..* | Reference(Condition | Observation) | Why is service needed?
description | | 0..1 | string | A textual description of the referral
supportingInfo | | 0..* | Reference(Any) | Additonal information to support referral or transfer of care request

note | | 0..* | Annotation | Comments made about referral request
relevantHistory | | 0..* | Reference(Provenance) | Key events in history of request
Documentation for this format

UML Diagram (Legend)

XML Template

```
<ReferralRequest xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Business identifier --></identifier>
 <definition><!-- 0..* Reference(ActivityDefinition|PlanDefinition) Instantiates protocol or definition --></definition>
 <basedOn><!-- 0..* Reference(ReferralRequest|CarePlan|ProcedureRequest) Request fulfilled by this request --></basedOn>
 <replaces><!-- 0..* Reference(ReferralRequest) Request(s) replaced by this request --></replaces>
 <groupIdentifier><!-- 0..1 Identifier Composite request this is part of --></groupIdentifier>
 <status value="[code]"/><!-- 1..1 draft | active | suspended | cancelled | completed | entered-in-error | unknown -->
 <intent value="[code]"/><!-- 1..1 proposal | plan | order -->
 <type><!-- 0..1 CodeableConcept Referral/Transition of care request type --></type>
 <priority value="[code]"/><!-- 0..1 Urgency of referral / transfer of care request -->
 <serviceRequested><!-- 0..* CodeableConcept Actions requested as part of the referral --></serviceRequested>
 <subject><!-- 1..1 Reference(Patient|Group) Patient referred to care or transfer --></subject>
 <context><!-- 0..1 Reference(Encounter|EpisodeOfCare) Originating encounter --></context>
 <occurrence[x]><!-- 0..1 dateTime|Period When the service(s) requested in the referral should occur --></occurrence[x]>
 <authoredOn value="[dateTime]"/><!-- 0..1 Date of creation/activation -->
 <requester>  <!-- 0..1 Who/what is requesting service -->
  <agent><!-- 1..1 Reference(Practitioner|Organization|Patient|RelatedPerson|
    Device) Individual making the request --></agent>
  <onBehalfOf><!--  0..1 Reference(Organization) Organization agent is acting for --></onBehalfOf>
 </requester>
 <specialty><!-- 0..1 CodeableConcept The clinical specialty (discipline) that the referral is requested for
    --></specialty>
 <recipient><!-- 0..* Reference(Practitioner|Organization|HealthcareService) Receiver of referral / transfer of care
    request --></recipient>
 <reasonCode><!-- 0..* CodeableConcept Reason for referral / transfer of care request --></reasonCode>
 <reasonReference><!-- 0..* Reference(Condition|Observation) Why is service needed? --></reasonReference>
 <description value="[string]"/><!-- 0..1 A textual description of the referral -->
 <supportingInfo><!-- 0..* Reference(Any) Additonal information to support referral or transfer of care request
    --></supportingInfo>
 <note><!-- 0..* Annotation Comments made about referral request --></note>
 <relevantHistory><!-- 0..* Reference(Provenance) Key events in history of request --></relevantHistory>
</ReferralRequest>
```

JSON Template

```
{
  "resourceType" : "ReferralRequest",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Business identifier
  "definition" : [{ Reference(ActivityDefinition|PlanDefinition) }], // Instantiates protocol or definition
  "basedOn" : [{ Reference(ReferralRequest|CarePlan|ProcedureRequest) }], // Request fulfilled by this request
  "replaces" : [{ Reference(ReferralRequest) }], // Request(s) replaced by this request
  "groupIdentifier" : { Identifier }, // Composite request this is part of
  "status" : "<code>", // R!  draft | active | suspended | cancelled | completed | entered-in-error | unknown
  "intent" : "<code>", // R!  proposal | plan | order
  "type" : { CodeableConcept }, // Referral/Transition of care request type
  "priority" : "<code>", // Urgency of referral / transfer of care request
  "serviceRequested" : [{ CodeableConcept }], // Actions requested as part of the referral
  "subject" : { Reference(Patient|Group) }, // R!  Patient referred to care or transfer
  "context" : { Reference(Encounter|EpisodeOfCare) }, // Originating encounter
  // occurrence[x]: When the service(s) requested in the referral should occur. One of these 2:
  "occurrenceDateTime" : "<dateTime>",
  "occurrencePeriod" : { Period },
  "authoredOn" : "<dateTime>", // Date of creation/activation
  "requester" : { // Who/what is requesting service
    "agent" : { Reference(Practitioner|Organization|Patient|RelatedPerson|
    Device) }, // R!  Individual making the request
    "onBehalfOf" : { Reference(Organization) } // C? Organization agent is acting for
  },
  "specialty" : { CodeableConcept }, // The clinical specialty (discipline) that the referral is requested for
  "recipient" : [{ Reference(Practitioner|Organization|HealthcareService) }], // Receiver of referral / transfer of care
     request
  "reasonCode" : [{ CodeableConcept }], // Reason for referral / transfer of care request
  "reasonReference" : [{ Reference(Condition|Observation) }], // Why is service needed?
  "description" : "<string>", // A textual description of the referral
  "supportingInfo" : [{ Reference(Any) }], // Additonal information to support referral or transfer of care request
  "note" : [{ Annotation }], // Comments made about referral request
  "relevantHistory" : [{ Reference(Provenance) }] // Key events in history of request
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:ReferralRequest;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
```

```
  fhir:ReferralRequest.identifier [ Identifier ], ... ; # 0..* Business identifier
  fhir:ReferralRequest.definition [ Reference(ActivityDefinition|PlanDefinition) ], ... ; # 0..* Instantiates protocol
    or definition
  fhir:ReferralRequest.basedOn [ Reference(ReferralRequest|CarePlan|ProcedureRequest) ], ... ; # 0..* Request fulfilled
    by this request
  fhir:ReferralRequest.replaces [ Reference(ReferralRequest) ], ... ; # 0..* Request(s) replaced by this request
  fhir:ReferralRequest.groupIdentifier [ Identifier ]; # 0..1 Composite request this is part of
  fhir:ReferralRequest.status [ code ]; # 1..1 draft | active | suspended | cancelled | completed | entered-in-error |
    unknown
  fhir:ReferralRequest.intent [ code ]; # 1..1 proposal | plan | order
  fhir:ReferralRequest.type [ CodeableConcept ]; # 0..1 Referral/Transition of care request type
  fhir:ReferralRequest.priority [ code ]; # 0..1 Urgency of referral / transfer of care request
  fhir:ReferralRequest.serviceRequested [ CodeableConcept ], ... ; # 0..* Actions requested as part of the referral
  fhir:ReferralRequest.subject [ Reference(Patient|Group) ]; # 1..1 Patient referred to care or transfer
  fhir:ReferralRequest.context [ Reference(Encounter|EpisodeOfCare) ]; # 0..1 Originating encounter
  # ReferralRequest.occurrence[x] : 0..1 When the service(s) requested in the referral should occur. One of these 2
    fhir:ReferralRequest.occurrenceDateTime [ dateTime ]
    fhir:ReferralRequest.occurrencePeriod [ Period ]
  fhir:ReferralRequest.authoredOn [ dateTime ]; # 0..1 Date of creation/activation
  fhir:ReferralRequest.requester [ # 0..1 Who/what is requesting service
    fhir:ReferralRequest.requester.agent [ Reference(Practitioner|Organization|Patient|RelatedPerson|Device) ]; # 1..1
    Individual making the request
    fhir:ReferralRequest.requester.onBehalfOf [ Reference(Organization) ]; # 0..1 Organization agent is acting for
  ];
  fhir:ReferralRequest.specialty [ CodeableConcept ]; # 0..1 The clinical specialty (discipline) that the referral is
    requested for
  fhir:ReferralRequest.recipient [ Reference(Practitioner|Organization|HealthcareService) ], ... ; # 0..* Receiver of
    referral / transfer of care request
  fhir:ReferralRequest.reasonCode [ CodeableConcept ], ... ; # 0..* Reason for referral / transfer of care request
  fhir:ReferralRequest.reasonReference [ Reference(Condition|Observation) ], ... ; # 0..* Why is service needed?
  fhir:ReferralRequest.description [ string ]; # 0..1 A textual description of the referral
  fhir:ReferralRequest.supportingInfo [ Reference(Any) ], ... ; # 0..* Additonal information to support referral or
    transfer of care request
  fhir:ReferralRequest.note [ Annotation ], ... ; # 0..* Comments made about referral request
  fhir:ReferralRequest.relevantHistory [ Reference(Provenance) ], ... ; # 0..* Key events in history of request
]
```

Changes since DSTU2

ReferralRequest
ReferralRequest.definition | Added Element
ReferralRequest.basedOn | Added Element
ReferralRequest.replaces | Added Element
ReferralRequest.groupIdentifier | Added Element
ReferralRequest.status | Change value set from http://hl7.org/fhir/ValueSet/referralstatus to http://hl7.org/fhir/ValueSet/request-status
ReferralRequest.intent | Added Element
ReferralRequest.priority | Type changed from CodeableConcept to code Add Binding http://hl7.org/fhir/ValueSet/request-priority (required)
ReferralRequest.subject | Renamed from patient to subject Min Cardinality changed from 0 to 1 Add Reference(Group)
ReferralRequest.context | Renamed from encounter to context Add Reference(EpisodeOfCare)
ReferralRequest.occurrence[x] | Added Element
ReferralRequest.authoredOn | Renamed from date to authoredOn
ReferralRequest.requester | Remove Reference(Practitioner), Remove Reference(Organization), Remove Reference(Patient)
ReferralRequest.requester.agent | Added Element
ReferralRequest.requester.onBehalfOf | Added Element
ReferralRequest.recipient | Add Reference(HealthcareService)
ReferralRequest.reasonCode | Renamed from reason to reasonCode Max Cardinality changed from 1 to *
ReferralRequest.reasonReference | Added Element
ReferralRequest.supportingInfo | Renamed from supportingInformation to supportingInfo
ReferralRequest.note | Added Element
ReferralRequest.relevantHistory | Added Element
ReferralRequest.dateSent | deleted
ReferralRequest.fulfillmentTime | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R2 <--> R3 Conversion Maps (status = 2 tests that all execute ok. 2 fail round-trip testing and 2 r3 resources are invalid (2 errors).).

Structure

Name | Flags | Card. | Type | Description & Constraints
ReferralRequest | | | DomainResource | A request for referral or transfer of care Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | ? | 0..* | Identifier | Business identifier
definition | ? | 0..* | Reference(ActivityDefinition | PlanDefinition) | Instantiates protocol or definition
basedOn | ? | 0..* | Reference(ReferralRequest | CarePlan | ProcedureRequest) | Request fulfilled by this request
replaces | ? | 0..* | Reference(ReferralRequest) | Request(s) replaced by this request
groupIdentifier | ? | 0..1 | Identifier | Composite request this is part of
status | ?!? | 1..1 | code | draft | active | suspended | cancelled | completed | entered-in-error | unknown RequestStatus (Required)
intent | ?!? | 1..1 | code | proposal | plan | order RequestIntent (Required)
type | ? | 0..1 | CodeableConcept | Referral/Transition of care request type SNOMED CT Patient Referral (Example)

priority | ? | 0..1 | code | Urgency of referral / transfer of care request RequestPriority (Required)
serviceRequested | ? | 0..* | CodeableConcept | Actions requested as part of the referral Practice Setting Code Value Set (Example)
subject | ? | 1..1 | Reference(Patient | Group) | Patient referred to care or transfer
context | ? | 0..1 | Reference(Encounter | EpisodeOfCare) | Originating encounter
occurrence[x] | ? | 0..1 | | When the service(s) requested in the referral should occur
occurrenceDateTime | | | dateTime
occurrencePeriod | | | Period
authoredOn | ? | 0..1 | dateTime | Date of creation/activation
requester | ?I | 0..1 | BackboneElement | Who/what is requesting service + onBehalfOf can only be specified if agent is practitioner or device
agent | ? | 1..1 | Reference(Practitioner | Organization | Patient | RelatedPerson | Device) | Individual making the request
onBehalfOf | ?I | 0..1 | Reference(Organization) | Organization agent is acting for
specialty | | 0..1 | CodeableConcept | The clinical specialty (discipline) that the referral is requested for PractitionerSpecialty (Example)
recipient | ? | 0..* | Reference(Practitioner | Organization | HealthcareService) | Receiver of referral / transfer of care request
reasonCode | ? | 0..* | CodeableConcept | Reason for referral / transfer of care request SNOMED CT Clinical Findings (Example)
reasonReference | ? | 0..* | Reference(Condition | Observation) | Why is service needed?
description | | 0..1 | string | A textual description of the referral
supportingInfo | | 0..* | Reference(Any) | Additonal information to support referral or transfer of care request
note | | 0..* | Annotation | Comments made about referral request
relevantHistory | | 0..* | Reference(Provenance) | Key events in history of request
Documentation for this format

UML Diagram (Legend)

XML Template

```
<ReferralRequest xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Business identifier --></identifier>
 <definition><!-- 0..* Reference(ActivityDefinition|PlanDefinition) Instantiates protocol or definition --></definition>
 <basedOn><!-- 0..* Reference(ReferralRequest|CarePlan|ProcedureRequest) Request fulfilled by this request --></basedOn>
 <replaces><!-- 0..* Reference(ReferralRequest) Request(s) replaced by this request --></replaces>
 <groupIdentifier><!-- 0..1 Identifier Composite request this is part of --></groupIdentifier>
 <status value="[code]"/><!-- 1..1 draft | active | suspended | cancelled | completed | entered-in-error | unknown -->
 <intent value="[code]"/><!-- 1..1 proposal | plan | order -->
 <type><!-- 0..1 CodeableConcept Referral/Transition of care request type --></type>
 <priority value="[code]"/><!-- 0..1 Urgency of referral / transfer of care request -->
 <serviceRequested><!-- 0..* CodeableConcept Actions requested as part of the referral --></serviceRequested>
 <subject><!-- 1..1 Reference(Patient|Group) Patient referred to care or transfer --></subject>
 <context><!-- 0..1 Reference(Encounter|EpisodeOfCare) Originating encounter --></context>
 <occurrence[x]><!-- 0..1 dateTime|Period When the service(s) requested in the referral should occur --></occurrence[x]>
 <authoredOn value="[dateTime]"/><!-- 0..1 Date of creation/activation -->
 <requester>  <!-- 0..1 Who/what is requesting service -->
  <agent><!-- 1..1 Reference(Practitioner|Organization|Patient|RelatedPerson|
    Device) Individual making the request --></agent>
  <onBehalfOf><!--  0..1 Reference(Organization) Organization agent is acting for --></onBehalfOf>
 </requester>
 <specialty><!-- 0..1 CodeableConcept The clinical specialty (discipline) that the referral is requested for
    --></specialty>
 <recipient><!-- 0..* Reference(Practitioner|Organization|HealthcareService) Receiver of referral / transfer of care
    request --></recipient>
 <reasonCode><!-- 0..* CodeableConcept Reason for referral / transfer of care request --></reasonCode>
 <reasonReference><!-- 0..* Reference(Condition|Observation) Why is service needed? --></reasonReference>
 <description value="[string]"/><!-- 0..1 A textual description of the referral -->
 <supportingInfo><!-- 0..* Reference(Any) Additonal information to support referral or transfer of care request
    --></supportingInfo>
 <note><!-- 0..* Annotation Comments made about referral request --></note>
 <relevantHistory><!-- 0..* Reference(Provenance) Key events in history of request --></relevantHistory>
</ReferralRequest>
```

JSON Template

```
{
  "resourceType" : "ReferralRequest",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Business identifier
  "definition" : [{ Reference(ActivityDefinition|PlanDefinition) }], // Instantiates protocol or definition
  "basedOn" : [{ Reference(ReferralRequest|CarePlan|ProcedureRequest) }], // Request fulfilled by this request
  "replaces" : [{ Reference(ReferralRequest) }], // Request(s) replaced by this request
  "groupIdentifier" : { Identifier }, // Composite request this is part of
  "status" : "<code>", // R!  draft | active | suspended | cancelled | completed | entered-in-error | unknown
  "intent" : "<code>", // R!  proposal | plan | order
  "type" : { CodeableConcept }, // Referral/Transition of care request type
  "priority" : "<code>", // Urgency of referral / transfer of care request
  "serviceRequested" : [{ CodeableConcept }], // Actions requested as part of the referral
  "subject" : { Reference(Patient|Group) }, // R!  Patient referred to care or transfer
  "context" : { Reference(Encounter|EpisodeOfCare) }, // Originating encounter
  // occurrence[x]: When the service(s) requested in the referral should occur. One of these 2:
  "occurrenceDateTime" : "<dateTime>",
  "occurrencePeriod" : { Period },
  "authoredOn" : "<dateTime>", // Date of creation/activation
  "requester" : { // Who/what is requesting service
    "agent" : { Reference(Practitioner|Organization|Patient|RelatedPerson|
    Device) }, // R!  Individual making the request
    "onBehalfOf" : { Reference(Organization) } // C? Organization agent is acting for
```

```
  },
  "specialty" : { CodeableConcept }, // The clinical specialty (discipline) that the referral is requested for
  "recipient" : [{ Reference(Practitioner|Organization|HealthcareService) }], // Receiver of referral / transfer of care
     request
  "reasonCode" : [{ CodeableConcept }], // Reason for referral / transfer of care request
  "reasonReference" : [{ Reference(Condition|Observation) }], // Why is service needed?
  "description" : "<string>", // A textual description of the referral
  "supportingInfo" : [{ Reference(Any) }], // Additonal information to support referral or transfer of care request
  "note" : [{ Annotation }], // Comments made about referral request
  "relevantHistory" : [{ Reference(Provenance) }] // Key events in history of request
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:ReferralRequest;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:ReferralRequest.identifier [ Identifier ], ... ; # 0..* Business identifier
  fhir:ReferralRequest.definition [ Reference(ActivityDefinition|PlanDefinition) ], ... ; # 0..* Instantiates protocol
    or definition
  fhir:ReferralRequest.basedOn [ Reference(ReferralRequest|CarePlan|ProcedureRequest) ], ... ; # 0..* Request fulfilled
    by this request
  fhir:ReferralRequest.replaces [ Reference(ReferralRequest) ], ... ; # 0..* Request(s) replaced by this request
  fhir:ReferralRequest.groupIdentifier [ Identifier ]; # 0..1 Composite request this is part of
  fhir:ReferralRequest.status [ code ]; # 1..1 draft | active | suspended | cancelled | completed | entered-in-error |
    unknown
  fhir:ReferralRequest.intent [ code ]; # 1..1 proposal | plan | order
  fhir:ReferralRequest.type [ CodeableConcept ]; # 0..1 Referral/Transition of care request type
  fhir:ReferralRequest.priority [ code ]; # 0..1 Urgency of referral / transfer of care request
  fhir:ReferralRequest.serviceRequested [ CodeableConcept ], ... ; # 0..* Actions requested as part of the referral
  fhir:ReferralRequest.subject [ Reference(Patient|Group) ]; # 1..1 Patient referred to care or transfer
  fhir:ReferralRequest.context [ Reference(Encounter|EpisodeOfCare) ]; # 0..1 Originating encounter
  # ReferralRequest.occurrence[x] : 0..1 When the service(s) requested in the referral should occur. One of these 2
    fhir:ReferralRequest.occurrenceDateTime [ dateTime ]
    fhir:ReferralRequest.occurrencePeriod [ Period ]
  fhir:ReferralRequest.authoredOn [ dateTime ]; # 0..1 Date of creation/activation
  fhir:ReferralRequest.requester [ # 0..1 Who/what is requesting service
    fhir:ReferralRequest.requester.agent [ Reference(Practitioner|Organization|Patient|RelatedPerson|Device) ]; # 1..1
    Individual making the request
    fhir:ReferralRequest.requester.onBehalfOf [ Reference(Organization) ]; # 0..1 Organization agent is acting for
  ];
  fhir:ReferralRequest.specialty [ CodeableConcept ]; # 0..1 The clinical specialty (discipline) that the referral is
    requested for
  fhir:ReferralRequest.recipient [ Reference(Practitioner|Organization|HealthcareService) ], ... ; # 0..* Receiver of
    referral / transfer of care request
  fhir:ReferralRequest.reasonCode [ CodeableConcept ], ... ; # 0..* Reason for referral / transfer of care request
  fhir:ReferralRequest.reasonReference [ Reference(Condition|Observation) ], ... ; # 0..* Why is service needed?
  fhir:ReferralRequest.description [ string ]; # 0..1 A textual description of the referral
  fhir:ReferralRequest.supportingInfo [ Reference(Any) ], ... ; # 0..* Additonal information to support referral or
    transfer of care request
  fhir:ReferralRequest.note [ Annotation ], ... ; # 0..* Comments made about referral request
  fhir:ReferralRequest.relevantHistory [ Reference(Provenance) ], ... ; # 0..* Key events in history of request
]
```

Changes since DSTU2

ReferralRequest

ReferralRequest.definition | Added Element

ReferralRequest.basedOn | Added Element

ReferralRequest.replaces | Added Element

ReferralRequest.groupIdentifier | Added Element

ReferralRequest.status | Change value set from http://hl7.org/fhir/ValueSet/referralstatus to http://hl7.org/fhir/ValueSet/request-status

ReferralRequest.intent | Added Element

ReferralRequest.priority | Type changed from CodeableConcept to code Add Binding http://hl7.org/fhir/ValueSet/request-priority (required)

ReferralRequest.subject | Renamed from patient to subject Min Cardinality changed from 0 to 1 Add Reference(Group)

ReferralRequest.context | Renamed from encounter to context Add Reference(EpisodeOfCare)

ReferralRequest.occurrence[x] | Added Element

ReferralRequest.authoredOn | Renamed from date to authoredOn

ReferralRequest.requester | Remove Reference(Practitioner), Remove Reference(Organization), Remove Reference(Patient)

ReferralRequest.requester.agent | Added Element

ReferralRequest.requester.onBehalfOf | Added Element

ReferralRequest.recipient | Add Reference(HealthcareService)

ReferralRequest.reasonCode | Renamed from reason to reasonCode Max Cardinality changed from 1 to *

ReferralRequest.reasonReference | Added Element

ReferralRequest.supportingInfo | Renamed from supportingInformation to supportingInfo

ReferralRequest.note | Added Element

ReferralRequest.relevantHistory | Added Element

ReferralRequest.dateSent | deleted

ReferralRequest.fulfillmentTime | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R2 <--> R3 Conversion Maps (status = 2 tests that all execute ok. 2 fail round-trip testing and 2 r3 resources are invalid (2 errors).).

Alternate definitions: Master Definition (XML, JSON), XML Schema/Schematron (for) + JSON Schema, ShEx (for Turtle)

12.14.3.1 Terminology Bindings

Path | Definition | Type | Reference
ReferralRequest.status | The status of the referral. | Required | RequestStatus
ReferralRequest.intent | Codes identifying the stage lifecycle stage of a request | Required | RequestIntent
ReferralRequest.type | Codes for types of referral; e.g. consult, transfer, temporary transfer. | Example | SNOMED CT Patient Referral
ReferralRequest.priority | Codes indicating the relative priority of the referral. | Required | RequestPriority
ReferralRequest.serviceRequested | Codes indicating the types of services that might be requested as part of a referral. | Example | Practice Setting Code Value Set
ReferralRequest.specialty | Codes indicating the types of capability the referred to service provider must have. | Example | PractitionerSpecialty
ReferralRequest.reasonCode | Codes indicating why the referral is being requested. | Example | SNOMED CT Clinical Findings

12.14.3.2 Constraints

* rfr-1: On ReferralRequest.requester: onBehalfOf can only be specified if agent is practitioner or device (expression on ReferralRequest.requester: (agent.resolve() is Device) or (agent.resolve() is Practitioner) or onBehalfOf.exists().not())

12.14.4 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
authored-on | date | Creation or activation date | ReferralRequest.authoredOn
based-on | reference | Request being fulfilled | ReferralRequest.basedOn (ReferralRequest, CarePlan, ProcedureRequest)
context | reference | Part of encounter or episode of care | ReferralRequest.context (EpisodeOfCare, Encounter)
definition | reference | Instantiates protocol or definition | ReferralRequest.definition (PlanDefinition, ActivityDefinition)
encounter | reference | Originating encounter | ReferralRequest.context (Encounter)
group-identifier | token | Part of common request | ReferralRequest.groupIdentifier
identifier | token | Business identifier | ReferralRequest.identifier
intent | token | Proposal, plan or order | ReferralRequest.intent
occurrence-date | date | When the service(s) requested in the referral should occur | ReferralRequest.occurrence
patient | reference | Who the referral is about | ReferralRequest.subject (Patient) | 31 Resources
priority | token | The priority assigned to the referral | ReferralRequest.priority
recipient | reference | The person that the referral was sent to | ReferralRequest.recipient (Practitioner, Organization, HealthcareService)
replaces | reference | Request(s) replaced by this request | ReferralRequest.replaces (ReferralRequest)
requester | reference | Individual making the request | ReferralRequest.requester.agent (Practitioner, Organization, Device, Patient, RelatedPerson)
service | token | Actions requested as part of the referral | ReferralRequest.serviceRequested
specialty | token | The specialty that the referral is for | ReferralRequest.specialty
status | token | The status of the referral | ReferralRequest.status
subject | reference | Patient referred to care or transfer | ReferralRequest.subject (Group, Patient)
type | token | The type of the referral | ReferralRequest.type | 6 Resources

12.14.5 Resource ReferralRequest - Examples

URL: https://hl7.org/fhir/STU3/referralrequest-examples.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

12.14.5 Resource ReferralRequest - Examples

Patient Care Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Patient, Practitioner

Example List:

Example of referralrequest | example | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

12.14.6 Resource ReferralRequest - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/referralrequest-definitions.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

12.14.6 Resource ReferralRequest - Detailed Descriptions

Patient Care Work Group | Maturity Level: 1 | Trial Use | Compartments: Patient, Practitioner

Detailed Descriptions for the elements in the ReferralRequest resource.

ReferralRequest
Definition | Used to record and send details about a request for referral service or transfer of a patient to the care of another provider or provider organization.
Control | 1..1
Alternate Names | ReferralRequest TransferOfCare Request
ReferralRequest.identifier
Definition | Business identifier that uniquely identifies the referral/care transfer request instance.
Note | This is a business identifer, not a resource identifier (see discussion)
Control | 0..*
Type | Identifier
Summary | true
ReferralRequest.definition
Definition | A protocol, guideline, orderset or other definition that is adhered to in whole or in part by this request.
Control | 0..*
Type | Reference(ActivityDefinition | PlanDefinition)
Summary | true
ReferralRequest.basedOn
Definition | Indicates any plans, proposals or orders that this request is intended to satisfy - in whole or in part.
Control | 0..*
Type | Reference(ReferralRequest | CarePlan | ProcedureRequest)
Alternate Names | fulfills
Summary | true
ReferralRequest.replaces
Definition | Completed or terminated request(s) whose function is taken by this new request.
Control | 0..*
Type | Reference(ReferralRequest)
Requirements | Allows tracing the continuation of a therapy or administrative process instantiated through multiple requests.
Alternate Names | supersedes; prior; renewed order
Summary | true
Comments | The replacement could be because the initial request was immediately rejected (due to an issue) or because the previous request was completed, but the need for the action described by the request remains ongoing.
ReferralRequest.groupIdentifier
Definition | The business identifier of the logical "grouping" request/order that this referral is a part of.
Control | 0..1
Type | Identifier
Requirements | Allows multiple orders to be collected as part of a single requisition.
Alternate Names | grouperId; requisition
Summary | true
ReferralRequest.status
Definition | The status of the authorization/intention reflected by the referral request record.
Control | 1..1
Terminology Binding | RequestStatus (Required)
Type | code
Is Modifier | true
Requirements | Workflow status is handled by the Task resource.
Summary | true
Comments | This element is labeled as a modifier because the status contains codes that mark the resource as not currently valid.
ReferralRequest.intent
Definition | Distinguishes the "level" of authorization/demand implicit in this request.
Control | 1..1
Terminology Binding | RequestIntent (Required)
Type | code
Is Modifier | true
Requirements | The same resource structure is used when capturing proposals/recommendations, plans and actual requests.
Alternate Names | category
Summary | true
ReferralRequest.type
Definition | An indication of the type of referral (or where applicable the type of transfer of care) request.
Control | 0..1
Terminology Binding | SNOMED CT Patient Referral (Example)
Type | CodeableConcept
Summary | true
ReferralRequest.priority
Definition | An indication of the urgency of referral (or where applicable the type of transfer of care) request.
Control | 0..1
Terminology Binding | RequestPriority (Required)
Type | code
Summary | true
ReferralRequest.serviceRequested
Definition | The service(s) that is/are requested to be provided to the patient. For example: cardiac pacemaker insertion.
Control | 0..*
Terminology Binding | Practice Setting Code Value Set (Example)

Type | CodeableConcept
Summary | true
ReferralRequest.subject
Definition | The patient who is the subject of a referral or transfer of care request.
Control | 1..1
Type | Reference(Patient | Group)
Summary | true
Comments | Referral of family, group or community is to be catered for by profiles.
ReferralRequest.context
Definition | The encounter at which the request for referral or transfer of care is initiated.
Control | 0..1
Type | Reference(Encounter | EpisodeOfCare)
Alternate Names | encounter
Summary | true
ReferralRequest.occurrence[x]
Definition | The period of time within which the services identified in the referral/transfer of care is specified or required to occur.
Control | 0..1
Type | dateTime|Period
[x] Note | See Choice of Data Types for further information about how to use [x]
Requirements | Use cases: (1) to indicate that the requested service is not to happen before a specified date, and saving the start date in Period.start; (2) to indicate that the requested service must happen before a specified date, and saving the end date in Period.end; (3) to indicate that the requested service must happen during the specified dates ("start" and "end" values).
Alternate Names | timing
Summary | true
Comments | When the occurrenceDateTime is used, then it is indicating that the requested service must happen before the specified date.
ReferralRequest.authoredOn
Definition | Date/DateTime of creation for draft requests and date of activation for active requests.
Control | 0..1
Type | dateTime
Alternate Names | createdOn
Summary | true
ReferralRequest.requester
Definition | The individual who initiated the request and has responsibility for its activation.
Control | 0..1
Alternate Names | author
Summary | true
Invariants | Defined on this element rfr-1: onBehalfOf can only be specified if agent is practitioner or device (expression : (agent.resolve() is Device) or (agent.resolve() is Practitioner) or onBehalfOf.exists().not(), xpath: contains(f:agent/f:reference/@value, '/Practitioner/') or contains(f:agent/f:reference/@value, '/Device/') or not(exists(f:onBehalfOf)))
ReferralRequest.requester.agent
Definition | The device, practitioner, etc. who initiated the request.
Control | 1..1
Type | Reference(Practitioner | Organization | Patient | RelatedPerson | Device)
Summary | true
ReferralRequest.requester.onBehalfOf
Definition | The organization the device or practitioner was acting on behalf of.
Control | 0..1
Type | Reference(Organization)
Requirements | Practitioners and Devices can be associated with multiple organizations. This element indicates which organization they were acting on behalf of when authoring the request.
Summary | true
Invariants | Affect this element rfr-1: onBehalfOf can only be specified if agent is practitioner or device (expression : (agent.resolve() is Device) or (agent.resolve() is Practitioner) or onBehalfOf.exists().not(), xpath: contains(f:agent/f:reference/@value, '/Practitioner/') or contains(f:agent/f:reference/@value, '/Device/') or not(exists(f:onBehalfOf)))
ReferralRequest.specialty
Definition | Indication of the clinical domain or discipline to which the referral or transfer of care request is sent. For example: Cardiology Gastroenterology Diabetology.
Control | 0..1
Terminology Binding | PractitionerSpecialty (Example)
Type | CodeableConcept
ReferralRequest.recipient
Definition | The healthcare provider(s) or provider organization(s) who/which is to receive the referral/transfer of care request.
Control | 0..*
Type | Reference(Practitioner | Organization | HealthcareService)
Summary | true
Comments | There will be a primary receiver. But the request can be received by any number of "copied to" providers or organizations.
ReferralRequest.reasonCode
Definition | Description of clinical condition indicating why referral/transfer of care is requested. For example: Pathological Anomalies, Disabled (physical or mental), Behavioral Management.
Control | 0..*
Terminology Binding | SNOMED CT Clinical Findings (Example)
Type | CodeableConcept
Summary | true

ReferralRequest.reasonReference
Definition | Indicates another resource whose existence justifies this request.
Control | 0..*
Type | Reference(Condition | Observation)
Summary | true
ReferralRequest.description
Definition | The reason element gives a short description of why the referral is being made, the description expands on this to support a more complete clinical summary.
Control | 0..1
Type | string
ReferralRequest.supportingInfo
Definition | Any additional (administrative, financial or clinical) information required to support request for referral or transfer of care. For example: Presenting problems/chief complaints Medical History Family History Alerts Allergy/Intolerance and Adverse Reactions Medications Observations/Assessments (may include cognitive and fundtional assessments) Diagnostic Reports Care Plan.
Control | 0..*
Type | Reference(Any)
ReferralRequest.note
Definition | Comments made about the referral request by any of the participants.
Control | 0..*
Type | Annotation
ReferralRequest.relevantHistory
Definition | Links to Provenance records for past versions of this resource or fulfilling request or event resources that identify key state transitions or updates that are likely to be relevant to a user looking at the current version of the resource.
Control | 0..*
Type | Reference(Provenance)
Alternate Names | eventHistory
Comments | This element does not point to the Provenance associated with the current version of the resource - as it would be created after this version existed. The Provenance for the current version can be retrieved with a _revinclude.

12.14.7 Resource ReferralRequest - Mappings

URL: https://hl7.org/fhir/STU3/referralrequest-mappings.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

12.14.7 Resource ReferralRequest - Mappings

Patient Care Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Patient, Practitioner

Mappings:

RIM Mapping

HL7 v2 Mapping

Mappings for the referralrequest resource.

12.14.7.1 Mappings for RIM Mapping (http://hl7.org/v3)

ReferralRequest
identifier
definition | .outboundRelationship[typeCode=DEFN].target
basedOn
replaces | .outboundRelationship[typeCode=RPLC].target
groupIdentifier
status
intent
type
priority
serviceRequested
subject
context
occurrence[x]
authoredOn
requester | .participation[typeCode=AUT].role
agent | .player
onBehalfOf | .scoper
specialty
recipient
reasonCode
reasonReference | .outboundRelationship[typeCode=RSON].target
description
supportingInfo
note
relevantHistory | .inboundRelationship(typeCode=SUBJ].source[classCode=CACT, moodCode=EVN]

12.14.7.2 Mappings for HL7 v2 Mapping (http://hl7.org/v2)

ReferralRequest
identifier | RF1-6 / RF1-11
definition
basedOn
replaces
groupIdentifier
status | RF1-1
intent
type | RF1-10
priority | RF1-2
serviceRequested | PR1-3 / OBR-4
subject | PID-3
context | PV1-19
occurrence[x] | OBR-7/OBR-8
authoredOn | RF1-7 / RF1-9?
requester
agent
onBehalfOf
specialty | RF1-3
recipient | Practitioner: PRD-2/PRD-7 where PRD-3 = RT; Organization: PRD-10 where PRD-3 = RT
reasonCode | RF1-10
reasonReference
description
supportingInfo
note
relevantHistory

12.14.8 Resource ReferralRequest - Design Notes

URL: https://hl7.org/fhir/STU3/referralrequest-explanations.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

12.14.8 Resource ReferralRequest - Design Notes

to do

12.14.9 Resource ReferralRequest - Extensions & Profiles

URL: https://hl7.org/fhir/STU3/referralrequest-profiles.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

12.14.9 Resource ReferralRequest - Extensions & Profiles

Patient Care Work Group | Maturity Level: N/A | Ballot Status: Informative

This table lists profiles and extensions for the ReferralRequest resource. For background information, see Profiling Resources and Extensibility. Additional profiles and resources may be found in published Implementation Guides , or in the Conformance resource registry .

12.14.9.1 Profiles

No Profiles defined for this resource

12.14.9.2 Extensions

referralrequest-reasonRefused | reasonRefused | for ReferralRequest HL7 Extensions

Extensions that reference this resource:

condition-basedOn | basedOn | for Condition HL7 Extensions

12.14.9.3 Search Extensions

No Search Extensions defined for this resource

12.14.10 ReferralRequest HL7 Extensions

URL: https://hl7.org/fhir/STU3/referralrequest-extensions.html

Defines common extensions used with or related to the ReferralRequest resource

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

12.14.10 ReferralRequest HL7 Extensions

(No assigned work group) | Maturity Level: N/A | Ballot Status: Informative

12.14.10.1 Content

Extensions:
referralrequest-reasonRefused | reasonRefused : The reason the referral request was refused. Only applicable when status = refused.

12.14.11 Resource ReferralRequest - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/referralrequest-version-maps.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

12.14.11 Resource ReferralRequest - Detailed Descriptions

Patient Care Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Patient, Practitioner

R2 : R3 Converson maps for ReferralRequest.

Functional status for this map: 2 tests that all execute ok. 2 fail round-trip testing and 2 r3 resources are invalid (2 errors). (see documentation)

12.14.11.1 R2 to R3

```
map "http://hl7.org/fhir/StructureMap/ReferralRequest2to3" = "R2 to R3 Conversions for ReferralRequest"


conceptmap "status" {
  prefix s = "http://hl7.org/fhir/referralstatus"
  prefix t = "http://hl7.org/fhir/referralstatus"

  s:draft = t:draft
  s:requested = t:active
  s:active = t:active
  s:cancelled = t:cancelled
  s:accepted = t:active
  s:rejected = t:cancelled
  s:completed = t:completed
}

uses "http://hl7.org/fhir/DSTU2/StructureDefinition/ReferralRequest" alias ReferralRequestR2 as source
uses "http://hl7.org/fhir/StructureDefinition/ReferralRequest" alias ReferralRequest as target

imports "http://hl7.org/fhir/StructureMap/*2to3"

group for type+types  ReferralRequest extends DomainResource
  input src : ReferralRequestR2 as source
  input tgt : ReferralRequest as target

  "ReferralRequest-status" : for src.status as v make tgt.status = translate(v, "#status", "code")
  "ReferralRequest-identifier" : for src.identifier make tgt.identifier
  "ReferralRequest-date" : for src.date make tgt.authoredOn
  "ReferralRequest-type" : for src.type make tgt.type
  "ReferralRequest-specialty" : for src.specialty make tgt.specialty
  "ReferralRequest-priority" : for src.priority as v then {
    "test" : for v.coding as v1 where v1.system = 'http://hl7.org/fhir/diagnostic-order-priority' then {
      "t2" : for v1.code make tgt.priority
    }
  }
  "ReferralRequest-patient" : for src.patient make tgt.subject
  "ReferralRequest-requester" : for src.requester as v make tgt.requester as t, t.agent = v
  "ReferralRequest-recipient" : for src.recipient make tgt.recipient
  "ReferralRequest-encounter" : for src.encounter make tgt.context
  // drop dateSent
  "ReferralRequest-reason" : for src.reason make tgt.reasonCode
  "ReferralRequest-description" : for src.description make tgt.description
  "ReferralRequest-serviceRequested" : for src.serviceRequested make tgt.serviceRequested
  "ReferralRequest-supportingInformation" : for src.supportingInformation make tgt.supportingInfo
  "ReferralRequest-fulfillmentTime" : for src.fulfillmentTime make tgt.occurrence
endgroup
```

12.14.11.2 R3 to R2

```
map "http://hl7.org/fhir/StructureMap/ReferralRequest3to2" = "R3 to R2 Conversion for ReferralRequest"

uses "http://hl7.org/fhir/StructureDefinition/ReferralRequest" alias ReferralRequest as source
uses "http://hl7.org/fhir/DSTU2/StructureDefinition/ReferralRequest" alias ReferralRequestR2 as target

imports "http://hl7.org/fhir/StructureMap/*3to2"

group for type+types  ReferralRequest extends DomainResource
  input src : ReferralRequest as source
  input tgt : ReferralRequestR2 as target

  "ReferralRequest-status" : for src.status check $this != 'entered-in-error' make tgt.status
```

```
  "ReferralRequest-identifier" : for src.identifier make tgt.identifier
  "ReferralRequest-date" : for src.authoredOn make tgt.date
  "ReferralRequest-type" : for src.type make tgt.type
  "ReferralRequest-specialty" : for src.specialty make tgt.specialty
  "ReferralRequest-priority" : for src.priority as v make tgt.priority = cc('http://hl7.org/fhir/diagnostic-order-
    priority', v)
  "ReferralRequest-patient" : for src.subject make tgt.patient
  "ReferralRequest-requester" : for src.requester as v then {
    "ReferralRequest-requester-inner" : for v.agent make tgt.requester
  }
  "ReferralRequest-recipient" : for src.recipient make tgt.recipient
  "ReferralRequest-encounter" : for src.context make tgt.encounter
  "ReferralRequest-reason" : for src.reasonCode make tgt.reason
  "ReferralRequest-description" : for src.description make tgt.description
  "ReferralRequest-serviceRequested" : for src.serviceRequested make tgt.serviceRequested
  "ReferralRequest-supportingInformation" : for src.supportingInfo make tgt.supportingInformation
  "ReferralRequest-fulfillmentTime" : for src.occurrence make tgt.fulfillmentTime
endgroup
```

12.14.11.3 R3 Validation Errors

example | ERROR @ ReferralRequest: Profile http://hl7.org/fhir/StructureDefinition/ReferralRequest, Element 'ReferralRequest.intent': minimum required = 1, but only found 0 (src = InstanceValidator)

referralrequest-example | ERROR @ ReferralRequest: Profile http://hl7.org/fhir/StructureDefinition/ReferralRequest, Element 'ReferralRequest.intent': minimum required = 1, but only found 0 (src = InstanceValidator)

12.15 Resource ProcedureRequest - Content

URL: https://hl7.org/fhir/STU3/procedurerequest.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

Orders and Observations Work Group | Maturity Level: 3 | Trial Use | Compartments: Device, Encounter, Patient, Practitioner, RelatedPerson

A record of a request for diagnostic investigations, treatments, or operations to be performed.

12.15.1 Scope and Usage

This resource is a request resource from a FHIR workflow perspective - see Workflow.

ProcedureRequest is a record of a request for a procedure to be planned, proposed, or performed, as distinguished by the ProcedureRequest.intent field value, with or on a patient. Examples of procedures include diagnostic tests/studies, endoscopic procedures, counseling, biopsies, therapies (e.g., physio-, social-, psychological-), (exploratory) surgeries or procedures, exercises, and other clinical interventions. Procedures may be performed by a healthcare professional, a friend or relative or in some cases by the patient themselves. The procedure will lead to either a Procedure or DiagnosticReport, that in turn may reference one or more Observations, that summarizes the performance of the procedures and associated documentation such as observations, images, findings that are relevant to the treatment/management of the subject.

The principal intention of ProcedureRequest is to support ordering procedures for one patient (which includes non-human patients in veterinary medicine). However, in many contexts, healthcare related processes include performing diagnostic investigations on groups of subjects, devices involved in the provision of healthcare, and even environmental locations such as ducts, bodies of water, etc. ProcedureRequest supports all these usages. The procedure request may represent an order that is entered by a practitioner in a CPOE system as well as a proposal made by a clinical decision support (CDS) system based on a patient's clinical record and context of care. Planned procedures referenced by a CarePlan may also be represented by this resource.

The general work flow that this resource facilitates is that a clinical system creates a procedure request. The procedure request is then accessed by or exchanged with a system, perhaps via intermediaries, that represents an organization (e.g., diagnostic or imaging service, surgical team, physical therapy department) that can perform the procedure. The organization receiving the procedure request will, after it accepts the request, update the request as the work is performed, and then finally issue a report that references the requests that it fulfilled.

The ProcedureRequest resource allows requesting only a single procedure. If a workflow requires requesting multiple procedures simultaneously, this is done using multiple instances of this resource. These instances can be linked in different ways, depending on the needs of the workflow. For guidance, refer to the Request pattern

12.15.2 Boundaries and Relationships

ProcedureRequest, ReferralRequest, and CommunicationRequest are closely related. In fact, for some services, it may be appropriate to use any one of these resources to request that the procedure be performed. Which one is used may be driven by organization practice and by context. When it is unclear which to use, the following principles may be helpful:

* ProcedureRequest is typically used when the requesting clinician has and wishes to exercise the authority (and expertise) to decide exactly what action will be done.

* A ReferralRequest is used when the requesting practitioner is seeking another practitioner or organization to use their own expertise and/or authority to determine the specific action to take.

* A CommunicationRequest is a request to merely disclose information whereas a ProcedureRequest is used when an action is considered to be training or counseling - i.e. when the process will involve verification of the patient's comprehension or an

attempt to change the patient's mental state.

Irrespective of this guidance, systems should be prepared for some degree of overlap between these resources and be prepared to execute searches against multiple resources in cases where differentiation cannot be guaranteed. In some workflows more than one type of resource might exist. For example, upon receiving a ReferralRequest a practitioner might initiate a ProcedureRequest.

This resource is referenced by CarePlan, ClinicalImpression, DiagnosticReport, Goal, ImagingStudy, Media, MedicationRequest, MedicationStatement, Observation, Procedure, QuestionnaireResponse, ReferralRequest and Specimen

12.15.3 Resource Content

* Structure

* UML

* XML

* JSON

* Turtle

* R2 Diff

* All

Structure

Name | Flags | Card. | Type | Description & Constraints
ProcedureRequest | | | DomainResource | A request for a procedure or diagnostic to be performed Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | ? | 0..* | Identifier | Identifiers assigned to this order
definition | ? | 0..* | Reference(ActivityDefinition | PlanDefinition) | Protocol or definition
basedOn | ? | 0..* | Reference(Any) | What request fulfills
replaces | ? | 0..* | Reference(Any) | What request replaces
requisition | ? | 0..1 | Identifier | Composite Request ID
status | ?!? | 1..1 | code | draft | active | suspended | completed | entered-in-error | cancelled RequestStatus (Required)
intent | ?!? | 1..1 | code | proposal | plan | order + RequestIntent (Required)
priority | ? | 0..1 | code | routine | urgent | asap | stat RequestPriority (Required)
doNotPerform | ?!? | 0..1 | boolean | True if procedure should not be performed
category | ? | 0..* | CodeableConcept | Classification of procedure Procedure Category Codes (SNOMED CT) (Example)
code | ? | 1..1 | CodeableConcept | What is being requested/ordered Procedure Codes (SNOMED CT) (Example)
subject | ? | 1..1 | Reference(Patient | Group | Location | Device) | Individual the service is ordered for
context | ? | 0..1 | Reference(Encounter | EpisodeOfCare) | Encounter or Episode during which request was created
occurrence[x] | ? | 0..1 | | When procedure should occur
occurrenceDateTime | | | dateTime
occurrencePeriod | | | Period
occurrenceTiming | | | Timing
asNeeded[x] | ? | 0..1 | | Preconditions for procedure or diagnostic SNOMED CT Medication As Needed Reason Codes (Example)
asNeededBoolean | | | boolean
asNeededCodeableConcept | | | CodeableConcept
authoredOn | ? | 0..1 | dateTime | Date request signed
requester | ? | 0..1 | BackboneElement | Who/what is requesting procedure or diagnostic
agent | ? | 1..1 | Reference(Device | Practitioner | Organization) | Individual making the request
onBehalfOf | ? | 0..1 | Reference(Organization) | Organization agent is acting for
performerType | ? | 0..1 | CodeableConcept | Performer role Participant Roles (Example)
performer | ? | 0..1 | Reference(Practitioner | Organization | Patient | Device | RelatedPerson | HealthcareService) | Requested perfomer
reasonCode | ? | 0..* | CodeableConcept | Explanation/Justification for test Procedure Reason Codes (Example)
reasonReference | ? | 0..* | Reference(Condition | Observation) | Explanation/Justification for test
supportingInfo | | 0..* | Reference(Any) | Additional clinical information
specimen | ? | 0..* | Reference(Specimen) | Procedure Samples
bodySite | ? | 0..* | CodeableConcept | Location on Body SNOMED CT Body Structures (Example)
note | | 0..* | Annotation | Comments
relevantHistory | | 0..* | Reference(Provenance) | Request provenance
Documentation for this format

UML Diagram (Legend)

XML Template

```
<ProcedureRequest xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Identifiers assigned to this order --></identifier>
 <definition><!-- 0..* Reference(ActivityDefinition|PlanDefinition) Protocol or definition --></definition>
 <basedOn><!-- 0..* Reference(Any) What request fulfills --></basedOn>
 <replaces><!-- 0..* Reference(Any) What request replaces --></replaces>
 <requisition><!-- 0..1 Identifier Composite Request ID --></requisition>
 <status value="[code]"/><!-- 1..1 draft | active | suspended | completed | entered-in-error | cancelled -->
 <intent value="[code]"/><!-- 1..1 proposal | plan | order + -->
 <priority value="[code]"/><!-- 0..1 routine | urgent | asap | stat -->
 <doNotPerform value="[boolean]"/><!-- 0..1 True if procedure should not be performed -->
 <category><!-- 0..* CodeableConcept Classification of procedure --></category>
```

```
 <code><!-- 1..1 CodeableConcept What is being requested/ordered --></code>
 <subject><!-- 1..1 Reference(Patient|Group|Location|Device) Individual the service is ordered for --></subject>
 <context><!-- 0..1 Reference(Encounter|EpisodeOfCare) Encounter or Episode during which request was created
    --></context>
 <occurrence[x]><!-- 0..1 dateTime|Period|Timing When procedure should occur --></occurrence[x]>
 <asNeeded[x]><!-- 0..1 boolean|CodeableConcept Preconditions for procedure or diagnostic --></asNeeded[x]>
 <authoredOn value="[dateTime]"/><!-- 0..1 Date request signed -->
 <requester>  <!-- 0..1 Who/what is requesting procedure or diagnostic -->
  <agent><!-- 1..1 Reference(Device|Practitioner|Organization) Individual making the request --></agent>
  <onBehalfOf><!-- 0..1 Reference(Organization) Organization agent is acting for --></onBehalfOf>
 </requester>
 <performerType><!-- 0..1 CodeableConcept Performer role --></performerType>
 <performer><!-- 0..1 Reference(Practitioner|Organization|Patient|Device|
   RelatedPerson|HealthcareService) Requested perfomer --></performer>
 <reasonCode><!-- 0..* CodeableConcept Explanation/Justification for test --></reasonCode>
 <reasonReference><!-- 0..* Reference(Condition|Observation) Explanation/Justification for test --></reasonReference>
 <supportingInfo><!-- 0..* Reference(Any) Additional clinical information --></supportingInfo>
 <specimen><!-- 0..* Reference(Specimen) Procedure Samples --></specimen>
 <bodySite><!-- 0..* CodeableConcept Location on Body --></bodySite>
 <note><!-- 0..* Annotation Comments --></note>
 <relevantHistory><!-- 0..* Reference(Provenance) Request provenance --></relevantHistory>
</ProcedureRequest>
```

JSON Template

```
{
  "resourceType" : "ProcedureRequest",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Identifiers assigned to this order
  "definition" : [{ Reference(ActivityDefinition|PlanDefinition) }], // Protocol or definition
  "basedOn" : [{ Reference(Any) }], // What request fulfills
  "replaces" : [{ Reference(Any) }], // What request replaces
  "requisition" : { Identifier }, // Composite Request ID
  "status" : "<code>", // R!  draft | active | suspended | completed | entered-in-error | cancelled
  "intent" : "<code>", // R!  proposal | plan | order +
  "priority" : "<code>", // routine | urgent | asap | stat
  "doNotPerform" : <boolean>, // True if procedure should not be performed
  "category" : [{ CodeableConcept }], // Classification of procedure
  "code" : { CodeableConcept }, // R!  What is being requested/ordered
  "subject" : { Reference(Patient|Group|Location|Device) }, // R!  Individual the service is ordered for
  "context" : { Reference(Encounter|EpisodeOfCare) }, // Encounter or Episode during which request was created
  // occurrence[x]: When procedure should occur. One of these 3:
  "occurrenceDateTime" : "<dateTime>",
  "occurrencePeriod" : { Period },
  "occurrenceTiming" : { Timing },
  // asNeeded[x]: Preconditions for procedure or diagnostic. One of these 2:
  "asNeededBoolean" : <boolean>,
  "asNeededCodeableConcept" : { CodeableConcept },
  "authoredOn" : "<dateTime>", // Date request signed
  "requester" : { // Who/what is requesting procedure or diagnostic
    "agent" : { Reference(Device|Practitioner|Organization) }, // R!  Individual making the request
    "onBehalfOf" : { Reference(Organization) } // Organization agent is acting for
  },
  "performerType" : { CodeableConcept }, // Performer role
  "performer" : { Reference(Practitioner|Organization|Patient|Device|
   RelatedPerson|HealthcareService) }, // Requested perfomer
  "reasonCode" : [{ CodeableConcept }], // Explanation/Justification for test
  "reasonReference" : [{ Reference(Condition|Observation) }], // Explanation/Justification for test
  "supportingInfo" : [{ Reference(Any) }], // Additional clinical information
  "specimen" : [{ Reference(Specimen) }], // Procedure Samples
  "bodySite" : [{ CodeableConcept }], // Location on Body
  "note" : [{ Annotation }], // Comments
  "relevantHistory" : [{ Reference(Provenance) }] // Request provenance
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:ProcedureRequest;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:ProcedureRequest.identifier [ Identifier ], ... ; # 0..* Identifiers assigned to this order
  fhir:ProcedureRequest.definition [ Reference(ActivityDefinition|PlanDefinition) ], ... ; # 0..* Protocol or definition
  fhir:ProcedureRequest.basedOn [ Reference(Any) ], ... ; # 0..* What request fulfills
  fhir:ProcedureRequest.replaces [ Reference(Any) ], ... ; # 0..* What request replaces
  fhir:ProcedureRequest.requisition [ Identifier ]; # 0..1 Composite Request ID
  fhir:ProcedureRequest.status [ code ]; # 1..1 draft | active | suspended | completed | entered-in-error | cancelled
  fhir:ProcedureRequest.intent [ code ]; # 1..1 proposal | plan | order +
  fhir:ProcedureRequest.priority [ code ]; # 0..1 routine | urgent | asap | stat
  fhir:ProcedureRequest.doNotPerform [ boolean ]; # 0..1 True if procedure should not be performed
  fhir:ProcedureRequest.category [ CodeableConcept ], ... ; # 0..* Classification of procedure
  fhir:ProcedureRequest.code [ CodeableConcept ]; # 1..1 What is being requested/ordered
  fhir:ProcedureRequest.subject [ Reference(Patient|Group|Location|Device) ]; # 1..1 Individual the service is ordered
    for
  fhir:ProcedureRequest.context [ Reference(Encounter|EpisodeOfCare) ]; # 0..1 Encounter or Episode during which request
     was created
```

```
  # ProcedureRequest.occurrence[x] : 0..1 When procedure should occur. One of these 3
    fhir:ProcedureRequest.occurrenceDateTime [ dateTime ]
    fhir:ProcedureRequest.occurrencePeriod [ Period ]
    fhir:ProcedureRequest.occurrenceTiming [ Timing ]
  # ProcedureRequest.asNeeded[x] : 0..1 Preconditions for procedure or diagnostic. One of these 2
    fhir:ProcedureRequest.asNeededBoolean [ boolean ]
    fhir:ProcedureRequest.asNeededCodeableConcept [ CodeableConcept ]
  fhir:ProcedureRequest.authoredOn [ dateTime ]; # 0..1 Date request signed
  fhir:ProcedureRequest.requester [ # 0..1 Who/what is requesting procedure or diagnostic
    fhir:ProcedureRequest.requester.agent [ Reference(Device|Practitioner|Organization) ]; # 1..1 Individual making the
    request
    fhir:ProcedureRequest.requester.onBehalfOf [ Reference(Organization) ]; # 0..1 Organization agent is acting for
  ];
  fhir:ProcedureRequest.performerType [ CodeableConcept ]; # 0..1 Performer role
  fhir:ProcedureRequest.performer [ Reference(Practitioner|Organization|Patient|Device|RelatedPerson|HealthcareService)
    ]; # 0..1 Requested perfomer
  fhir:ProcedureRequest.reasonCode [ CodeableConcept ], ... ; # 0..* Explanation/Justification for test
  fhir:ProcedureRequest.reasonReference [ Reference(Condition|Observation) ], ... ; # 0..* Explanation/Justification for
     test
  fhir:ProcedureRequest.supportingInfo [ Reference(Any) ], ... ; # 0..* Additional clinical information
  fhir:ProcedureRequest.specimen [ Reference(Specimen) ], ... ; # 0..* Procedure Samples
  fhir:ProcedureRequest.bodySite [ CodeableConcept ], ... ; # 0..* Location on Body
  fhir:ProcedureRequest.note [ Annotation ], ... ; # 0..* Comments
  fhir:ProcedureRequest.relevantHistory [ Reference(Provenance) ], ... ; # 0..* Request provenance
]
```

Changes since DSTU2

ProcedureRequest
ProcedureRequest.definition | Added Element
ProcedureRequest.basedOn | Added Element
ProcedureRequest.replaces | Added Element
ProcedureRequest.requisition | Added Element
ProcedureRequest.status | Min Cardinality changed from 0 to 1 Change value set from http://hl7.org/fhir/ValueSet/procedure-request-status to http://hl7.org/fhir/ValueSet/request-status
ProcedureRequest.intent | Added Element
ProcedureRequest.priority | Change value set from http://hl7.org/fhir/ValueSet/procedure-request-priority to http://hl7.org/fhir/ValueSet/request-priority
ProcedureRequest.doNotPerform | Added Element
ProcedureRequest.category | Added Element
ProcedureRequest.subject | Add Reference(Location), Add Reference(Device)
ProcedureRequest.context | Renamed from encounter to context Add Reference(EpisodeOfCare)
ProcedureRequest.occurrence[x] | Added Element
ProcedureRequest.authoredOn | Renamed from orderedOn to authoredOn
ProcedureRequest.requester | Added Element
ProcedureRequest.requester.agent | Added Element
ProcedureRequest.requester.onBehalfOf | Added Element
ProcedureRequest.performerType | Added Element
ProcedureRequest.performer | Add Reference(Device), Add Reference(HealthcareService)
ProcedureRequest.reasonCode | Added Element
ProcedureRequest.reasonReference | Added Element
ProcedureRequest.supportingInfo | Added Element
ProcedureRequest.specimen | Added Element
ProcedureRequest.note | Renamed from notes to note
ProcedureRequest.relevantHistory | Added Element
ProcedureRequest.reason[x] | deleted
ProcedureRequest.scheduled[x] | deleted
ProcedureRequest.orderer | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R2 <--> R3 Conversion Maps (status = 6 tests that all execute ok. 5 fail round-trip testing and 6 r3 resources are invalid (9 errors).).

Structure

Name | Flags | Card. | Type | Description & Constraints
ProcedureRequest | | | DomainResource | A request for a procedure or diagnostic to be performed Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | ? | 0..* | Identifier | Identifiers assigned to this order
definition | ? | 0..* | Reference(ActivityDefinition | PlanDefinition) | Protocol or definition
basedOn | ? | 0..* | Reference(Any) | What request fulfills
replaces | ? | 0..* | Reference(Any) | What request replaces
requisition | ? | 0..1 | Identifier | Composite Request ID
status | ?!? | 1..1 | code | draft | active | suspended | completed | entered-in-error | cancelled RequestStatus (Required)
intent | ?!? | 1..1 | code | proposal | plan | order + RequestIntent (Required)
priority | ? | 0..1 | code | routine | urgent | asap | stat RequestPriority (Required)
doNotPerform | ?!? | 0..1 | boolean | True if procedure should not be performed
category | ? | 0..* | CodeableConcept | Classification of procedure Procedure Category Codes (SNOMED CT) (Example)
code | ? | 1..1 | CodeableConcept | What is being requested/ordered Procedure Codes (SNOMED CT) (Example)

subject | ? | 1..1 | Reference(Patient | Group | Location | Device) | Individual the service is ordered for
context | ? | 0..1 | Reference(Encounter | EpisodeOfCare) | Encounter or Episode during which request was created
occurrence[x] | ? | 0..1 | | When procedure should occur
occurrenceDateTime | | | dateTime
occurrencePeriod | | | Period
occurrenceTiming | | | Timing
asNeeded[x] | ? | 0..1 | | Preconditions for procedure or diagnostic SNOMED CT Medication As Needed Reason Codes (Example)
asNeededBoolean | | | boolean
asNeededCodeableConcept | | | CodeableConcept
authoredOn | ? | 0..1 | dateTime | Date request signed
requester | ? | 0..1 | BackboneElement | Who/what is requesting procedure or diagnostic
agent | ? | 1..1 | Reference(Device | Practitioner | Organization) | Individual making the request
onBehalfOf | ? | 0..1 | Reference(Organization) | Organization agent is acting for
performerType | ? | 0..1 | CodeableConcept | Performer role Participant Roles (Example)
performer | ? | 0..1 | Reference(Practitioner | Organization | Patient | Device | RelatedPerson | HealthcareService) | Requested perfomer
reasonCode | ? | 0..* | CodeableConcept | Explanation/Justification for test Procedure Reason Codes (Example)
reasonReference | ? | 0..* | Reference(Condition | Observation) | Explanation/Justification for test
supportingInfo | | 0..* | Reference(Any) | Additional clinical information
specimen | ? | 0..* | Reference(Specimen) | Procedure Samples
bodySite | ? | 0..* | CodeableConcept | Location on Body SNOMED CT Body Structures (Example)
note | | 0..* | Annotation | Comments
relevantHistory | | 0..* | Reference(Provenance) | Request provenance
Documentation for this format

UML Diagram (Legend)

XML Template

```
<ProcedureRequest xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Identifiers assigned to this order --></identifier>
 <definition><!-- 0..* Reference(ActivityDefinition|PlanDefinition) Protocol or definition --></definition>
 <basedOn><!-- 0..* Reference(Any) What request fulfills --></basedOn>
 <replaces><!-- 0..* Reference(Any) What request replaces --></replaces>
 <requisition><!-- 0..1 Identifier Composite Request ID --></requisition>
 <status value="[code]"/><!-- 1..1 draft | active | suspended | completed | entered-in-error | cancelled -->
 <intent value="[code]"/><!-- 1..1 proposal | plan | order + -->
 <priority value="[code]"/><!-- 0..1 routine | urgent | asap | stat -->
 <doNotPerform value="[boolean]"/><!-- 0..1 True if procedure should not be performed -->
 <category><!-- 0..* CodeableConcept Classification of procedure --></category>
 <code><!-- 1..1 CodeableConcept What is being requested/ordered --></code>
 <subject><!-- 1..1 Reference(Patient|Group|Location|Device) Individual the service is ordered for --></subject>
 <context><!-- 0..1 Reference(Encounter|EpisodeOfCare) Encounter or Episode during which request was created
    --></context>
 <occurrence[x]><!-- 0..1 dateTime|Period|Timing When procedure should occur --></occurrence[x]>
 <asNeeded[x]><!-- 0..1 boolean|CodeableConcept Preconditions for procedure or diagnostic --></asNeeded[x]>
 <authoredOn value="[dateTime]"/><!-- 0..1 Date request signed -->
 <requester>  <!-- 0..1 Who/what is requesting procedure or diagnostic -->
  <agent><!-- 1..1 Reference(Device|Practitioner|Organization) Individual making the request --></agent>
  <onBehalfOf><!-- 0..1 Reference(Organization) Organization agent is acting for --></onBehalfOf>
 </requester>
 <performerType><!-- 0..1 CodeableConcept Performer role --></performerType>
 <performer><!-- 0..1 Reference(Practitioner|Organization|Patient|Device|
   RelatedPerson|HealthcareService) Requested perfomer --></performer>
 <reasonCode><!-- 0..* CodeableConcept Explanation/Justification for test --></reasonCode>
 <reasonReference><!-- 0..* Reference(Condition|Observation) Explanation/Justification for test --></reasonReference>
 <supportingInfo><!-- 0..* Reference(Any) Additional clinical information --></supportingInfo>
 <specimen><!-- 0..* Reference(Specimen) Procedure Samples --></specimen>
 <bodySite><!-- 0..* CodeableConcept Location on Body --></bodySite>
 <note><!-- 0..* Annotation Comments --></note>
 <relevantHistory><!-- 0..* Reference(Provenance) Request provenance --></relevantHistory>
</ProcedureRequest>
```

JSON Template

```
{
  "resourceType" : "ProcedureRequest",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Identifiers assigned to this order
  "definition" : [{ Reference(ActivityDefinition|PlanDefinition) }], // Protocol or definition
  "basedOn" : [{ Reference(Any) }], // What request fulfills
  "replaces" : [{ Reference(Any) }], // What request replaces
  "requisition" : { Identifier }, // Composite Request ID
  "status" : "<code>", // R!  draft | active | suspended | completed | entered-in-error | cancelled
  "intent" : "<code>", // R!  proposal | plan | order +
  "priority" : "<code>", // routine | urgent | asap | stat
  "doNotPerform" : <boolean>, // True if procedure should not be performed
  "category" : [{ CodeableConcept }], // Classification of procedure
  "code" : { CodeableConcept }, // R!  What is being requested/ordered
  "subject" : { Reference(Patient|Group|Location|Device) }, // R!  Individual the service is ordered for
  "context" : { Reference(Encounter|EpisodeOfCare) }, // Encounter or Episode during which request was created
  // occurrence[x]: When procedure should occur. One of these 3:
  "occurrenceDateTime" : "<dateTime>",
  "occurrencePeriod" : { Period },
```

```
  "occurrenceTiming" : { Timing },
  // asNeeded[x]: Preconditions for procedure or diagnostic. One of these 2:
  "asNeededBoolean" : <boolean>,
  "asNeededCodeableConcept" : { CodeableConcept },
  "authoredOn" : "<dateTime>", // Date request signed
  "requester" : { // Who/what is requesting procedure or diagnostic
    "agent" : { Reference(Device|Practitioner|Organization) }, // R!  Individual making the request
    "onBehalfOf" : { Reference(Organization) } // Organization agent is acting for
  },
  "performerType" : { CodeableConcept }, // Performer role
  "performer" : { Reference(Practitioner|Organization|Patient|Device|
   RelatedPerson|HealthcareService) }, // Requested perfomer
  "reasonCode" : [{ CodeableConcept }], // Explanation/Justification for test
  "reasonReference" : [{ Reference(Condition|Observation) }], // Explanation/Justification for test
  "supportingInfo" : [{ Reference(Any) }], // Additional clinical information
  "specimen" : [{ Reference(Specimen) }], // Procedure Samples
  "bodySite" : [{ CodeableConcept }], // Location on Body
  "note" : [{ Annotation }], // Comments
  "relevantHistory" : [{ Reference(Provenance) }] // Request provenance
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:ProcedureRequest;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:ProcedureRequest.identifier [ Identifier ], ... ; # 0..* Identifiers assigned to this order
  fhir:ProcedureRequest.definition [ Reference(ActivityDefinition|PlanDefinition) ], ... ; # 0..* Protocol or definition
  fhir:ProcedureRequest.basedOn [ Reference(Any) ], ... ; # 0..* What request fulfills
  fhir:ProcedureRequest.replaces [ Reference(Any) ], ... ; # 0..* What request replaces
  fhir:ProcedureRequest.requisition [ Identifier ]; # 0..1 Composite Request ID
  fhir:ProcedureRequest.status [ code ]; # 1..1 draft | active | suspended | completed | entered-in-error | cancelled
  fhir:ProcedureRequest.intent [ code ]; # 1..1 proposal | plan | order +
  fhir:ProcedureRequest.priority [ code ]; # 0..1 routine | urgent | asap | stat
  fhir:ProcedureRequest.doNotPerform [ boolean ]; # 0..1 True if procedure should not be performed
  fhir:ProcedureRequest.category [ CodeableConcept ], ... ; # 0..* Classification of procedure
  fhir:ProcedureRequest.code [ CodeableConcept ]; # 1..1 What is being requested/ordered
  fhir:ProcedureRequest.subject [ Reference(Patient|Group|Location|Device) ]; # 1..1 Individual the service is ordered
    for
  fhir:ProcedureRequest.context [ Reference(Encounter|EpisodeOfCare) ]; # 0..1 Encounter or Episode during which request
     was created
  # ProcedureRequest.occurrence[x] : 0..1 When procedure should occur. One of these 3
    fhir:ProcedureRequest.occurrenceDateTime [ dateTime ]
    fhir:ProcedureRequest.occurrencePeriod [ Period ]
    fhir:ProcedureRequest.occurrenceTiming [ Timing ]
  # ProcedureRequest.asNeeded[x] : 0..1 Preconditions for procedure or diagnostic. One of these 2
    fhir:ProcedureRequest.asNeededBoolean [ boolean ]
    fhir:ProcedureRequest.asNeededCodeableConcept [ CodeableConcept ]
  fhir:ProcedureRequest.authoredOn [ dateTime ]; # 0..1 Date request signed
  fhir:ProcedureRequest.requester [ # 0..1 Who/what is requesting procedure or diagnostic
    fhir:ProcedureRequest.requester.agent [ Reference(Device|Practitioner|Organization) ]; # 1..1 Individual making the
    request
    fhir:ProcedureRequest.requester.onBehalfOf [ Reference(Organization) ]; # 0..1 Organization agent is acting for
  ];
  fhir:ProcedureRequest.performerType [ CodeableConcept ]; # 0..1 Performer role
  fhir:ProcedureRequest.performer [ Reference(Practitioner|Organization|Patient|Device|RelatedPerson|HealthcareService)
    ]; # 0..1 Requested perfomer
  fhir:ProcedureRequest.reasonCode [ CodeableConcept ], ... ; # 0..* Explanation/Justification for test
  fhir:ProcedureRequest.reasonReference [ Reference(Condition|Observation) ], ... ; # 0..* Explanation/Justification for
     test
  fhir:ProcedureRequest.supportingInfo [ Reference(Any) ], ... ; # 0..* Additional clinical information
  fhir:ProcedureRequest.specimen [ Reference(Specimen) ], ... ; # 0..* Procedure Samples
  fhir:ProcedureRequest.bodySite [ CodeableConcept ], ... ; # 0..* Location on Body
  fhir:ProcedureRequest.note [ Annotation ], ... ; # 0..* Comments
  fhir:ProcedureRequest.relevantHistory [ Reference(Provenance) ], ... ; # 0..* Request provenance
]
```

Changes since DSTU2

ProcedureRequest

ProcedureRequest.definition | Added Element

ProcedureRequest.basedOn | Added Element

ProcedureRequest.replaces | Added Element

ProcedureRequest.requisition | Added Element

ProcedureRequest.status | Min Cardinality changed from 0 to 1 Change value set from http://hl7.org/fhir/ValueSet/procedure-request-status to http://hl7.org/fhir/ValueSet/request-status

ProcedureRequest.intent | Added Element

ProcedureRequest.priority | Change value set from http://hl7.org/fhir/ValueSet/procedure-request-priority to http://hl7.org/fhir/ValueSet/request-priority

ProcedureRequest.doNotPerform | Added Element

ProcedureRequest.category | Added Element

ProcedureRequest.subject | Add Reference(Location), Add Reference(Device)

ProcedureRequest.context | Renamed from encounter to context Add Reference(EpisodeOfCare)

ProcedureRequest.occurrence[x] | Added Element
ProcedureRequest.authoredOn | Renamed from orderedOn to authoredOn
ProcedureRequest.requester | Added Element
ProcedureRequest.requester.agent | Added Element
ProcedureRequest.requester.onBehalfOf | Added Element
ProcedureRequest.performerType | Added Element
ProcedureRequest.performer | Add Reference(Device), Add Reference(HealthcareService)
ProcedureRequest.reasonCode | Added Element
ProcedureRequest.reasonReference | Added Element
ProcedureRequest.supportingInfo | Added Element
ProcedureRequest.specimen | Added Element
ProcedureRequest.note | Renamed from notes to note
ProcedureRequest.relevantHistory | Added Element
ProcedureRequest.reason[x] | deleted
ProcedureRequest.scheduled[x] | deleted
ProcedureRequest.orderer | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R2 <--> R3 Conversion Maps (status = 6 tests that all execute ok. 5 fail round-trip testing and 6 r3 resources are invalid (9 errors).).

Alternate definitions: Master Definition (XML, JSON), XML Schema/Schematron (for) + JSON Schema, ShEx (for Turtle)

12.15.3.1 Terminology Bindings

Path | Definition | Type | Reference
ProcedureRequest.status | The status of a procedure or diagnostic order. | Required | RequestStatus
ProcedureRequest.intent | The kind of procedure or diagnostic request | Required | RequestIntent
ProcedureRequest.priority | Identifies the level of importance to be assigned to actioning the request | Required | RequestPriority
ProcedureRequest.category | Classification of the procedure | Example | Procedure Category Codes (SNOMED CT)
ProcedureRequest.code | Codes for tests/services that can be performed by procedure or diagnostic services. For laboratory, LOINC is (preferred)[http://hl7.org/fhir/STU3/terminologies.html#preferred] and a valueset using LOINC Order codes is available [here](valueset-diagnostic-requests.html). | Example | Procedure Codes (SNOMED CT)
ProcedureRequest.asNeeded[x] | A coded concept identifying the pre-condition that should hold prior to performing a procedure. For example "pain", "on flare-up", etc. | Example | SNOMED CT Medication As Needed Reason Codes
ProcedureRequest.performerType | Indicates specific responsibility of an individual within the care team, such as "Primary physician", "Team coordinator", "Caregiver", etc. | Example | Participant Roles
ProcedureRequest.reasonCode | Diagnosis or problem codes justifying the reason for requesting the procedure or diagnostic investigation. | Example | Procedure Reason Codes
ProcedureRequest.bodySite | Codes describing anatomical locations. May include laterality. | Example | SNOMED CT Body Structures

12.15.4 Notes:

* Many procedure requests will create a need to specify a specimen, body site, or body system. The request code will often have this information embedded in it - for example, 'serum glucose' or 'chest xray'. Alternatively, the specimen or bodysite element may be used to specify it.

* The ProcedureRequest should only reference the Specimen resource directly when the diagnostic investigation is requested on already existing specimens. Conversely, if the request is entered first with an uncollected specimen, the Specimen resource will reference the DiagnosticRequest resource when it is created.

* The reasonCode element is often for billing purposes. It may relate to the resources referred to in supportinginfo element and may be used to decide how a procedure or diagnostic investigation will be performed, or even if it will be performed at all

12.15.5 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
authored | date | Date request signed | ProcedureRequest.authoredOn
based-on | reference | What request fulfills | ProcedureRequest.basedOn (Any)
body-site | token | Where procedure is going to be done | ProcedureRequest.bodySite
code | token | What is being requested/ordered | ProcedureRequest.code | 8 Resources
context | reference | Encounter or Episode during which request was created | ProcedureRequest.context (EpisodeOfCare, Encounter)
definition | reference | Protocol or definition | ProcedureRequest.definition (PlanDefinition, ActivityDefinition)
encounter | reference | An encounter in which this request is made | ProcedureRequest.context (Encounter) | 11 Resources
identifier | token | Identifiers assigned to this order | ProcedureRequest.identifier | 25 Resources
intent | token | proposal | plan | order + | ProcedureRequest.intent
occurrence | date | When procedure should occur | ProcedureRequest.occurrence
patient | reference | Search by subject - a patient | ProcedureRequest.subject (Patient) | 30 Resources
performer | reference | Requested perfomer | ProcedureRequest.performer (Practitioner, Organization, Device, Patient, HealthcareService, RelatedPerson)
performer-type | token | Performer role | ProcedureRequest.performerType
priority | token | routine | urgent | asap | stat | ProcedureRequest.priority

replaces | reference | What request replaces | ProcedureRequest.replaces (Any)
requester | reference | Individual making the request | ProcedureRequest.requester.agent (Practitioner, Organization, Device)
requisition | token | Composite Request ID | ProcedureRequest.requisition
specimen | reference | Specimen to be tested | ProcedureRequest.specimen (Specimen)
status | token | draft | active | suspended | completed | entered-in-error | cancelled | ProcedureRequest.status
subject | reference | Search by subject | ProcedureRequest.subject (Group, Device, Patient, Location)

12.15.6 Resource ProcedureRequest - Examples

URL: https://hl7.org/fhir/STU3/procedurerequest-examples.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

12.15.6 Resource ProcedureRequest - Examples

Orders and Observations Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Device, Encounter, Patient, Practitioner, RelatedPerson

An introduction to this page as well as proper examples are to be submitted shortly.

Example List:

Lipid Panel Order | lipid | XML | JSON | Turtle
CT Scan order | di | XML | JSON | Turtle
Free T4 Add on Order | ft4 | XML | JSON | Turtle
An example of a Head CT procedure | example | XML | JSON | Turtle
Example with sub-requests derived from the request | subrequest | XML | JSON | Turtle
An example of an order for home physiotherapy | physiotherapy | XML | JSON | Turtle
An example of an order to not turn a patient | do-not-turn | XML | JSON | Turtle
Part of an exercise plan | benchpress | XML | JSON | Turtle
Example of an order for an implant | example-implant | XML | JSON | Turtle
Example of an order for anambulation procedure | ambulation | XML | JSON | Turtle
Example of an order for a biopsy procedure that was part of a colonoscopy | colon-biopsy | XML | JSON | Turtle
Example of an order for a colonoscopy procedure with complication | colonoscopy | XML | JSON | Turtle
Example of an order for an appendectomy procedure that is primarily narrative | appendectomy-narrative | XML | JSON | Turtle
Example of an order for an obstetric procedure | ob | XML | JSON | Turtle
Example of an order for an physical therapy evaluation procedure | physical-therapy | XML | JSON | Turtle
Example of an order for patient education | education | XML | JSON | Turtle
PGx Phenotype example | example-pgx | XML | JSON | Turtle | for Profile ProcedureRequest-Genetics
Order for ABCB4 gene mutation analysis | og-example1 | XML | JSON | Turtle | for Profile ProcedureRequest-Genetics

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

12.15.7 Resource ProcedureRequest - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/procedurerequest-definitions.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

12.15.7 Resource ProcedureRequest - Detailed Descriptions

Orders and Observations Work Group | Maturity Level: 3 | Trial Use | Compartments: Device, Encounter, Patient, Practitioner, RelatedPerson

Detailed Descriptions for the elements in the ProcedureRequest resource.

ProcedureRequest
Definition | A record of a request for diagnostic investigations, treatments, or operations to be performed.
Control | 1..1
Alternate Names | diagnostic request
ProcedureRequest.identifier
Definition | Identifiers assigned to this order instance by the orderer and/or the receiver and/or order fulfiller.
Note | This is a business identifer, not a resource identifier (see discussion)
Control | 0..*
Type | Identifier
Summary | true
Comments | The identifier.type element is used to distinguish between the identifiers assigned by the orderer (known as the 'Placer' in HL7 v2) and the producer of the observations in response to the order (known as the 'Filler' in HL7 v2). For further discussion and examples see the resource notes section below.
ProcedureRequest.definition
Definition | Protocol or definition followed by this request.
Control | 0..*
Type | Reference(ActivityDefinition | PlanDefinition)
Alternate Names | protocol
Summary | true
ProcedureRequest.basedOn

Definition | Plan/proposal/order fulfilled by this request.
Control | 0..*
Type | Reference(Any)
Alternate Names | fulfills
Summary | true
ProcedureRequest.replaces
Definition | The request takes the place of the referenced completed or terminated request(s).
Control | 0..*
Type | Reference(Any)
Summary | true
ProcedureRequest.requisition
Definition | A shared identifier common to all procedure or diagnostic requests that were authorized more or less simultaneously by a single author, representing the composite or group identifier.
Control | 0..1
Type | Identifier
Requirements | Some business processes need to know if multiple items were ordered as part of the same "requisition" for billing or other purposes.
Alternate Names | grouperId; groupIdentifier
Summary | true
Comments | Requests are linked either by a "basedOn" relationship (i.e. one request is fulfilling another) or by having a common requisition. Requests that are part of the same requisition are generally treated independently from the perspective of changing their state or maintaining them after initial creation.
ProcedureRequest.status
Definition | The status of the order.
Control | 1..1
Terminology Binding | RequestStatus (Required)
Type | code
Is Modifier | true
Summary | true
Comments | The status is generally fully in the control of the requester - they determine whether the order is draft or active and, after it has been activated, competed, cancelled or suspended. States relating to the activities of the performer are reflected on either the corresponding event (see Event Pattern for general discussion) or using the Task resource. This element is labeled as a modifier because the status contains codes that mark the resource as not currently valid.
ProcedureRequest.intent
Definition | Whether the request is a proposal, plan, an original order or a reflex order.
Control | 1..1
Terminology Binding | RequestIntent (Required)
Type | code
Is Modifier | true
Summary | true
Comments | This element is labeled as a modifier because the intent alters when and how the resource is actually applicable.
ProcedureRequest.priority
Definition | Indicates how quickly the ProcedureRequest should be addressed with respect to other requests.
Control | 0..1
Terminology Binding | RequestPriority (Required)
Type | code
Meaning if Missing | If missing, this task should be performed with normal priority
Summary | true
ProcedureRequest.doNotPerform
Definition | Set this to true if the record is saying that the procedure should NOT be performed.
Control | 0..1
Type | boolean
Is Modifier | true
Default Value | false
Requirements | Used for do not ambulate, do not elevate head of bed, do not flush NG tube, do not take blood pressure on a certain arm, etc.
Summary | true
Comments | This element is labeled as a modifier because it indicates that a procedure shouldn't happen, instead of a request for it to happen. In general, only the code and timeframe will be present, though occasional additional qualifiers such as body site or even performer could be included to narrow the scope of the prohibition. If the ProcedureRequest.code and ProcedureRequest.doNotPerform both contain negation, that will reinforce prohibition and should not have a double negative interpretation.
ProcedureRequest.category
Definition | A code that classifies the procedure for searching, sorting and display purposes (e.g. "Surgical Procedure").
Control | 0..*
Terminology Binding | Procedure Category Codes (SNOMED CT) (Example)
Type | CodeableConcept
Requirements | Used for filtering what procedure request are retrieved and displayed.
Summary | true
Comments | There may be multiple axis of categorization depending on the context or use case for retrieving or displaying the resource. The level of granularity is defined by the category concepts in the value set.
ProcedureRequest.code
Definition | A code that identifies a particular procedure, diagnostic investigation, or panel of investigations, that have been requested.
Control | 1..1
Terminology Binding | Procedure Codes (SNOMED CT) (Example)
Type | CodeableConcept

Summary | true
Comments | Many laboratory and radiology procedure codes embed the specimen/organ system in the test ordeer name, for example, serum or serum/plasma glucose, or a chest xray. The specimen may not be recorded separately from the test code.
ProcedureRequest.subject
Definition | On whom or what the procedure or diagnostic is to be performed. This is usually a human patient, but can also be requested on animals, groups of humans or animals, devices such as dialysis machines, or even locations (typically for environmental scans).
Control | 1..1
Type | Reference(Patient | Group | Location | Device)
Summary | true
ProcedureRequest.context
Definition | An encounter or episode of care that provides additional information about the healthcare context in which this request is made.
Control | 0..1
Type | Reference(Encounter | EpisodeOfCare)
Alternate Names | encounter
Summary | true
ProcedureRequest.occurrence[x]
Definition | The date/time at which the diagnostic testing should occur.
Control | 0..1
Type | dateTime|Period|Timing
[x] Note | See Choice of Data Types for further information about how to use [x]
Alternate Names | schedule
Summary | true
ProcedureRequest.asNeeded[x]
Definition | If a CodeableConcept is present, it indicates the pre-condition for performing the procedure. For example "pain", "on flare-up", etc.
Control | 0..1
Terminology Binding | SNOMED CT Medication As Needed Reason Codes (Example)
Type | boolean|CodeableConcept
[x] Note | See Choice of Data Types for further information about how to use [x]
Summary | true
ProcedureRequest.authoredOn
Definition | When the request transitioned to being actionable.
Control | 0..1
Type | dateTime
Alternate Names | orderedOn
Summary | true
ProcedureRequest.requester
Definition | The individual who initiated the request and has responsibility for its activation.
Control | 0..1
Alternate Names | author; orderer
Summary | true
Comments | This not the dispatcher, but rather who is the authorizer.
ProcedureRequest.requester.agent
Definition | The device, practitioner or organization who initiated the request.
Control | 1..1
Type | Reference(Device | Practitioner | Organization)
Summary | true
ProcedureRequest.requester.onBehalfOf
Definition | The organization the device or practitioner was acting on behalf of.
Control | 0..1
Type | Reference(Organization)
Requirements | Practitioners and Devices can be associated with multiple organizations. This element indicates which organization they were acting on behalf of when authoring the request.
Summary | true
ProcedureRequest.performerType
Definition | Desired type of performer for doing the diagnostic testing.
Control | 0..1
Terminology Binding | Participant Roles (Example)
Type | CodeableConcept
Summary | true
Comments | this is a role, not a participation type. I.e. does not describe the task, but describes the capacity. For example, “compounding pharmacy” or “psychiatrist” or “internal referral”.
ProcedureRequest.performer
Definition | The desired perfomer for doing the diagnostic testing. For example, the surgeon, dermatopathologist, endoscopist, etc.
Control | 0..1
Type | Reference(Practitioner | Organization | Patient | Device | RelatedPerson | HealthcareService)
Summary | true
Comments | If needed, use an extension for listing alternative performers and/or roles and/or preference.
ProcedureRequest.reasonCode
Definition | An explanation or justification for why this diagnostic investigation is being requested in coded or textual form. This is often for billing purposes. May relate to the resources referred to in supportingInformation.
Control | 0..*
Terminology Binding | Procedure Reason Codes (Example)
Type | CodeableConcept

Summary | true
Comments | This may be used to decide how the diagnostic investigation will be performed, or even if it will be performed at all. Use CodeableConcept text element if the data is free (uncoded) text as shown in the CT Scan example.
ProcedureRequest.reasonReference
Definition | Indicates another resource that provides a justification for why this diagnostic investigation is being requested. May relate to the resources referred to in supportingInformation.
Control | 0..*
Type | Reference(Condition | Observation)
Summary | true
Comments | This may be used to decide how the diagnostic investigation will be performed, or even if it will be performed at all. Use CodeableConcept text element if the data is free (uncoded) text as shown in the CT Scan example.
ProcedureRequest.supportingInfo
Definition | Additional clinical information about the patient or specimen that may influence the procedure or diagnostics or their interpretations. This information includes diagnosis, clinical findings and other observations. In laboratory ordering these are typically referred to as "ask at order entry questions (AOEs)". This includes observations explicitly requested by the producer (filler) to provide context or supporting information needed to complete the order. For example, reporting the amount of inspired oxygen for blood gas measurements.
Control | 0..*
Type | Reference(Any)
Alternate Names | Ask at order entry question; AOE
ProcedureRequest.specimen
Definition | One or more specimens that the laboratory procedure will use.
Control | 0..*
Type | Reference(Specimen)
Summary | true
Comments | Many diagnostic procedures need a specimen, but the request itself is not actually about the specimen. This element is for when the diagnostic is requested on already existing specimens and the request points to the specimen it applies to. Conversely, If the request is entered first with an unknown specimen, The Specimen resource references to the ProcedureRequest.
ProcedureRequest.bodySite
Definition | Anatomic location where the procedure should be performed. This is the target site.
Control | 0..*
Terminology Binding | SNOMED CT Body Structures (Example)
Type | CodeableConcept
Requirements | Knowing where the procedure is performed is important for tracking if multiple sites are possible.
Alternate Names | location
Summary | true
Comments | Only used if not implicit in the code found in ProcedureRequest.type. If the use case requires BodySite to be handled as a separate resource instead of an inline coded element (e.g. to identify and track separately) then use the standard extension procedurerequest-targetBodySite.
ProcedureRequest.note
Definition | Any other notes and comments made about the service request. For example, letting provider know that "patient hates needles" or other provider instructions.
Control | 0..*
Type | Annotation
ProcedureRequest.relevantHistory
Definition | Key events in the history of the request.
Control | 0..*
Type | Reference(Provenance)
Comments | This may not include provenances for all versions of the request – only those deemed “relevant” or important. This SHALL NOT include the Provenance associated with this current version of the resource. (If that provenance is deemed to be a “relevant” change, it will need to be added as part of a later update. Until then, it can be queried directly as the Provenance that points to this version using _revinclude All Provenances should have some historical version of this Request as their subject.

12.15.8 Resource ProcedureRequest - Mappings

URL: https://hl7.org/fhir/STU3/procedurerequest-mappings.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

12.15.8 Resource ProcedureRequest - Mappings

Orders and Observations Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Device, Encounter, Patient, Practitioner, RelatedPerson

Mappings:

RIM Mapping

HL7 v2 Mapping

Quality Improvement and Clinical Knowledge (QUICK)

Workflow Mapping

Mappings for the procedurerequest resource.

12.15.8.1 Mappings for RIM Mapping (http://hl7.org/v3)

ProcedureRequest | Act[moodCode<=INT]
identifier | .identifier

definition | .outboundRelationship[typeCode=DEFN].target
basedOn | .outboundRelationship[typeCode=FLFS].target
replaces | .outboundRelationship[typeCode=RPLC].target
requisition | .inboundRelationship(typeCode=COMP].source[moodCode=INT].identifier
status | .status
intent | .moodCode (nuances beyond PRP/PLAN/RQO would need to be elsewhere)
priority | .priorityCode
doNotPerform | .actionNegationInd
category | .outboundRelationship[typeCode="COMP].target[classCode="LIST", moodCode="INT"].code
code | .code
subject | .participation[typeCode=SBJ].role
context | .inboundRelationship(typeCode=COMP].source[classCode<=PCPR, moodCode=EVN]
occurrence[x] | .effectiveTime
asNeeded[x] | boolean: precondition.negationInd (inversed - so negationInd = true means asNeeded=false CodeableConcept: precondition.observationEventCriterion[code="Assertion"].value
authoredOn | .participation[typeCode=AUT].time
requester | .participation[typeCode=AUT].role
agent | .player
onBehalfOf | .scoper
performerType | .participation[typeCode=PRF].role[scoper.determinerCode=KIND].code
performer | .participation[typeCode=PRF].role[scoper.determinerCode=INSTANCE]
reasonCode | .reasonCode
reasonReference | .outboundRelationship[typeCode=RSON].target
supportingInfo | .outboundRelationship[typeCode=PERT].target
specimen | .participation[typeCode=SPC].role
bodySite | targetSiteCode
note | .inboundRelationship(typeCode=SUBJ].source[classCode=ANNGEN, moodCode=EVN].value[xsi:type=ST]
relevantHistory | .inboundRelationship(typeCode=SUBJ].source[classCode=CACT, moodCode=EVN]

12.15.8.2 Mappings for HL7 v2 Mapping (http://hl7.org/v2)

ProcedureRequest | ORC
identifier | ORC.2, ORC.3
definition | Varies by domain
basedOn | ORC.8 (plus others)
replaces | Handled by message location of ORC (ORC.1=RO or RU)
requisition | ORC.4
status | ORC.5
intent | N/A
priority | TQ1.9
doNotPerform
category
code | Varies by domain
subject | Accompanying PID segment
context | Accompanying PV1
occurrence[x] | Accompanying TQ1/TQ2 segments
asNeeded[x]
authoredOn | ORC.9
requester | ORC.12
agent | ORC.12
onBehalfOf | N/A
performerType | PRT
performer | PRT
reasonCode | ORC.16
reasonReference | ORC.16
supportingInfo | Accompanying segments
specimen | SPM
bodySite | SPM
note | NTE
relevantHistory | N/A

12.15.8.3 Mappings for Quality Improvement and Clinical Knowledge (QUICK) (http://wiki.siframework.org/CQF)

ProcedureRequest
identifier | ClinicalStatement.identifier
definition
basedOn
replaces
requisition
status | Action.currentStatus
intent
priority

doNotPerform
category
code | Procedure.procedureCode
subject | ClinicalStatement.subject
context | ClinicalStatement.encounter
occurrence[x] | Procedure.procedureSchedule
asNeeded[x] | Proposal.prnReason.reason
authoredOn | Proposal.proposedAtTime
requester | ClinicalStatement.statementAuthor
agent | ClinicalStatement.statementAuthor
onBehalfOf
performerType
performer
reasonCode
reasonReference
supportingInfo
specimen
bodySite | Procedure.targetBodySite
note | ClinicalStatement.additionalText
relevantHistory

12.15.8.4 Mappings for Workflow Mapping (http://hl7.org/fhir/workflow)

ProcedureRequest | Request
identifier | Request.identifier
definition | Request.definition
basedOn | Request.basedOn
replaces | Request.replaces
requisition
status | Request.status
intent | Request.intent
priority | Request.priority
doNotPerform
category
code | Request.code
subject | Request.subject
context | Request.context
occurrence[x] | Request.occurrence[x]
asNeeded[x]
authoredOn | Request.authoredOn
requester | Request.requester
agent | Request.requester.agent
onBehalfOf | Request.requester.onBehalfOf
performerType | Request.performerType
performer | Request.Performer
reasonCode | Request.reasonCod
reasonReference | Request.reasonReference
supportingInfo | Request.supportingInformation
specimen
bodySite
note | Request.note
relevantHistory | Request.relevantHistory

12.15.9 Resource ProcedureRequest - Design Notes

URL: https://hl7.org/fhir/STU3/procedurerequest-explanations.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

12.15.9 Resource ProcedureRequest - Design Notes

to do

12.15.10 Resource ProcedureRequest - Extensions & Profiles

URL: https://hl7.org/fhir/STU3/procedurerequest-profiles.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

12.15.10 Resource ProcedureRequest - Extensions & Profiles

Orders and Observations Work Group | Maturity Level: N/A | Ballot Status: Informative

This table lists profiles and extensions for the ProcedureRequest resource. For background information, see Profiling Resources and Extensibility. Additional profiles and resources may be found in published Implementation Guides , or in the Conformance resource registry .

12.15.10.1 Profiles

ProcedureRequest-Genetics | Describes how the ProcedureRequest resource is used to for genetics | for ProcedureRequest-Genetics

12.15.10.2 Extensions

procedurerequest-approachBodySite | approachBodySite | for Procedure Request HL7 Extensions
procedurerequest-authorizedBy | authorizedBy | for Procedure Request HL7 Extensions
procedurerequest-precondition | precondition | for Procedure Request HL7 Extensions
procedurerequest-questionnaireRequest | questionnaireRequest | for Procedure Request HL7 Extensions
procedurerequest-reasonRefused | reasonRefused | for Procedure Request HL7 Extensions
procedurerequest-reasonRejected | reasonRejected | for Procedure Request HL7 Extensions
procedurerequest-targetBodySite | targetBodySite | for Procedure Request HL7 Extensions

Extensions that reference this resource:

condition-basedOn | basedOn | for Condition HL7 Extensions

12.15.10.3 Search Extensions

No Search Extensions defined for this resource

12.15.11 ProcedureRequest-Genetics

URL: https://hl7.org/fhir/STU3/procedurerequest-genetic.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

12.15.11 ProcedureRequest-Genetics

(No assigned work group) | Maturity Level: N/A | Ballot Status: Informative

12.15.11.1 Content

Profiles:
ProcedureRequest-Genetics | Describes how the ProcedureRequest resource is used to for genetics
Extensions:
procedurerequest-geneticsItem | Item : The specific diagnostic investigations that are requested as part of this request. Sometimes, there can only be one item per request, but in most contexts, more than one investigation can be requested.
Examples:
PGx | PGx Phenotype example
Order-genetics-1 | Order for ABCB4 gene mutation analysis

12.15.11.2 StructureDefinition: ProcedureRequest-Genetics

URL: https://hl7.org/fhir/STU3/procedurerequest-genetics.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

12.15.11.2 StructureDefinition: ProcedureRequest-Genetics

(No assigned work group) | Maturity Level: 1 | Informative | Use Context: Any

The official URL for this profile is:

`http://hl7.org/fhir/StructureDefinition/procedurerequest-genetics`

Describes how the ProcedureRequest resource is used to for genetics

This profile was published on Tue, Oct 20, 2015 00:00+1100 as a draft by Health Level Seven International (Clinical Genomics).

12.15.11.2.1 Formal Views of Profile Content

Description of Profiles, Differentials, Snapshots, and how the XML and JSON presentations work.

* Text Summary

* Differential Table

* Snapshot Table

* XML Template

* JSON Template

* All

This structure is derived from ProcedureRequest.

Summary

Extensions

This structure refers to these extensions:

* http://hl7.org/fhir/StructureDefinition/procedurerequest-geneticsItem

This structure is derived from ProcedureRequest.

Name | Flags | Card. | Type | Description & Constraints
ProcedureRequest | | 0..*
procedurerequest-geneticsItem | | 0..* | (Complex) | The items the orderer requested URL: http://hl7.org/fhir/StructureDefinition/procedurerequest-geneticsItem
Documentation for this format

Name | Flags | Card. | Type | Description & Constraints
ProcedureRequest | I | 0..* | | A request for a procedure or diagnostic to be performed
id | ? | 0..1 | id | Logical id of this artifact
meta | ? | 0..1 | Meta | Metadata about the resource
implicitRules | ?!? | 0..1 | uri | A set of rules under which this content was created
language | | 0..1 | code | Language of the resource content Binding: Common Languages (extensible)
text | I | 0..1 | Narrative | Text summary of the resource, for human interpretation
contained | | 0..* | Resource | Contained, inline Resources
procedurerequest-geneticsItem | I | 0..* | (Complex) | The items the orderer requested URL: http://hl7.org/fhir/StructureDefinition/procedurerequest-geneticsItem
modifierExtension | ?! | 0..* | Extension | Extensions that cannot be ignored
identifier | ? | 0..* | Identifier | Identifiers assigned to this order
definition | ? | 0..* | Reference(ActivityDefinition), Reference(PlanDefinition) | Protocol or definition
basedOn | ? | 0..* | Reference(Resource) | What request fulfills
replaces | ? | 0..* | Reference(Resource) | What request replaces
requisition | ? | 0..1 | Identifier | Composite Request ID
status | ?!? | 1..1 | code | draft | active | suspended | completed | entered-in-error | cancelled Binding: RequestStatus (required)
intent | ?!? | 1..1 | code | proposal | plan | order + Binding: RequestIntent (required)
priority | ? | 0..1 | code | routine | urgent | asap | stat Binding: RequestPriority (required)
doNotPerform | ?!? | 0..1 | boolean | True if procedure should not be performed
category | ? | 0..* | CodeableConcept | Classification of procedure Binding: Procedure Category Codes (SNOMED CT) (example)
code | ? | 1..1 | CodeableConcept | What is being requested/ordered Binding: Procedure Codes (SNOMED CT) (example)
subject | ? | 1..1 | Reference(Patient), Reference(Group), Reference(Location), Reference(Device) | Individual the service is ordered for
context | ? | 0..1 | Reference(Encounter), Reference(EpisodeOfCare) | Encounter or Episode during which request was created
occurrence[x] | ? | 0..1 | dateTime, Period, Timing | When procedure should occur
asNeeded[x] | ? | 0..1 | boolean, CodeableConcept | Preconditions for procedure or diagnostic Binding: SNOMED CT Medication As Needed Reason Codes (example)
authoredOn | ? | 0..1 | dateTime | Date request signed
requester | ?I | 0..1 | BackboneElement | Who/what is requesting procedure or diagnostic
id | | 0..1 | string | xml:id (or equivalent in JSON)
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?!? | 0..* | Extension | Extensions that cannot be ignored
agent | ? | 1..1 | Reference(Device), Reference(Practitioner), Reference(Organization) | Individual making the request
onBehalfOf | ? | 0..1 | Reference(Organization) | Organization agent is acting for
performerType | ? | 0..1 | CodeableConcept | Performer role Binding: Participant Roles (example)
performer | ? | 0..1 | Reference(Practitioner), Reference(Organization), Reference(Patient), Reference(Device), Reference(RelatedPerson), Reference(HealthcareService) | Requested perfomer
reasonCode | ? | 0..* | CodeableConcept | Explanation/Justification for test Binding: Procedure Reason Codes (example)
reasonReference | ? | 0..* | Reference(Condition), Reference(Observation) | Explanation/Justification for test
supportingInfo | | 0..* | Reference(Resource) | Additional clinical information
specimen | ? | 0..* | Reference(Specimen) | Procedure Samples
bodySite | ? | 0..* | CodeableConcept | Location on Body Binding: SNOMED CT Body Structures (example)
note | | 0..* | Annotation | Comments
relevantHistory | | 0..* | Reference(Provenance) | Request provenance
Documentation for this format

```
<!-- ProcedureRequest-Genetics -->

<ProcedureRequest xmlns="http://hl7.org/fhir"
>
  <!-- from Element: extension -->
 <id value="[id]"/><!-- 0..1 Logical id of this artifact -->
 <meta><!-- 0..1 Meta Metadata about the resource --></meta>
 <implicitRules value="[uri]"/><!-- 0..1 A set of rules under which this content was created -->
 <language value="[code]"/><!-- 0..1 Language of the resource content  -->
 <text><!-- 0..1 Narrative
     Text summary of the resource, for human interpretation --></text>
 <contained><!-- 0..* Resource
     Contained, inline Resources --></contained>
 <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
 <identifier><!-- 0..* Identifier
     Identifiers assigned to this order --></identifier>
 <definition><!-- 0..* Reference(ActivityDefinition)|Reference(PlanDefinition)
     Protocol or definition --></definition>
 <basedOn><!-- 0..* Reference(Resource) What request fulfills --></basedOn>
 <replaces><!-- 0..* Reference(Resource) What request replaces --></replaces>
```

```
 <requisition><!-- 0..1 Identifier
     Composite Request ID --></requisition>
 <status value="[code]"/><!-- 1..1 draft | active | suspended | completed | entered-in-error | cancelled  -->
 <intent value="[code]"/><!-- 1..1 proposal | plan | order +  -->
 <priority value="[code]"/><!-- 0..1 routine | urgent | asap | stat  -->
 <doNotPerform value="[boolean]"/><!-- 0..1 True if procedure should not be performed -->
 <category><!-- 0..* CodeableConcept Classification of procedure  --></category>
 <code><!-- 1..1 CodeableConcept What is being requested/ordered  --></code>
 <subject><!-- 1..1 Reference(Patient)|Reference(Group)|Reference(Location)|Reference(Device)
     Individual the service is ordered for --></subject>
 <context><!-- 0..1 Reference(Encounter)|Reference(EpisodeOfCare)
     Encounter or Episode during which request was created --></context>
 <occurrence[x]><!-- 0..1 dateTime|Period|Timing
     When procedure should occur --></occurrence[x]>
 <asNeeded[x]><!-- 0..1 boolean|CodeableConcept Preconditions for procedure or diagnostic  --></asNeeded[x]>
 <authoredOn value="[dateTime]"/><!-- 0..1 Date request signed -->
 <requester>  0..1 BackboneElement  <!--  0..1 Who/what is requesting procedure or diagnostic -->
  <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
  <agent><!-- 1..1 Reference(Device)|Reference(Practitioner)|Reference(Organization)
      Individual making the request --></agent>
  <onBehalfOf><!-- 0..1 Reference(Organization)
      Organization agent is acting for --></onBehalfOf>
 </requester>
 <performerType><!-- 0..1 CodeableConcept Performer role  --></performerType>
 <performer><!-- 0..1 Reference(Practitioner)|Reference(Organization)|Reference(Patient)|Reference(Device)|
   Reference(RelatedPerson)|Reference(HealthcareService) Requested perfomer --></performer>
 <reasonCode><!-- 0..* CodeableConcept Explanation/Justification for test  --></reasonCode>
 <reasonReference><!-- 0..* Reference(Condition)|Reference(Observation)
     Explanation/Justification for test --></reasonReference>
 <supportingInfo><!-- 0..* Reference(Resource)
     Additional clinical information --></supportingInfo>
 <specimen><!-- 0..* Reference(Specimen) Procedure Samples --></specimen>
 <bodySite><!-- 0..* CodeableConcept Location on Body  --></bodySite>
 <note><!-- 0..* Annotation Comments --></note>
 <relevantHistory><!-- 0..* Reference(Provenance)
     Request provenance --></relevantHistory>
</ProcedureRequest>
```

```
{ // ProcedureRequest-Genetics
  // from Element: extension
    "meta" : { Meta }, //Metadata about the resource
    "implicitRules" : "<uri>", //A set of rules under which this content was created
    "language" : "<code>", //Language of the resource content
    "text" : { Narrative }, //Text summary of the resource, for human interpretation
    "contained" : [{ Resource }], //Contained, inline Resources
    "extension" : [ //  sliced by value:url  in the specified orderOpen
      { // The items the orderer requested // C?
        // from Element: extension
      }
    ],
    "modifierExtension" : [{ Extension }], //Extensions that cannot be ignored
    "identifier" : [{ Identifier }], //Identifiers assigned to this order
// value[x]: Protocol or definition. One of these 2:
    "definition" : [{ Reference(ActivityDefinition) }], //Protocol or definition
    "definition" : [{ Reference(PlanDefinition) }], //Protocol or definition
    "basedOn" : [{ Reference(Resource) }], //What request fulfills
    "replaces" : [{ Reference(Resource) }], //What request replaces
    "requisition" : { Identifier }, //Composite Request ID
    "status" : "<code>", // R! draft | active | suspended | completed | entered-in-error | cancelled
    "intent" : "<code>", // R! proposal | plan | order +
    "priority" : "<code>", //routine | urgent | asap | stat
    "doNotPerform" : <boolean>, //True if procedure should not be performed
    "category" : [{ CodeableConcept }], //Classification of procedure
    "code" : { CodeableConcept }, // R! What is being requested/ordered
// value[x]: Individual the service is ordered for. One of these 4:
    "subject" : { Reference(Patient) }, // R! Individual the service is ordered for
    "subject" : { Reference(Group) }, // R! Individual the service is ordered for
    "subject" : { Reference(Location) }, // R! Individual the service is ordered for
    "subject" : { Reference(Device) }, // R! Individual the service is ordered for
// value[x]: Encounter or Episode during which request was created. One of these 2:
    "context" : { Reference(Encounter) }, //Encounter or Episode during which request was created
    "context" : { Reference(EpisodeOfCare) }, //Encounter or Episode during which request was created
// value[x]: When procedure should occur. One of these 3:
    "occurrenceDateTime" : "<dateTime>", //When procedure should occur
    "occurrencePeriod" : { Period }, //When procedure should occur
    "occurrenceTiming" : { Timing }, //When procedure should occur
// value[x]: Preconditions for procedure or diagnostic. One of these 2:
    "asNeededBoolean" : <boolean>, //Preconditions for procedure or diagnostic
    "asNeededCodeableConcept" : { CodeableConcept }, //Preconditions for procedure or diagnostic
    "authoredOn" : "<dateTime>", //Date request signed
    "requester" : { BackboneElement }, // C?Who/what is requesting procedure or diagnostic
    "performerType" : { CodeableConcept }, //Performer role
// value[x]: Requested perfomer. One of these 6:
    "performer" : { Reference(Practitioner) } //Requested perfomer
    "performer" : { Reference(Organization) }, //Requested perfomer
    "performer" : { Reference(Patient) }, //Requested perfomer
    "performer" : { Reference(Device) }, //Requested perfomer
    "performer" : { Reference(RelatedPerson) }, //Requested perfomer
    "performer" : { Reference(HealthcareService) }, //Requested perfomer
    "reasonCode" : [{ CodeableConcept }], //Explanation/Justification for test
```

```
// value[x]: Explanation/Justification for test. One of these 2:
    "reasonReference" : [{ Reference(Condition) }], //Explanation/Justification for test
    "reasonReference" : [{ Reference(Observation) }], //Explanation/Justification for test
    "supportingInfo" : [{ Reference(Resource) }], //Additional clinical information
    "specimen" : [{ Reference(Specimen) }], //Procedure Samples
    "bodySite" : [{ CodeableConcept }], //Location on Body
    "note" : [{ Annotation }], //Comments
    "relevantHistory" : [{ Reference(Provenance) }], //Request provenance
  }
```

This structure is derived from ProcedureRequest.

Summary

Extensions

This structure refers to these extensions:

* http://hl7.org/fhir/StructureDefinition/procedurerequest-geneticsItem

Differential View

This structure is derived from ProcedureRequest.

Name | Flags | Card. | Type | Description & Constraints
ProcedureRequest | | 0..*
procedurerequest-geneticsItem | | 0..* | (Complex) | The items the orderer requested URL: http://hl7.org/fhir/StructureDefinition/procedurerequest-geneticsItem
Documentation for this format

Snapshot View

Name | Flags | Card. | Type | Description & Constraints
ProcedureRequest | I | 0..* | | A request for a procedure or diagnostic to be performed
id | ? | 0..1 | id | Logical id of this artifact
meta | ? | 0..1 | Meta | Metadata about the resource
implicitRules | ?!? | 0..1 | uri | A set of rules under which this content was created
language | | 0..1 | code | Language of the resource content Binding: Common Languages (extensible)
text | I | 0..1 | Narrative | Text summary of the resource, for human interpretation
contained | | 0..* | Resource | Contained, inline Resources
procedurerequest-geneticsItem | I | 0..* | (Complex) | The items the orderer requested URL: http://hl7.org/fhir/StructureDefinition/procedurerequest-geneticsItem
modifierExtension | ?! | 0..* | Extension | Extensions that cannot be ignored
identifier | ? | 0..* | Identifier | Identifiers assigned to this order
definition | ? | 0..* | Reference(ActivityDefinition), Reference(PlanDefinition) | Protocol or definition
basedOn | ? | 0..* | Reference(Resource) | What request fulfills
replaces | ? | 0..* | Reference(Resource) | What request replaces
requisition | ? | 0..1 | Identifier | Composite Request ID
status | ?!? | 1..1 | code | draft | active | suspended | completed | entered-in-error | cancelled Binding: RequestStatus (required)
intent | ?!? | 1..1 | code | proposal | plan | order + Binding: RequestIntent (required)
priority | ? | 0..1 | code | routine | urgent | asap | stat Binding: RequestPriority (required)
doNotPerform | ?!? | 0..1 | boolean | True if procedure should not be performed
category | ? | 0..* | CodeableConcept | Classification of procedure Binding: Procedure Category Codes (SNOMED CT) (example)
code | ? | 1..1 | CodeableConcept | What is being requested/ordered Binding: Procedure Codes (SNOMED CT) (example)
subject | ? | 1..1 | Reference(Patient), Reference(Group), Reference(Location), Reference(Device) | Individual the service is ordered for
context | ? | 0..1 | Reference(Encounter), Reference(EpisodeOfCare) | Encounter or Episode during which request was created
occurrence[x] | ? | 0..1 | dateTime, Period, Timing | When procedure should occur
asNeeded[x] | ? | 0..1 | boolean, CodeableConcept | Preconditions for procedure or diagnostic Binding: SNOMED CT Medication As Needed Reason Codes (example)
authoredOn | ? | 0..1 | dateTime | Date request signed
requester | ?I | 0..1 | BackboneElement | Who/what is requesting procedure or diagnostic
id | | 0..1 | string | xml:id (or equivalent in JSON)
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?!? | 0..* | Extension | Extensions that cannot be ignored
agent | ? | 1..1 | Reference(Device), Reference(Practitioner), Reference(Organization) | Individual making the request
onBehalfOf | ? | 0..1 | Reference(Organization) | Organization agent is acting for
performerType | ? | 0..1 | CodeableConcept | Performer role Binding: Participant Roles (example)
performer | ? | 0..1 | Reference(Practitioner), Reference(Organization), Reference(Patient), Reference(Device), Reference(RelatedPerson), Reference(HealthcareService) | Requested perfomer
reasonCode | ? | 0..* | CodeableConcept | Explanation/Justification for test Binding: Procedure Reason Codes (example)
reasonReference | ? | 0..* | Reference(Condition), Reference(Observation) | Explanation/Justification for test
supportingInfo | | 0..* | Reference(Resource) | Additional clinical information
specimen | ? | 0..* | Reference(Specimen) | Procedure Samples
bodySite | ? | 0..* | CodeableConcept | Location on Body Binding: SNOMED CT Body Structures (example)
note | | 0..* | Annotation | Comments
relevantHistory | | 0..* | Reference(Provenance) | Request provenance
Documentation for this format

XML Template

```
<!-- ProcedureRequest-Genetics -->

<ProcedureRequest xmlns="http://hl7.org/fhir"
```

```
>
  <!-- from Element: extension -->
 <id value="[id]"/><!-- 0..1 Logical id of this artifact -->
 <meta><!-- 0..1 Meta Metadata about the resource --></meta>
 <implicitRules value="[uri]"/><!-- 0..1 A set of rules under which this content was created -->
 <language value="[code]"/><!-- 0..1 Language of the resource content  -->
 <text><!-- 0..1 Narrative
     Text summary of the resource, for human interpretation --></text>
 <contained><!-- 0..* Resource
     Contained, inline Resources --></contained>
 <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
 <identifier><!-- 0..* Identifier
     Identifiers assigned to this order --></identifier>
 <definition><!-- 0..* Reference(ActivityDefinition)|Reference(PlanDefinition)
     Protocol or definition --></definition>
 <basedOn><!-- 0..* Reference(Resource) What request fulfills --></basedOn>
 <replaces><!-- 0..* Reference(Resource) What request replaces --></replaces>
 <requisition><!-- 0..1 Identifier
     Composite Request ID --></requisition>
 <status value="[code]"/><!-- 1..1 draft | active | suspended | completed | entered-in-error | cancelled  -->
 <intent value="[code]"/><!-- 1..1 proposal | plan | order +  -->
 <priority value="[code]"/><!-- 0..1 routine | urgent | asap | stat  -->
 <doNotPerform value="[boolean]"/><!-- 0..1 True if procedure should not be performed -->
 <category><!-- 0..* CodeableConcept Classification of procedure  --></category>
 <code><!-- 1..1 CodeableConcept What is being requested/ordered  --></code>
 <subject><!-- 1..1 Reference(Patient)|Reference(Group)|Reference(Location)|Reference(Device)
     Individual the service is ordered for --></subject>
 <context><!-- 0..1 Reference(Encounter)|Reference(EpisodeOfCare)
     Encounter or Episode during which request was created --></context>
 <occurrence[x]><!-- 0..1 dateTime|Period|Timing
     When procedure should occur --></occurrence[x]>
 <asNeeded[x]><!-- 0..1 boolean|CodeableConcept Preconditions for procedure or diagnostic  --></asNeeded[x]>
 <authoredOn value="[dateTime]"/><!-- 0..1 Date request signed -->
 <requester>  0..1 BackboneElement  <!--  0..1 Who/what is requesting procedure or diagnostic -->
  <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
  <agent><!-- 1..1 Reference(Device)|Reference(Practitioner)|Reference(Organization)
      Individual making the request --></agent>
  <onBehalfOf><!-- 0..1 Reference(Organization)
      Organization agent is acting for --></onBehalfOf>
 </requester>
 <performerType><!-- 0..1 CodeableConcept Performer role  --></performerType>
 <performer><!-- 0..1 Reference(Practitioner)|Reference(Organization)|Reference(Patient)|Reference(Device)|
   Reference(RelatedPerson)|Reference(HealthcareService) Requested perfomer --></performer>
 <reasonCode><!-- 0..* CodeableConcept Explanation/Justification for test  --></reasonCode>
 <reasonReference><!-- 0..* Reference(Condition)|Reference(Observation)
     Explanation/Justification for test --></reasonReference>
 <supportingInfo><!-- 0..* Reference(Resource)
     Additional clinical information --></supportingInfo>
 <specimen><!-- 0..* Reference(Specimen) Procedure Samples --></specimen>
 <bodySite><!-- 0..* CodeableConcept Location on Body  --></bodySite>
 <note><!-- 0..* Annotation Comments --></note>
 <relevantHistory><!-- 0..* Reference(Provenance)
     Request provenance --></relevantHistory>
</ProcedureRequest>
```

JSON Template

```
{ // ProcedureRequest-Genetics
  // from Element: extension
    "meta" : { Meta }, //Metadata about the resource
    "implicitRules" : "<uri>", //A set of rules under which this content was created
    "language" : "<code>", //Language of the resource content
    "text" : { Narrative }, //Text summary of the resource, for human interpretation
    "contained" : [{ Resource }], //Contained, inline Resources
    "extension" : [ //  sliced by value:url  in the specified orderOpen
      { // The items the orderer requested // C?
        // from Element: extension
      }
    ],
    "modifierExtension" : [{ Extension }], //Extensions that cannot be ignored
    "identifier" : [{ Identifier }], //Identifiers assigned to this order
// value[x]: Protocol or definition. One of these 2:
    "definition" : [{ Reference(ActivityDefinition) }], //Protocol or definition
    "definition" : [{ Reference(PlanDefinition) }], //Protocol or definition
    "basedOn" : [{ Reference(Resource) }], //What request fulfills
    "replaces" : [{ Reference(Resource) }], //What request replaces
    "requisition" : { Identifier }, //Composite Request ID
    "status" : "<code>", // R! draft | active | suspended | completed | entered-in-error | cancelled
    "intent" : "<code>", // R! proposal | plan | order +
    "priority" : "<code>", //routine | urgent | asap | stat
    "doNotPerform" : <boolean>, //True if procedure should not be performed
    "category" : [{ CodeableConcept }], //Classification of procedure
    "code" : { CodeableConcept }, // R! What is being requested/ordered
// value[x]: Individual the service is ordered for. One of these 4:
    "subject" : { Reference(Patient) }, // R! Individual the service is ordered for
    "subject" : { Reference(Group) }, // R! Individual the service is ordered for
    "subject" : { Reference(Location) }, // R! Individual the service is ordered for
    "subject" : { Reference(Device) }, // R! Individual the service is ordered for
// value[x]: Encounter or Episode during which request was created. One of these 2:
    "context" : { Reference(Encounter) }, //Encounter or Episode during which request was created
```

```
    "context" : { Reference(EpisodeOfCare) }, //Encounter or Episode during which request was created
// value[x]: When procedure should occur. One of these 3:
    "occurrenceDateTime" : "<dateTime>", //When procedure should occur
    "occurrencePeriod" : { Period }, //When procedure should occur
    "occurrenceTiming" : { Timing }, //When procedure should occur
// value[x]: Preconditions for procedure or diagnostic. One of these 2:
    "asNeededBoolean" : <boolean>, //Preconditions for procedure or diagnostic
    "asNeededCodeableConcept" : { CodeableConcept }, //Preconditions for procedure or diagnostic
    "authoredOn" : "<dateTime>", //Date request signed
    "requester" : { BackboneElement }, // C?Who/what is requesting procedure or diagnostic
    "performerType" : { CodeableConcept }, //Performer role
// value[x]: Requested perfomer. One of these 6:
    "performer" : { Reference(Practitioner) } //Requested perfomer
    "performer" : { Reference(Organization) }, //Requested perfomer
    "performer" : { Reference(Patient) }, //Requested perfomer
    "performer" : { Reference(Device) }, //Requested perfomer
    "performer" : { Reference(RelatedPerson) }, //Requested perfomer
    "performer" : { Reference(HealthcareService) }, //Requested perfomer
    "reasonCode" : [{ CodeableConcept }], //Explanation/Justification for test
// value[x]: Explanation/Justification for test. One of these 2:
    "reasonReference" : [{ Reference(Condition) }], //Explanation/Justification for test
    "reasonReference" : [{ Reference(Observation) }], //Explanation/Justification for test
    "supportingInfo" : [{ Reference(Resource) }], //Additional clinical information
    "specimen" : [{ Reference(Specimen) }], //Procedure Samples
    "bodySite" : [{ CodeableConcept }], //Location on Body
    "note" : [{ Annotation }], //Comments
    "relevantHistory" : [{ Reference(Provenance) }], //Request provenance
  }
```

Other representations of profile: Schematron

12.15.11.2.2 Terminology Bindings

Path | Name | Conformance | ValueSet

ProcedureRequest.language | Common Languages | extensible | Common Languages

ProcedureRequest.status | RequestStatus | required | RequestStatus

ProcedureRequest.intent | RequestIntent | required | RequestIntent

ProcedureRequest.priority | RequestPriority | required | RequestPriority

ProcedureRequest.category | Procedure Category Codes (SNOMED CT) | example | Procedure Category Codes (SNOMED CT)

ProcedureRequest.code | Procedure Codes (SNOMED CT) | example | Procedure Codes (SNOMED CT)

ProcedureRequest.asNeeded[x] | SNOMED CT Medication As Needed Reason Codes | example | SNOMED CT Medication As Needed Reason Codes

ProcedureRequest.performerType | Participant Roles | example | Participant Roles

ProcedureRequest.reasonCode | Procedure Reason Codes | example | Procedure Reason Codes

ProcedureRequest.bodySite | SNOMED CT Body Structures | example | SNOMED CT Body Structures

12.15.12 Procedure Request HL7 Extensions

URL: https://hl7.org/fhir/STU3/procedurerequest-extensions.html

Defines common extensions used with or related to the Procedure Request resource

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

12.15.12 Procedure Request HL7 Extensions

(No assigned work group) | Maturity Level: N/A | Ballot Status: Informative

12.15.12.1 Content

Extensions:

procedurerequest-reasonRefused | reasonRefused : The reason the request was refused. Applies only if the status is refused.

procedurerequest-authorizedBy | authorizedBy : The practitioner who authorized the procedure request, usually the same as the orderer when the orderer is a practitioner.

procedurerequest-approachBodySite | approachBodySite : The requested approach body site used for this procedure. Multiple locations are allowed.

procedurerequest-targetBodySite | targetBodySite : The requested target body site for this procedure. Multiple locations are allowed.

procedurerequest-reasonRejected | reasonRejected : The coded reason why the diagnostic order was rejected.

procedurerequest-precondition | precondition : The condition or state of the patient, prior or during the diagnostic procedure or test, for example, fasting, at-rest, or post-operative. This captures circumstances that may influence the measured value and have bearing on the interpretation of the result.

procedurerequest-questionnaireRequest | questionnaireRequest : Reference to a specific Questionnaire Resource as an ordered item. Allows for ordering a specific questionnaire to be completed.

12.15.13 Resource ProcedureRequest - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/procedurerequest-version-maps.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

12.15.13 Resource ProcedureRequest - Detailed Descriptions

Orders and Observations Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Device, Encounter, Patient, Practitioner, RelatedPerson

R2 : R3 Converson maps for ProcedureRequest.

Functional status for this map: 6 tests that all execute ok. 5 fail round-trip testing and 6 r3 resources are invalid (9 errors). (see documentation)

12.15.13.1 R2 to R3

```
map "http://hl7.org/fhir/StructureMap/ProcedureRequest2to3" = "R2 to R3 Conversions for ProcedureRequest"

uses "http://hl7.org/fhir/DSTU2/StructureDefinition/ProcedureRequest" alias ProcedureRequestR2 as source
uses "http://hl7.org/fhir/StructureDefinition/ProcedureRequest" alias ProcedureRequest as target

imports "http://hl7.org/fhir/StructureMap/*2to3"

group for type+types  ProcedureRequest extends DomainResource
  input src : ProcedureRequestR2 as source
  input tgt : ProcedureRequest as target

  "ProcedureRequest-identifier" : for src.identifier make tgt.identifier
  "ProcedureRequest-subject" : for src.subject make tgt.subject
  "ProcedureRequest-code" : for src.code make tgt.code
  "ProcedureRequest-bodySite" : for src.bodySite make tgt.bodySite
  "ProcedureRequest-reason1" : for src.reason as v where v.is(CodeableConcept) make tgt.reasonCode = v
  "ProcedureRequest-reason2" : for src.reason as v where v.is(Reference) make tgt.reasonReference = v
  "ProcedureRequest-scheduled" : for src.scheduled make tgt.occurrence
  "ProcedureRequest-encounter" : for src.encounter make tgt.context
  "ProcedureRequest-performer" : for src.performer make tgt.performer
  "ProcedureRequest-status" : for src.status make tgt.status
  "ProcedureRequest-notes" : for src.notes make tgt.note
  "ProcedureRequest-asNeeded" : for src.asNeeded make tgt.asNeeded
  "ProcedureRequest-orderedOn" : for src.orderedOn make tgt.authoredOn
  "ProcedureRequest-orderer" : for src.orderer as v make tgt.requester as d, d.agent = v
  "ProcedureRequest-priority" : for src.priority make tgt.priority


endgroup
```

12.15.13.2 R3 to R2

```
map "http://hl7.org/fhir/StructureMap/ProcedureRequest3to2" = "R3 to R2 Conversion for ProcedureRequest"

uses "http://hl7.org/fhir/StructureDefinition/ProcedureRequest" alias ProcedureRequest as source
uses "http://hl7.org/fhir/DSTU2/StructureDefinition/ProcedureRequest" alias ProcedureRequestR2 as target

imports "http://hl7.org/fhir/StructureMap/*3to2"

group for type+types  ProcedureRequest extends DomainResource
  input src : ProcedureRequest as source
  input tgt : ProcedureRequestR2 as target

  "ProcedureRequest-identifier" : for src.identifier make tgt.identifier
  "ProcedureRequest-subject" : for src.subject make tgt.subject
  "ProcedureRequest-code" : for src.code make tgt.code
  "ProcedureRequest-bodySite" : for src.bodySite make tgt.bodySite
  "ProcedureRequest-reason1" : for src.reasonCode make tgt.reason
  "ProcedureRequest-reason2" : for src.reasonReference make tgt.reason
  "ProcedureRequest-scheduled" : for src.occurrence make tgt.scheduled
  "ProcedureRequest-encounter" : for src.context make tgt.encounter
  "ProcedureRequest-performer" : for src.performer make tgt.performer
  "ProcedureRequest-status" : for src.status make tgt.status
  "ProcedureRequest-notes" : for src.note make tgt.notes
  "ProcedureRequest-asNeeded" : for src.asNeeded make tgt.asNeeded
  "ProcedureRequest-orderedOn" : for src.authoredOn make tgt.orderedOn
  "ProcedureRequest-orderer" : for src.requester as v then {
    "ProcedureRequest-orderer-inner" : for v.agent make tgt.orderer
  }
  "ProcedureRequest-priority" : for src.priority make tgt.priority

endgroup
```

12.15.13.3 R3 Validation Errors

example | ERROR @ ProcedureRequest: Profile http://hl7.org/fhir/StructureDefinition/ProcedureRequest, Element 'ProcedureRequest.status': minimum required = 1, but only found 0 (src = InstanceValidator) ERROR @ ProcedureRequest: Profile http://hl7.org/fhir/StructureDefinition/ProcedureRequest, Element 'ProcedureRequest.intent': minimum required = 1, but only found 0 (src = InstanceValidator)

procedurerequest-example | ERROR @ ProcedureRequest: Profile http://hl7.org/fhir/StructureDefinition/ProcedureRequest, Element 'ProcedureRequest.intent': minimum required = 1, but only found 0 (src = InstanceValidator)

diagnosticorder-example | ERROR @ ProcedureRequest.extension[1]: The extension http://hl7.org/fhir/StructureDefinition/diagnosticorder-reason is unknown, and not allowed here (src = InstanceValidator) ERROR @ ProcedureRequest.extension[2]: The extension http://hl7.org/fhir/StructureDefinition/diagnosticorder-reasonRejected is unknown, and not allowed here (src = InstanceValidator) ERROR @ ProcedureRequest.extension[3]: The extension http://hl7.org/fhir/diagnosticorder-r2-marker is unknown, and not allowed here (src = InstanceValidator)

di | ERROR @ ProcedureRequest.extension: The extension http://hl7.org/fhir/diagnosticorder-r2-marker is unknown, and not allowed here (src = InstanceValidator)

ft4 | ERROR @ ProcedureRequest.extension: The extension http://hl7.org/fhir/diagnosticorder-r2-marker is unknown, and not allowed here (src = InstanceValidator)

do-uslab-example1 | ERROR @ ProcedureRequest.extension: The extension http://hl7.org/fhir/diagnosticorder-r2-marker is unknown, and not allowed here (src = InstanceValidator)

12.16 Resource NutritionOrder - Content

URL: https://hl7.org/fhir/STU3/nutritionorder.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

Orders and Observations Work Group | Maturity Level: 2 | Trial Use | Compartments: Encounter, Patient, Practitioner

A request to supply a diet, formula feeding (enteral) or oral nutritional supplement to a patient/resident.

12.16.1 Scope and Usage

This resource is a request resource from a FHIR workflow perspective - see Workflow. It is the intent of the Orders and Observation Workgroup to align this resource with the workflow pattern for request resources.

The NutritionOrder resource describes a request for oral diets (including general diets such as General Healthy diet, or therapeutic diets such as Consistent Carbohydrate, 2 gram Sodium, or Fluid Restricted), oral nutrition supplements (such as nutritionally complete pre-packed drinks), enteral nutrition (tube feedings) and infant formula which govern the distribution of food and nutritional products used to feed patients within an in-patient setting. It does not cover orders for parenteral (IV) nutrition which are typically filled by pharmacy. These nutrition orders are combined with information on a patient's food allergies and intolerances, and ethnic or cultural food preferences (e.g. Kosher or Vegetarian) to inform healthcare personnel about the type, texture, and/or quantity of foods that the patient should receive or consume.

Enteral orders are distinguished from supplements because they have some unique attributes and typically include administration information whereas oral nutritional supplements may simply be supplied (e.g. home health or outpatient settings). In a simple case, the requestor may designate type of product, product name and the route of administration along with free text instructions without a having to complete the additional structured details.

This resource is intended to be used by providers from a variety of specialties such as physicians, dietitian/nutritionists, or speech therapists. One provider may simply order a base element oral diet such as General Healthful diet. Another provider, based on scope of practice, may use other elements to communicate additional therapeutic needs or patient preferences. The optionality included gives an ordering provider the capability to write a simple order for an oral diet, nutritional supplement or formula with minimal requirements beyond that of specifying the diet, supplement or formula product, but also supports the ability to provide more detailed information that may be further augmented by a dietitian or nutrition specialist. For example, a physician may order a 2 g sodium diet. A speech therapist, based on the results of a swallowing evaluation, then orders a mechanically altered texture with nectar thick liquids.

12.16.2 Boundaries and Relationships

The NutritionOrder resource is used for requesting oral diets, oral nutrition supplements and enteral feedings in an in-patient setting. The MedicationRequest resource should be used for requesting parenteral (IV) nutrition and prescribing dietary supplements such as vitamin or mineral supplements.

The Nutrition Order is a record of the request for the supply of a diet, oral supplement or enteral formulas for a patient. However, to initiate the request requires the use of the Task resource and its associated workflow with the Nutrition Order referenced from Task.basedOn, or by using the Nutrition Task resource in the context of a messaging or service workflow where the request is explicit or implicit.

This resource is referenced by CarePlan, DiagnosticReport, Goal and Observation

12.16.3 Resource Content

* Structure

* UML

* XML

* JSON

* Turtle

* R2 Diff

* All

Structure

Name | Flags | Card. | Type | Description & Constraints

NutritionOrder | I | | DomainResource | Diet, formula or nutritional supplement request + Nutrition Order SHALL contain either Oral Diet , Supplement, or Enteral Formula class Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension

identifier | | 0..* | Identifier | Identifiers assigned to this order

status | ?!? | 0..1 | code | proposed | draft | planned | requested | active | on-hold | completed | cancelled | entered-in-error NutritionOrderStatus (Required)

patient | ? | 1..1 | Reference(Patient) | The person who requires the diet, formula or nutritional supplement

encounter | | 0..1 | Reference(Encounter) | The encounter associated with this nutrition order

dateTime | ? | 1..1 | dateTime | Date and time the nutrition order was requested
orderer | ? | 0..1 | Reference(Practitioner) | Who ordered the diet, formula or nutritional supplement
allergyIntolerance | | 0..* | Reference(AllergyIntolerance) | List of the patient's food and nutrition-related allergies and intolerances
foodPreferenceModifier | | 0..* | CodeableConcept | Order-specific modifier about the type of food that should be given Diet (Example)
excludeFoodModifier | | 0..* | CodeableConcept | Order-specific modifier about the type of food that should not be given Food Type Codes (Example)
oralDiet | I | 0..1 | BackboneElement | Oral diet components
type | ? | 0..* | CodeableConcept | Type of oral diet or diet restrictions that describe what can be consumed orally Diet Codes (Example)
schedule | | 0..* | Timing | Scheduled frequency of diet
nutrient | | 0..* | BackboneElement | Required nutrient modifications
modifier | | 0..1 | CodeableConcept | Type of nutrient that is being modified Nutrient Modifier Codes (Example)
amount | | 0..1 | SimpleQuantity | Quantity of the specified nutrient
texture | | 0..* | BackboneElement | Required texture modifications
modifier | | 0..1 | CodeableConcept | Code to indicate how to alter the texture of the foods, e.g. pureed Texture Modifier Codes (Example)
foodType | | 0..1 | CodeableConcept | Concepts that are used to identify an entity that is ingested for nutritional purposes Texture Modified Food Type Codes (Example)
fluidConsistencyType | | 0..* | CodeableConcept | The required consistency of fluids and liquids provided to the patient Fluid Consistency Type Codes (Example)
instruction | ? | 0..1 | string | Instructions or additional information about the oral diet
supplement | I | 0..* | BackboneElement | Supplement components
type | ? | 0..1 | CodeableConcept | Type of supplement product requested Supplement Type Codes (Example)
productName | | 0..1 | string | Product or brand name of the nutritional supplement
schedule | | 0..* | Timing | Scheduled frequency of supplement
quantity | | 0..1 | SimpleQuantity | Amount of the nutritional supplement
instruction | ? | 0..1 | string | Instructions or additional information about the oral supplement
enteralFormula | I | 0..1 | BackboneElement | Enteral formula components
baseFormulaType | ? | 0..1 | CodeableConcept | Type of enteral or infant formula Enteral Formula Type Codes (Example)
baseFormulaProductName | | 0..1 | string | Product or brand name of the enteral or infant formula
additiveType | | 0..1 | CodeableConcept | Type of modular component to add to the feeding Enteral Formula Additive Type Code (Example)
additiveProductName | | 0..1 | string | Product or brand name of the modular additive
caloricDensity | | 0..1 | SimpleQuantity | Amount of energy per specified volume that is required
routeofAdministration | | 0..1 | CodeableConcept | How the formula should enter the patient's gastrointestinal tract Enteral Route Codes (Extensible)
administration | | 0..* | BackboneElement | Formula feeding instruction as structured data
schedule | | 0..1 | Timing | Scheduled frequency of enteral feeding
quantity | | 0..1 | SimpleQuantity | The volume of formula to provide
rate[x] | | 0..1 | | Speed with which the formula is provided per period of time
rateQuantity | | | SimpleQuantity
rateRatio | | | Ratio
maxVolumeToDeliver | | 0..1 | SimpleQuantity | Upper limit on formula volume per unit of time
administrationInstruction | ? | 0..1 | string | Formula feeding instructions expressed as text
Documentation for this format

UML Diagram (Legend)

XML Template

```
<NutritionOrder xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Identifiers assigned to this order --></identifier>
 <status value="[code]"/><!-- 0..1 proposed | draft | planned | requested | active | on-hold | completed | cancelled |
    entered-in-error -->
 <patient><!-- 1..1 Reference(Patient) The person who requires the diet, formula or nutritional supplement --></patient>
 <encounter><!-- 0..1 Reference(Encounter) The encounter associated with this nutrition order --></encounter>
 <dateTime value="[dateTime]"/><!-- 1..1 Date and time the nutrition order was requested -->
 <orderer><!-- 0..1 Reference(Practitioner) Who ordered the diet, formula or nutritional supplement --></orderer>
 <allergyIntolerance><!-- 0..* Reference(AllergyIntolerance) List of the patient's food and nutrition-related allergies
    and intolerances --></allergyIntolerance>
 <foodPreferenceModifier><!-- 0..* CodeableConcept Order-specific modifier about the type of food that should be given
    --></foodPreferenceModifier>
 <excludeFoodModifier><!-- 0..* CodeableConcept Order-specific modifier about the type of food that should not be given
    --></excludeFoodModifier>
 <oralDiet>  <!--  0..1 Oral diet components -->
  <type><!-- 0..* CodeableConcept Type of oral diet or diet restrictions that describe what can be consumed orally
    --></type>
  <schedule><!-- 0..* Timing Scheduled frequency of diet --></schedule>
  <nutrient>  <!-- 0..* Required  nutrient modifications -->
   <modifier><!-- 0..1 CodeableConcept Type of nutrient that is being modified --></modifier>
   <amount><!-- 0..1 Quantity(SimpleQuantity) Quantity of the specified nutrient --></amount>
  </nutrient>
  <texture>  <!-- 0..* Required  texture modifications -->
   <modifier><!-- 0..1 CodeableConcept Code to indicate how to alter the texture of the foods, e.g. pureed
    --></modifier>
   <foodType><!-- 0..1 CodeableConcept Concepts that are used to identify an entity that is ingested for nutritional
    purposes --></foodType>
  </texture>
  <fluidConsistencyType><!-- 0..* CodeableConcept The required consistency of fluids and liquids provided to the patient
    --></fluidConsistencyType>
  <instruction value="[string]"/><!-- 0..1 Instructions or additional information about the oral diet -->
 </oralDiet>
 <supplement>  <!--  0..* Supplement components -->
  <type><!-- 0..1 CodeableConcept Type of supplement product requested --></type>
```

```
  <productName value="[string]"/><!-- 0..1 Product or brand name of the nutritional supplement -->
  <schedule><!-- 0..* Timing Scheduled frequency of supplement --></schedule>
  <quantity><!-- 0..1 Quantity(SimpleQuantity) Amount of the nutritional supplement --></quantity>
  <instruction value="[string]"/><!-- 0..1 Instructions or additional information about the oral supplement -->
 </supplement>
 <enteralFormula>  <!--  0..1 Enteral formula components -->
  <baseFormulaType><!-- 0..1 CodeableConcept Type of enteral or infant formula --></baseFormulaType>
  <baseFormulaProductName value="[string]"/><!-- 0..1 Product or brand name of the enteral or infant formula -->
  <additiveType><!-- 0..1 CodeableConcept Type of modular component to add to the feeding --></additiveType>
  <additiveProductName value="[string]"/><!-- 0..1 Product or brand name of the modular additive -->
  <caloricDensity><!-- 0..1 Quantity(SimpleQuantity) Amount of energy per specified volume that is required
    --></caloricDensity>
  <routeofAdministration><!-- 0..1 CodeableConcept How the formula should enter the patient's gastrointestinal tract
    --></routeofAdministration>
  <administration>  <!-- 0..* Formula feeding instruction as structured data -->
   <schedule><!-- 0..1 Timing Scheduled frequency of enteral feeding --></schedule>
   <quantity><!-- 0..1 Quantity(SimpleQuantity) The volume of formula to provide --></quantity>
   <rate[x]><!-- 0..1 Quantity(SimpleQuantity)|Ratio Speed with which the formula is provided per period of time
    --></rate[x]>
  </administration>
  <maxVolumeToDeliver><!-- 0..1 Quantity(SimpleQuantity) Upper limit on formula volume per unit of time
    --></maxVolumeToDeliver>
  <administrationInstruction value="[string]"/><!-- 0..1 Formula feeding instructions expressed as text -->
 </enteralFormula>
</NutritionOrder>
```

JSON Template

```
{
  "resourceType" : "NutritionOrder",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Identifiers assigned to this order
  "status" : "<code>", // proposed | draft | planned | requested | active | on-hold | completed | cancelled | entered-
    in-error
  "patient" : { Reference(Patient) }, // R!  The person who requires the diet, formula or nutritional supplement
  "encounter" : { Reference(Encounter) }, // The encounter associated with this nutrition order
  "dateTime" : "<dateTime>", // R!  Date and time the nutrition order was requested
  "orderer" : { Reference(Practitioner) }, // Who ordered the diet, formula or nutritional supplement
  "allergyIntolerance" : [{ Reference(AllergyIntolerance) }], // List of the patient's food and nutrition-related
    allergies and intolerances
  "foodPreferenceModifier" : [{ CodeableConcept }], // Order-specific modifier about the type of food that should be
    given
  "excludeFoodModifier" : [{ CodeableConcept }], // Order-specific modifier about the type of food that should not be
    given
  "oralDiet" : { // C? Oral diet components
    "type" : [{ CodeableConcept }], // Type of oral diet or diet restrictions that describe what can be consumed orally
    "schedule" : [{ Timing }], // Scheduled frequency of diet
    "nutrient" : [{ // Required  nutrient modifications
      "modifier" : { CodeableConcept }, // Type of nutrient that is being modified
      "amount" : { Quantity(SimpleQuantity) } // Quantity of the specified nutrient
    }],
    "texture" : [{ // Required  texture modifications
      "modifier" : { CodeableConcept }, // Code to indicate how to alter the texture of the foods, e.g. pureed
      "foodType" : { CodeableConcept } // Concepts that are used to identify an entity that is ingested for nutritional
    purposes
    }],
    "fluidConsistencyType" : [{ CodeableConcept }], // The required consistency of fluids and liquids provided to the
    patient
    "instruction" : "<string>" // Instructions or additional information about the oral diet
  },
  "supplement" : [{ // C? Supplement components
    "type" : { CodeableConcept }, // Type of supplement product requested
    "productName" : "<string>", // Product or brand name of the nutritional supplement
    "schedule" : [{ Timing }], // Scheduled frequency of supplement
    "quantity" : { Quantity(SimpleQuantity) }, // Amount of the nutritional supplement
    "instruction" : "<string>" // Instructions or additional information about the oral supplement
  }],
  "enteralFormula" : { // C? Enteral formula components
    "baseFormulaType" : { CodeableConcept }, // Type of enteral or infant formula
    "baseFormulaProductName" : "<string>", // Product or brand name of the enteral or infant formula
    "additiveType" : { CodeableConcept }, // Type of modular component to add to the feeding
    "additiveProductName" : "<string>", // Product or brand name of the modular additive
    "caloricDensity" : { Quantity(SimpleQuantity) }, // Amount of energy per specified volume that is required
    "routeofAdministration" : { CodeableConcept }, // How the formula should enter the patient's gastrointestinal tract
    "administration" : [{ // Formula feeding instruction as structured data
      "schedule" : { Timing }, // Scheduled frequency of enteral feeding
      "quantity" : { Quantity(SimpleQuantity) }, // The volume of formula to provide
      // rate[x]: Speed with which the formula is provided per period of time. One of these 2:
      "rateQuantity" : { Quantity(SimpleQuantity) }
      "rateRatio" : { Ratio }
    }],
    "maxVolumeToDeliver" : { Quantity(SimpleQuantity) }, // Upper limit on formula volume per unit of time
    "administrationInstruction" : "<string>" // Formula feeding instructions expressed as text
  }
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .
```

```
[ a fhir:NutritionOrder;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:NutritionOrder.identifier [ Identifier ], ... ; # 0..* Identifiers assigned to this order
  fhir:NutritionOrder.status [ code ]; # 0..1 proposed | draft | planned | requested | active | on-hold | completed |
    cancelled | entered-in-error
  fhir:NutritionOrder.patient [ Reference(Patient) ]; # 1..1 The person who requires the diet, formula or nutritional
    supplement
  fhir:NutritionOrder.encounter [ Reference(Encounter) ]; # 0..1 The encounter associated with this nutrition order
  fhir:NutritionOrder.dateTime [ dateTime ]; # 1..1 Date and time the nutrition order was requested
  fhir:NutritionOrder.orderer [ Reference(Practitioner) ]; # 0..1 Who ordered the diet, formula or nutritional
    supplement
  fhir:NutritionOrder.allergyIntolerance [ Reference(AllergyIntolerance) ], ... ; # 0..* List of the patient's food and
    nutrition-related allergies and intolerances
  fhir:NutritionOrder.foodPreferenceModifier [ CodeableConcept ], ... ; # 0..* Order-specific modifier about the type of
     food that should be given
  fhir:NutritionOrder.excludeFoodModifier [ CodeableConcept ], ... ; # 0..* Order-specific modifier about the type of
    food that should not be given
  fhir:NutritionOrder.oralDiet [ # 0..1 Oral diet components
    fhir:NutritionOrder.oralDiet.type [ CodeableConcept ], ... ; # 0..* Type of oral diet or diet restrictions that
    describe what can be consumed orally
    fhir:NutritionOrder.oralDiet.schedule [ Timing ], ... ; # 0..* Scheduled frequency of diet
    fhir:NutritionOrder.oralDiet.nutrient [ # 0..* Required  nutrient modifications
      fhir:NutritionOrder.oralDiet.nutrient.modifier [ CodeableConcept ]; # 0..1 Type of nutrient that is being modified
      fhir:NutritionOrder.oralDiet.nutrient.amount [ Quantity(SimpleQuantity) ]; # 0..1 Quantity of the specified
    nutrient
    ], ...;
    fhir:NutritionOrder.oralDiet.texture [ # 0..* Required  texture modifications
      fhir:NutritionOrder.oralDiet.texture.modifier [ CodeableConcept ]; # 0..1 Code to indicate how to alter the
    texture of the foods, e.g. pureed
      fhir:NutritionOrder.oralDiet.texture.foodType [ CodeableConcept ]; # 0..1 Concepts that are used to identify an
    entity that is ingested for nutritional purposes
    ], ...;
    fhir:NutritionOrder.oralDiet.fluidConsistencyType [ CodeableConcept ], ... ; # 0..* The required consistency of
    fluids and liquids provided to the patient
    fhir:NutritionOrder.oralDiet.instruction [ string ]; # 0..1 Instructions or additional information about the oral
    diet
  ];
  fhir:NutritionOrder.supplement [ # 0..* Supplement components
    fhir:NutritionOrder.supplement.type [ CodeableConcept ]; # 0..1 Type of supplement product requested
    fhir:NutritionOrder.supplement.productName [ string ]; # 0..1 Product or brand name of the nutritional supplement
    fhir:NutritionOrder.supplement.schedule [ Timing ], ... ; # 0..* Scheduled frequency of supplement
    fhir:NutritionOrder.supplement.quantity [ Quantity(SimpleQuantity) ]; # 0..1 Amount of the nutritional supplement
    fhir:NutritionOrder.supplement.instruction [ string ]; # 0..1 Instructions or additional information about the oral
    supplement
  ], ...;
  fhir:NutritionOrder.enteralFormula [ # 0..1 Enteral formula components
    fhir:NutritionOrder.enteralFormula.baseFormulaType [ CodeableConcept ]; # 0..1 Type of enteral or infant formula
    fhir:NutritionOrder.enteralFormula.baseFormulaProductName [ string ]; # 0..1 Product or brand name of the enteral or
     infant formula
    fhir:NutritionOrder.enteralFormula.additiveType [ CodeableConcept ]; # 0..1 Type of modular component to add to the
    feeding
    fhir:NutritionOrder.enteralFormula.additiveProductName [ string ]; # 0..1 Product or brand name of the modular
    additive
    fhir:NutritionOrder.enteralFormula.caloricDensity [ Quantity(SimpleQuantity) ]; # 0..1 Amount of energy per
    specified volume that is required
    fhir:NutritionOrder.enteralFormula.routeofAdministration [ CodeableConcept ]; # 0..1 How the formula should enter
    the patient's gastrointestinal tract
    fhir:NutritionOrder.enteralFormula.administration [ # 0..* Formula feeding instruction as structured data
      fhir:NutritionOrder.enteralFormula.administration.schedule [ Timing ]; # 0..1 Scheduled frequency of enteral
    feeding
      fhir:NutritionOrder.enteralFormula.administration.quantity [ Quantity(SimpleQuantity) ]; # 0..1 The volume of
    formula to provide
      # NutritionOrder.enteralFormula.administration.rate[x] : 0..1 Speed with which the formula is provided per period
    of time. One of these 2
        fhir:NutritionOrder.enteralFormula.administration.rateSimpleQuantity [ Quantity(SimpleQuantity) ]
        fhir:NutritionOrder.enteralFormula.administration.rateRatio [ Ratio ]
    ], ...;
    fhir:NutritionOrder.enteralFormula.maxVolumeToDeliver [ Quantity(SimpleQuantity) ]; # 0..1 Upper limit on formula
    volume per unit of time
    fhir:NutritionOrder.enteralFormula.administrationInstruction [ string ]; # 0..1 Formula feeding instructions
    expressed as text
  ];
]
```

Changes since DSTU2

NutritionOrder

NutritionOrder.status | Change value set from http://hl7.org/fhir/ValueSet/nutrition-order-status to http://hl7.org/fhir/ValueSet/nutrition-request-status

See the Full Difference for further information

This analysis is available as XML or JSON.

Structure

Name | Flags | Card. | Type | Description & Constraints

NutritionOrder | I | | DomainResource | Diet, formula or nutritional supplement request + Nutrition Order SHALL contain either Oral Diet , Supplement, or Enteral Formula class Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension

identifier | | 0..* | Identifier | Identifiers assigned to this order
status | ?!? | 0..1 | code | proposed | draft | planned | requested | active | on-hold | completed | cancelled | entered-in-error NutritionOrderStatus (Required)
patient | ? | 1..1 | Reference(Patient) | The person who requires the diet, formula or nutritional supplement
encounter | | 0..1 | Reference(Encounter) | The encounter associated with this nutrition order
dateTime | ? | 1..1 | dateTime | Date and time the nutrition order was requested
orderer | ? | 0..1 | Reference(Practitioner) | Who ordered the diet, formula or nutritional supplement
allergyIntolerance | | 0..* | Reference(AllergyIntolerance) | List of the patient's food and nutrition-related allergies and intolerances
foodPreferenceModifier | | 0..* | CodeableConcept | Order-specific modifier about the type of food that should be given Diet (Example)
excludeFoodModifier | | 0..* | CodeableConcept | Order-specific modifier about the type of food that should not be given Food Type Codes (Example)
oralDiet | I | 0..1 | BackboneElement | Oral diet components
type | ? | 0..* | CodeableConcept | Type of oral diet or diet restrictions that describe what can be consumed orally Diet Codes (Example)
schedule | | 0..* | Timing | Scheduled frequency of diet
nutrient | | 0..* | BackboneElement | Required nutrient modifications
modifier | | 0..1 | CodeableConcept | Type of nutrient that is being modified Nutrient Modifier Codes (Example)
amount | | 0..1 | SimpleQuantity | Quantity of the specified nutrient
texture | | 0..* | BackboneElement | Required texture modifications
modifier | | 0..1 | CodeableConcept | Code to indicate how to alter the texture of the foods, e.g. pureed Texture Modifier Codes (Example)
foodType | | 0..1 | CodeableConcept | Concepts that are used to identify an entity that is ingested for nutritional purposes Texture Modified Food Type Codes (Example)
fluidConsistencyType | | 0..* | CodeableConcept | The required consistency of fluids and liquids provided to the patient Fluid Consistency Type Codes (Example)
instruction | ? | 0..1 | string | Instructions or additional information about the oral diet
supplement | I | 0..* | BackboneElement | Supplement components
type | ? | 0..1 | CodeableConcept | Type of supplement product requested Supplement Type Codes (Example)
productName | | 0..1 | string | Product or brand name of the nutritional supplement
schedule | | 0..* | Timing | Scheduled frequency of supplement
quantity | | 0..1 | SimpleQuantity | Amount of the nutritional supplement
instruction | ? | 0..1 | string | Instructions or additional information about the oral supplement
enteralFormula | I | 0..1 | BackboneElement | Enteral formula components
baseFormulaType | ? | 0..1 | CodeableConcept | Type of enteral or infant formula Enteral Formula Type Codes (Example)
baseFormulaProductName | | 0..1 | string | Product or brand name of the enteral or infant formula
additiveType | | 0..1 | CodeableConcept | Type of modular component to add to the feeding Enteral Formula Additive Type Code (Example)
additiveProductName | | 0..1 | string | Product or brand name of the modular additive
caloricDensity | | 0..1 | SimpleQuantity | Amount of energy per specified volume that is required
routeofAdministration | | 0..1 | CodeableConcept | How the formula should enter the patient's gastrointestinal tract Enteral Route Codes (Extensible)
administration | | 0..* | BackboneElement | Formula feeding instruction as structured data
schedule | | 0..1 | Timing | Scheduled frequency of enteral feeding
quantity | | 0..1 | SimpleQuantity | The volume of formula to provide
rate[x] | | 0..1 | | Speed with which the formula is provided per period of time
rateQuantity | | | SimpleQuantity
rateRatio | | | Ratio
maxVolumeToDeliver | | 0..1 | SimpleQuantity | Upper limit on formula volume per unit of time
administrationInstruction | ? | 0..1 | string | Formula feeding instructions expressed as text
Documentation for this format

UML Diagram (Legend)

XML Template

```
<NutritionOrder xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Identifiers assigned to this order --></identifier>
 <status value="[code]"/><!-- 0..1 proposed | draft | planned | requested | active | on-hold | completed | cancelled |
    entered-in-error -->
 <patient><!-- 1..1 Reference(Patient) The person who requires the diet, formula or nutritional supplement --></patient>
 <encounter><!-- 0..1 Reference(Encounter) The encounter associated with this nutrition order --></encounter>
 <dateTime value="[dateTime]"/><!-- 1..1 Date and time the nutrition order was requested -->
 <orderer><!-- 0..1 Reference(Practitioner) Who ordered the diet, formula or nutritional supplement --></orderer>
 <allergyIntolerance><!-- 0..* Reference(AllergyIntolerance) List of the patient's food and nutrition-related allergies
    and intolerances --></allergyIntolerance>
 <foodPreferenceModifier><!-- 0..* CodeableConcept Order-specific modifier about the type of food that should be given
    --></foodPreferenceModifier>
 <excludeFoodModifier><!-- 0..* CodeableConcept Order-specific modifier about the type of food that should not be given
    --></excludeFoodModifier>
 <oralDiet>  <!--  0..1 Oral diet components -->
  <type><!-- 0..* CodeableConcept Type of oral diet or diet restrictions that describe what can be consumed orally
    --></type>
  <schedule><!-- 0..* Timing Scheduled frequency of diet --></schedule>
  <nutrient>  <!-- 0..* Required  nutrient modifications -->
   <modifier><!-- 0..1 CodeableConcept Type of nutrient that is being modified --></modifier>
   <amount><!-- 0..1 Quantity(SimpleQuantity) Quantity of the specified nutrient --></amount>
  </nutrient>
  <texture>  <!-- 0..* Required  texture modifications -->
   <modifier><!-- 0..1 CodeableConcept Code to indicate how to alter the texture of the foods, e.g. pureed
    --></modifier>
   <foodType><!-- 0..1 CodeableConcept Concepts that are used to identify an entity that is ingested for nutritional
    purposes --></foodType>
  </texture>
```

```
  <fluidConsistencyType><!-- 0..* CodeableConcept The required consistency of fluids and liquids provided to the patient
     --></fluidConsistencyType>
  <instruction value="[string]"/><!-- 0..1 Instructions or additional information about the oral diet -->
 </oralDiet>
 <supplement>  <!--  0..* Supplement components -->
  <type><!-- 0..1 CodeableConcept Type of supplement product requested --></type>
  <productName value="[string]"/><!-- 0..1 Product or brand name of the nutritional supplement -->
  <schedule><!-- 0..* Timing Scheduled frequency of supplement --></schedule>
  <quantity><!-- 0..1 Quantity(SimpleQuantity) Amount of the nutritional supplement --></quantity>
  <instruction value="[string]"/><!-- 0..1 Instructions or additional information about the oral supplement -->
 </supplement>
 <enteralFormula>  <!--  0..1 Enteral formula components -->
  <baseFormulaType><!-- 0..1 CodeableConcept Type of enteral or infant formula --></baseFormulaType>
  <baseFormulaProductName value="[string]"/><!-- 0..1 Product or brand name of the enteral or infant formula -->
  <additiveType><!-- 0..1 CodeableConcept Type of modular component to add to the feeding --></additiveType>
  <additiveProductName value="[string]"/><!-- 0..1 Product or brand name of the modular additive -->
  <caloricDensity><!-- 0..1 Quantity(SimpleQuantity) Amount of energy per specified volume that is required
     --></caloricDensity>
  <routeofAdministration><!-- 0..1 CodeableConcept How the formula should enter the patient's gastrointestinal tract
     --></routeofAdministration>
  <administration>  <!-- 0..* Formula feeding instruction as structured data -->
   <schedule><!-- 0..1 Timing Scheduled frequency of enteral feeding --></schedule>
   <quantity><!-- 0..1 Quantity(SimpleQuantity) The volume of formula to provide --></quantity>
   <rate[x]><!-- 0..1 Quantity(SimpleQuantity)|Ratio Speed with which the formula is provided per period of time
     --></rate[x]>
  </administration>
  <maxVolumeToDeliver><!-- 0..1 Quantity(SimpleQuantity) Upper limit on formula volume per unit of time
     --></maxVolumeToDeliver>
  <administrationInstruction value="[string]"/><!-- 0..1 Formula feeding instructions expressed as text -->
 </enteralFormula>
</NutritionOrder>
```

JSON Template

```
{
  "resourceType" : "NutritionOrder",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Identifiers assigned to this order
  "status" : "<code>", // proposed | draft | planned | requested | active | on-hold | completed | cancelled | entered-
    in-error
  "patient" : { Reference(Patient) }, // R!  The person who requires the diet, formula or nutritional supplement
  "encounter" : { Reference(Encounter) }, // The encounter associated with this nutrition order
  "dateTime" : "<dateTime>", // R!  Date and time the nutrition order was requested
  "orderer" : { Reference(Practitioner) }, // Who ordered the diet, formula or nutritional supplement
  "allergyIntolerance" : [{ Reference(AllergyIntolerance) }], // List of the patient's food and nutrition-related
    allergies and intolerances
  "foodPreferenceModifier" : [{ CodeableConcept }], // Order-specific modifier about the type of food that should be
    given
  "excludeFoodModifier" : [{ CodeableConcept }], // Order-specific modifier about the type of food that should not be
    given
  "oralDiet" : { // C? Oral diet components
    "type" : [{ CodeableConcept }], // Type of oral diet or diet restrictions that describe what can be consumed orally
    "schedule" : [{ Timing }], // Scheduled frequency of diet
    "nutrient" : [{ // Required  nutrient modifications
      "modifier" : { CodeableConcept }, // Type of nutrient that is being modified
      "amount" : { Quantity(SimpleQuantity) } // Quantity of the specified nutrient
    }],
    "texture" : [{ // Required  texture modifications
      "modifier" : { CodeableConcept }, // Code to indicate how to alter the texture of the foods, e.g. pureed
      "foodType" : { CodeableConcept } // Concepts that are used to identify an entity that is ingested for nutritional
    purposes
    }],
    "fluidConsistencyType" : [{ CodeableConcept }], // The required consistency of fluids and liquids provided to the
    patient
    "instruction" : "<string>" // Instructions or additional information about the oral diet
  },
  "supplement" : [{ // C? Supplement components
    "type" : { CodeableConcept }, // Type of supplement product requested
    "productName" : "<string>", // Product or brand name of the nutritional supplement
    "schedule" : [{ Timing }], // Scheduled frequency of supplement
    "quantity" : { Quantity(SimpleQuantity) }, // Amount of the nutritional supplement
    "instruction" : "<string>" // Instructions or additional information about the oral supplement
  }],
  "enteralFormula" : { // C? Enteral formula components
    "baseFormulaType" : { CodeableConcept }, // Type of enteral or infant formula
    "baseFormulaProductName" : "<string>", // Product or brand name of the enteral or infant formula
    "additiveType" : { CodeableConcept }, // Type of modular component to add to the feeding
    "additiveProductName" : "<string>", // Product or brand name of the modular additive
    "caloricDensity" : { Quantity(SimpleQuantity) }, // Amount of energy per specified volume that is required
    "routeofAdministration" : { CodeableConcept }, // How the formula should enter the patient's gastrointestinal tract
    "administration" : [{ // Formula feeding instruction as structured data
      "schedule" : { Timing }, // Scheduled frequency of enteral feeding
      "quantity" : { Quantity(SimpleQuantity) }, // The volume of formula to provide
      // rate[x]: Speed with which the formula is provided per period of time. One of these 2:
      "rateQuantity" : { Quantity(SimpleQuantity) }
      "rateRatio" : { Ratio }
    }],
    "maxVolumeToDeliver" : { Quantity(SimpleQuantity) }, // Upper limit on formula volume per unit of time
    "administrationInstruction" : "<string>" // Formula feeding instructions expressed as text
```

```
  }
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:NutritionOrder;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:NutritionOrder.identifier [ Identifier ], ... ; # 0..* Identifiers assigned to this order
  fhir:NutritionOrder.status [ code ]; # 0..1 proposed | draft | planned | requested | active | on-hold | completed |
    cancelled | entered-in-error
  fhir:NutritionOrder.patient [ Reference(Patient) ]; # 1..1 The person who requires the diet, formula or nutritional
    supplement
  fhir:NutritionOrder.encounter [ Reference(Encounter) ]; # 0..1 The encounter associated with this nutrition order
  fhir:NutritionOrder.dateTime [ dateTime ]; # 1..1 Date and time the nutrition order was requested
  fhir:NutritionOrder.orderer [ Reference(Practitioner) ]; # 0..1 Who ordered the diet, formula or nutritional
    supplement
  fhir:NutritionOrder.allergyIntolerance [ Reference(AllergyIntolerance) ], ... ; # 0..* List of the patient's food and
    nutrition-related allergies and intolerances
  fhir:NutritionOrder.foodPreferenceModifier [ CodeableConcept ], ... ; # 0..* Order-specific modifier about the type of
     food that should be given
  fhir:NutritionOrder.excludeFoodModifier [ CodeableConcept ], ... ; # 0..* Order-specific modifier about the type of
    food that should not be given
  fhir:NutritionOrder.oralDiet [ # 0..1 Oral diet components
    fhir:NutritionOrder.oralDiet.type [ CodeableConcept ], ... ; # 0..* Type of oral diet or diet restrictions that
    describe what can be consumed orally
    fhir:NutritionOrder.oralDiet.schedule [ Timing ], ... ; # 0..* Scheduled frequency of diet
    fhir:NutritionOrder.oralDiet.nutrient [ # 0..* Required  nutrient modifications
      fhir:NutritionOrder.oralDiet.nutrient.modifier [ CodeableConcept ]; # 0..1 Type of nutrient that is being modified
      fhir:NutritionOrder.oralDiet.nutrient.amount [ Quantity(SimpleQuantity) ]; # 0..1 Quantity of the specified
    nutrient
    ], ...;
    fhir:NutritionOrder.oralDiet.texture [ # 0..* Required  texture modifications
      fhir:NutritionOrder.oralDiet.texture.modifier [ CodeableConcept ]; # 0..1 Code to indicate how to alter the
    texture of the foods, e.g. pureed
      fhir:NutritionOrder.oralDiet.texture.foodType [ CodeableConcept ]; # 0..1 Concepts that are used to identify an
    entity that is ingested for nutritional purposes
    ], ...;
    fhir:NutritionOrder.oralDiet.fluidConsistencyType [ CodeableConcept ], ... ; # 0..* The required consistency of
    fluids and liquids provided to the patient
    fhir:NutritionOrder.oralDiet.instruction [ string ]; # 0..1 Instructions or additional information about the oral
    diet
  ];
  fhir:NutritionOrder.supplement [ # 0..* Supplement components
    fhir:NutritionOrder.supplement.type [ CodeableConcept ]; # 0..1 Type of supplement product requested
    fhir:NutritionOrder.supplement.productName [ string ]; # 0..1 Product or brand name of the nutritional supplement
    fhir:NutritionOrder.supplement.schedule [ Timing ], ... ; # 0..* Scheduled frequency of supplement
    fhir:NutritionOrder.supplement.quantity [ Quantity(SimpleQuantity) ]; # 0..1 Amount of the nutritional supplement
    fhir:NutritionOrder.supplement.instruction [ string ]; # 0..1 Instructions or additional information about the oral
    supplement
  ], ...;
  fhir:NutritionOrder.enteralFormula [ # 0..1 Enteral formula components
    fhir:NutritionOrder.enteralFormula.baseFormulaType [ CodeableConcept ]; # 0..1 Type of enteral or infant formula
    fhir:NutritionOrder.enteralFormula.baseFormulaProductName [ string ]; # 0..1 Product or brand name of the enteral or
     infant formula
    fhir:NutritionOrder.enteralFormula.additiveType [ CodeableConcept ]; # 0..1 Type of modular component to add to the
    feeding
    fhir:NutritionOrder.enteralFormula.additiveProductName [ string ]; # 0..1 Product or brand name of the modular
    additive
    fhir:NutritionOrder.enteralFormula.caloricDensity [ Quantity(SimpleQuantity) ]; # 0..1 Amount of energy per
    specified volume that is required
    fhir:NutritionOrder.enteralFormula.routeofAdministration [ CodeableConcept ]; # 0..1 How the formula should enter
    the patient's gastrointestinal tract
    fhir:NutritionOrder.enteralFormula.administration [ # 0..* Formula feeding instruction as structured data
      fhir:NutritionOrder.enteralFormula.administration.schedule [ Timing ]; # 0..1 Scheduled frequency of enteral
    feeding
      fhir:NutritionOrder.enteralFormula.administration.quantity [ Quantity(SimpleQuantity) ]; # 0..1 The volume of
    formula to provide
      # NutritionOrder.enteralFormula.administration.rate[x] : 0..1 Speed with which the formula is provided per period
    of time. One of these 2
        fhir:NutritionOrder.enteralFormula.administration.rateSimpleQuantity [ Quantity(SimpleQuantity) ]
        fhir:NutritionOrder.enteralFormula.administration.rateRatio [ Ratio ]
    ], ...;
    fhir:NutritionOrder.enteralFormula.maxVolumeToDeliver [ Quantity(SimpleQuantity) ]; # 0..1 Upper limit on formula
    volume per unit of time
    fhir:NutritionOrder.enteralFormula.administrationInstruction [ string ]; # 0..1 Formula feeding instructions
    expressed as text
  ];
]
```

Changes since DSTU2

NutritionOrder

NutritionOrder.status | Change value set from http://hl7.org/fhir/ValueSet/nutrition-order-status to http://hl7.org/fhir/ValueSet/nutrition-request-status

See the Full Difference for further information

This analysis is available as XML or JSON.

Alternate definitions: Master Definition (XML, JSON), XML Schema/Schematron (for) + JSON Schema, ShEx (for Turtle)

12.16.3.1 Terminology Bindings

Path | Definition | Type | Reference

NutritionOrder.status | Codes specifying the state of the request. Describes the lifecycle of the nutrition order. | Required | NutritionOrderStatus

NutritionOrder.foodPreferenceModifier | Medical, cultural or ethical food preferences to help with catering requirements | Example | Diet

NutritionOrder.excludeFoodModifier | Codes used to indicate the type of food that should NOT be given to the patient. | Example | Food Type Codes

NutritionOrder.oralDiet.type | Codes used to indicate the type of diet being ordered for a patient. | Example | Diet Codes

NutritionOrder.oralDiet.nutrient.modifier | Codes for types of nutrient that is being modified such as carbohydrate or sodium. | Example | Nutrient Modifier Codes

NutritionOrder.oralDiet.texture.modifier | Codes for food consistency types or texture modifications to apply to foods. | Example | Texture Modifier Codes

NutritionOrder.oralDiet.texture.foodType | Codes for types of foods that are texture modified. | Example | Texture Modified Food Type Codes

NutritionOrder.oralDiet.fluidConsistencyType | Codes used to represent the consistency of fluids and liquids provided to the patient. | Example | Fluid Consistency Type Codes

NutritionOrder.supplement.type | Codes for nutritional supplements to be provided to the patient | Example | Supplement Type Codes

NutritionOrder.enteralFormula.baseFormulaType | Codes for type of enteral formula to be administered to patient. | Example | Enteral Formula Type Codes

NutritionOrder.enteralFormula.additiveType | Codes for the type of modular component such as protein, carbohydrate or fiber to be provided in addition to or mixed with the base formula. | Example | Enteral Formula Additive Type Code

NutritionOrder.enteralFormula.routeofAdministration | Codes specifying the route of administration of enteral formula. | Extensible | Enteral Route Codes

12.16.3.2 Constraints

* nor-1: Nutrition Order SHALL contain either Oral Diet , Supplement, or Enteral Formula class (expression : oralDiet.exists() or supplement.exists() or enteralFormula.exists())

12.16.4 Notes:

12.16.4.1 Enteral continuous vs. intermittent tube feedings

Tube feedings can be administered via continuous drip using a pump or via intermittent feedings, using gravity drip or a pump. The examples Nutrition Order Enteral Bolus Feeding Example and Nutrition Order Enteral Continuous Feeding Example show how this resource can be used to order both kinds of enteral feeding using the structured data elements. The continuous feeding typically specifies rate of administration and a maximum volume of delivery using the enteralFormula.rate and enteralFormula.maxVolumeToDeliver elements. On the other hand, the intermittent feeding typically specifies the amount and frequency of administration using the enteralFormula.quantity and enteralFormula.schedule elements. In both cases, to vary the rate or quantity over time the enteralFormula.administration element can be repeated.

12.16.4.2 Note about the examples

The examples associated with this resource demonstrate the core elements and do not necessarily reflect real world implementations that may be constrained by future profiles for a given implementation or setting. For example, a renal diet is often comprised of pre-coordinated components including common nutrient modifications such as protein, potassium and phosphorus to assist with the speed of entry of complex diet orders.

12.16.5 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common

additive | token | Type of module component to add to the feeding | NutritionOrder.enteralFormula.additiveType

datetime | date | Return nutrition orders requested on this date | NutritionOrder.dateTime

encounter | reference | Return nutrition orders with this encounter identifier | NutritionOrder.encounter (Encounter) | 12 Resources

formula | token | Type of enteral or infant formula | NutritionOrder.enteralFormula.baseFormulaType

identifier | token | Return nutrition orders with this external identifier | NutritionOrder.identifier | 26 Resources

oraldiet | token | Type of diet that can be consumed orally (i.e., take via the mouth). | NutritionOrder.oralDiet.type

patient | reference | The identity of the person who requires the diet, formula or nutritional supplement | NutritionOrder.patient (Patient) | 31 Resources

provider | reference | The identify of the provider who placed the nutrition order | NutritionOrder.orderer (Practitioner)

status | token | Status of the nutrition order. | NutritionOrder.status

supplement | token | Type of supplement product requested | NutritionOrder.supplement.type

12.16.6 Resource NutritionOrder - Examples

URL: https://hl7.org/fhir/STU3/nutritionorder-examples.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

12.16.6 Resource NutritionOrder - Examples

Orders and Observations Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Encounter, Patient, Practitioner

Example List:

Nutrition Order Pureed Diet and Energy/Protein Supplement Simple Example | pureeddiet-simple | XML | JSON | Turtle
Nutrition Order Low Sodium and Fluid Restricted Diet Example | cardiacdiet | XML | JSON | Turtle
Nutrition Order Diabetic Diet Example | diabeticdiet | XML | JSON | Turtle
Nutrition Order Texture Modified Diet Example | texturemodified | XML | JSON | Turtle
Nutrition Order Pureed Diet Example | pureeddiet | XML | JSON | Turtle
Nutrition Order Renal Diet Example | renaldiet | XML | JSON | Turtle
Nutrition Order Low Fat and Fiber Restricted Diet Example | fiberrestricteddiet | XML | JSON | Turtle
Nutrition Order Diabetic Supplement Example | diabeticsupplement | XML | JSON | Turtle
Nutrition Order Protein Supplement Example | proteinsupplement | XML | JSON | Turtle
Nutrition Order Energy/Protein Supplement Example | energysupplement | XML | JSON | Turtle
Nutrition Order Enteral Bolus Feeding Example | enteralbolus | XML | JSON | Turtle
Nutrition Order Enteral Continuous Feeding Example | enteralcontinuous | XML | JSON | Turtle
Nutrition Order Infant Enteral Feeding Example | infantenteral | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

12.16.7 Resource NutritionOrder - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/nutritionorder-definitions.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

12.16.7 Resource NutritionOrder - Detailed Descriptions

Orders and Observations Work Group | Maturity Level: 2 | Trial Use | Compartments: Encounter, Patient, Practitioner

Detailed Descriptions for the elements in the NutritionOrder resource.

NutritionOrder
Definition | A request to supply a diet, formula feeding (enteral) or oral nutritional supplement to a patient/resident.
Control | 1..1
Alternate Names | Nutrition Order; Diet Order; Diet; Nutritional Supplement; Enteral Nutrition
Comments | Referenced by an Order Request (workflow).
Invariants | Defined on this element nor-1: Nutrition Order SHALL contain either Oral Diet , Supplement, or Enteral Formula class (expression : oralDiet.exists() or supplement.exists() or enteralFormula.exists(), xpath: exists(f:oralDiet) or exists(f:supplement) or exists(f:enteralFormula))
NutritionOrder.identifier
Definition | Identifiers assigned to this order by the order sender or by the order receiver.
Note | This is a business identifer, not a resource identifier (see discussion)
Control | 0..*
Type | Identifier
Comments | The Identifier.type element can be to indicate filler vs. placer if needed. This is explained in further detail here.
NutritionOrder.status
Definition | The workflow status of the nutrition order/request.
Control | 0..1
Terminology Binding | NutritionOrderStatus (Required)
Type | code
Is Modifier | true
Summary | true
Comments | Typically the system placing the order sets the status to "requested". Thereafter, the order is maintained by the receiver that updates the status as the request is handled. This element is labeled as a modifier because the status contains codes that mark the resource as not currently valid.
NutritionOrder.patient
Definition | The person (patient) who needs the nutrition order for an oral diet, nutritional supplement and/or enteral or formula feeding.
Control | 1..1
Type | Reference(Patient)
Summary | true
NutritionOrder.encounter
Definition | An encounter that provides additional information about the healthcare context in which this request is made.
Control | 0..1
Type | Reference(Encounter)
NutritionOrder.dateTime
Definition | The date and time that this nutrition order was requested.
Control | 1..1
Type | dateTime
Summary | true
NutritionOrder.orderer
Definition | The practitioner that holds legal responsibility for ordering the diet, nutritional supplement, or formula feedings.
Control | 0..1
Type | Reference(Practitioner)
Summary | true
NutritionOrder.allergyIntolerance

Definition | A link to a record of allergies or intolerances which should be included in the nutrition order.
Control | 0..*
Type | Reference(AllergyIntolerance)
Comments | Information on a patient's food allergies and intolerances to inform healthcare personnel about the type of foods that the patient shouldn't receive or consume.
NutritionOrder.foodPreferenceModifier
Definition | This modifier is used to convey order-specific modifiers about the type of food that should be given. These can be derived from patient allergies, intolerances, or preferences such as Halal, Vegan or Kosher. This modifier applies to the entire nutrition order inclusive of the oral diet, nutritional supplements and enteral formula feedings.
Control | 0..*
Terminology Binding | Diet (Example)
Type | CodeableConcept
Comments | Information on a patient's food preferences that inform healthcare personnel about the food that the patient should receive or consume.
NutritionOrder.excludeFoodModifier
Definition | This modifier is used to convey order-specific modifiers about the type of food that should NOT be given. These can be derived from patient allergies, intolerances, or preferences such as No Red Meat, No Soy or No Wheat or Gluten-Free. While it should not be necessary to repeat allergy or intolerance information captured in the referenced AllergyIntolerance resource in the excludeFoodModifier, this element may be used to convey additional specificity related to foods that should be eliminated from the patient's diet for any reason. This modifier applies to the entire nutrition order inclusive of the oral diet, nutritional supplements and enteral formula feedings.
Control | 0..*
Terminology Binding | Food Type Codes (Example)
Type | CodeableConcept
Comments | Information on a patient's food allergies, intolerances and preferences to inform healthcare personnel about the type of foods that the patient shouldn't receive or consume.
NutritionOrder.oralDiet
Definition | Diet given orally in contrast to enteral (tube) feeding.
Control | 0..1
Invariants | Affect this element nor-1: Nutrition Order SHALL contain either Oral Diet , Supplement, or Enteral Formula class (expression : oralDiet.exists() or supplement.exists() or enteralFormula.exists(), xpath: exists(f:oralDiet) or exists(f:supplement) or exists(f:enteralFormula))
NutritionOrder.oralDiet.type
Definition | The kind of diet or dietary restriction such as fiber restricted diet or diabetic diet.
Control | 0..*
Terminology Binding | Diet Codes (Example)
Type | CodeableConcept
Summary | true
NutritionOrder.oralDiet.schedule
Definition | The time period and frequency at which the diet should be given. The diet should be given for the combination of all schedules if more than one schedule is present.
Control | 0..*
Type | Timing
Alternate Names | Frequency
NutritionOrder.oralDiet.nutrient
Definition | Class that defines the quantity and type of nutrient modifications (for example carbohydrate, fiber or sodium) required for the oral diet.
Control | 0..*
NutritionOrder.oralDiet.nutrient.modifier
Definition | The nutrient that is being modified such as carbohydrate or sodium.
Control | 0..1
Terminology Binding | Nutrient Modifier Codes (Example)
Type | CodeableConcept
NutritionOrder.oralDiet.nutrient.amount
Definition | The quantity of the specified nutrient to include in diet.
Control | 0..1
Type | SimpleQuantity
NutritionOrder.oralDiet.texture
Definition | Class that describes any texture modifications required for the patient to safely consume various types of solid foods.
Control | 0..*
NutritionOrder.oralDiet.texture.modifier
Definition | Any texture modifications (for solid foods) that should be made, e.g. easy to chew, chopped, ground, and pureed.
Control | 0..1
Terminology Binding | Texture Modifier Codes (Example)
Type | CodeableConcept
Comments | Coupled with the foodType (Meat).
NutritionOrder.oralDiet.texture.foodType
Definition | The food type(s) (e.g. meats, all foods) that the texture modification applies to. This could be all foods types.
Control | 0..1
Terminology Binding | Texture Modified Food Type Codes (Example)
Type | CodeableConcept
Comments | Coupled with the textureModifier; could be (All Foods).
NutritionOrder.oralDiet.fluidConsistencyType
Definition | The required consistency (e.g. honey-thick, nectar-thick, thin, thickened.) of liquids or fluids served to the patient.
Control | 0..*
Terminology Binding | Fluid Consistency Type Codes (Example)
Type | CodeableConcept

NutritionOrder.oralDiet.instruction
Definition | Free text or additional instructions or information pertaining to the oral diet.
Control | 0..1
Type | string
Summary | true
Comments | Free text dosage instructions can be used for cases where the instructions are too complex to code.
NutritionOrder.supplement
Definition | Oral nutritional products given in order to add further nutritional value to the patient's diet.
Control | 0..*
Invariants | Affect this element nor-1: Nutrition Order SHALL contain either Oral Diet , Supplement, or Enteral Formula class (expression : oralDiet.exists() or supplement.exists() or enteralFormula.exists(), xpath: exists(f:oralDiet) or exists(f:supplement) or exists(f:enteralFormula))
NutritionOrder.supplement.type
Definition | The kind of nutritional supplement product required such as a high protein or pediatric clear liquid supplement.
Control | 0..1
Terminology Binding | Supplement Type Codes (Example)
Type | CodeableConcept
Summary | true
NutritionOrder.supplement.productName
Definition | The product or brand name of the nutritional supplement such as "Acme Protein Shake".
Control | 0..1
Type | string
NutritionOrder.supplement.schedule
Definition | The time period and frequency at which the supplement(s) should be given. The supplement should be given for the combination of all schedules if more than one schedule is present.
Control | 0..*
Type | Timing
Alternate Names | Frequency
NutritionOrder.supplement.quantity
Definition | The amount of the nutritional supplement to be given.
Control | 0..1
Type | SimpleQuantity
NutritionOrder.supplement.instruction
Definition | Free text or additional instructions or information pertaining to the oral supplement.
Control | 0..1
Type | string
Summary | true
Comments | Free text dosage instructions can be used for cases where the instructions are too complex to code.
NutritionOrder.enteralFormula
Definition | Feeding provided through the gastrointestinal tract via a tube, catheter, or stoma that delivers nutrition distal to the oral cavity.
Control | 0..1
Invariants | Affect this element nor-1: Nutrition Order SHALL contain either Oral Diet , Supplement, or Enteral Formula class (expression : oralDiet.exists() or supplement.exists() or enteralFormula.exists(), xpath: exists(f:oralDiet) or exists(f:supplement) or exists(f:enteralFormula))
NutritionOrder.enteralFormula.baseFormulaType
Definition | The type of enteral or infant formula such as an adult standard formula with fiber or a soy-based infant formula.
Control | 0..1
Terminology Binding | Enteral Formula Type Codes (Example)
Type | CodeableConcept
Summary | true
NutritionOrder.enteralFormula.baseFormulaProductName
Definition | The product or brand name of the enteral or infant formula product such as "ACME Adult Standard Formula".
Control | 0..1
Type | string
NutritionOrder.enteralFormula.additiveType
Definition | Indicates the type of modular component such as protein, carbohydrate, fat or fiber to be provided in addition to or mixed with the base formula.
Control | 0..1
Terminology Binding | Enteral Formula Additive Type Code (Example)
Type | CodeableConcept
NutritionOrder.enteralFormula.additiveProductName
Definition | The product or brand name of the type of modular component to be added to the formula.
Control | 0..1
Type | string
NutritionOrder.enteralFormula.caloricDensity
Definition | The amount of energy (calories) that the formula should provide per specified volume, typically per mL or fluid oz. For example, an infant may require a formula that provides 24 calories per fluid ounce or an adult may require an enteral formula that provides 1.5 calorie/mL.
Control | 0..1
Type | SimpleQuantity
NutritionOrder.enteralFormula.routeofAdministration
Definition | The route or physiological path of administration into the patient's gastrointestinal tract for purposes of providing the formula feeding, e.g. nasogastric tube.
Control | 0..1
Terminology Binding | Enteral Route Codes (Extensible)
Type | CodeableConcept

NutritionOrder.enteralFormula.administration
Definition | Formula administration instructions as structured data. This repeating structure allows for changing the administration rate or volume over time for both bolus and continuous feeding. An example of this would be an instruction to increase the rate of continuous feeding every 2 hours.
Control | 0..*
Comments | See implementation notes below for further discussion on how to order continuous vs bolus enteral feeding using this resource.
NutritionOrder.enteralFormula.administration.schedule
Definition | The time period and frequency at which the enteral formula should be delivered to the patient.
Control | 0..1
Type | Timing
Alternate Names | Frequency
NutritionOrder.enteralFormula.administration.quantity
Definition | The volume of formula to provide to the patient per the specified administration schedule.
Control | 0..1
Type | SimpleQuantity
NutritionOrder.enteralFormula.administration.rate[x]
Definition | The rate of administration of formula via a feeding pump, e.g. 60 mL per hour, according to the specified schedule.
Control | 0..1
Type | SimpleQuantity|Ratio
[x] Note | See Choice of Data Types for further information about how to use [x]
Comments | Ratio is used when the quantity value in the denominator is not "1", otherwise use Quantity. For example, the Ratio datatype is used for "200 mL/4 hrs" versus the Quantity datatype for "50 mL/hr".
NutritionOrder.enteralFormula.maxVolumeToDeliver
Definition | The maximum total quantity of formula that may be administered to a subject over the period of time, e.g. 1440 mL over 24 hours.
Control | 0..1
Type | SimpleQuantity
NutritionOrder.enteralFormula.administrationInstruction
Definition | Free text formula administration, feeding instructions or additional instructions or information.
Control | 0..1
Type | string
Summary | true
Comments | Free text dosage instructions can be used for cases where the instructions are too complex to code.

12.16.8 Resource NutritionOrder - Mappings

URL: https://hl7.org/fhir/STU3/nutritionorder-mappings.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

12.16.8 Resource NutritionOrder - Mappings

Orders and Observations Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Encounter, Patient, Practitioner

Mappings:

RIM Mapping

Workflow Mapping

Mappings for the nutritionorder resource.

12.16.8.1 Mappings for RIM Mapping (http://hl7.org/v3)

NutritionOrder | SupplyRequest
identifier | .id
status | .statusCode
patient | .participation[typeCode=SBJ].role
encounter | .inboundRelationship[typeCode=COMP].source[classCode=ENC, moodCode=EVN]
dateTime | author.time
orderer | .participation[typeCode=AUT].role
allergyIntolerance | .outboundRelationship[typeCode=RSON].target[classCode=OBS, moodCode=EVN]
foodPreferenceModifier | .outboundRelationship [typeCode =COMP], target [classCode=PROC, moodCode=EVN, code=NutritionModificationType]
excludeFoodModifier | N/A
oralDiet | oralDiet .outboundRelationship [typeCode =COMP], target [classCode=SPLY, moodCode=RQO, code=OralDietSupplyType]
type | .code
schedule | .expectedUseTime
nutrient | .participation{typeCode=PRD].role{classCode=MANU), manufacturedFoodKind
modifier | .code
amount | .quantity
texture | .inboundRelationship [typeCode =COMP], target [classCode=PROC, moodCode=EVN, code=DietTextureModificationType]
modifier | .code
foodType | .participation{typeCode=PRD].role{classCode=MANU), manufacturedFoodKind; .code
fluidConsistencyType | .code (Not differentiated from dietTextureModificationType in HL7 v3)
instruction | .text
supplement | .outboundRelationship [typeCode =COMP], target [classCode=SPLY, moodCode=RQO, code=SUPPLEMENT]
type | .participation{typeCode=PRD].role{classCode=MANU), manufacturedFoodKind; .code (NutritionalSupplementType)
productName | .id

schedule | .effectiveTime
quantity | .quantity
instruction | .text
enteralFormula | .outboundRelationship [typeCode =COMP], target [classCode=SPLY, moodCode=RQO, code=FORMULA]
baseFormulaType | .participation{typeCode=PRD].role{classCode=MANU), manufacturedFoodKind; .code (NutritionalFormulaType)
baseFormulaProductName | .id
additiveType | N/A (not differentiated from NutritionalFormulaType in HL7 v3)
additiveProductName | N/A
caloricDensity | .quantity (caloricDensity)
routeofAdministration | .routeCode
administration | .outboundRelationship [typeCode =COMP], target [classCode=SBADM, moodCode=RQO]
schedule | .effectiveUseTime
quantity | .doseQuantity
rate[x] | .rateQuantity
maxVolumeToDeliver | .maxDoseQuantity
administrationInstruction | N/A

12.16.8.2 Mappings for Workflow Mapping (http://hl7.org/fhir/workflow)

NutritionOrder | Request
identifier | Request.identifier
status | Request.status
patient | Request.subject.reference(Patient)
encounter | Request.context.reference(Encounter)
dateTime | Request.authoredOn
orderer | Request.requester
allergyIntolerance
foodPreferenceModifier
excludeFoodModifier
oralDiet
type
schedule
nutrient
modifier
amount
texture
modifier
foodType
fluidConsistencyType
instruction
supplement
type
productName
schedule
quantity
instruction
enteralFormula
baseFormulaType
baseFormulaProductName
additiveType
additiveProductName
caloricDensity
routeofAdministration
administration
schedule
quantity
rate[x]
maxVolumeToDeliver
administrationInstruction

12.16.9 Resource NutritionOrder - Design Notes

URL: https://hl7.org/fhir/STU3/nutritionorder-explanations.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

12.16.9 Resource NutritionOrder - Design Notes

to do

12.16.10 Resource NutritionOrder - Extensions & Profiles

URL: https://hl7.org/fhir/STU3/nutritionorder-profiles.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

12.16.10 Resource NutritionOrder - Extensions & Profiles

Orders and Observations Work Group | Maturity Level: N/A | Ballot Status: Informative

This table lists profiles and extensions for the NutritionOrder resource. For background information, see Profiling Resources and Extensibility. Additional profiles and resources may be found in published Implementation Guides , or in the Conformance resource registry .

12.16.10.1 Profiles

No Profiles defined for this resource

12.16.10.2 Extensions

No Extensions defined for this resource

Extensions that reference this resource:

No Extensions refer to this resource

12.16.10.3 Search Extensions

No Search Extensions defined for this resource

12.17 Resource VisionPrescription - Content

URL: https://hl7.org/fhir/STU3/visionprescription.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

Financial Management Work Group | Maturity Level: 1 | Trial Use | Compartments: Encounter, Patient, Practitioner

An authorization for the supply of glasses and/or contact lenses to a patient.

12.17.1 Scope and Usage

This resource covers all prescriptions for glasses and contact lenses for a patient.

12.17.2 Boundaries and Relationships

This resource is referenced by CarePlan, Claim and ExplanationOfBenefit

12.17.3 Resource Content

* Structure
* UML
* XML
* JSON
* Turtle
* R2 Diff
* All

Structure

Name | Flags | Card. | Type | Description & Constraints
VisionPrescription | | | DomainResource | Prescription for vision correction products for a patient Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | | 0..* | Identifier | Business identifier
status | ?!? | 0..1 | code | active | cancelled | draft | entered-in-error Financial Resource Status Codes (Required)
patient | | 0..1 | Reference(Patient) | Who prescription is for
encounter | | 0..1 | Reference(Encounter) | Created during encounter / admission / stay
dateWritten | | 0..1 | dateTime | When prescription was authorized
prescriber | | 0..1 | Reference(Practitioner) | Who authorizes the vision product
reason[x] | | 0..1 | | Reason or indication for writing the prescription
reasonCodeableConcept | | | CodeableConcept
reasonReference | | | Reference(Condition)
dispense | | 0..* | BackboneElement | Vision supply authorization
product | | 0..1 | CodeableConcept | Product to be supplied Example Vision Prescription Product Codes (Example)
eye | | 0..1 | code | right | left VisionEyes (Required)
sphere | | 0..1 | decimal | Lens sphere
cylinder | | 0..1 | decimal | Lens cylinder
axis | | 0..1 | integer | Lens axis
prism | | 0..1 | decimal | Lens prism
base | | 0..1 | code | up | down | in | out VisionBase (Required)

add | | 0..1 | decimal | Lens add
power | | 0..1 | decimal | Contact lens power
backCurve | | 0..1 | decimal | Contact lens back curvature
diameter | | 0..1 | decimal | Contact lens diameter
duration | | 0..1 | SimpleQuantity | Lens wear duration
color | | 0..1 | string | Color required
brand | | 0..1 | string | Brand required
note | | 0..* | Annotation | Notes for coatings
Documentation for this format

UML Diagram (Legend)

XML Template

```
<VisionPrescription xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Business identifier --></identifier>
 <status value="[code]"/><!-- 0..1 active | cancelled | draft | entered-in-error -->
 <patient><!-- 0..1 Reference(Patient) Who prescription is for --></patient>
 <encounter><!-- 0..1 Reference(Encounter) Created during encounter / admission / stay --></encounter>
 <dateWritten value="[dateTime]"/><!-- 0..1 When prescription was authorized -->
 <prescriber><!-- 0..1 Reference(Practitioner) Who authorizes the vision product --></prescriber>
 <reason[x]><!-- 0..1 CodeableConcept|Reference(Condition) Reason or indication for writing the prescription
    --></reason[x]>
 <dispense>  <!-- 0..* Vision supply authorization -->
  <product><!-- 0..1 CodeableConcept Product to be supplied --></product>
  <eye value="[code]"/><!-- 0..1 right | left -->
  <sphere value="[decimal]"/><!-- 0..1 Lens sphere -->
  <cylinder value="[decimal]"/><!-- 0..1 Lens cylinder -->
  <axis value="[integer]"/><!-- 0..1 Lens axis -->
  <prism value="[decimal]"/><!-- 0..1 Lens prism -->
  <base value="[code]"/><!-- 0..1 up | down | in | out -->
  <add value="[decimal]"/><!-- 0..1 Lens add -->
  <power value="[decimal]"/><!-- 0..1 Contact lens power -->
  <backCurve value="[decimal]"/><!-- 0..1 Contact lens back curvature -->
  <diameter value="[decimal]"/><!-- 0..1 Contact lens diameter -->
  <duration><!-- 0..1 Quantity(SimpleQuantity) Lens wear duration --></duration>
  <color value="[string]"/><!-- 0..1 Color required -->
  <brand value="[string]"/><!-- 0..1 Brand required -->
  <note><!-- 0..* Annotation Notes for coatings --></note>
 </dispense>
</VisionPrescription>
```

JSON Template

```
{
  "resourceType" : "VisionPrescription",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Business identifier
  "status" : "<code>", // active | cancelled | draft | entered-in-error
  "patient" : { Reference(Patient) }, // Who prescription is for
  "encounter" : { Reference(Encounter) }, // Created during encounter / admission / stay
  "dateWritten" : "<dateTime>", // When prescription was authorized
  "prescriber" : { Reference(Practitioner) }, // Who authorizes the vision product
  // reason[x]: Reason or indication for writing the prescription. One of these 2:
  "reasonCodeableConcept" : { CodeableConcept },
  "reasonReference" : { Reference(Condition) },
  "dispense" : [{ // Vision supply authorization
    "product" : { CodeableConcept }, // Product to be supplied
    "eye" : "<code>", // right | left
    "sphere" : <decimal>, // Lens sphere
    "cylinder" : <decimal>, // Lens cylinder
    "axis" : <integer>, // Lens axis
    "prism" : <decimal>, // Lens prism
    "base" : "<code>", // up | down | in | out
    "add" : <decimal>, // Lens add
    "power" : <decimal>, // Contact lens power
    "backCurve" : <decimal>, // Contact lens back curvature
    "diameter" : <decimal>, // Contact lens diameter
    "duration" : { Quantity(SimpleQuantity) }, // Lens wear duration
    "color" : "<string>", // Color required
    "brand" : "<string>", // Brand required
    "note" : [{ Annotation }] // Notes for coatings
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:VisionPrescription;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:VisionPrescription.identifier [ Identifier ], ... ; # 0..* Business identifier
  fhir:VisionPrescription.status [ code ]; # 0..1 active | cancelled | draft | entered-in-error
```

```
  fhir:VisionPrescription.patient [ Reference(Patient) ]; # 0..1 Who prescription is for
  fhir:VisionPrescription.encounter [ Reference(Encounter) ]; # 0..1 Created during encounter / admission / stay
  fhir:VisionPrescription.dateWritten [ dateTime ]; # 0..1 When prescription was authorized
  fhir:VisionPrescription.prescriber [ Reference(Practitioner) ]; # 0..1 Who authorizes the vision product
  # VisionPrescription.reason[x] : 0..1 Reason or indication for writing the prescription. One of these 2
    fhir:VisionPrescription.reasonCodeableConcept [ CodeableConcept ]
    fhir:VisionPrescription.reasonReference [ Reference(Condition) ]
  fhir:VisionPrescription.dispense [ # 0..* Vision supply authorization
    fhir:VisionPrescription.dispense.product [ CodeableConcept ]; # 0..1 Product to be supplied
    fhir:VisionPrescription.dispense.eye [ code ]; # 0..1 right | left
    fhir:VisionPrescription.dispense.sphere [ decimal ]; # 0..1 Lens sphere
    fhir:VisionPrescription.dispense.cylinder [ decimal ]; # 0..1 Lens cylinder
    fhir:VisionPrescription.dispense.axis [ integer ]; # 0..1 Lens axis
    fhir:VisionPrescription.dispense.prism [ decimal ]; # 0..1 Lens prism
    fhir:VisionPrescription.dispense.base [ code ]; # 0..1 up | down | in | out
    fhir:VisionPrescription.dispense.add [ decimal ]; # 0..1 Lens add
    fhir:VisionPrescription.dispense.power [ decimal ]; # 0..1 Contact lens power
    fhir:VisionPrescription.dispense.backCurve [ decimal ]; # 0..1 Contact lens back curvature
    fhir:VisionPrescription.dispense.diameter [ decimal ]; # 0..1 Contact lens diameter
    fhir:VisionPrescription.dispense.duration [ Quantity(SimpleQuantity) ]; # 0..1 Lens wear duration
    fhir:VisionPrescription.dispense.color [ string ]; # 0..1 Color required
    fhir:VisionPrescription.dispense.brand [ string ]; # 0..1 Brand required
    fhir:VisionPrescription.dispense.note [ Annotation ], ... ; # 0..* Notes for coatings
  ], ...;
]
```

Changes since DSTU2

VisionPrescription

VisionPrescription.status | Added Element

VisionPrescription.dispense.product | Min Cardinality changed from 1 to 0 Type changed from Coding to CodeableConcept

VisionPrescription.dispense.note | Added Element

VisionPrescription.dispense.notes | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R2 <--> R3 Conversion Maps (status = 2 tests that all execute ok. 2 fail round-trip testing and all r3 resources are valid.).

Structure

Name | Flags | Card. | Type | Description & Constraints

VisionPrescription | | | DomainResource | Prescription for vision correction products for a patient Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension

identifier | | 0..* | Identifier | Business identifier

status | ?!? | 0..1 | code | active | cancelled | draft | entered-in-error Financial Resource Status Codes (Required)

patient | | 0..1 | Reference(Patient) | Who prescription is for

encounter | | 0..1 | Reference(Encounter) | Created during encounter / admission / stay

dateWritten | | 0..1 | dateTime | When prescription was authorized

prescriber | | 0..1 | Reference(Practitioner) | Who authorizes the vision product

reason[x] | | 0..1 | | Reason or indication for writing the prescription

reasonCodeableConcept | | | CodeableConcept

reasonReference | | | Reference(Condition)

dispense | | 0..* | BackboneElement | Vision supply authorization

product | | 0..1 | CodeableConcept | Product to be supplied Example Vision Prescription Product Codes (Example)

eye | | 0..1 | code | right | left VisionEyes (Required)

sphere | | 0..1 | decimal | Lens sphere

cylinder | | 0..1 | decimal | Lens cylinder

axis | | 0..1 | integer | Lens axis

prism | | 0..1 | decimal | Lens prism

base | | 0..1 | code | up | down | in | out VisionBase (Required)

add | | 0..1 | decimal | Lens add

power | | 0..1 | decimal | Contact lens power

backCurve | | 0..1 | decimal | Contact lens back curvature

diameter | | 0..1 | decimal | Contact lens diameter

duration | | 0..1 | SimpleQuantity | Lens wear duration

color | | 0..1 | string | Color required

brand | | 0..1 | string | Brand required

note | | 0..* | Annotation | Notes for coatings

Documentation for this format

UML Diagram (Legend)

XML Template

```
<VisionPrescription xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Business identifier --></identifier>
 <status value="[code]"/><!-- 0..1 active | cancelled | draft | entered-in-error -->
 <patient><!-- 0..1 Reference(Patient) Who prescription is for --></patient>
 <encounter><!-- 0..1 Reference(Encounter) Created during encounter / admission / stay --></encounter>
 <dateWritten value="[dateTime]"/><!-- 0..1 When prescription was authorized -->
 <prescriber><!-- 0..1 Reference(Practitioner) Who authorizes the vision product --></prescriber>
```

```
 <reason[x]><!-- 0..1 CodeableConcept|Reference(Condition) Reason or indication for writing the prescription
    --></reason[x]>
 <dispense>  <!-- 0..* Vision supply authorization -->
  <product><!-- 0..1 CodeableConcept Product to be supplied --></product>
  <eye value="[code]"/><!-- 0..1 right | left -->
  <sphere value="[decimal]"/><!-- 0..1 Lens sphere -->
  <cylinder value="[decimal]"/><!-- 0..1 Lens cylinder -->
  <axis value="[integer]"/><!-- 0..1 Lens axis -->
  <prism value="[decimal]"/><!-- 0..1 Lens prism -->
  <base value="[code]"/><!-- 0..1 up | down | in | out -->
  <add value="[decimal]"/><!-- 0..1 Lens add -->
  <power value="[decimal]"/><!-- 0..1 Contact lens power -->
  <backCurve value="[decimal]"/><!-- 0..1 Contact lens back curvature -->
  <diameter value="[decimal]"/><!-- 0..1 Contact lens diameter -->
  <duration><!-- 0..1 Quantity(SimpleQuantity) Lens wear duration --></duration>
  <color value="[string]"/><!-- 0..1 Color required -->
  <brand value="[string]"/><!-- 0..1 Brand required -->
  <note><!-- 0..* Annotation Notes for coatings --></note>
 </dispense>
</VisionPrescription>
```

JSON Template

```
{
  "resourceType" : "VisionPrescription",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Business identifier
  "status" : "<code>", // active | cancelled | draft | entered-in-error
  "patient" : { Reference(Patient) }, // Who prescription is for
  "encounter" : { Reference(Encounter) }, // Created during encounter / admission / stay
  "dateWritten" : "<dateTime>", // When prescription was authorized
  "prescriber" : { Reference(Practitioner) }, // Who authorizes the vision product
  // reason[x]: Reason or indication for writing the prescription. One of these 2:
  "reasonCodeableConcept" : { CodeableConcept },
  "reasonReference" : { Reference(Condition) },
  "dispense" : [{ // Vision supply authorization
    "product" : { CodeableConcept }, // Product to be supplied
    "eye" : "<code>", // right | left
    "sphere" : <decimal>, // Lens sphere
    "cylinder" : <decimal>, // Lens cylinder
    "axis" : <integer>, // Lens axis
    "prism" : <decimal>, // Lens prism
    "base" : "<code>", // up | down | in | out
    "add" : <decimal>, // Lens add
    "power" : <decimal>, // Contact lens power
    "backCurve" : <decimal>, // Contact lens back curvature
    "diameter" : <decimal>, // Contact lens diameter
    "duration" : { Quantity(SimpleQuantity) }, // Lens wear duration
    "color" : "<string>", // Color required
    "brand" : "<string>", // Brand required
    "note" : [{ Annotation }] // Notes for coatings
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:VisionPrescription;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:VisionPrescription.identifier [ Identifier ], ... ; # 0..* Business identifier
  fhir:VisionPrescription.status [ code ]; # 0..1 active | cancelled | draft | entered-in-error
  fhir:VisionPrescription.patient [ Reference(Patient) ]; # 0..1 Who prescription is for
  fhir:VisionPrescription.encounter [ Reference(Encounter) ]; # 0..1 Created during encounter / admission / stay
  fhir:VisionPrescription.dateWritten [ dateTime ]; # 0..1 When prescription was authorized
  fhir:VisionPrescription.prescriber [ Reference(Practitioner) ]; # 0..1 Who authorizes the vision product
  # VisionPrescription.reason[x] : 0..1 Reason or indication for writing the prescription. One of these 2
    fhir:VisionPrescription.reasonCodeableConcept [ CodeableConcept ]
    fhir:VisionPrescription.reasonReference [ Reference(Condition) ]
  fhir:VisionPrescription.dispense [ # 0..* Vision supply authorization
    fhir:VisionPrescription.dispense.product [ CodeableConcept ]; # 0..1 Product to be supplied
    fhir:VisionPrescription.dispense.eye [ code ]; # 0..1 right | left
    fhir:VisionPrescription.dispense.sphere [ decimal ]; # 0..1 Lens sphere
    fhir:VisionPrescription.dispense.cylinder [ decimal ]; # 0..1 Lens cylinder
    fhir:VisionPrescription.dispense.axis [ integer ]; # 0..1 Lens axis
    fhir:VisionPrescription.dispense.prism [ decimal ]; # 0..1 Lens prism
    fhir:VisionPrescription.dispense.base [ code ]; # 0..1 up | down | in | out
    fhir:VisionPrescription.dispense.add [ decimal ]; # 0..1 Lens add
    fhir:VisionPrescription.dispense.power [ decimal ]; # 0..1 Contact lens power
    fhir:VisionPrescription.dispense.backCurve [ decimal ]; # 0..1 Contact lens back curvature
    fhir:VisionPrescription.dispense.diameter [ decimal ]; # 0..1 Contact lens diameter
    fhir:VisionPrescription.dispense.duration [ Quantity(SimpleQuantity) ]; # 0..1 Lens wear duration
    fhir:VisionPrescription.dispense.color [ string ]; # 0..1 Color required
    fhir:VisionPrescription.dispense.brand [ string ]; # 0..1 Brand required
    fhir:VisionPrescription.dispense.note [ Annotation ], ... ; # 0..* Notes for coatings
  ], ...;
```

]

Changes since DSTU2

VisionPrescription
VisionPrescription.status | Added Element
VisionPrescription.dispense.product | Min Cardinality changed from 1 to 0 Type changed from Coding to CodeableConcept
VisionPrescription.dispense.note | Added Element
VisionPrescription.dispense.notes | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R2 <--> R3 Conversion Maps (status = 2 tests that all execute ok. 2 fail round-trip testing and all r3 resources are valid.).

Alternate definitions: Master Definition (XML, JSON), XML Schema/Schematron (for) + JSON Schema, ShEx (for Turtle)

12.17.3.1 Terminology Bindings

Path | Definition | Type | Reference
VisionPrescription.status | A code specifying the state of the resource instance. | Required | Financial Resource Status Codes
VisionPrescription.dispense.product | A coded concept describing the vision products. | Example | Example Vision Prescription Product Codes
VisionPrescription.dispense.eye | A coded concept listing the eye codes. | Required | VisionEyes
VisionPrescription.dispense.base | A coded concept listing the base codes. | Required | VisionBase

12.17.4 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
datewritten | date | Return prescriptions written on this date | VisionPrescription.dateWritten
encounter | reference | Return prescriptions with this encounter identifier | VisionPrescription.encounter (Encounter) | 12 Resources
identifier | token | Return prescriptions with this external identifier | VisionPrescription.identifier | 26 Resources
patient | reference | The identity of a patient to list dispenses for | VisionPrescription.patient (Patient) | 31 Resources
prescriber | reference | Who authorizes the vision product | VisionPrescription.prescriber (Practitioner)

12.17.5 Resource VisionPrescription - Examples

URL: https://hl7.org/fhir/STU3/visionprescription-examples.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

12.17.5 Resource VisionPrescription - Examples

Financial Management Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Encounter, Patient, Practitioner

Example List:

General Glasses Example | 33123 | XML | JSON | Turtle
General Contacts Example | 33124 | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

12.17.6 Resource VisionPrescription - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/visionprescription-definitions.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

12.17.6 Resource VisionPrescription - Detailed Descriptions

Financial Management Work Group | Maturity Level: 1 | Trial Use | Compartments: Encounter, Patient, Practitioner

Detailed Descriptions for the elements in the VisionPrescription resource.

VisionPrescription
Definition | An authorization for the supply of glasses and/or contact lenses to a patient.
Control | 1..1
VisionPrescription.identifier
Definition | Business identifier which may be used by other parties to reference or identify the prescription.
Note | This is a business identifer, not a resource identifier (see discussion)
Control | 0..*
Type | Identifier
VisionPrescription.status
Definition | The status of the resource instance.
Control | 0..1
Terminology Binding | Financial Resource Status Codes (Required)
Type | code

Is Modifier | true
Summary | true
VisionPrescription.patient
Definition | A link to a resource representing the person to whom the vision products will be supplied.
Control | 0..1
Type | Reference(Patient)
VisionPrescription.encounter
Definition | A link to a resource that identifies the particular occurrence of contact between patient and health care provider.
Control | 0..1
Type | Reference(Encounter)
VisionPrescription.dateWritten
Definition | The date (and perhaps time) when the prescription was written.
Control | 0..1
Type | dateTime
VisionPrescription.prescriber
Definition | The healthcare professional responsible for authorizing the prescription.
Control | 0..1
Type | Reference(Practitioner)
VisionPrescription.reason[x]
Definition | Can be the reason or the indication for writing the prescription.
Control | 0..1
Type | CodeableConcept|Reference(Condition)
[x] Note | See Choice of Data Types for further information about how to use [x]
VisionPrescription.dispense
Definition | Deals with details of the dispense part of the supply specification.
Control | 0..*
VisionPrescription.dispense.product
Definition | Identifies the type of vision correction product which is required for the patient.
Control | 0..1
Terminology Binding | Example Vision Prescription Product Codes (Example)
Type | CodeableConcept
VisionPrescription.dispense.eye
Definition | The eye for which the lens applies.
Control | 0..1
Terminology Binding | VisionEyes (Required)
Type | code
VisionPrescription.dispense.sphere
Definition | Lens power measured in diopters (0.25 units).
Control | 0..1
Type | decimal
VisionPrescription.dispense.cylinder
Definition | Power adjustment for astigmatism measured in diopters (0.25 units).
Control | 0..1
Type | decimal
VisionPrescription.dispense.axis
Definition | Adjustment for astigmatism measured in integer degrees.
Control | 0..1
Type | integer
VisionPrescription.dispense.prism
Definition | Amount of prism to compensate for eye alignment in fractional units.
Control | 0..1
Type | decimal
VisionPrescription.dispense.base
Definition | The relative base, or reference lens edge, for the prism.
Control | 0..1
Terminology Binding | VisionBase (Required)
Type | code
VisionPrescription.dispense.add
Definition | Power adjustment for multifocal lenses measured in diopters (0.25 units).
Control | 0..1
Type | decimal
VisionPrescription.dispense.power
Definition | Contact lens power measured in diopters (0.25 units).
Control | 0..1
Type | decimal
VisionPrescription.dispense.backCurve
Definition | Back curvature measured in millimeters.
Control | 0..1
Type | decimal
VisionPrescription.dispense.diameter
Definition | Contact lens diameter measured in millimeters.
Control | 0..1

Type | decimal
VisionPrescription.dispense.duration
Definition | The recommended maximum wear period for the lens.
Control | 0..1
Type | SimpleQuantity
VisionPrescription.dispense.color
Definition | Special color or pattern.
Control | 0..1
Type | string
VisionPrescription.dispense.brand
Definition | Brand recommendations or restrictions.
Control | 0..1
Type | string
VisionPrescription.dispense.note
Definition | Notes for special requirements such as coatings and lens materials.
Control | 0..*
Type | Annotation

12.17.7 Resource VisionPrescription - Mappings

URL: https://hl7.org/fhir/STU3/visionprescription-mappings.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

12.17.7 Resource VisionPrescription - Mappings

Financial Management Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Encounter, Patient, Practitioner

Mappings:

RIM Mapping

HL7 v2 Mapping

Workflow Mapping

Mappings for the visionprescription resource.

12.17.7.1 Mappings for RIM Mapping (http://hl7.org/v3)

VisionPrescription | CombinedVisionRequest
identifier | id
status
patient | subject.role
encounter | componentOf.patientEncounter
dateWritten | author.time
prescriber | author.role
reason[x] | reason.observation or reason.observation[code=ASSERTION].value
dispense | component.supplyEvent
product | consumable.role
eye
sphere
cylinder
axis
prism
base
add
power
backCurve
diameter
duration
color
brand
note

12.17.7.2 Mappings for HL7 v2 Mapping (http://hl7.org/v2)

VisionPrescription
identifier | ORC-2-Placer Order Number / ORC-3-Filler Order Number
status
patient | PID-3-Patient ID List
encounter | PV1-19-Visit Number
dateWritten | RXE-32-Original Order Date/Time / ORC-9-Date/Time of Transaction
prescriber | RXE-13-Ordering Provider's DEA Number / RXO-14-Ordering Provider's DEA Number / RXE-14-Pharmacist/Treatment Supplier's Verifier ID / RXO-15-Pharmacist/Treatment Supplier's Verifier ID / ORC-12-Ordering Provider / PRT-5-Participation Person: PRT-4-Participation='OP' (all but last deprecated)
reason[x] | ORC-16-Order Control Code Reason

dispense
product | RXE-2-Give Code / RXO-1-Requested Give Code / RXD-2-Dispense/Give Code
eye
sphere
cylinder
axis
prism
base
add
power
backCurve
diameter
duration
color
brand
note

12.17.7.3 Mappings for Workflow Mapping (http://hl7.org/fhir/workflow)

VisionPrescription | Request
identifier | Request.identifier
status | Request.status
patient | Request.subject
encounter | Request.context
dateWritten | Request.authoredOn
prescriber | Request.requester.onBehalfOf
reason[x]
dispense
product
eye
sphere
cylinder
axis
prism
base
add
power
backCurve
diameter
duration
color
brand
note

12.17.8 Resource VisionPrescription - Design Notes

URL: https://hl7.org/fhir/STU3/visionprescription-explanations.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

12.17.8 Resource VisionPrescription - Design Notes

to do

12.17.9 Resource VisionPrescription - Extensions & Profiles

URL: https://hl7.org/fhir/STU3/visionprescription-profiles.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

12.17.9 Resource VisionPrescription - Extensions & Profiles

Financial Management Work Group | Maturity Level: N/A | Ballot Status: Informative

This table lists profiles and extensions for the VisionPrescription resource. For background information, see Profiling Resources and Extensibility. Additional profiles and resources may be found in published Implementation Guides , or in the Conformance resource registry .

12.17.9.1 Profiles

No Profiles defined for this resource

12.17.9.2 Extensions

No Extensions defined for this resource

Extensions that reference this resource:

No Extensions refer to this resource

12.17.9.3 Search Extensions

No Search Extensions defined for this resource

12.17.10 Resource VisionPrescription - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/visionprescription-version-maps.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

12.17.10 Resource VisionPrescription - Detailed Descriptions

Financial Management Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Encounter, Patient, Practitioner

R2 : R3 Converson maps for VisionPrescription.

Functional status for this map: 2 tests that all execute ok. 2 fail round-trip testing and all r3 resources are valid. (see documentation)

12.17.10.1 R2 to R3

```
map "http://hl7.org/fhir/StructureMap/VisionPrescription2to3" = "R2 to R3 Conversions for VisionPrescription"

uses "http://hl7.org/fhir/DSTU2/StructureDefinition/VisionPrescription" alias VisionPrescriptionR2 as source
uses "http://hl7.org/fhir/StructureDefinition/VisionPrescription" alias VisionPrescription as target

imports "http://hl7.org/fhir/StructureMap/*2to3"

group for type+types  VisionPrescription extends DomainResource
  input src : VisionPrescriptionR2 as source
  input tgt : VisionPrescription as target

  "VisionPrescription-identifier" : for src.identifier make tgt.identifier
  "VisionPrescription-dateWritten" : for src.dateWritten make tgt.dateWritten
  "VisionPrescription-patient" : for src.patient make tgt.patient
  "VisionPrescription-prescriber" : for src.prescriber make tgt.prescriber
  "VisionPrescription-encounter" : for src.encounter make tgt.encounter
  "VisionPrescription-reason" : for src.reason make tgt.reason
  "VisionPrescription-dispense" : for src.dispense as vs0 make tgt.dispense as vt0 then vspDispense(vs0, vt0)
endgroup

group vspDispense extends BackboneElement
  input src as source
  input tgt as target

  "VisionPrescription.dispense-product" : for src.product as v make tgt.product as d, d.coding as v
  "VisionPrescription.dispense-eye" : for src.eye make tgt.eye
  "VisionPrescription.dispense-sphere" : for src.sphere make tgt.sphere
  "VisionPrescription.dispense-cylinder" : for src.cylinder make tgt.cylinder
  "VisionPrescription.dispense-axis" : for src.axis make tgt.axis
  "VisionPrescription.dispense-prism" : for src.prism make tgt.prism
  "VisionPrescription.dispense-base" : for src.base make tgt.base
  "VisionPrescription.dispense-add" : for src.add make tgt.add
  "VisionPrescription.dispense-power" : for src.power make tgt.power
  "VisionPrescription.dispense-backCurve" : for src.backCurve make tgt.backCurve
  "VisionPrescription.dispense-diameter" : for src.diameter make tgt.diameter
  "VisionPrescription.dispense-duration" : for src.duration make tgt.duration
  "VisionPrescription.dispense-color" : for src.color make tgt.color
  "VisionPrescription.dispense-brand" : for src.brand make tgt.brand
  "VisionPrescription.dispense-note" : for src.notes as v make tgt.note as n, n.text = v
endgroup
```

12.17.10.2 R3 to R2

```
map "http://hl7.org/fhir/StructureMap/VisionPrescription3to2" = "R3 to R2 Conversion for VisionPrescription"

uses "http://hl7.org/fhir/StructureDefinition/VisionPrescription" alias VisionPrescription as source
uses "http://hl7.org/fhir/DSTU2/StructureDefinition/VisionPrescription" alias VisionPrescriptionR2 as target

imports "http://hl7.org/fhir/StructureMap/*3to2"

group for type+types  VisionPrescription extends DomainResource
  input src : VisionPrescription as source
  input tgt : VisionPrescriptionR2 as target

  "VisionPrescription-identifier" : for src.identifier make tgt.identifier
  "VisionPrescription-dateWritten" : for src.dateWritten make tgt.dateWritten
  "VisionPrescription-patient" : for src.patient make tgt.patient
  "VisionPrescription-prescriber" : for src.prescriber make tgt.prescriber
  "VisionPrescription-encounter" : for src.encounter make tgt.encounter
  "VisionPrescription-reason" : for src.reason make tgt.reason
  "VisionPrescription-dispense" : for src.dispense as vs0 make tgt.dispense as vt0 then vspDispense(vs0, vt0)
endgroup
```

```
group vspDispense extends BackboneElement
  input src as source
  input tgt as target

  "VisionPrescription.dispense-product" : for src.product as v then vpsProduct(v, tgt)
  "VisionPrescription.dispense-eye1" : for src.eye as v where coding.code = 'left' make tgt.eye = "left"
  "VisionPrescription.dispense-eye2" : for src.eye as v where coding.code = 'right' make tgt.eye = "right"
  "VisionPrescription.dispense-sphere" : for src.sphere make tgt.sphere
  "VisionPrescription.dispense-cylinder" : for src.cylinder make tgt.cylinder
  "VisionPrescription.dispense-axis" : for src.axis make tgt.axis
  "VisionPrescription.dispense-prism" : for src.prism make tgt.prism
  "VisionPrescription.dispense-base1" : for src.base make tgt.base
  "VisionPrescription.dispense-add" : for src.add make tgt.add
  "VisionPrescription.dispense-power" : for src.power make tgt.power
  "VisionPrescription.dispense-backCurve" : for src.backCurve make tgt.backCurve
  "VisionPrescription.dispense-diameter" : for src.diameter make tgt.diameter
  "VisionPrescription.dispense-duration" : for src.duration make tgt.duration
  "VisionPrescription.dispense-color" : for src.color make tgt.color
  "VisionPrescription.dispense-brand" : for src.brand make tgt.brand
  "VisionPrescription.dispense-note" : for src.note as n then vspNote(n, tgt)
endgroup


group vpsProduct extends Element
  input src as source
  input tgt as target

  "VisionPrescription.dispense.product-coding" : for src.coding make tgt.product
endgroup

group vspNote extends Element
  input src as source
  input tgt as target

  "VisionPrescription.dispense.note-text" : for src.text make tgt.notes
endgroup
```

12.17.10.3 R3 Validation Errors

12.18 Resource ActivityDefinition - Content

URL: https://hl7.org/fhir/STU3/activitydefinition.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

Clinical Decision Support Work Group | Maturity Level: 2 | Trial Use | Compartments: Not linked to any defined compartments

This resource allows for the definition of some activity to be performed, independent of a particular patient, practitioner, or other performance context.

12.18.1 Scope and Usage

This resource is a definition resource from a FHIR workflow perspective - see Workflow, specifically Definition.

An ActivityDefinition is a shareable, consumable description of some activity to be performed. It may be used to specify actions to be taken as part of a workflow, order set, or protocol, or it may be used independently as part of a catalog of activities such as orderables.

For more information on how activity definitions can be used to construct request resources, see the Realizing an ActivityDefinition topic below.

12.18.2 Boundaries and Relationships

In general, an activity definition is simply a conceptual description of some specific action that should be taken. An instance of an ActivityDefinition does not indicate that any action has been performed (as an event resource does), nor does it indicate the actual intent to carry out any particular action (as a request resource does).

Instead, an activity definition provides a reusable template that can be used to construct specific request resources such as ProcedureRequest and MedicationRequest. This is different than a DataElement in that ActivityDefinitions are generally broader than data elements, which are typically used to describe more granular elements, such as a blood pressure, or the answer to a particular question.

Note that this is conceptually similar to the Task resource as well, with the distinction being that ActivityDefinition represents the description of a task in the abstract, while the Task resource is used to track a specific instance of a task as it moves through the steps of a workflow.

This resource is referenced by CarePlan, Communication, DeviceRequest, MedicationAdministration, MedicationRequest, MessageDefinition, PlanDefinition, Procedure, ProcedureRequest, ReferralRequest and Task

12.18.3 Resource Content

* Structure

* UML

* XML
* JSON
* Turtle
* R2 Diff
* All

Structure

Name | Flags | Card. | Type | Description & Constraints
ActivityDefinition | | | DomainResource | The definition of a specific activity to be taken, independent of any particular patient or context Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
url | ? | 0..1 | uri | Logical URI to reference this activity definition (globally unique)
identifier | ? | 0..* | Identifier | Additional identifier for the activity definition
version | ? | 0..1 | string | Business version of the activity definition
name | ? | 0..1 | string | Name for this activity definition (computer friendly)
title | ? | 0..1 | string | Name for this activity definition (human friendly)
status | ?!? | 1..1 | code | draft | active | retired | unknown PublicationStatus (Required)
experimental | ?!? | 0..1 | boolean | For testing purposes, not real usage
date | ? | 0..1 | dateTime | Date this was last changed
publisher | ? | 0..1 | string | Name of the publisher (organization or individual)
description | ? | 0..1 | markdown | Natural language description of the activity definition
purpose | | 0..1 | markdown | Why this activity definition is defined
usage | | 0..1 | string | Describes the clinical usage of the asset
approvalDate | | 0..1 | date | When the activity definition was approved by publisher
lastReviewDate | | 0..1 | date | When the activity definition was last reviewed
effectivePeriod | ? | 0..1 | Period | When the activity definition is expected to be used
useContext | ? | 0..* | UsageContext | Context the content is intended to support
jurisdiction | ? | 0..* | CodeableConcept | Intended jurisdiction for activity definition (if applicable) Jurisdiction ValueSet (Extensible)
topic | | 0..* | CodeableConcept | E.g. Education, Treatment, Assessment, etc DefinitionTopic (Example)
contributor | | 0..* | Contributor | A content contributor
contact | ? | 0..* | ContactDetail | Contact details for the publisher
copyright | | 0..1 | markdown | Use and/or publishing restrictions
relatedArtifact | | 0..* | RelatedArtifact | Additional documentation, citations, etc
library | | 0..* | Reference(Library) | Logic used by the asset
kind | | 0..1 | code | Kind of resource ResourceType (Required)
code | | 0..1 | CodeableConcept | Detail type of activity Procedure Codes (SNOMED CT) (Example)
timing[x] | | 0..1 | | When activity is to occur
timingTiming | | | Timing
timingDateTime | | | dateTime
timingPeriod | | | Period
timingRange | | | Range
location | | 0..1 | Reference(Location) | Where it should happen
participant | | 0..* | BackboneElement | Who should participate in the action
type | | 1..1 | code | patient | practitioner | related-person ActionParticipantType (Required)
role | | 0..1 | CodeableConcept | E.g. Nurse, Surgeon, Parent, etc ActionParticipantRole (Example)
product[x] | | 0..1 | | What's administered/supplied SNOMED CT Medication Codes (Example)
productReference | | | Reference(Medication | Substance)
productCodeableConcept | | | CodeableConcept
quantity | | 0..1 | SimpleQuantity | How much is administered/consumed/supplied
dosage | | 0..* | Dosage | Detailed dosage instructions
bodySite | | 0..* | CodeableConcept | What part of body to perform on SNOMED CT Body Structures (Example)
transform | | 0..1 | Reference(StructureMap) | Transform to apply the template
dynamicValue | | 0..* | BackboneElement | Dynamic aspects of the definition
description | | 0..1 | string | Natural language description of the dynamic value
path | | 0..1 | string | The path to the element to be set dynamically
language | | 0..1 | string | Language of the expression
expression | | 0..1 | string | An expression that provides the dynamic value for the customization
Documentation for this format

UML Diagram (Legend)

XML Template

```
<ActivityDefinition xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <url value="[uri]"/><!-- 0..1 Logical URI to reference this activity definition (globally unique) -->
 <identifier><!-- 0..* Identifier Additional identifier for the activity definition --></identifier>
 <version value="[string]"/><!-- 0..1 Business version of the activity definition -->
 <name value="[string]"/><!-- 0..1 Name for this activity definition (computer friendly) -->
 <title value="[string]"/><!-- 0..1 Name for this activity definition (human friendly) -->
 <status value="[code]"/><!-- 1..1 draft | active | retired | unknown -->
 <experimental value="[boolean]"/><!-- 0..1 For testing purposes, not real usage -->
 <date value="[dateTime]"/><!-- 0..1 Date this was last changed -->
 <publisher value="[string]"/><!-- 0..1 Name of the publisher (organization or individual) -->
 <description value="[markdown]"/><!-- 0..1 Natural language description of the activity definition -->
```

```
 <purpose value="[markdown]"/><!-- 0..1 Why this activity definition is defined -->
 <usage value="[string]"/><!-- 0..1 Describes the clinical usage of the asset -->
 <approvalDate value="[date]"/><!-- 0..1 When the activity definition was approved by publisher -->
 <lastReviewDate value="[date]"/><!-- 0..1 When the activity definition was last reviewed -->
 <effectivePeriod><!-- 0..1 Period When the activity definition is expected to be used --></effectivePeriod>
 <useContext><!-- 0..* UsageContext Context the content is intended to support --></useContext>
 <jurisdiction><!-- 0..* CodeableConcept Intended jurisdiction for activity definition (if applicable)
    --></jurisdiction>
 <topic><!-- 0..* CodeableConcept E.g. Education, Treatment, Assessment, etc --></topic>
 <contributor><!-- 0..* Contributor A content contributor --></contributor>
 <contact><!-- 0..* ContactDetail Contact details for the publisher --></contact>
 <copyright value="[markdown]"/><!-- 0..1 Use and/or publishing restrictions -->
 <relatedArtifact><!-- 0..* RelatedArtifact Additional documentation, citations, etc --></relatedArtifact>
 <library><!-- 0..* Reference(Library) Logic used by the asset --></library>
 <kind value="[code]"/><!-- 0..1 Kind of resource -->
 <code><!-- 0..1 CodeableConcept Detail type of activity --></code>
 <timing[x]><!-- 0..1 Timing|dateTime|Period|Range When activity is to occur --></timing[x]>
 <location><!-- 0..1 Reference(Location) Where it should happen --></location>
 <participant>  <!-- 0..* Who should participate in the action -->
  <type value="[code]"/><!-- 1..1 patient | practitioner | related-person -->
  <role><!-- 0..1 CodeableConcept E.g. Nurse, Surgeon, Parent, etc --></role>
 </participant>
 <product[x]><!-- 0..1 Reference(Medication|Substance)|CodeableConcept What's administered/supplied --></product[x]>
 <quantity><!-- 0..1 Quantity(SimpleQuantity) How much is administered/consumed/supplied --></quantity>
 <dosage><!-- 0..* Dosage Detailed dosage instructions --></dosage>
 <bodySite><!-- 0..* CodeableConcept What part of body to perform on --></bodySite>
 <transform><!-- 0..1 Reference(StructureMap) Transform to apply the template --></transform>
 <dynamicValue>  <!-- 0..* Dynamic aspects of the definition -->
  <description value="[string]"/><!-- 0..1 Natural language description of the dynamic value -->
  <path value="[string]"/><!-- 0..1 The path to the element to be set dynamically -->
  <language value="[string]"/><!-- 0..1 Language of the expression -->
  <expression value="[string]"/><!-- 0..1 An expression that provides the dynamic value for the customization -->
 </dynamicValue>
</ActivityDefinition>
```

JSON Template

```
{
  "resourceType" : "ActivityDefinition",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "url" : "<uri>", // Logical URI to reference this activity definition (globally unique)
  "identifier" : [{ Identifier }], // Additional identifier for the activity definition
  "version" : "<string>", // Business version of the activity definition
  "name" : "<string>", // Name for this activity definition (computer friendly)
  "title" : "<string>", // Name for this activity definition (human friendly)
  "status" : "<code>", // R!  draft | active | retired | unknown
  "experimental" : <boolean>, // For testing purposes, not real usage
  "date" : "<dateTime>", // Date this was last changed
  "publisher" : "<string>", // Name of the publisher (organization or individual)
  "description" : "<markdown>", // Natural language description of the activity definition
  "purpose" : "<markdown>", // Why this activity definition is defined
  "usage" : "<string>", // Describes the clinical usage of the asset
  "approvalDate" : "<date>", // When the activity definition was approved by publisher
  "lastReviewDate" : "<date>", // When the activity definition was last reviewed
  "effectivePeriod" : { Period }, // When the activity definition is expected to be used
  "useContext" : [{ UsageContext }], // Context the content is intended to support
  "jurisdiction" : [{ CodeableConcept }], // Intended jurisdiction for activity definition (if applicable)
  "topic" : [{ CodeableConcept }], // E.g. Education, Treatment, Assessment, etc
  "contributor" : [{ Contributor }], // A content contributor
  "contact" : [{ ContactDetail }], // Contact details for the publisher
  "copyright" : "<markdown>", // Use and/or publishing restrictions
  "relatedArtifact" : [{ RelatedArtifact }], // Additional documentation, citations, etc
  "library" : [{ Reference(Library) }], // Logic used by the asset
  "kind" : "<code>", // Kind of resource
  "code" : { CodeableConcept }, // Detail type of activity
  // timing[x]: When activity is to occur. One of these 4:
  "timingTiming" : { Timing },
  "timingDateTime" : "<dateTime>",
  "timingPeriod" : { Period },
  "timingRange" : { Range },
  "location" : { Reference(Location) }, // Where it should happen
  "participant" : [{ // Who should participate in the action
    "type" : "<code>", // R!  patient | practitioner | related-person
    "role" : { CodeableConcept } // E.g. Nurse, Surgeon, Parent, etc
  }],
  // product[x]: What's administered/supplied. One of these 2:
  "productReference" : { Reference(Medication|Substance) },
  "productCodeableConcept" : { CodeableConcept },
  "quantity" : { Quantity(SimpleQuantity) }, // How much is administered/consumed/supplied
  "dosage" : [{ Dosage }], // Detailed dosage instructions
  "bodySite" : [{ CodeableConcept }], // What part of body to perform on
  "transform" : { Reference(StructureMap) }, // Transform to apply the template
  "dynamicValue" : [{ // Dynamic aspects of the definition
    "description" : "<string>", // Natural language description of the dynamic value
    "path" : "<string>", // The path to the element to be set dynamically
    "language" : "<string>", // Language of the expression
    "expression" : "<string>" // An expression that provides the dynamic value for the customization
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:ActivityDefinition;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:ActivityDefinition.url [ uri ]; # 0..1 Logical URI to reference this activity definition (globally unique)
  fhir:ActivityDefinition.identifier [ Identifier ], ... ; # 0..* Additional identifier for the activity definition
  fhir:ActivityDefinition.version [ string ]; # 0..1 Business version of the activity definition
  fhir:ActivityDefinition.name [ string ]; # 0..1 Name for this activity definition (computer friendly)
  fhir:ActivityDefinition.title [ string ]; # 0..1 Name for this activity definition (human friendly)
  fhir:ActivityDefinition.status [ code ]; # 1..1 draft | active | retired | unknown
  fhir:ActivityDefinition.experimental [ boolean ]; # 0..1 For testing purposes, not real usage
  fhir:ActivityDefinition.date [ dateTime ]; # 0..1 Date this was last changed
  fhir:ActivityDefinition.publisher [ string ]; # 0..1 Name of the publisher (organization or individual)
  fhir:ActivityDefinition.description [ markdown ]; # 0..1 Natural language description of the activity definition
  fhir:ActivityDefinition.purpose [ markdown ]; # 0..1 Why this activity definition is defined
  fhir:ActivityDefinition.usage [ string ]; # 0..1 Describes the clinical usage of the asset
  fhir:ActivityDefinition.approvalDate [ date ]; # 0..1 When the activity definition was approved by publisher
  fhir:ActivityDefinition.lastReviewDate [ date ]; # 0..1 When the activity definition was last reviewed
  fhir:ActivityDefinition.effectivePeriod [ Period ]; # 0..1 When the activity definition is expected to be used
  fhir:ActivityDefinition.useContext [ UsageContext ], ... ; # 0..* Context the content is intended to support
  fhir:ActivityDefinition.jurisdiction [ CodeableConcept ], ... ; # 0..* Intended jurisdiction for activity definition
    (if applicable)
  fhir:ActivityDefinition.topic [ CodeableConcept ], ... ; # 0..* E.g. Education, Treatment, Assessment, etc
  fhir:ActivityDefinition.contributor [ Contributor ], ... ; # 0..* A content contributor
  fhir:ActivityDefinition.contact [ ContactDetail ], ... ; # 0..* Contact details for the publisher
  fhir:ActivityDefinition.copyright [ markdown ]; # 0..1 Use and/or publishing restrictions
  fhir:ActivityDefinition.relatedArtifact [ RelatedArtifact ], ... ; # 0..* Additional documentation, citations, etc
  fhir:ActivityDefinition.library [ Reference(Library) ], ... ; # 0..* Logic used by the asset
  fhir:ActivityDefinition.kind [ code ]; # 0..1 Kind of resource
  fhir:ActivityDefinition.code [ CodeableConcept ]; # 0..1 Detail type of activity
  # ActivityDefinition.timing[x] : 0..1 When activity is to occur. One of these 4
    fhir:ActivityDefinition.timingTiming [ Timing ]
    fhir:ActivityDefinition.timingDateTime [ dateTime ]
    fhir:ActivityDefinition.timingPeriod [ Period ]
    fhir:ActivityDefinition.timingRange [ Range ]
  fhir:ActivityDefinition.location [ Reference(Location) ]; # 0..1 Where it should happen
  fhir:ActivityDefinition.participant [ # 0..* Who should participate in the action
    fhir:ActivityDefinition.participant.type [ code ]; # 1..1 patient | practitioner | related-person
    fhir:ActivityDefinition.participant.role [ CodeableConcept ]; # 0..1 E.g. Nurse, Surgeon, Parent, etc
  ], ...;
  # ActivityDefinition.product[x] : 0..1 What's administered/supplied. One of these 2
    fhir:ActivityDefinition.productReference [ Reference(Medication|Substance) ]
    fhir:ActivityDefinition.productCodeableConcept [ CodeableConcept ]
  fhir:ActivityDefinition.quantity [ Quantity(SimpleQuantity) ]; # 0..1 How much is administered/consumed/supplied
  fhir:ActivityDefinition.dosage [ Dosage ], ... ; # 0..* Detailed dosage instructions
  fhir:ActivityDefinition.bodySite [ CodeableConcept ], ... ; # 0..* What part of body to perform on
  fhir:ActivityDefinition.transform [ Reference(StructureMap) ]; # 0..1 Transform to apply the template
  fhir:ActivityDefinition.dynamicValue [ # 0..* Dynamic aspects of the definition
    fhir:ActivityDefinition.dynamicValue.description [ string ]; # 0..1 Natural language description of the dynamic
    value
    fhir:ActivityDefinition.dynamicValue.path [ string ]; # 0..1 The path to the element to be set dynamically
    fhir:ActivityDefinition.dynamicValue.language [ string ]; # 0..1 Language of the expression
    fhir:ActivityDefinition.dynamicValue.expression [ string ]; # 0..1 An expression that provides the dynamic value for
     the customization
  ], ...;
]
```

Changes since DSTU2

This resource did not exist in Release 2

This analysis is available as XML or JSON.

Structure

Name | Flags | Card. | Type | Description & Constraints

ActivityDefinition | | | DomainResource | The definition of a specific activity to be taken, independent of any particular patient or context Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension

url | ? | 0..1 | uri | Logical URI to reference this activity definition (globally unique)

identifier | ? | 0..* | Identifier | Additional identifier for the activity definition

version | ? | 0..1 | string | Business version of the activity definition

name | ? | 0..1 | string | Name for this activity definition (computer friendly)

title | ? | 0..1 | string | Name for this activity definition (human friendly)

status | ?!? | 1..1 | code | draft | active | retired | unknown PublicationStatus (Required)

experimental | ?!? | 0..1 | boolean | For testing purposes, not real usage

date | ? | 0..1 | dateTime | Date this was last changed

publisher | ? | 0..1 | string | Name of the publisher (organization or individual)

description | ? | 0..1 | markdown | Natural language description of the activity definition

purpose | | 0..1 | markdown | Why this activity definition is defined

usage | | 0..1 | string | Describes the clinical usage of the asset

approvalDate | | 0..1 | date | When the activity definition was approved by publisher

lastReviewDate | | 0..1 | date | When the activity definition was last reviewed
effectivePeriod | ? | 0..1 | Period | When the activity definition is expected to be used
useContext | ? | 0..* | UsageContext | Context the content is intended to support
jurisdiction | ? | 0..* | CodeableConcept | Intended jurisdiction for activity definition (if applicable) Jurisdiction ValueSet (Extensible)
topic | | 0..* | CodeableConcept | E.g. Education, Treatment, Assessment, etc DefinitionTopic (Example)
contributor | | 0..* | Contributor | A content contributor
contact | ? | 0..* | ContactDetail | Contact details for the publisher
copyright | | 0..1 | markdown | Use and/or publishing restrictions
relatedArtifact | | 0..* | RelatedArtifact | Additional documentation, citations, etc
library | | 0..* | Reference(Library) | Logic used by the asset
kind | | 0..1 | code | Kind of resource ResourceType (Required)
code | | 0..1 | CodeableConcept | Detail type of activity Procedure Codes (SNOMED CT) (Example)
timing[x] | | 0..1 | | When activity is to occur
timingTiming | | | Timing
timingDateTime | | | dateTime
timingPeriod | | | Period
timingRange | | | Range
location | | 0..1 | Reference(Location) | Where it should happen
participant | | 0..* | BackboneElement | Who should participate in the action
type | | 1..1 | code | patient | practitioner | related-person ActionParticipantType (Required)
role | | 0..1 | CodeableConcept | E.g. Nurse, Surgeon, Parent, etc ActionParticipantRole (Example)
product[x] | | 0..1 | | What's administered/supplied SNOMED CT Medication Codes (Example)
productReference | | | Reference(Medication | Substance)
productCodeableConcept | | | CodeableConcept
quantity | | 0..1 | SimpleQuantity | How much is administered/consumed/supplied
dosage | | 0..* | Dosage | Detailed dosage instructions
bodySite | | 0..* | CodeableConcept | What part of body to perform on SNOMED CT Body Structures (Example)
transform | | 0..1 | Reference(StructureMap) | Transform to apply the template
dynamicValue | | 0..* | BackboneElement | Dynamic aspects of the definition
description | | 0..1 | string | Natural language description of the dynamic value
path | | 0..1 | string | The path to the element to be set dynamically
language | | 0..1 | string | Language of the expression
expression | | 0..1 | string | An expression that provides the dynamic value for the customization
Documentation for this format

UML Diagram (Legend)

XML Template

```
<ActivityDefinition xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <url value="[uri]"/><!-- 0..1 Logical URI to reference this activity definition (globally unique) -->
 <identifier><!-- 0..* Identifier Additional identifier for the activity definition --></identifier>
 <version value="[string]"/><!-- 0..1 Business version of the activity definition -->
 <name value="[string]"/><!-- 0..1 Name for this activity definition (computer friendly) -->
 <title value="[string]"/><!-- 0..1 Name for this activity definition (human friendly) -->
 <status value="[code]"/><!-- 1..1 draft | active | retired | unknown -->
 <experimental value="[boolean]"/><!-- 0..1 For testing purposes, not real usage -->
 <date value="[dateTime]"/><!-- 0..1 Date this was last changed -->
 <publisher value="[string]"/><!-- 0..1 Name of the publisher (organization or individual) -->
 <description value="[markdown]"/><!-- 0..1 Natural language description of the activity definition -->
 <purpose value="[markdown]"/><!-- 0..1 Why this activity definition is defined -->
 <usage value="[string]"/><!-- 0..1 Describes the clinical usage of the asset -->
 <approvalDate value="[date]"/><!-- 0..1 When the activity definition was approved by publisher -->
 <lastReviewDate value="[date]"/><!-- 0..1 When the activity definition was last reviewed -->
 <effectivePeriod><!-- 0..1 Period When the activity definition is expected to be used --></effectivePeriod>
 <useContext><!-- 0..* UsageContext Context the content is intended to support --></useContext>
 <jurisdiction><!-- 0..* CodeableConcept Intended jurisdiction for activity definition (if applicable)
    --></jurisdiction>
 <topic><!-- 0..* CodeableConcept E.g. Education, Treatment, Assessment, etc --></topic>
 <contributor><!-- 0..* Contributor A content contributor --></contributor>
 <contact><!-- 0..* ContactDetail Contact details for the publisher --></contact>
 <copyright value="[markdown]"/><!-- 0..1 Use and/or publishing restrictions -->
 <relatedArtifact><!-- 0..* RelatedArtifact Additional documentation, citations, etc --></relatedArtifact>
 <library><!-- 0..* Reference(Library) Logic used by the asset --></library>
 <kind value="[code]"/><!-- 0..1 Kind of resource -->
 <code><!-- 0..1 CodeableConcept Detail type of activity --></code>
 <timing[x]><!-- 0..1 Timing|dateTime|Period|Range When activity is to occur --></timing[x]>
 <location><!-- 0..1 Reference(Location) Where it should happen --></location>
 <participant>  <!-- 0..* Who should participate in the action -->
  <type value="[code]"/><!-- 1..1 patient | practitioner | related-person -->
  <role><!-- 0..1 CodeableConcept E.g. Nurse, Surgeon, Parent, etc --></role>
 </participant>
 <product[x]><!-- 0..1 Reference(Medication|Substance)|CodeableConcept What's administered/supplied --></product[x]>
 <quantity><!-- 0..1 Quantity(SimpleQuantity) How much is administered/consumed/supplied --></quantity>
 <dosage><!-- 0..* Dosage Detailed dosage instructions --></dosage>
 <bodySite><!-- 0..* CodeableConcept What part of body to perform on --></bodySite>
 <transform><!-- 0..1 Reference(StructureMap) Transform to apply the template --></transform>
 <dynamicValue>  <!-- 0..* Dynamic aspects of the definition -->
  <description value="[string]"/><!-- 0..1 Natural language description of the dynamic value -->
  <path value="[string]"/><!-- 0..1 The path to the element to be set dynamically -->
```

```
  <language value="[string]"/><!-- 0..1 Language of the expression -->
  <expression value="[string]"/><!-- 0..1 An expression that provides the dynamic value for the customization -->
 </dynamicValue>
</ActivityDefinition>
```

JSON Template

```
{
  "resourceType" : "ActivityDefinition",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "url" : "<uri>", // Logical URI to reference this activity definition (globally unique)
  "identifier" : [{ Identifier }], // Additional identifier for the activity definition
  "version" : "<string>", // Business version of the activity definition
  "name" : "<string>", // Name for this activity definition (computer friendly)
  "title" : "<string>", // Name for this activity definition (human friendly)
  "status" : "<code>", // R!  draft | active | retired | unknown
  "experimental" : <boolean>, // For testing purposes, not real usage
  "date" : "<dateTime>", // Date this was last changed
  "publisher" : "<string>", // Name of the publisher (organization or individual)
  "description" : "<markdown>", // Natural language description of the activity definition
  "purpose" : "<markdown>", // Why this activity definition is defined
  "usage" : "<string>", // Describes the clinical usage of the asset
  "approvalDate" : "<date>", // When the activity definition was approved by publisher
  "lastReviewDate" : "<date>", // When the activity definition was last reviewed
  "effectivePeriod" : { Period }, // When the activity definition is expected to be used
  "useContext" : [{ UsageContext }], // Context the content is intended to support
  "jurisdiction" : [{ CodeableConcept }], // Intended jurisdiction for activity definition (if applicable)
  "topic" : [{ CodeableConcept }], // E.g. Education, Treatment, Assessment, etc
  "contributor" : [{ Contributor }], // A content contributor
  "contact" : [{ ContactDetail }], // Contact details for the publisher
  "copyright" : "<markdown>", // Use and/or publishing restrictions
  "relatedArtifact" : [{ RelatedArtifact }], // Additional documentation, citations, etc
  "library" : [{ Reference(Library) }], // Logic used by the asset
  "kind" : "<code>", // Kind of resource
  "code" : { CodeableConcept }, // Detail type of activity
  // timing[x]: When activity is to occur. One of these 4:
  "timingTiming" : { Timing },
  "timingDateTime" : "<dateTime>",
  "timingPeriod" : { Period },
  "timingRange" : { Range },
  "location" : { Reference(Location) }, // Where it should happen
  "participant" : [{ // Who should participate in the action
    "type" : "<code>", // R!  patient | practitioner | related-person
    "role" : { CodeableConcept } // E.g. Nurse, Surgeon, Parent, etc
  }],
  // product[x]: What's administered/supplied. One of these 2:
  "productReference" : { Reference(Medication|Substance) },
  "productCodeableConcept" : { CodeableConcept },
  "quantity" : { Quantity(SimpleQuantity) }, // How much is administered/consumed/supplied
  "dosage" : [{ Dosage }], // Detailed dosage instructions
  "bodySite" : [{ CodeableConcept }], // What part of body to perform on
  "transform" : { Reference(StructureMap) }, // Transform to apply the template
  "dynamicValue" : [{ // Dynamic aspects of the definition
    "description" : "<string>", // Natural language description of the dynamic value
    "path" : "<string>", // The path to the element to be set dynamically
    "language" : "<string>", // Language of the expression
    "expression" : "<string>" // An expression that provides the dynamic value for the customization
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:ActivityDefinition;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:ActivityDefinition.url [ uri ]; # 0..1 Logical URI to reference this activity definition (globally unique)
  fhir:ActivityDefinition.identifier [ Identifier ], ... ; # 0..* Additional identifier for the activity definition
  fhir:ActivityDefinition.version [ string ]; # 0..1 Business version of the activity definition
  fhir:ActivityDefinition.name [ string ]; # 0..1 Name for this activity definition (computer friendly)
  fhir:ActivityDefinition.title [ string ]; # 0..1 Name for this activity definition (human friendly)
  fhir:ActivityDefinition.status [ code ]; # 1..1 draft | active | retired | unknown
  fhir:ActivityDefinition.experimental [ boolean ]; # 0..1 For testing purposes, not real usage
  fhir:ActivityDefinition.date [ dateTime ]; # 0..1 Date this was last changed
  fhir:ActivityDefinition.publisher [ string ]; # 0..1 Name of the publisher (organization or individual)
  fhir:ActivityDefinition.description [ markdown ]; # 0..1 Natural language description of the activity definition
  fhir:ActivityDefinition.purpose [ markdown ]; # 0..1 Why this activity definition is defined
  fhir:ActivityDefinition.usage [ string ]; # 0..1 Describes the clinical usage of the asset
  fhir:ActivityDefinition.approvalDate [ date ]; # 0..1 When the activity definition was approved by publisher
  fhir:ActivityDefinition.lastReviewDate [ date ]; # 0..1 When the activity definition was last reviewed
  fhir:ActivityDefinition.effectivePeriod [ Period ]; # 0..1 When the activity definition is expected to be used
  fhir:ActivityDefinition.useContext [ UsageContext ], ... ; # 0..* Context the content is intended to support
  fhir:ActivityDefinition.jurisdiction [ CodeableConcept ], ... ; # 0..* Intended jurisdiction for activity definition
    (if applicable)
  fhir:ActivityDefinition.topic [ CodeableConcept ], ... ; # 0..* E.g. Education, Treatment, Assessment, etc
  fhir:ActivityDefinition.contributor [ Contributor ], ... ; # 0..* A content contributor
```

```
  fhir:ActivityDefinition.contact [ ContactDetail ], ... ; # 0..* Contact details for the publisher
  fhir:ActivityDefinition.copyright [ markdown ]; # 0..1 Use and/or publishing restrictions
  fhir:ActivityDefinition.relatedArtifact [ RelatedArtifact ], ... ; # 0..* Additional documentation, citations, etc
  fhir:ActivityDefinition.library [ Reference(Library) ], ... ; # 0..* Logic used by the asset
  fhir:ActivityDefinition.kind [ code ]; # 0..1 Kind of resource
  fhir:ActivityDefinition.code [ CodeableConcept ]; # 0..1 Detail type of activity
  # ActivityDefinition.timing[x] : 0..1 When activity is to occur. One of these 4
    fhir:ActivityDefinition.timingTiming [ Timing ]
    fhir:ActivityDefinition.timingDateTime [ dateTime ]
    fhir:ActivityDefinition.timingPeriod [ Period ]
    fhir:ActivityDefinition.timingRange [ Range ]
  fhir:ActivityDefinition.location [ Reference(Location) ]; # 0..1 Where it should happen
  fhir:ActivityDefinition.participant [ # 0..* Who should participate in the action
    fhir:ActivityDefinition.participant.type [ code ]; # 1..1 patient | practitioner | related-person
    fhir:ActivityDefinition.participant.role [ CodeableConcept ]; # 0..1 E.g. Nurse, Surgeon, Parent, etc
  ], ...;
  # ActivityDefinition.product[x] : 0..1 What's administered/supplied. One of these 2
    fhir:ActivityDefinition.productReference [ Reference(Medication|Substance) ]
    fhir:ActivityDefinition.productCodeableConcept [ CodeableConcept ]
  fhir:ActivityDefinition.quantity [ Quantity(SimpleQuantity) ]; # 0..1 How much is administered/consumed/supplied
  fhir:ActivityDefinition.dosage [ Dosage ], ... ; # 0..* Detailed dosage instructions
  fhir:ActivityDefinition.bodySite [ CodeableConcept ], ... ; # 0..* What part of body to perform on
  fhir:ActivityDefinition.transform [ Reference(StructureMap) ]; # 0..1 Transform to apply the template
  fhir:ActivityDefinition.dynamicValue [ # 0..* Dynamic aspects of the definition
    fhir:ActivityDefinition.dynamicValue.description [ string ]; # 0..1 Natural language description of the dynamic
    value
    fhir:ActivityDefinition.dynamicValue.path [ string ]; # 0..1 The path to the element to be set dynamically
    fhir:ActivityDefinition.dynamicValue.language [ string ]; # 0..1 Language of the expression
    fhir:ActivityDefinition.dynamicValue.expression [ string ]; # 0..1 An expression that provides the dynamic value for
     the customization
  ], ...;
]
```

Changes since DSTU2

This resource did not exist in Release 2

This analysis is available as XML or JSON.

Alternate definitions: Master Definition (XML, JSON), XML Schema/Schematron (for) + JSON Schema, ShEx (for Turtle)

12.18.3.1 Terminology Bindings

Path	Definition	Type	Reference
ActivityDefinition.status	The lifecycle status of a Value Set or Concept Map.	Required	PublicationStatus
ActivityDefinition.jurisdiction	Countries and regions within which this artifact is targeted for use	Extensible	Jurisdiction ValueSet
ActivityDefinition.topic	High-level categorization of the definition, used for searching, sorting, and filtering	Example	DefinitionTopic
ActivityDefinition.kind	The kind of activity the definition is describing	Required	ResourceType
ActivityDefinition.code	Detailed type of the activity; e.g. CBC	Example	Procedure Codes (SNOMED CT)
ActivityDefinition.participant.type	The type of participant in the activity	Required	ActionParticipantType
ActivityDefinition.participant.role	Defines roles played by participants for the action	Example	ActionParticipantRole
ActivityDefinition.product[x]	Code describing the type of substance or medication	Example	SNOMED CT Medication Codes
ActivityDefinition.bodySite	A code that identifies the anatomical location	Example	SNOMED CT Body Structures

12.18.3.2 Realizing an ActivityDefinition

An ActivityDefinition resource provides a description, or template, of an action to performed. These actions can be purely text-based descriptions of the action to be performed, only interpretable by a human user, or they can be structured definitions with enough information to construct a resource to represent the request directly. This process of converting the ActivityDefinition into a specific request resource in a particular context is performed with the $apply operation.

In the simplest case, the process maps the elements specified in the ActivityDefinition to the corresponding elements on the request resource of the appropriate type, using the kind element of the definition to determine the type of resource to be created.

More dynamic scenarios can be achieved in one of two ways, either by providing dynamicValue expressions, or by specifying a StructureMap that transforms the definition into the appropriate request resource.

Note that systems will likely vary widely in the degree of support they provide for the $apply operation. In particular, the fact that a system supports applying definitions of one category, does not imply that it supports applying definitions for all categories. For example, a service focused on medication order sets may have sophisticated support for MedicationRequest activities, but no support at all for the ProcedureRequest activities.

The following diagram illustrates the relationship between the PlanDefinition and ActivityDefinition resources, as well as a typical realization to CarePlan, RequestGroup, and Request resources:

Note to implementers: There is some overlap between the content that can be specified in PlanDefinition.actionDefinition and the ActivityDefinition resource. Part of the work for FHIR Release 4 will be to resolve the overlap and this will likely result in some minor changes to the structures.

12.18.3.3 Profiling ActivityDefinition

Because the ActivityDefinition resource can be used to describe many different types of request resources, profiles of the resource will be useful in communicating additional constraints and expectations about how the resource should be used in a particular context. For example, a profile of ActivityDefinition may specify that it is intended to describe medication orders

within an order set. Such a profile would indicate that the quantity element should not be used, but the dosage element must be.

12.18.4 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
composed-of | reference | What resource is being referenced | ActivityDefinition.relatedArtifact.where(type='composed-of').resource (Any)
date | date | The activity definition publication date | ActivityDefinition.date
depends-on | reference | What resource is being referenced | ActivityDefinition.relatedArtifact.where(type='depends-on').resource | ActivityDefinition.library (Library, Any)
derived-from | reference | What resource is being referenced | ActivityDefinition.relatedArtifact.where(type='derived-from').resource (Any)
description | string | The description of the activity definition | ActivityDefinition.description
effective | date | The time during which the activity definition is intended to be in use | ActivityDefinition.effectivePeriod
identifier | token | External identifier for the activity definition | ActivityDefinition.identifier
jurisdiction | token | Intended jurisdiction for the activity definition | ActivityDefinition.jurisdiction
name | string | Computationally friendly name of the activity definition | ActivityDefinition.name
predecessor | reference | What resource is being referenced | ActivityDefinition.relatedArtifact.where(type='predecessor').resource (Any)
publisher | string | Name of the publisher of the activity definition | ActivityDefinition.publisher
status | token | The current status of the activity definition | ActivityDefinition.status
successor | reference | What resource is being referenced | ActivityDefinition.relatedArtifact.where(type='successor').resource (Any)
title | string | The human-friendly name of the activity definition | ActivityDefinition.title
topic | token | Topics associated with the module | ActivityDefinition.topic
url | uri | The uri that identifies the activity definition | ActivityDefinition.url
version | token | The business version of the activity definition | ActivityDefinition.version

12.18.5 Resource ActivityDefinition - Examples

URL: https://hl7.org/fhir/STU3/activitydefinition-examples.html

These examples illustrate the use of the ActivityDefinition resource to represent activities on various artifacts includes a Suicide Risk Assessment Order Set, and a Zika Virus Infection Management workflow.

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

12.18.5 Resource ActivityDefinition - Examples

Clinical Decision Support Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Not linked to any defined compartments

Example List:

Activity definition to refer a patient to a mental health integrated care program | referralPrimaryCareMentalHealth | XML | JSON | Turtle
Activity definition to prescribe citalopram to a patient | citalopramPrescription | XML | JSON | Turtle
Definition for administering a Zika Virus Exposure assessment | administer-zika-virus-exposure-assessment | XML | JSON | Turtle
Definition for ordering a Serum Zika and Dengue Virus IgM Antibody Diagnostic | serum-zika-dengue-virus-igm | XML | JSON | Turtle
Definition for providing mosquito prevention advice | provide-mosquito-prevention-advice | XML | JSON | Turtle
Definition for heart valve replacement procedure request | heart-valve-replacement | XML | JSON | Turtle
Definition for blood collection tubes supply | blood-tubes-supply | XML | JSON | Turtle
Example of a predecessor/successor relationship | referralPrimaryCareMentalHealth-initial | XML | JSON | Turtle
Definition for ordering a Serum Dengue Virus IgM Antibody Diagnostic | serum-dengue-virus-igm | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

12.18.6 Resource ActivityDefinition - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/activitydefinition-definitions.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

12.18.6 Resource ActivityDefinition - Detailed Descriptions

Clinical Decision Support Work Group | Maturity Level: 2 | Trial Use | Compartments: Not linked to any defined compartments

Detailed Descriptions for the elements in the ActivityDefinition resource.

ActivityDefinition
Definition | This resource allows for the definition of some activity to be performed, independent of a particular patient, practitioner, or other performance context.
Control | 1..1
ActivityDefinition.url
Definition | An absolute URI that is used to identify this activity definition when it is referenced in a specification, model, design or an instance. This SHALL be a URL, SHOULD be globally unique, and SHOULD be an address at which this activity definition is (or will be) published. The URL SHOULD include the major version of the activity definition. For more information see Technical and Business Versions.
Control | 0..1

Type | uri
Requirements | Allows the activity definition to be referenced by a single globally unique identifier.
Summary | true
Comments | Can be a urn:uuid: or a urn:oid:, but real http: addresses are preferred. Multiple instances may share the same url if they have a distinct version.
ActivityDefinition.identifier
Definition | A formal identifier that is used to identify this activity definition when it is represented in other formats, or referenced in a specification, model, design or an instance.
Note | This is a business identifer, not a resource identifier (see discussion)
Control | 0..*
Type | Identifier
Requirements | Allows externally provided and/or usable business identifiers to be easily associated with the module.
Summary | true
Comments | Typically, this is used for identifiers that can go in an HL7 V3 II (instance identifier) data type, e.g., to identify this activity definition outside of FHIR, where it is not possible to use the logical URI.
ActivityDefinition.version
Definition | The identifier that is used to identify this version of the activity definition when it is referenced in a specification, model, design or instance. This is an arbitrary value managed by the activity definition author and is not expected to be globally unique. For example, it might be a timestamp (e.g. yyyymmdd) if a managed version is not available. There is also no expectation that versions can be placed in a lexicographical sequence. To provide a version consistent with the Decision Support Service specification, use the format Major.Minor.Revision (e.g. 1.0.0). For more information on versioning knowledge assets, refer to the Decision Support Service specification. Note that a version is required for non-experimental active assets.
Note | This is a business versionId, not a resource version id (see discussion)
Control | 0..1
Type | string
Summary | true
Comments | There may be different activity definition instances that have the same identifier but different versions. The version can be appended to the url in a reference to allow a refrence to a particular business version of the activity definition with the format [url]|[version].
ActivityDefinition.name
Definition | A natural language name identifying the activity definition. This name should be usable as an identifier for the module by machine processing applications such as code generation.
Control | 0..1
Type | string
Requirements | Support human navigation and code generation.
Summary | true
Comments | The name is not expected to be globally unique. The name should be a simple alpha-numeric type name to ensure that it is computable friendly.
ActivityDefinition.title
Definition | A short, descriptive, user-friendly title for the activity definition.
Control | 0..1
Type | string
Summary | true
Comments | This name does not need to be machine-processing friendly and may contain punctuation, white-space, etc.
ActivityDefinition.status
Definition | The status of this activity definition. Enables tracking the life-cycle of the content.
Control | 1..1
Terminology Binding | PublicationStatus (Required)
Type | code
Is Modifier | true
Summary | true
Comments | Allows filtering of activity definitions that are appropriate for use vs. not. This is labeled as "Is Modifier" because applications should not use a retired activity definition without due consideration.
ActivityDefinition.experimental
Definition | A boolean value to indicate that this activity definition is authored for testing purposes (or education/evaluation/marketing), and is not intended to be used for genuine usage.
Control | 0..1
Type | boolean
Is Modifier | true
Requirements | Enables experimental content to be developed following the same lifecycle that would be used for a production-level activity definition.
Summary | true
Comments | Allows filtering of activity definition that are appropriate for use vs. not. This is labeled as "Is Modifier" because applications should not use an experimental activity definition in production.
ActivityDefinition.date
Definition | The date (and optionally time) when the activity definition was published. The date must change if and when the business version changes and it must change if the status code changes. In addition, it should change when the substantive content of the activity definition changes.
Control | 0..1
Type | dateTime
Alternate Names | Revision Date
Summary | true
Comments | Note that this is not the same as the resource last-modified-date, since the resource may be a secondary representation of the activity definition. Additional specific dates may be added as extensions or be found by consulting Provenances associated with past versions of the resource.
ActivityDefinition.publisher

Definition | The name of the individual or organization that published the activity definition.
Control | 0..1
Type | string
Requirements | Helps establish the "authority/credibility" of the activity definition. May also allow for contact.
Summary | true
Comments | Usually an organization, but may be an individual. The publisher (or steward) of the activity definition is the organization or individual primarily responsible for the maintenance and upkeep of the activity definition. This is not necessarily the same individual or organization that developed and initially authored the content. The publisher is the primary point of contact for questions or issues with the activity definition. This item SHOULD be populated unless the information is available from context.
ActivityDefinition.description
Definition | A free text natural language description of the activity definition from a consumer's perspective.
Control | 0..1
Type | markdown
Summary | true
Comments | This description can be used to capture details such as why the activity definition was built, comments about misuse, instructions for clinical use and interpretation, literature references, examples from the paper world, etc. It is not a rendering of the activity definition as conveyed in the 'text' field of the resource itself. This item SHOULD be populated unless the information is available from context (e.g. the language of the profile is presumed to be the predominant language in the place the profile was created).
ActivityDefinition.purpose
Definition | Explaination of why this activity definition is needed and why it has been designed as it has.
Control | 0..1
Type | markdown
Comments | This element does not describe the usage of the activity definition Instead it provides traceability of "why" the resource is either needed or "why" it is defined as it is. This may be used to point to source materials or specifications that drove the structure of this activity definition.
ActivityDefinition.usage
Definition | A detailed description of how the asset is used from a clinical perspective.
Control | 0..1
Type | string
ActivityDefinition.approvalDate
Definition | The date on which the resource content was approved by the publisher. Approval happens once when the content is officially approved for usage.
Control | 0..1
Type | date
Comments | The 'date' element may be more recent than the approval date because of minor changes / editorial corrections.
ActivityDefinition.lastReviewDate
Definition | The date on which the resource content was last reviewed. Review happens periodically after approval, but doesn't change the original approval date.
Control | 0..1
Type | date
Requirements | Gives a sense of how "current" the content is. Resources that have not been reviewed in a long time may have a risk of being less appropriate/relevant.
Comments | If specified, this is usually after the approval date.
ActivityDefinition.effectivePeriod
Definition | The period during which the activity definition content was or is planned to be in active use.
Control | 0..1
Type | Period
Requirements | Allows establishing a transition before a resource comes into effect and also allows for a sunsetting process when new versions of a the activity definition are or are expected to be used instead.
Summary | true
Comments | The effective period for a activity definition determines when the content is applicable for usage and is independent of publication and review dates. For example, a measure intended to be used for the year 2016 might be published in 2015.
ActivityDefinition.useContext
Definition | The content was developed with a focus and intent of supporting the contexts that are listed. These terms may be used to assist with indexing and searching for appropriate activity definition instances.
Control | 0..*
Type | UsageContext
Requirements | Assist in searching for appropriate content.
Summary | true
Comments | When multiple useContexts are specified, there is no expectation whether all or any of the contexts apply.
ActivityDefinition.jurisdiction
Definition | A legal or geographic region in which the activity definition is intended to be used.
Control | 0..*
Terminology Binding | Jurisdiction ValueSet (Extensible)
Type | CodeableConcept
Summary | true
Comments | It may be possible for the activity definition to be used in jurisdictions other than those for which it was originally designed or intended.
ActivityDefinition.topic
Definition | Descriptive topics related to the content of the activity. Topics provide a high-level categorization of the activity that can be useful for filtering and searching.
Control | 0..*
Terminology Binding | DefinitionTopic (Example)
Type | CodeableConcept
Requirements | Repositories must be able to determine how to categorize the asset so that it can be found by topical searches.

ActivityDefinition.contributor
Definition | A contributor to the content of the asset, including authors, editors, reviewers, and endorsers.
Control | 0..*
Type | Contributor
Requirements | Consumers of the content must be able to quickly determine who contributed to the content of the knowledge asset.
ActivityDefinition.contact
Definition | Contact details to assist a user in finding and communicating with the publisher.
Control | 0..*
Type | ContactDetail
Summary | true
Comments | May be a web site, an email address, a telephone number, etc.
ActivityDefinition.copyright
Definition | A copyright statement relating to the activity definition and/or its contents. Copyright statements are generally legal restrictions on the use and publishing of the activity definition.
Control | 0..1
Type | markdown
Requirements | Consumers must be able to determine any legal restrictions on the use of the activity definition and/or its content.
Alternate Names | License; Restrictions
ActivityDefinition.relatedArtifact
Definition | Related artifacts such as additional documentation, justification, or bibliographic references.
Control | 0..*
Type | RelatedArtifact
Requirements | Assets must be able to provide enough information for consumers of the content (and/or interventions or results produced by the content) to be able to determine and understand the justification for and evidence in support of the content.
Comments | Each related artifact is either an attachment, or a reference to another resource, but not both.
ActivityDefinition.library
Definition | A reference to a Library resource containing any formal logic used by the asset.
Control | 0..*
Type | Reference(Library)
ActivityDefinition.kind
Definition | A description of the kind of resource the activity definition is representing. For example, a MedicationRequest, a ProcedureRequest, or a CommunicationRequest. Typically, but not always, this is a Request resource.
Control | 0..1
Terminology Binding | ResourceType (Required)
Type | code
Comments | May determine what types of extensions are permitted.
ActivityDefinition.code
Definition | Detailed description of the type of activity; e.g. What lab test, what procedure, what kind of encounter.
Control | 0..1
Terminology Binding | Procedure Codes (SNOMED CT) (Example)
Type | CodeableConcept
Requirements | Allows matching performed to planned as well as validation against protocols.
Comments | Tends to be less relevant for activities involving particular products.
ActivityDefinition.timing[x]
Definition | The period, timing or frequency upon which the described activity is to occur.
Control | 0..1
Type | Timing|dateTime|Period|Range
[x] Note | See Choice of Data Types for further information about how to use [x]
Requirements | Allows prompting for activities and detection of missed planned activities.
ActivityDefinition.location
Definition | Identifies the facility where the activity will occur; e.g. home, hospital, specific clinic, etc.
Control | 0..1
Type | Reference(Location)
Requirements | Helps in planning of activity.
Alternate Names | facility
Comments | May reference a specific clinical location or may just identify a type of location.
ActivityDefinition.participant
Definition | Indicates who should participate in performing the action described.
Control | 0..*
ActivityDefinition.participant.type
Definition | The type of participant in the action.
Control | 1..1
Terminology Binding | ActionParticipantType (Required)
Type | code
ActivityDefinition.participant.role
Definition | The role the participant should play in performing the described action.
Control | 0..1
Terminology Binding | ActionParticipantRole (Example)
Type | CodeableConcept
ActivityDefinition.product[x]
Definition | Identifies the food, drug or other product being consumed or supplied in the activity.
Control | 0..1

Terminology Binding | SNOMED CT Medication Codes (Example)
Type | Reference(Medication | Substance)|CodeableConcept
[x] Note | See Choice of Data Types for further information about how to use [x]
ActivityDefinition.quantity
Definition | Identifies the quantity expected to be consumed at once (per dose, per meal, etc.).
Control | 0..1
Type | SimpleQuantity
Alternate Names | dose
ActivityDefinition.dosage
Definition | Provides detailed dosage instructions in the same way that they are described for MedicationRequest resources.
Control | 0..*
Type | Dosage
Comments | If a dosage instruction is used, the definition should not specify timing or quantity.
ActivityDefinition.bodySite
Definition | Indicates the sites on the subject's body where the procedure should be performed (I.e. the target sites).
Control | 0..*
Terminology Binding | SNOMED CT Body Structures (Example)
Type | CodeableConcept
Requirements | Knowing where the procedure is made is important for tracking if multiple sites are possible.
Comments | Only used if not implicit in the code found in ProcedureRequest.type.
ActivityDefinition.transform
Definition | A reference to a StructureMap resource that defines a transform that can be executed to produce the intent resource using the ActivityDefinition instance as the input.
Control | 0..1
Type | Reference(StructureMap)
ActivityDefinition.dynamicValue
Definition | Dynamic values that will be evaluated to produce values for elements of the resulting resource. For example, if the dosage of a medication must be computed based on the patient's weight, a dynamic value would be used to specify an expression that calculated the weight, and the path on the intent resource that would contain the result.
Control | 0..*
ActivityDefinition.dynamicValue.description
Definition | A brief, natural language description of the intended semantics of the dynamic value.
Control | 0..1
Type | string
ActivityDefinition.dynamicValue.path
Definition | The path to the element to be customized. This is the path on the resource that will hold the result of the calculation defined by the expression.
Control | 0..1
Type | string
ActivityDefinition.dynamicValue.language
Definition | The media type of the language for the expression.
Control | 0..1
Type | string
ActivityDefinition.dynamicValue.expression
Definition | An expression specifying the value of the customized element.
Control | 0..1
Type | string
Comments | The expression may be inlined, or may be a reference to a named expression within a logic library referenced by the library element.

12.18.7 Resource ActivityDefinition - Mappings

URL: https://hl7.org/fhir/STU3/activitydefinition-mappings.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

12.18.7 Resource ActivityDefinition - Mappings

Clinical Decision Support Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Not linked to any defined compartments

Mappings:

RIM Mapping

Workflow Mapping

Mappings for the activitydefinition resource.

12.18.7.1 Mappings for RIM Mapping (http://hl7.org/v3)

ActivityDefinition | Act[classCode=ACT; moodCode=DEFN]
url | .identifier[scope=BUSN;reliability=ISS]
identifier | .identifier
version | N/A (to add?)
name | N/A
title | .title
status | .status

experimental | N/A (to add?)
date | .participation[typeCode=AUT].time
publisher | .participation[typeCode=AUT].role
description | .text
purpose | .reasonCode.text
usage | N/A (to add?)
approvalDate | .outboundRelationship[typeCode="SUBJ"].act[classCode=CACT;moodCode=EVN;code="approval"].effectiveTime
lastReviewDate | .outboundRelationship[typeCode="SUBJ";
subsetCode="RECENT"].act[classCode=CACT;moodCode=EVN;code="review"].effectiveTime
effectivePeriod | N/A (to add?)
useContext | N/A (to add?)
jurisdiction | N/A (to add?)
topic | N/A (to add?)
contributor | .participation[typeCode=AUT,VRF] {No support yet for editor nor clear whether VRF best corresponds to reviewer or endorser}
contact | .participation[typeCode=CALLBCK].role
copyright | N/A (to add?)
relatedArtifact | .outboundRelationship[typeCode=DOC,RSON,PREV, DRIV, USE, COMP] {successor would be PREV w/ inversionInd=true; No support for citation}
library | .outboundRelationship[typeCode=USE].target[classCode=GROUPER,moodCode=EVN]
kind | .classCode
code | .code
timing[x] | .effectiveTime
location | .participation[typeCode=LOC].role
participant | .participation[typeCode=PFM]
type | .role.classCode
role | .role.code
product[x] | .participation[typeCode=CSM].role
quantity | .quantity {for classCode=SPLY]
dosage | .outboundRelationship[typeCode=COMP].target[classCode=SBADM; moodCode=DEFN]
bodySite | .targetSiteCode {for classCode=PROC, SBADM}
transform | N/A
dynamicValue | N/A
description | N/A
path | N/A
language | N/A
expression | N/A

12.18.7.2 Mappings for Workflow Mapping (http://hl7.org/fhir/workflow)

ActivityDefinition | Definition
url | Definition.url
identifier | Definition.identifier
version | Definition.version
name
title | Definition.title
status | Definitition.status (different ValueSet)
experimental | Definition.experimental
date | Definition.date
publisher | Definition.publisher (as string instead of CodeableConcept)
description | Definition.description
purpose | Definition.purpose
usage
approvalDate | Definition.approvalDate
lastReviewDate | Definition.lastReviewDate
effectivePeriod | Definition.effectivePeriod
useContext | Definition.useContext
jurisdiction | Definition.jurisdiction
topic
contributor
contact | Definition.contact
copyright | Definition.copyright
relatedArtifact
library
kind
code | Definition.subjectCodeableConcept
timing[x]
location
participant
type
role
product[x]
quantity
dosage

bodySite | Procedure.targetBodySite
transform
dynamicValue
description
path
language
expression

12.18.8 Resource ActivityDefinition - Design Notes

URL: https://hl7.org/fhir/STU3/activitydefinition-explanations.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

12.18.8 Resource ActivityDefinition - Design Notes

to do

12.18.9 Resource ActivityDefinition - Extensions & Profiles

URL: https://hl7.org/fhir/STU3/activitydefinition-profiles.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

12.18.9 Resource ActivityDefinition - Extensions & Profiles

Clinical Decision Support Work Group | Maturity Level: N/A | Ballot Status: Informative

This table lists profiles and extensions for the ActivityDefinition resource. For background information, see Profiling Resources and Extensibility. Additional profiles and resources may be found in published Implementation Guides , or in the Conformance resource registry .

12.18.9.1 Profiles

No Profiles defined for this resource

12.18.9.2 Extensions

No Extensions defined for this resource

Extensions that reference this resource:

communicationrequest-definition | definition | for CommunicationRequest HL7 Extensions
event-definition | definition | for Event Pattern HL7 Extensions

12.18.9.3 Search Extensions

No Search Extensions defined for this resource

12.18.10 Resource ActivityDefinition - Operations

URL: https://hl7.org/fhir/STU3/activitydefinition-operations.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

12.18.10 Resource ActivityDefinition - Operations

Clinical Decision Support Work Group | Maturity Level: 2 | Trial Use | Compartments: Not linked to any defined compartments

This resource has 2 operations associated with it:

$apply | Apply
$data-requirements | Data Requirements

For more information about operations, including how they are invoked, see Operations.

12.18.10.1 Apply

The apply operation realizes a definition in a specific context

The official URL for this operation definition is

```
http://hl7.org/fhir/OperationDefinition/ActivityDefinition-apply
```

Formal Definition (as a OperationDefinition).

URL: [base]/ActivityDefinition/[id]/$apply

This is not an idempotent operation

In Parameters:
Name | Cardinality | Type | Binding | Profile | Documentation
patient | 1..1 | Reference(Patient) | | | The patient that is the target of the activity to be applied

encounter | 0..1 | Reference(Encounter) | | | The encounter in context, if any
practitioner | 0..1 | Reference(Practitioner) | | | The practitioner in context
organization | 0..1 | Reference(Organization) | | | The organization in context
userType | 0..1 | CodeableConcept | | | The type of user initiating the request, e.g. patient, healthcare provider, or specific type of healthcare provider (physician, nurse, etc.)
userLanguage | 0..1 | CodeableConcept | | | Preferred language of the person using the system
userTaskContext | 0..1 | CodeableConcept | | | The task the system user is performing, e.g. laboratory results review, medication list review, etc. This information can be used to tailor decision support outputs, such as recommended information resources
setting | 0..1 | CodeableConcept | | | The current setting of the request (inpatient, outpatient, etc)
settingContext | 0..1 | CodeableConcept | | | Additional detail about the setting of the request, if any
Out Parameters:
Name | Cardinality | Type | Binding | Profile | Documentation
return | 1..1 | Any | | | The resource that is the result of applying the definition

The result of invoking this operation is a resource of the type specified by the activity definition, with all the definitions resolved as appropriate for the type of resource. Any dynamicValue elements will be evaluated and the resulting value used for the values of the appropriate elements of the returned resource. If the ActivityDefinition includes library references, those libraries will be available to the evaluated expressions. If those libraries have parameters, those parameters will be bound by name to the parameters given to the operation

12.18.10.1.1 Examples

Apply a citalopram prescription activity definition to a patient (Request):

```
GET [base]/ActivityDefinition/citalopramPrescription/$apply?patient=124
```

Returned on successful application (Response):

```
HTTP/1.1 200 OK

<?xml version="1.0" encoding="UTF-8"?>
<MedicationRequest xmlns="http://hl7.org/fhir">
  <!-- This example demonstrates the result of an $apply operation for the citalopramPrescription activity definition.
    -->
  <id value="apply-operation-response"/>
  <text>
  <status value="generated"/>
    <div xmlns="http://www.w3.org/1999/xhtml">
    Generated...
    </div>
  </text>
  <contained>
    <Medication>
      <id value="citalopramMedication"/>
      <code>
        <coding>
          <system value="http://www.nlm.nih.gov/research/umls/rxnorm"/>
          <code value="200371"/>
        </coding>
        <text value="citalopram"/>
      </code>
        <form>
          <coding>
            <system value="http://snomed.info/sct"/>
            <code value="385055001"/>
            <display value="Tablet dose form"/>
          </coding>
          <text value="Tablet dose form"/>
        </form>
        <ingredient>
          <itemReference>
            <reference value="#citalopramSubstance"/>
          </itemReference>
          <amount>
            <numerator>
              <value value="20"/>
              <unit value="mg"/>
            </numerator>
            <denominator>
              <value value="1"/>
              <unit value="{tbl}"/>
            </denominator>
          </amount>
        </ingredient>
    </Medication>
  </contained>
  <contained>
    <Substance>
      <id value="citalopramSubstance"/>
      <code>
        <coding>
          <system value="http://www.nlm.nih.gov/research/umls/rxnorm"/>
          <code value="2556"/>
        </coding>
        <text value="citalopram"/>
      </code>
    </Substance>
```

```
  </contained>
  <definition>
    <reference value="ActivityDefinition/citalopramPrescription"/>
  </definition>
  <status value="draft"/>
  <intent value="proposal"/>
  <medicationReference>
    <reference value="#citalopramMedication"/>
  </medicationReference>
  <subject>
    <reference value="Patient/124"/>
  </subject>
  <dosageInstruction>
  <text value="1 tablet oral 1 time daily"/>
  <timing>
    <repeat>
      <frequency value="1"/>
      <period value="1"/>
      <periodUnit value="d"/>
    </repeat>
  </timing>
  <route>
    <coding>
      <code value="26643006"/>
      <display value="Oral route (qualifier value)"/>
    </coding>
    <text value="Oral route (qualifier value)"/>
  </route>
  <doseQuantity>
    <value value="1"/>
    <unit value="{tbl}"/>
  </doseQuantity>
  </dosageInstruction>
  <dispenseRequest>
    <numberOfRepeatsAllowed value="3"/>
    <quantity>
      <value value="30"/>
      <unit value="{tbl}"/>
    </quantity>
  </dispenseRequest>
</MedicationRequest>
```

12.18.10.2 Data Requirements

The data-requirements operation aggregates and returns the parameters and data requirements for the activity definition and all its dependencies as a single module definition library

The official URL for this operation definition is

```
http://hl7.org/fhir/OperationDefinition/ActivityDefinition-data-requirements
```

Formal Definition (as a OperationDefinition).

URL: [base]/ActivityDefinition/[id]/$data-requirements

This is not an idempotent operation

Out Parameters:

Name | Cardinality | Type | Binding | Profile | Documentation

return | 1..1 | Library | | | The result of the requirements gathering represented as a module-definition Library that describes the aggregate parameters, data requirements, and dependencies of the activity definition Note: as this the only out parameter, it is a resource, and it has the name 'return', the result of this operation is returned directly as a resource

The effect of invoking this operation is to determine the aggregate set of data requirements and dependencies for the activity definition. The result is a Library resource with a type of module-definition that contains all the parameter definitions and data requirements of the activity definition and any libraries referenced by it. Implementations SHOULD aggregate data requirements intelligently (i.e. by collapsing overlapping data requirements)

12.18.10.2.1 Examples

Request data requirements for the administer zika virus exposure assessment activity definition (Request):

```
GET [base]/ActivityDefinition/administer-zika-virus-exposure-assessment/$data-requirements
```

Returned on successful evaluation (Response):

```
HTTP/1.1 200 OK

{
  "resourceType": "Library",
  "type": {
    "coding": [
      {
        "code": "module-definition"
      }
    ]
  },
  "relatedArtifact": [
    {
      "type": "depends-on",
      "display": "Zika Affected Areas",
```

```
      "resource": {
        "reference": "ValueSet/zika-affected-areas"
      }
    },
    {
      "type": "depends-on",
      "display": "Pregnancy",
      "resource": {
        "identifier": "urn:oid:X.Y.Z",
        "display": "Pregnancy value set"
      }
    },
    {
      "type": "depends-on",
      "display": "Zika Exposure Assessment",
      "resource": {
        "identifier": "urn:oid:X.Y.Z",
        "display": "Zika Virus Exposure Assessment value set"
      }
    },
    {
      "type": "depends-on",
      "display": "Zika Virus Signs and Symptoms",
      "url": "https://phinvads.cdc.gov/vads/ViewValueSet.action?oid=2.16.840.1.114222.4.11.7459",
      "resource": {
        "identifier": "urn:oid:2.16.840.1.114222.4.11.7459",
        "display": "Zika Virus Signs and Symptoms value set"
      }
    },
  ]
  "dataRequirement": [
    {
      "type": "Condition",
      "codeFilter": [
        {
          "path": "code",
          "valueSetString": "Pregnancy"
        }
      ]
    },
    {
      "type": "Patient"
    },
    {
      "type": "Observation",
      "codeFilter": [
        {
          "path": "code",
          "valueSetString": [
            "Zika Exposure Assessment"
          ]
        },
        {
          "path": "status",
          "valueCode": [
            "final"
          ]
        }
      ]
    },
    {
      "type": "Condition",
      "codeFilter": [
        {
          "path": "category",
          "valueCode": [
            "diagnosis"
          ]
        },
        {
          "path": "clinicalStatus",
          "valueCode": [
            "confirmed"
          ]
        },
        {
          "path": "code",
          "valueSetString": "Zika Virus Signs and Symptoms"
        }
      ]
    }
  ]
}
```

12.19 Resource PlanDefinition - Content

URL: https://hl7.org/fhir/STU3/plandefinition.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

Clinical Decision Support Work Group | Maturity Level: 2 | Trial Use | Compartments: Not linked to any defined compartments

This resource allows for the definition of various types of plans as a sharable, consumable, and executable artifact. The resource is general enough to support the description of a broad range of clinical artifacts such as clinical decision support rules, order sets and protocols.

12.19.1 Scope and Usage

This resource is a definition resource from a FHIR workflow perspective - see Workflow, specifically Definition.

A plan definition is a pre-defined group of actions to be taken in particular circumstances, often including conditional elements, options, and other decision points. The resource is flexible enough to be used to represent a variety of workflows, as well as clinical decision support and quality improvement assets, including order sets, protocols, and decision support rules.

PlanDefinitions can contain hierarchical groups of action definitions, where each specific action definition describes an activity to be performed (in terms of an ActivityDefinition resource), and each group defines additional behavior, relationships, and applicable conditions between the actions in the overall definition.

In addition to describing what should take place, each action in a plan definition can specify when and whether the action should take place. For when the action should be taken, the triggerDefinition element specifies the action should be taken in response to some trigger occurring (such as a particular point in a workflow being reached, or as the result of a prescription being ordered). For whether the action should be taken, the condition element can be used to provide an expression that evaluates to true or false to indicate the applicability of the action to the specific context.

The process of applying a PlanDefinition to a particular context typically produces request resources representing the actions that should be performed, typically grouped within a CarePlan and/or RequestGroup to capture relationships between the resulting request resources.

Each ActivityDefinition is used to construct a specific resource, based on the definition of the activity and combined with contextual information for the particular patient that the plan definition is being applied to.

As with the ActivityDefinition, a PlanDefinition may provide information about how to transform the activity to a specific intent resource, either by specifying a StructureMap that can be used to perform the transformation completely, or by specifying values for specific elements of the resulting resource using dynamicValue elements in the action.

Note that these mechanisms are provided on both the ActivityDefinition and the PlanDefinition to allow both reusable transformation descriptions, as well as customization of those descriptions within specific contexts. As such, the transform descriptions specified on the PlanDefinition override transform descriptions defined on the ActivityDefinition.

Dynamic values within the definitions can be provided by specifying the expression directly, or by referencing an expression defined within a library. For more information on how to reference expressions within resources, refer to the Using Expressions topic.

As an example, the Low Suicide Risk example order set from the Clinical Decision Support Knowledge Artifact Specification can be represented using the PlanDefinition and ActivityDefinition structures: Low Suicide Risk Example Order Set.

In addition to the representation of PlanDefinitions, the $apply operation allows PlanDefinitions to be realized for a specific context such as a patient, practitioner, or institution. For Order Sets specifically, this operation is expected to place the orders defined by the order set, consistent with the service functional requirements defined by the Order Set specification .

12.19.2 Boundaries and Relationships

The PlanDefinition resource is used to describe series, sequences, or groups of actions to be taken, while the ActivityDefinition resource is used to define each specific step or activity to be performed.

As the name implies, the PlanDefinition resource is strictly definitional. It does not represent the intention to take any action, nor does it represent that any actions have been taken. Rather, the resource provides a definition that can be applied in the appropriate circumstances. When the plan definition is applied, the result will in general be a set of actions that should be (or potentially even have been) performed.

Note that the PlanDefinition still has action-level information, as well as a reference to an ActivityDefinition. The action-level information defined in the PlanDefinition itself is used to describe how the actions are related to each other within the plan, where the ActivityDefinition contains only information about the activity itself.

The following diagram illustrates the relationship between the PlanDefinition and ActivityDefinition resources, as well as a typical realization to CarePlan, RequestGroup, and Request resources:

Note to implementers: There is some overlap between the content that can be specified in PlanDefinition.actionDefinition and the ActivityDefinition resource. Part of the work for FHIR Release 4 will be to resolve the overlap and this will likely result in some minor changes to the structures.

This resource is referenced by CarePlan, Communication, DeviceRequest, FamilyMemberHistory, Flag, MedicationAdministration, MedicationRequest, MessageDefinition, Procedure, ProcedureRequest, ReferralRequest and ResearchStudy

12.19.3 Resource Content

* Structure
* UML
* XML
* JSON
* Turtle
* R2 Diff
* All

Structure

Name | Flags | Card. | Type | Description & Constraints
PlanDefinition | | | DomainResource | The definition of a plan for a series of actions, independent of any specific patient or context Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
url | ? | 0..1 | uri | Logical URI to reference this plan definition (globally unique)
identifier | ? | 0..* | Identifier | Additional identifier for the plan definition
version | ? | 0..1 | string | Business version of the plan definition
name | ? | 0..1 | string | Name for this plan definition (computer friendly)
title | ? | 0..1 | string | Name for this plan definition (human friendly)
type | ? | 0..1 | CodeableConcept | order-set | protocol | eca-rule PlanDefinitionType (Extensible)
status | ?!? | 1..1 | code | draft | active | retired | unknown PublicationStatus (Required)
experimental | ?!? | 0..1 | boolean | For testing purposes, not real usage
date | ? | 0..1 | dateTime | Date this was last changed
publisher | ? | 0..1 | string | Name of the publisher (organization or individual)
description | ? | 0..1 | markdown | Natural language description of the plan definition
purpose | | 0..1 | markdown | Why this plan definition is defined
usage | | 0..1 | string | Describes the clinical usage of the asset
approvalDate | | 0..1 | date | When the plan definition was approved by publisher
lastReviewDate | | 0..1 | date | When the plan definition was last reviewed
effectivePeriod | ? | 0..1 | Period | When the plan definition is expected to be used
useContext | ? | 0..* | UsageContext | Context the content is intended to support
jurisdiction | ? | 0..* | CodeableConcept | Intended jurisdiction for plan definition (if applicable) Jurisdiction ValueSet (Extensible)
topic | | 0..* | CodeableConcept | E.g. Education, Treatment, Assessment, etc DefinitionTopic (Example)
contributor | | 0..* | Contributor | A content contributor
contact | ? | 0..* | ContactDetail | Contact details for the publisher
copyright | | 0..1 | markdown | Use and/or publishing restrictions
relatedArtifact | | 0..* | RelatedArtifact | Related artifacts for the asset
library | | 0..* | Reference(Library) | Logic used by the plan definition
goal | | 0..* | BackboneElement | What the plan is trying to accomplish
category | | 0..1 | CodeableConcept | E.g. Treatment, dietary, behavioral, etc GoalCategory (Example)
description | | 1..1 | CodeableConcept | Code or text describing the goal SNOMED CT Clinical Findings (Example)
priority | | 0..1 | CodeableConcept | high-priority | medium-priority | low-priority GoalPriority (Preferred)
start | | 0..1 | CodeableConcept | When goal pursuit begins GoalStartEvent (Example)
addresses | | 0..* | CodeableConcept | What does the goal address Condition/Problem/Diagnosis Codes (Example)
documentation | | 0..* | RelatedArtifact | Supporting documentation for the goal
target | | 0..* | BackboneElement | Target outcome for the goal
measure | | 0..1 | CodeableConcept | The parameter whose value is to be tracked LOINC Codes (Example)
detail[x] | | 0..1 | | The target value to be achieved
detailQuantity | | | Quantity
detailRange | | | Range
detailCodeableConcept | | | CodeableConcept
due | | 0..1 | Duration | Reach goal within
action | | 0..* | BackboneElement | Action defined by the plan
label | | 0..1 | string | User-visible label for the action (e.g. 1. or A.)
title | | 0..1 | string | User-visible title
description | | 0..1 | string | Short description of the action
textEquivalent | | 0..1 | string | Static text equivalent of the action, used if the dynamic aspects cannot be interpreted by the receiving system
code | | 0..* | CodeableConcept | Code representing the meaning of the action or sub-actions
reason | | 0..* | CodeableConcept | Why the action should be performed
documentation | | 0..* | RelatedArtifact | Supporting documentation for the intended performer of the action
goalId | | 0..* | id | What goals this action supports
triggerDefinition | | 0..* | TriggerDefinition | When the action should be triggered
condition | | 0..* | BackboneElement | Whether or not the action is applicable
kind | | 1..1 | code | applicability | start | stop ActionConditionKind (Required)
description | | 0..1 | string | Natural language description of the condition
language | | 0..1 | string | Language of the expression
expression | | 0..1 | string | Boolean-valued expression
input | | 0..* | DataRequirement | Input data requirements
output | | 0..* | DataRequirement | Output data definition
relatedAction | | 0..* | BackboneElement | Relationship to another action
actionId | | 1..1 | id | What action is this related to

relationship | | 1..1 | code | before-start | before | before-end | concurrent-with-start | concurrent | concurrent-with-end | after-start | after | after-end ActionRelationshipType (Required)
offset[x] | | 0..1 | | Time offset for the relationship
offsetDuration | | | Duration
offsetRange | | | Range
timing[x] | | 0..1 | | When the action should take place
timingDateTime | | | dateTime
timingPeriod | | | Period
timingDuration | | | Duration
timingRange | | | Range
timingTiming | | | Timing
participant | | 0..* | BackboneElement | Who should participate in the action
type | | 1..1 | code | patient | practitioner | related-person ActionParticipantType (Required)
role | | 0..1 | CodeableConcept | E.g. Nurse, Surgeon, Parent, etc ActionParticipantRole (Example)
type | | 0..1 | Coding | create | update | remove | fire-event ActionType (Required)
groupingBehavior | | 0..1 | code | visual-group | logical-group | sentence-group ActionGroupingBehavior (Required)
selectionBehavior | | 0..1 | code | any | all | all-or-none | exactly-one | at-most-one | one-or-more ActionSelectionBehavior (Required)
requiredBehavior | | 0..1 | code | must | could | must-unless-documented ActionRequiredBehavior (Required)
precheckBehavior | | 0..1 | code | yes | no ActionPrecheckBehavior (Required)
cardinalityBehavior | | 0..1 | code | single | multiple ActionCardinalityBehavior (Required)
definition | | 0..1 | Reference(ActivityDefinition | PlanDefinition) | Description of the activity to be performed
transform | | 0..1 | Reference(StructureMap) | Transform to apply the template
dynamicValue | | 0..* | BackboneElement | Dynamic aspects of the definition
description | | 0..1 | string | Natural language description of the dynamic value
path | | 0..1 | string | The path to the element to be set dynamically
language | | 0..1 | string | Language of the expression
expression | | 0..1 | string | An expression that provides the dynamic value for the customization
action | | 0..* | see action | A sub-action
Documentation for this format

UML Diagram (Legend)

XML Template

```
<PlanDefinition xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <url value="[uri]"/><!-- 0..1 Logical URI to reference this plan definition (globally unique) -->
 <identifier><!-- 0..* Identifier Additional identifier for the plan definition --></identifier>
 <version value="[string]"/><!-- 0..1 Business version of the plan definition -->
 <name value="[string]"/><!-- 0..1 Name for this plan definition (computer friendly) -->
 <title value="[string]"/><!-- 0..1 Name for this plan definition (human friendly) -->
 <type><!-- 0..1 CodeableConcept order-set | protocol | eca-rule --></type>
 <status value="[code]"/><!-- 1..1 draft | active | retired | unknown -->
 <experimental value="[boolean]"/><!-- 0..1 For testing purposes, not real usage -->
 <date value="[dateTime]"/><!-- 0..1 Date this was last changed -->
 <publisher value="[string]"/><!-- 0..1 Name of the publisher (organization or individual) -->
 <description value="[markdown]"/><!-- 0..1 Natural language description of the plan definition -->
 <purpose value="[markdown]"/><!-- 0..1 Why this plan definition is defined -->
 <usage value="[string]"/><!-- 0..1 Describes the clinical usage of the asset -->
 <approvalDate value="[date]"/><!-- 0..1 When the plan definition was approved by publisher -->
 <lastReviewDate value="[date]"/><!-- 0..1 When the plan definition was last reviewed -->
 <effectivePeriod><!-- 0..1 Period When the plan definition is expected to be used --></effectivePeriod>
 <useContext><!-- 0..* UsageContext Context the content is intended to support --></useContext>
 <jurisdiction><!-- 0..* CodeableConcept Intended jurisdiction for plan definition (if applicable) --></jurisdiction>
 <topic><!-- 0..* CodeableConcept E.g. Education, Treatment, Assessment, etc --></topic>
 <contributor><!-- 0..* Contributor A content contributor --></contributor>
 <contact><!-- 0..* ContactDetail Contact details for the publisher --></contact>
 <copyright value="[markdown]"/><!-- 0..1 Use and/or publishing restrictions -->
 <relatedArtifact><!-- 0..* RelatedArtifact Related artifacts for the asset --></relatedArtifact>
 <library><!-- 0..* Reference(Library) Logic used by the plan definition --></library>
 <goal>  <!-- 0..* What the plan is trying to accomplish -->
  <category><!-- 0..1 CodeableConcept E.g. Treatment, dietary, behavioral, etc --></category>
  <description><!-- 1..1 CodeableConcept Code or text describing the goal --></description>
  <priority><!-- 0..1 CodeableConcept high-priority | medium-priority | low-priority --></priority>
  <start><!-- 0..1 CodeableConcept When goal pursuit begins --></start>
  <addresses><!-- 0..* CodeableConcept What does the goal address --></addresses>
  <documentation><!-- 0..* RelatedArtifact Supporting documentation for the goal --></documentation>
  <target>  <!-- 0..* Target outcome for the goal -->
   <measure><!-- 0..1 CodeableConcept The parameter whose value is to be tracked --></measure>
   <detail[x]><!-- 0..1 Quantity|Range|CodeableConcept The target value to be achieved --></detail[x]>
   <due><!-- 0..1 Duration Reach goal within --></due>
  </target>
 </goal>
 <action>  <!-- 0..* Action defined by the plan -->
  <label value="[string]"/><!-- 0..1 User-visible label for the action (e.g. 1. or A.) -->
  <title value="[string]"/><!-- 0..1 User-visible title -->
  <description value="[string]"/><!-- 0..1 Short description of the action -->
  <textEquivalent value="[string]"/><!-- 0..1 Static text equivalent of the action, used if the dynamic aspects cannot
    be interpreted by the receiving system -->
  <code><!-- 0..* CodeableConcept Code representing the meaning of the action or sub-actions --></code>
  <reason><!-- 0..* CodeableConcept Why the action should be performed --></reason>
  <documentation><!-- 0..* RelatedArtifact Supporting documentation for the intended performer of the action
    --></documentation>
```

```
  <goalId value="[id]"/><!-- 0..* What goals this action supports -->
  <triggerDefinition><!-- 0..* TriggerDefinition When the action should be triggered --></triggerDefinition>
  <condition>  <!-- 0..* Whether or not the action is applicable -->
   <kind value="[code]"/><!-- 1..1 applicability | start | stop -->
   <description value="[string]"/><!-- 0..1 Natural language description of the condition -->
   <language value="[string]"/><!-- 0..1 Language of the expression -->
   <expression value="[string]"/><!-- 0..1 Boolean-valued expression -->
  </condition>
  <input><!-- 0..* DataRequirement Input data requirements --></input>
  <output><!-- 0..* DataRequirement Output data definition --></output>
  <relatedAction>  <!-- 0..* Relationship to another action -->
   <actionId value="[id]"/><!-- 1..1 What action is this related to -->
   <relationship value="[code]"/><!-- 1..1 before-start | before | before-end | concurrent-with-start | concurrent |
    concurrent-with-end | after-start | after | after-end -->
   <offset[x]><!-- 0..1 Duration|Range Time offset for the relationship --></offset[x]>
  </relatedAction>
  <timing[x]><!-- 0..1 dateTime|Period|Duration|Range|Timing When the action should take place --></timing[x]>
  <participant>  <!-- 0..* Who should participate in the action -->
   <type value="[code]"/><!-- 1..1 patient | practitioner | related-person -->
   <role><!-- 0..1 CodeableConcept E.g. Nurse, Surgeon, Parent, etc --></role>
  </participant>
  <type><!-- 0..1 Coding create | update | remove | fire-event --></type>
  <groupingBehavior value="[code]"/><!-- 0..1 visual-group | logical-group | sentence-group -->
  <selectionBehavior value="[code]"/><!-- 0..1 any | all | all-or-none | exactly-one | at-most-one | one-or-more -->
  <requiredBehavior value="[code]"/><!-- 0..1 must | could | must-unless-documented -->
  <precheckBehavior value="[code]"/><!-- 0..1 yes | no -->
  <cardinalityBehavior value="[code]"/><!-- 0..1 single | multiple -->
  <definition><!-- 0..1 Reference(ActivityDefinition|PlanDefinition) Description of the activity to be performed
    --></definition>
  <transform><!-- 0..1 Reference(StructureMap) Transform to apply the template --></transform>
  <dynamicValue>  <!-- 0..* Dynamic aspects of the definition -->
   <description value="[string]"/><!-- 0..1 Natural language description of the dynamic value -->
   <path value="[string]"/><!-- 0..1 The path to the element to be set dynamically -->
   <language value="[string]"/><!-- 0..1 Language of the expression -->
   <expression value="[string]"/><!-- 0..1 An expression that provides the dynamic value for the customization -->
  </dynamicValue>
  <action><!-- 0..* Content as for PlanDefinition.action A sub-action --></action>
 </action>
</PlanDefinition>
```

JSON Template

```
{
  "resourceType" : "PlanDefinition",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "url" : "<uri>", // Logical URI to reference this plan definition (globally unique)
  "identifier" : [{ Identifier }], // Additional identifier for the plan definition
  "version" : "<string>", // Business version of the plan definition
  "name" : "<string>", // Name for this plan definition (computer friendly)
  "title" : "<string>", // Name for this plan definition (human friendly)
  "type" : { CodeableConcept }, // order-set | protocol | eca-rule
  "status" : "<code>", // R!  draft | active | retired | unknown
  "experimental" : <boolean>, // For testing purposes, not real usage
  "date" : "<dateTime>", // Date this was last changed
  "publisher" : "<string>", // Name of the publisher (organization or individual)
  "description" : "<markdown>", // Natural language description of the plan definition
  "purpose" : "<markdown>", // Why this plan definition is defined
  "usage" : "<string>", // Describes the clinical usage of the asset
  "approvalDate" : "<date>", // When the plan definition was approved by publisher
  "lastReviewDate" : "<date>", // When the plan definition was last reviewed
  "effectivePeriod" : { Period }, // When the plan definition is expected to be used
  "useContext" : [{ UsageContext }], // Context the content is intended to support
  "jurisdiction" : [{ CodeableConcept }], // Intended jurisdiction for plan definition (if applicable)
  "topic" : [{ CodeableConcept }], // E.g. Education, Treatment, Assessment, etc
  "contributor" : [{ Contributor }], // A content contributor
  "contact" : [{ ContactDetail }], // Contact details for the publisher
  "copyright" : "<markdown>", // Use and/or publishing restrictions
  "relatedArtifact" : [{ RelatedArtifact }], // Related artifacts for the asset
  "library" : [{ Reference(Library) }], // Logic used by the plan definition
  "goal" : [{ // What the plan is trying to accomplish
    "category" : { CodeableConcept }, // E.g. Treatment, dietary, behavioral, etc
    "description" : { CodeableConcept }, // R!  Code or text describing the goal
    "priority" : { CodeableConcept }, // high-priority | medium-priority | low-priority
    "start" : { CodeableConcept }, // When goal pursuit begins
    "addresses" : [{ CodeableConcept }], // What does the goal address
    "documentation" : [{ RelatedArtifact }], // Supporting documentation for the goal
    "target" : [{ // Target outcome for the goal
      "measure" : { CodeableConcept }, // The parameter whose value is to be tracked
      // detail[x]: The target value to be achieved. One of these 3:
      "detailQuantity" : { Quantity },
      "detailRange" : { Range },
      "detailCodeableConcept" : { CodeableConcept },
      "due" : { Duration } // Reach goal within
    }]
  }],
  "action" : [{ // Action defined by the plan
    "label" : "<string>", // User-visible label for the action (e.g. 1. or A.)
    "title" : "<string>", // User-visible title
    "description" : "<string>", // Short description of the action
```

```
    "textEquivalent" : "<string>", // Static text equivalent of the action, used if the dynamic aspects cannot be
    interpreted by the receiving system
    "code" : [{ CodeableConcept }], // Code representing the meaning of the action or sub-actions
    "reason" : [{ CodeableConcept }], // Why the action should be performed
    "documentation" : [{ RelatedArtifact }], // Supporting documentation for the intended performer of the action
    "goalId" : ["<id>"], // What goals this action supports
    "triggerDefinition" : [{ TriggerDefinition }], // When the action should be triggered
    "condition" : [{ // Whether or not the action is applicable
      "kind" : "<code>", // R!  applicability | start | stop
      "description" : "<string>", // Natural language description of the condition
      "language" : "<string>", // Language of the expression
      "expression" : "<string>" // Boolean-valued expression
    }],
    "input" : [{ DataRequirement }], // Input data requirements
    "output" : [{ DataRequirement }], // Output data definition
    "relatedAction" : [{ // Relationship to another action
      "actionId" : "<id>", // R!  What action is this related to
      "relationship" : "<code>", // R!  before-start | before | before-end | concurrent-with-start | concurrent |
    concurrent-with-end | after-start | after | after-end
      // offset[x]: Time offset for the relationship. One of these 2:
      "offsetDuration" : { Duration }
      "offsetRange" : { Range }
    }],
    // timing[x]: When the action should take place. One of these 5:
    "timingDateTime" : "<dateTime>",
    "timingPeriod" : { Period },
    "timingDuration" : { Duration },
    "timingRange" : { Range },
    "timingTiming" : { Timing },
    "participant" : [{ // Who should participate in the action
      "type" : "<code>", // R!  patient | practitioner | related-person
      "role" : { CodeableConcept } // E.g. Nurse, Surgeon, Parent, etc
    }],
    "type" : { Coding }, // create | update | remove | fire-event
    "groupingBehavior" : "<code>", // visual-group | logical-group | sentence-group
    "selectionBehavior" : "<code>", // any | all | all-or-none | exactly-one | at-most-one | one-or-more
    "requiredBehavior" : "<code>", // must | could | must-unless-documented
    "precheckBehavior" : "<code>", // yes | no
    "cardinalityBehavior" : "<code>", // single | multiple
    "definition" : { Reference(ActivityDefinition|PlanDefinition) }, // Description of the activity to be performed
    "transform" : { Reference(StructureMap) }, // Transform to apply the template
    "dynamicValue" : [{ // Dynamic aspects of the definition
      "description" : "<string>", // Natural language description of the dynamic value
      "path" : "<string>", // The path to the element to be set dynamically
      "language" : "<string>", // Language of the expression
      "expression" : "<string>" // An expression that provides the dynamic value for the customization
    }],
    "action" : [{ Content as for PlanDefinition.action }] // A sub-action
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:PlanDefinition;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:PlanDefinition.url [ uri ]; # 0..1 Logical URI to reference this plan definition (globally unique)
  fhir:PlanDefinition.identifier [ Identifier ], ... ; # 0..* Additional identifier for the plan definition
  fhir:PlanDefinition.version [ string ]; # 0..1 Business version of the plan definition
  fhir:PlanDefinition.name [ string ]; # 0..1 Name for this plan definition (computer friendly)
  fhir:PlanDefinition.title [ string ]; # 0..1 Name for this plan definition (human friendly)
  fhir:PlanDefinition.type [ CodeableConcept ]; # 0..1 order-set | protocol | eca-rule
  fhir:PlanDefinition.status [ code ]; # 1..1 draft | active | retired | unknown
  fhir:PlanDefinition.experimental [ boolean ]; # 0..1 For testing purposes, not real usage
  fhir:PlanDefinition.date [ dateTime ]; # 0..1 Date this was last changed
  fhir:PlanDefinition.publisher [ string ]; # 0..1 Name of the publisher (organization or individual)
  fhir:PlanDefinition.description [ markdown ]; # 0..1 Natural language description of the plan definition
  fhir:PlanDefinition.purpose [ markdown ]; # 0..1 Why this plan definition is defined
  fhir:PlanDefinition.usage [ string ]; # 0..1 Describes the clinical usage of the asset
  fhir:PlanDefinition.approvalDate [ date ]; # 0..1 When the plan definition was approved by publisher
  fhir:PlanDefinition.lastReviewDate [ date ]; # 0..1 When the plan definition was last reviewed
  fhir:PlanDefinition.effectivePeriod [ Period ]; # 0..1 When the plan definition is expected to be used
  fhir:PlanDefinition.useContext [ UsageContext ], ... ; # 0..* Context the content is intended to support
  fhir:PlanDefinition.jurisdiction [ CodeableConcept ], ... ; # 0..* Intended jurisdiction for plan definition (if
    applicable)
  fhir:PlanDefinition.topic [ CodeableConcept ], ... ; # 0..* E.g. Education, Treatment, Assessment, etc
  fhir:PlanDefinition.contributor [ Contributor ], ... ; # 0..* A content contributor
  fhir:PlanDefinition.contact [ ContactDetail ], ... ; # 0..* Contact details for the publisher
  fhir:PlanDefinition.copyright [ markdown ]; # 0..1 Use and/or publishing restrictions
  fhir:PlanDefinition.relatedArtifact [ RelatedArtifact ], ... ; # 0..* Related artifacts for the asset
  fhir:PlanDefinition.library [ Reference(Library) ], ... ; # 0..* Logic used by the plan definition
  fhir:PlanDefinition.goal [ # 0..* What the plan is trying to accomplish
    fhir:PlanDefinition.goal.category [ CodeableConcept ]; # 0..1 E.g. Treatment, dietary, behavioral, etc
    fhir:PlanDefinition.goal.description [ CodeableConcept ]; # 1..1 Code or text describing the goal
    fhir:PlanDefinition.goal.priority [ CodeableConcept ]; # 0..1 high-priority | medium-priority | low-priority
```

```
    fhir:PlanDefinition.goal.start [ CodeableConcept ]; # 0..1 When goal pursuit begins
    fhir:PlanDefinition.goal.addresses [ CodeableConcept ], ... ; # 0..* What does the goal address
    fhir:PlanDefinition.goal.documentation [ RelatedArtifact ], ... ; # 0..* Supporting documentation for the goal
    fhir:PlanDefinition.goal.target [ # 0..* Target outcome for the goal
      fhir:PlanDefinition.goal.target.measure [ CodeableConcept ]; # 0..1 The parameter whose value is to be tracked
      # PlanDefinition.goal.target.detail[x] : 0..1 The target value to be achieved. One of these 3
        fhir:PlanDefinition.goal.target.detailQuantity [ Quantity ]
        fhir:PlanDefinition.goal.target.detailRange [ Range ]
        fhir:PlanDefinition.goal.target.detailCodeableConcept [ CodeableConcept ]
      fhir:PlanDefinition.goal.target.due [ Duration ]; # 0..1 Reach goal within
    ], ...;
  ], ...;
  fhir:PlanDefinition.action [ # 0..* Action defined by the plan
    fhir:PlanDefinition.action.label [ string ]; # 0..1 User-visible label for the action (e.g. 1. or A.)
    fhir:PlanDefinition.action.title [ string ]; # 0..1 User-visible title
    fhir:PlanDefinition.action.description [ string ]; # 0..1 Short description of the action
    fhir:PlanDefinition.action.textEquivalent [ string ]; # 0..1 Static text equivalent of the action, used if the
    dynamic aspects cannot be interpreted by the receiving system
    fhir:PlanDefinition.action.code [ CodeableConcept ], ... ; # 0..* Code representing the meaning of the action or
    sub-actions
    fhir:PlanDefinition.action.reason [ CodeableConcept ], ... ; # 0..* Why the action should be performed
    fhir:PlanDefinition.action.documentation [ RelatedArtifact ], ... ; # 0..* Supporting documentation for the intended
     performer of the action
    fhir:PlanDefinition.action.goalId [ id ], ... ; # 0..* What goals this action supports
    fhir:PlanDefinition.action.triggerDefinition [ TriggerDefinition ], ... ; # 0..* When the action should be triggered
    fhir:PlanDefinition.action.condition [ # 0..* Whether or not the action is applicable
      fhir:PlanDefinition.action.condition.kind [ code ]; # 1..1 applicability | start | stop
      fhir:PlanDefinition.action.condition.description [ string ]; # 0..1 Natural language description of the condition
      fhir:PlanDefinition.action.condition.language [ string ]; # 0..1 Language of the expression
      fhir:PlanDefinition.action.condition.expression [ string ]; # 0..1 Boolean-valued expression
    ], ...;
    fhir:PlanDefinition.action.input [ DataRequirement ], ... ; # 0..* Input data requirements
    fhir:PlanDefinition.action.output [ DataRequirement ], ... ; # 0..* Output data definition
    fhir:PlanDefinition.action.relatedAction [ # 0..* Relationship to another action
      fhir:PlanDefinition.action.relatedAction.actionId [ id ]; # 1..1 What action is this related to
      fhir:PlanDefinition.action.relatedAction.relationship [ code ]; # 1..1 before-start | before | before-end |
    concurrent-with-start | concurrent | concurrent-with-end | after-start | after | after-end
      # PlanDefinition.action.relatedAction.offset[x] : 0..1 Time offset for the relationship. One of these 2
        fhir:PlanDefinition.action.relatedAction.offsetDuration [ Duration ]
        fhir:PlanDefinition.action.relatedAction.offsetRange [ Range ]
    ], ...;
    # PlanDefinition.action.timing[x] : 0..1 When the action should take place. One of these 5
      fhir:PlanDefinition.action.timingDateTime [ dateTime ]
      fhir:PlanDefinition.action.timingPeriod [ Period ]
      fhir:PlanDefinition.action.timingDuration [ Duration ]
      fhir:PlanDefinition.action.timingRange [ Range ]
      fhir:PlanDefinition.action.timingTiming [ Timing ]
    fhir:PlanDefinition.action.participant [ # 0..* Who should participate in the action
      fhir:PlanDefinition.action.participant.type [ code ]; # 1..1 patient | practitioner | related-person
      fhir:PlanDefinition.action.participant.role [ CodeableConcept ]; # 0..1 E.g. Nurse, Surgeon, Parent, etc
    ], ...;
    fhir:PlanDefinition.action.type [ Coding ]; # 0..1 create | update | remove | fire-event
    fhir:PlanDefinition.action.groupingBehavior [ code ]; # 0..1 visual-group | logical-group | sentence-group
    fhir:PlanDefinition.action.selectionBehavior [ code ]; # 0..1 any | all | all-or-none | exactly-one | at-most-one |
    one-or-more
    fhir:PlanDefinition.action.requiredBehavior [ code ]; # 0..1 must | could | must-unless-documented
    fhir:PlanDefinition.action.precheckBehavior [ code ]; # 0..1 yes | no
    fhir:PlanDefinition.action.cardinalityBehavior [ code ]; # 0..1 single | multiple
    fhir:PlanDefinition.action.definition [ Reference(ActivityDefinition|PlanDefinition) ]; # 0..1 Description of the
    activity to be performed
    fhir:PlanDefinition.action.transform [ Reference(StructureMap) ]; # 0..1 Transform to apply the template
    fhir:PlanDefinition.action.dynamicValue [ # 0..* Dynamic aspects of the definition
      fhir:PlanDefinition.action.dynamicValue.description [ string ]; # 0..1 Natural language description of the dynamic
     value
      fhir:PlanDefinition.action.dynamicValue.path [ string ]; # 0..1 The path to the element to be set dynamically
      fhir:PlanDefinition.action.dynamicValue.language [ string ]; # 0..1 Language of the expression
      fhir:PlanDefinition.action.dynamicValue.expression [ string ]; # 0..1 An expression that provides the dynamic
    value for the customization
    ], ...;
    fhir:PlanDefinition.action.action [ See PlanDefinition.action ], ... ; # 0..* A sub-action
  ], ...;
]
```

Changes since DSTU2

This resource did not exist in Release 2

This analysis is available as XML or JSON.

Structure

Name | Flags | Card. | Type | Description & Constraints

PlanDefinition | | | DomainResource | The definition of a plan for a series of actions, independent of any specific patient or context Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension

url | ? | 0..1 | uri | Logical URI to reference this plan definition (globally unique)

identifier | ? | 0..* | Identifier | Additional identifier for the plan definition

version | ? | 0..1 | string | Business version of the plan definition

name | ? | 0..1 | string | Name for this plan definition (computer friendly)

title | ? | 0..1 | string | Name for this plan definition (human friendly)

type | ? | 0..1 | CodeableConcept | order-set | protocol | eca-rule PlanDefinitionType (Extensible)
status | ?!? | 1..1 | code | draft | active | retired | unknown PublicationStatus (Required)
experimental | ?!? | 0..1 | boolean | For testing purposes, not real usage
date | ? | 0..1 | dateTime | Date this was last changed
publisher | ? | 0..1 | string | Name of the publisher (organization or individual)
description | ? | 0..1 | markdown | Natural language description of the plan definition
purpose | | 0..1 | markdown | Why this plan definition is defined
usage | | 0..1 | string | Describes the clinical usage of the asset
approvalDate | | 0..1 | date | When the plan definition was approved by publisher
lastReviewDate | | 0..1 | date | When the plan definition was last reviewed
effectivePeriod | ? | 0..1 | Period | When the plan definition is expected to be used
useContext | ? | 0..* | UsageContext | Context the content is intended to support
jurisdiction | ? | 0..* | CodeableConcept | Intended jurisdiction for plan definition (if applicable) Jurisdiction ValueSet (Extensible)
topic | | 0..* | CodeableConcept | E.g. Education, Treatment, Assessment, etc DefinitionTopic (Example)
contributor | | 0..* | Contributor | A content contributor
contact | ? | 0..* | ContactDetail | Contact details for the publisher
copyright | | 0..1 | markdown | Use and/or publishing restrictions
relatedArtifact | | 0..* | RelatedArtifact | Related artifacts for the asset
library | | 0..* | Reference(Library) | Logic used by the plan definition
goal | | 0..* | BackboneElement | What the plan is trying to accomplish
category | | 0..1 | CodeableConcept | E.g. Treatment, dietary, behavioral, etc GoalCategory (Example)
description | | 1..1 | CodeableConcept | Code or text describing the goal SNOMED CT Clinical Findings (Example)
priority | | 0..1 | CodeableConcept | high-priority | medium-priority | low-priority GoalPriority (Preferred)
start | | 0..1 | CodeableConcept | When goal pursuit begins GoalStartEvent (Example)
addresses | | 0..* | CodeableConcept | What does the goal address Condition/Problem/Diagnosis Codes (Example)
documentation | | 0..* | RelatedArtifact | Supporting documentation for the goal
target | | 0..* | BackboneElement | Target outcome for the goal
measure | | 0..1 | CodeableConcept | The parameter whose value is to be tracked LOINC Codes (Example)
detail[x] | | 0..1 | | The target value to be achieved
detailQuantity | | | Quantity
detailRange | | | Range
detailCodeableConcept | | | CodeableConcept
due | | 0..1 | Duration | Reach goal within
action | | 0..* | BackboneElement | Action defined by the plan
label | | 0..1 | string | User-visible label for the action (e.g. 1. or A.)
title | | 0..1 | string | User-visible title
description | | 0..1 | string | Short description of the action
textEquivalent | | 0..1 | string | Static text equivalent of the action, used if the dynamic aspects cannot be interpreted by the receiving system
code | | 0..* | CodeableConcept | Code representing the meaning of the action or sub-actions
reason | | 0..* | CodeableConcept | Why the action should be performed
documentation | | 0..* | RelatedArtifact | Supporting documentation for the intended performer of the action
goalId | | 0..* | id | What goals this action supports
triggerDefinition | | 0..* | TriggerDefinition | When the action should be triggered
condition | | 0..* | BackboneElement | Whether or not the action is applicable
kind | | 1..1 | code | applicability | start | stop ActionConditionKind (Required)
description | | 0..1 | string | Natural language description of the condition
language | | 0..1 | string | Language of the expression
expression | | 0..1 | string | Boolean-valued expression
input | | 0..* | DataRequirement | Input data requirements
output | | 0..* | DataRequirement | Output data definition
relatedAction | | 0..* | BackboneElement | Relationship to another action
actionId | | 1..1 | id | What action is this related to
relationship | | 1..1 | code | before-start | before | before-end | concurrent-with-start | concurrent | concurrent-with-end | after-start | after | after-end ActionRelationshipType (Required)
offset[x] | | 0..1 | | Time offset for the relationship
offsetDuration | | | Duration
offsetRange | | | Range
timing[x] | | 0..1 | | When the action should take place
timingDateTime | | | dateTime
timingPeriod | | | Period
timingDuration | | | Duration
timingRange | | | Range
timingTiming | | | Timing
participant | | 0..* | BackboneElement | Who should participate in the action
type | | 1..1 | code | patient | practitioner | related-person ActionParticipantType (Required)
role | | 0..1 | CodeableConcept | E.g. Nurse, Surgeon, Parent, etc ActionParticipantRole (Example)
type | | 0..1 | Coding | create | update | remove | fire-event ActionType (Required)
groupingBehavior | | 0..1 | code | visual-group | logical-group | sentence-group ActionGroupingBehavior (Required)
selectionBehavior | | 0..1 | code | any | all | all-or-none | exactly-one | at-most-one | one-or-more ActionSelectionBehavior (Required)
requiredBehavior | | 0..1 | code | must | could | must-unless-documented ActionRequiredBehavior (Required)
precheckBehavior | | 0..1 | code | yes | no ActionPrecheckBehavior (Required)
cardinalityBehavior | | 0..1 | code | single | multiple ActionCardinalityBehavior (Required)

definition | | 0..1 | Reference(ActivityDefinition | PlanDefinition) | Description of the activity to be performed
transform | | 0..1 | Reference(StructureMap) | Transform to apply the template
dynamicValue | | 0..* | BackboneElement | Dynamic aspects of the definition
description | | 0..1 | string | Natural language description of the dynamic value
path | | 0..1 | string | The path to the element to be set dynamically
language | | 0..1 | string | Language of the expression
expression | | 0..1 | string | An expression that provides the dynamic value for the customization
action | | 0..* | see action | A sub-action
Documentation for this format

UML Diagram (Legend)

XML Template

```
<PlanDefinition xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <url value="[uri]"/><!-- 0..1 Logical URI to reference this plan definition (globally unique) -->
 <identifier><!-- 0..* Identifier Additional identifier for the plan definition --></identifier>
 <version value="[string]"/><!-- 0..1 Business version of the plan definition -->
 <name value="[string]"/><!-- 0..1 Name for this plan definition (computer friendly) -->
 <title value="[string]"/><!-- 0..1 Name for this plan definition (human friendly) -->
 <type><!-- 0..1 CodeableConcept order-set | protocol | eca-rule --></type>
 <status value="[code]"/><!-- 1..1 draft | active | retired | unknown -->
 <experimental value="[boolean]"/><!-- 0..1 For testing purposes, not real usage -->
 <date value="[dateTime]"/><!-- 0..1 Date this was last changed -->
 <publisher value="[string]"/><!-- 0..1 Name of the publisher (organization or individual) -->
 <description value="[markdown]"/><!-- 0..1 Natural language description of the plan definition -->
 <purpose value="[markdown]"/><!-- 0..1 Why this plan definition is defined -->
 <usage value="[string]"/><!-- 0..1 Describes the clinical usage of the asset -->
 <approvalDate value="[date]"/><!-- 0..1 When the plan definition was approved by publisher -->
 <lastReviewDate value="[date]"/><!-- 0..1 When the plan definition was last reviewed -->
 <effectivePeriod><!-- 0..1 Period When the plan definition is expected to be used --></effectivePeriod>
 <useContext><!-- 0..* UsageContext Context the content is intended to support --></useContext>
 <jurisdiction><!-- 0..* CodeableConcept Intended jurisdiction for plan definition (if applicable) --></jurisdiction>
 <topic><!-- 0..* CodeableConcept E.g. Education, Treatment, Assessment, etc --></topic>
 <contributor><!-- 0..* Contributor A content contributor --></contributor>
 <contact><!-- 0..* ContactDetail Contact details for the publisher --></contact>
 <copyright value="[markdown]"/><!-- 0..1 Use and/or publishing restrictions -->
 <relatedArtifact><!-- 0..* RelatedArtifact Related artifacts for the asset --></relatedArtifact>
 <library><!-- 0..* Reference(Library) Logic used by the plan definition --></library>
 <goal>  <!-- 0..* What the plan is trying to accomplish -->
  <category><!-- 0..1 CodeableConcept E.g. Treatment, dietary, behavioral, etc --></category>
  <description><!-- 1..1 CodeableConcept Code or text describing the goal --></description>
  <priority><!-- 0..1 CodeableConcept high-priority | medium-priority | low-priority --></priority>
  <start><!-- 0..1 CodeableConcept When goal pursuit begins --></start>
  <addresses><!-- 0..* CodeableConcept What does the goal address --></addresses>
  <documentation><!-- 0..* RelatedArtifact Supporting documentation for the goal --></documentation>
  <target>  <!-- 0..* Target outcome for the goal -->
   <measure><!-- 0..1 CodeableConcept The parameter whose value is to be tracked --></measure>
   <detail[x]><!-- 0..1 Quantity|Range|CodeableConcept The target value to be achieved --></detail[x]>
   <due><!-- 0..1 Duration Reach goal within --></due>
  </target>
 </goal>
 <action>  <!-- 0..* Action defined by the plan -->
  <label value="[string]"/><!-- 0..1 User-visible label for the action (e.g. 1. or A.) -->
  <title value="[string]"/><!-- 0..1 User-visible title -->
  <description value="[string]"/><!-- 0..1 Short description of the action -->
  <textEquivalent value="[string]"/><!-- 0..1 Static text equivalent of the action, used if the dynamic aspects cannot
    be interpreted by the receiving system -->
  <code><!-- 0..* CodeableConcept Code representing the meaning of the action or sub-actions --></code>
  <reason><!-- 0..* CodeableConcept Why the action should be performed --></reason>
  <documentation><!-- 0..* RelatedArtifact Supporting documentation for the intended performer of the action
    --></documentation>
  <goalId value="[id]"/><!-- 0..* What goals this action supports -->
  <triggerDefinition><!-- 0..* TriggerDefinition When the action should be triggered --></triggerDefinition>
  <condition>  <!-- 0..* Whether or not the action is applicable -->
   <kind value="[code]"/><!-- 1..1 applicability | start | stop -->
   <description value="[string]"/><!-- 0..1 Natural language description of the condition -->
   <language value="[string]"/><!-- 0..1 Language of the expression -->
   <expression value="[string]"/><!-- 0..1 Boolean-valued expression -->
  </condition>
  <input><!-- 0..* DataRequirement Input data requirements --></input>
  <output><!-- 0..* DataRequirement Output data definition --></output>
  <relatedAction>  <!-- 0..* Relationship to another action -->
   <actionId value="[id]"/><!-- 1..1 What action is this related to -->
   <relationship value="[code]"/><!-- 1..1 before-start | before | before-end | concurrent-with-start | concurrent |
    concurrent-with-end | after-start | after | after-end -->
   <offset[x]><!-- 0..1 Duration|Range Time offset for the relationship --></offset[x]>
  </relatedAction>
  <timing[x]><!-- 0..1 dateTime|Period|Duration|Range|Timing When the action should take place --></timing[x]>
  <participant>  <!-- 0..* Who should participate in the action -->
   <type value="[code]"/><!-- 1..1 patient | practitioner | related-person -->
   <role><!-- 0..1 CodeableConcept E.g. Nurse, Surgeon, Parent, etc --></role>
  </participant>
  <type><!-- 0..1 Coding create | update | remove | fire-event --></type>
  <groupingBehavior value="[code]"/><!-- 0..1 visual-group | logical-group | sentence-group -->
  <selectionBehavior value="[code]"/><!-- 0..1 any | all | all-or-none | exactly-one | at-most-one | one-or-more -->
```

```
  <requiredBehavior value="[code]"/><!-- 0..1 must | could | must-unless-documented -->
  <precheckBehavior value="[code]"/><!-- 0..1 yes | no -->
  <cardinalityBehavior value="[code]"/><!-- 0..1 single | multiple -->
  <definition><!-- 0..1 Reference(ActivityDefinition|PlanDefinition) Description of the activity to be performed
    --></definition>
  <transform><!-- 0..1 Reference(StructureMap) Transform to apply the template --></transform>
  <dynamicValue>  <!-- 0..* Dynamic aspects of the definition -->
   <description value="[string]"/><!-- 0..1 Natural language description of the dynamic value -->
   <path value="[string]"/><!-- 0..1 The path to the element to be set dynamically -->
   <language value="[string]"/><!-- 0..1 Language of the expression -->
   <expression value="[string]"/><!-- 0..1 An expression that provides the dynamic value for the customization -->
  </dynamicValue>
  <action><!-- 0..* Content as for PlanDefinition.action A sub-action --></action>
 </action>
</PlanDefinition>
```

JSON Template

```
{
  "resourceType" : "PlanDefinition",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "url" : "<uri>", // Logical URI to reference this plan definition (globally unique)
  "identifier" : [{ Identifier }], // Additional identifier for the plan definition
  "version" : "<string>", // Business version of the plan definition
  "name" : "<string>", // Name for this plan definition (computer friendly)
  "title" : "<string>", // Name for this plan definition (human friendly)
  "type" : { CodeableConcept }, // order-set | protocol | eca-rule
  "status" : "<code>", // R!  draft | active | retired | unknown
  "experimental" : <boolean>, // For testing purposes, not real usage
  "date" : "<dateTime>", // Date this was last changed
  "publisher" : "<string>", // Name of the publisher (organization or individual)
  "description" : "<markdown>", // Natural language description of the plan definition
  "purpose" : "<markdown>", // Why this plan definition is defined
  "usage" : "<string>", // Describes the clinical usage of the asset
  "approvalDate" : "<date>", // When the plan definition was approved by publisher
  "lastReviewDate" : "<date>", // When the plan definition was last reviewed
  "effectivePeriod" : { Period }, // When the plan definition is expected to be used
  "useContext" : [{ UsageContext }], // Context the content is intended to support
  "jurisdiction" : [{ CodeableConcept }], // Intended jurisdiction for plan definition (if applicable)
  "topic" : [{ CodeableConcept }], // E.g. Education, Treatment, Assessment, etc
  "contributor" : [{ Contributor }], // A content contributor
  "contact" : [{ ContactDetail }], // Contact details for the publisher
  "copyright" : "<markdown>", // Use and/or publishing restrictions
  "relatedArtifact" : [{ RelatedArtifact }], // Related artifacts for the asset
  "library" : [{ Reference(Library) }], // Logic used by the plan definition
  "goal" : [{ // What the plan is trying to accomplish
    "category" : { CodeableConcept }, // E.g. Treatment, dietary, behavioral, etc
    "description" : { CodeableConcept }, // R!  Code or text describing the goal
    "priority" : { CodeableConcept }, // high-priority | medium-priority | low-priority
    "start" : { CodeableConcept }, // When goal pursuit begins
    "addresses" : [{ CodeableConcept }], // What does the goal address
    "documentation" : [{ RelatedArtifact }], // Supporting documentation for the goal
    "target" : [{ // Target outcome for the goal
      "measure" : { CodeableConcept }, // The parameter whose value is to be tracked
      // detail[x]: The target value to be achieved. One of these 3:
      "detailQuantity" : { Quantity },
      "detailRange" : { Range },
      "detailCodeableConcept" : { CodeableConcept },
      "due" : { Duration } // Reach goal within
    }]
  }],
  "action" : [{ // Action defined by the plan
    "label" : "<string>", // User-visible label for the action (e.g. 1. or A.)
    "title" : "<string>", // User-visible title
    "description" : "<string>", // Short description of the action
    "textEquivalent" : "<string>", // Static text equivalent of the action, used if the dynamic aspects cannot be
    interpreted by the receiving system
    "code" : [{ CodeableConcept }], // Code representing the meaning of the action or sub-actions
    "reason" : [{ CodeableConcept }], // Why the action should be performed
    "documentation" : [{ RelatedArtifact }], // Supporting documentation for the intended performer of the action
    "goalId" : ["<id>"], // What goals this action supports
    "triggerDefinition" : [{ TriggerDefinition }], // When the action should be triggered
    "condition" : [{ // Whether or not the action is applicable
      "kind" : "<code>", // R!  applicability | start | stop
      "description" : "<string>", // Natural language description of the condition
      "language" : "<string>", // Language of the expression
      "expression" : "<string>" // Boolean-valued expression
    }],
    "input" : [{ DataRequirement }], // Input data requirements
    "output" : [{ DataRequirement }], // Output data definition
    "relatedAction" : [{ // Relationship to another action
      "actionId" : "<id>", // R!  What action is this related to
      "relationship" : "<code>", // R!  before-start | before | before-end | concurrent-with-start | concurrent |
    concurrent-with-end | after-start | after | after-end
      // offset[x]: Time offset for the relationship. One of these 2:
      "offsetDuration" : { Duration }
      "offsetRange" : { Range }
    }],
    // timing[x]: When the action should take place. One of these 5:
```

```
    "timingDateTime" : "<dateTime>",
    "timingPeriod" : { Period },
    "timingDuration" : { Duration },
    "timingRange" : { Range },
    "timingTiming" : { Timing },
    "participant" : [{ // Who should participate in the action
      "type" : "<code>", // R!  patient | practitioner | related-person
      "role" : { CodeableConcept } // E.g. Nurse, Surgeon, Parent, etc
    }],
    "type" : { Coding }, // create | update | remove | fire-event
    "groupingBehavior" : "<code>", // visual-group | logical-group | sentence-group
    "selectionBehavior" : "<code>", // any | all | all-or-none | exactly-one | at-most-one | one-or-more
    "requiredBehavior" : "<code>", // must | could | must-unless-documented
    "precheckBehavior" : "<code>", // yes | no
    "cardinalityBehavior" : "<code>", // single | multiple
    "definition" : { Reference(ActivityDefinition|PlanDefinition) }, // Description of the activity to be performed
    "transform" : { Reference(StructureMap) }, // Transform to apply the template
    "dynamicValue" : [{ // Dynamic aspects of the definition
      "description" : "<string>", // Natural language description of the dynamic value
      "path" : "<string>", // The path to the element to be set dynamically
      "language" : "<string>", // Language of the expression
      "expression" : "<string>" // An expression that provides the dynamic value for the customization
    }],
    "action" : [{ Content as for PlanDefinition.action }] // A sub-action
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:PlanDefinition;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:PlanDefinition.url [ uri ]; # 0..1 Logical URI to reference this plan definition (globally unique)
  fhir:PlanDefinition.identifier [ Identifier ], ... ; # 0..* Additional identifier for the plan definition
  fhir:PlanDefinition.version [ string ]; # 0..1 Business version of the plan definition
  fhir:PlanDefinition.name [ string ]; # 0..1 Name for this plan definition (computer friendly)
  fhir:PlanDefinition.title [ string ]; # 0..1 Name for this plan definition (human friendly)
  fhir:PlanDefinition.type [ CodeableConcept ]; # 0..1 order-set | protocol | eca-rule
  fhir:PlanDefinition.status [ code ]; # 1..1 draft | active | retired | unknown
  fhir:PlanDefinition.experimental [ boolean ]; # 0..1 For testing purposes, not real usage
  fhir:PlanDefinition.date [ dateTime ]; # 0..1 Date this was last changed
  fhir:PlanDefinition.publisher [ string ]; # 0..1 Name of the publisher (organization or individual)
  fhir:PlanDefinition.description [ markdown ]; # 0..1 Natural language description of the plan definition
  fhir:PlanDefinition.purpose [ markdown ]; # 0..1 Why this plan definition is defined
  fhir:PlanDefinition.usage [ string ]; # 0..1 Describes the clinical usage of the asset
  fhir:PlanDefinition.approvalDate [ date ]; # 0..1 When the plan definition was approved by publisher
  fhir:PlanDefinition.lastReviewDate [ date ]; # 0..1 When the plan definition was last reviewed
  fhir:PlanDefinition.effectivePeriod [ Period ]; # 0..1 When the plan definition is expected to be used
  fhir:PlanDefinition.useContext [ UsageContext ], ... ; # 0..* Context the content is intended to support
  fhir:PlanDefinition.jurisdiction [ CodeableConcept ], ... ; # 0..* Intended jurisdiction for plan definition (if
    applicable)
  fhir:PlanDefinition.topic [ CodeableConcept ], ... ; # 0..* E.g. Education, Treatment, Assessment, etc
  fhir:PlanDefinition.contributor [ Contributor ], ... ; # 0..* A content contributor
  fhir:PlanDefinition.contact [ ContactDetail ], ... ; # 0..* Contact details for the publisher
  fhir:PlanDefinition.copyright [ markdown ]; # 0..1 Use and/or publishing restrictions
  fhir:PlanDefinition.relatedArtifact [ RelatedArtifact ], ... ; # 0..* Related artifacts for the asset
  fhir:PlanDefinition.library [ Reference(Library) ], ... ; # 0..* Logic used by the plan definition
  fhir:PlanDefinition.goal [ # 0..* What the plan is trying to accomplish
    fhir:PlanDefinition.goal.category [ CodeableConcept ]; # 0..1 E.g. Treatment, dietary, behavioral, etc
    fhir:PlanDefinition.goal.description [ CodeableConcept ]; # 1..1 Code or text describing the goal
    fhir:PlanDefinition.goal.priority [ CodeableConcept ]; # 0..1 high-priority | medium-priority | low-priority
    fhir:PlanDefinition.goal.start [ CodeableConcept ]; # 0..1 When goal pursuit begins
    fhir:PlanDefinition.goal.addresses [ CodeableConcept ], ... ; # 0..* What does the goal address
    fhir:PlanDefinition.goal.documentation [ RelatedArtifact ], ... ; # 0..* Supporting documentation for the goal
    fhir:PlanDefinition.goal.target [ # 0..* Target outcome for the goal
      fhir:PlanDefinition.goal.target.measure [ CodeableConcept ]; # 0..1 The parameter whose value is to be tracked
      # PlanDefinition.goal.target.detail[x] : 0..1 The target value to be achieved. One of these 3
        fhir:PlanDefinition.goal.target.detailQuantity [ Quantity ]
        fhir:PlanDefinition.goal.target.detailRange [ Range ]
        fhir:PlanDefinition.goal.target.detailCodeableConcept [ CodeableConcept ]
      fhir:PlanDefinition.goal.target.due [ Duration ]; # 0..1 Reach goal within
    ], ...;
  ], ...;
  fhir:PlanDefinition.action [ # 0..* Action defined by the plan
    fhir:PlanDefinition.action.label [ string ]; # 0..1 User-visible label for the action (e.g. 1. or A.)
    fhir:PlanDefinition.action.title [ string ]; # 0..1 User-visible title
    fhir:PlanDefinition.action.description [ string ]; # 0..1 Short description of the action
    fhir:PlanDefinition.action.textEquivalent [ string ]; # 0..1 Static text equivalent of the action, used if the
    dynamic aspects cannot be interpreted by the receiving system
    fhir:PlanDefinition.action.code [ CodeableConcept ], ... ; # 0..* Code representing the meaning of the action or
    sub-actions
    fhir:PlanDefinition.action.reason [ CodeableConcept ], ... ; # 0..* Why the action should be performed
    fhir:PlanDefinition.action.documentation [ RelatedArtifact ], ... ; # 0..* Supporting documentation for the intended
     performer of the action
    fhir:PlanDefinition.action.goalId [ id ], ... ; # 0..* What goals this action supports
```

```
    fhir:PlanDefinition.action.triggerDefinition [ TriggerDefinition ], ... ; # 0..* When the action should be triggered
    fhir:PlanDefinition.action.condition [ # 0..* Whether or not the action is applicable
      fhir:PlanDefinition.action.condition.kind [ code ]; # 1..1 applicability | start | stop
      fhir:PlanDefinition.action.condition.description [ string ]; # 0..1 Natural language description of the condition
      fhir:PlanDefinition.action.condition.language [ string ]; # 0..1 Language of the expression
      fhir:PlanDefinition.action.condition.expression [ string ]; # 0..1 Boolean-valued expression
    ], ...;
    fhir:PlanDefinition.action.input [ DataRequirement ], ... ; # 0..* Input data requirements
    fhir:PlanDefinition.action.output [ DataRequirement ], ... ; # 0..* Output data definition
    fhir:PlanDefinition.action.relatedAction [ # 0..* Relationship to another action
      fhir:PlanDefinition.action.relatedAction.actionId [ id ]; # 1..1 What action is this related to
      fhir:PlanDefinition.action.relatedAction.relationship [ code ]; # 1..1 before-start | before | before-end |
    concurrent-with-start | concurrent | concurrent-with-end | after-start | after | after-end
      # PlanDefinition.action.relatedAction.offset[x] : 0..1 Time offset for the relationship. One of these 2
        fhir:PlanDefinition.action.relatedAction.offsetDuration [ Duration ]
        fhir:PlanDefinition.action.relatedAction.offsetRange [ Range ]
    ], ...;
    # PlanDefinition.action.timing[x] : 0..1 When the action should take place. One of these 5
      fhir:PlanDefinition.action.timingDateTime [ dateTime ]
      fhir:PlanDefinition.action.timingPeriod [ Period ]
      fhir:PlanDefinition.action.timingDuration [ Duration ]
      fhir:PlanDefinition.action.timingRange [ Range ]
      fhir:PlanDefinition.action.timingTiming [ Timing ]
    fhir:PlanDefinition.action.participant [ # 0..* Who should participate in the action
      fhir:PlanDefinition.action.participant.type [ code ]; # 1..1 patient | practitioner | related-person
      fhir:PlanDefinition.action.participant.role [ CodeableConcept ]; # 0..1 E.g. Nurse, Surgeon, Parent, etc
    ], ...;
    fhir:PlanDefinition.action.type [ Coding ]; # 0..1 create | update | remove | fire-event
    fhir:PlanDefinition.action.groupingBehavior [ code ]; # 0..1 visual-group | logical-group | sentence-group
    fhir:PlanDefinition.action.selectionBehavior [ code ]; # 0..1 any | all | all-or-none | exactly-one | at-most-one |
    one-or-more
    fhir:PlanDefinition.action.requiredBehavior [ code ]; # 0..1 must | could | must-unless-documented
    fhir:PlanDefinition.action.precheckBehavior [ code ]; # 0..1 yes | no
    fhir:PlanDefinition.action.cardinalityBehavior [ code ]; # 0..1 single | multiple
    fhir:PlanDefinition.action.definition [ Reference(ActivityDefinition|PlanDefinition) ]; # 0..1 Description of the
    activity to be performed
    fhir:PlanDefinition.action.transform [ Reference(StructureMap) ]; # 0..1 Transform to apply the template
    fhir:PlanDefinition.action.dynamicValue [ # 0..* Dynamic aspects of the definition
      fhir:PlanDefinition.action.dynamicValue.description [ string ]; # 0..1 Natural language description of the dynamic
     value
      fhir:PlanDefinition.action.dynamicValue.path [ string ]; # 0..1 The path to the element to be set dynamically
      fhir:PlanDefinition.action.dynamicValue.language [ string ]; # 0..1 Language of the expression
      fhir:PlanDefinition.action.dynamicValue.expression [ string ]; # 0..1 An expression that provides the dynamic
    value for the customization
    ], ...;
    fhir:PlanDefinition.action.action [ See PlanDefinition.action ], ... ; # 0..* A sub-action
  ], ...;
]
```

Changes since DSTU2

This resource did not exist in Release 2

This analysis is available as XML or JSON.

Alternate definitions: Master Definition (XML, JSON), XML Schema/Schematron (for) + JSON Schema, ShEx (for Turtle)

12.19.3.1 Terminology Bindings

Path | Definition | Type | Reference

PlanDefinition.type | The type of PlanDefinition | Extensible | PlanDefinitionType

PlanDefinition.status | The lifecycle status of a Value Set or Concept Map. | Required | PublicationStatus

PlanDefinition.jurisdiction | Countries and regions within which this artifact is targeted for use | Extensible | Jurisdiction ValueSet

PlanDefinition.topic | High-level categorization of the definition, used for searching, sorting, and filtering | Example | DefinitionTopic

PlanDefinition.goal.category | Example codes for grouping goals for filtering or presentation | Example | GoalCategory

PlanDefinition.goal.description | Describes goals that can be achieved | Example | SNOMED CT Clinical Findings

PlanDefinition.goal.priority | Indicates the level of importance associated with reaching or sustaining a goal | Preferred | GoalPriority

PlanDefinition.goal.start | Identifies the types of events that might trigger the start of a goal | Example | GoalStartEvent

PlanDefinition.goal.addresses | Identifies problems, conditions, issues, or concerns that goals may address | Example | Condition/Problem/Diagnosis Codes

PlanDefinition.goal.target.measure | Identifies types of parameters that can be tracked to determine goal achievement | Example | LOINC Codes

PlanDefinition.action.condition.kind | Defines the kinds of conditions that can appear on actions | Required | ActionConditionKind

PlanDefinition.action.relatedAction.relationship | Defines the types of relationships between actions | Required | ActionRelationshipType

PlanDefinition.action.participant.type | The type of participant for the action | Required | ActionParticipantType

PlanDefinition.action.participant.role | Defines roles played by participants for the action | Example | ActionParticipantRole

PlanDefinition.action.type | The type of action to be performed | Required | ActionType

PlanDefinition.action.groupingBehavior | Defines organization behavior of a group | Required | ActionGroupingBehavior

PlanDefinition.action.selectionBehavior | Defines selection behavior of a group | Required | ActionSelectionBehavior

PlanDefinition.action.requiredBehavior | Defines requiredness behavior for selecting an action or an action group | Required | ActionRequiredBehavior

PlanDefinition.action.precheckBehavior | Defines selection frequency behavior for an action or group | Required | ActionPrecheckBehavior

PlanDefinition.action.cardinalityBehavior | Defines behavior for an action or a group for how many times that item may be repeated | Required | ActionCardinalityBehavior

12.19.4 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
composed-of | reference | What resource is being referenced | PlanDefinition.relatedArtifact.where(type='composed-of').resource (Any)
date | date | The plan definition publication date | PlanDefinition.date
depends-on | reference | What resource is being referenced | PlanDefinition.relatedArtifact.where(type='depends-on').resource | PlanDefinition.library (Library, Any)
derived-from | reference | What resource is being referenced | PlanDefinition.relatedArtifact.where(type='derived-from').resource (Any)
description | string | The description of the plan definition | PlanDefinition.description
effective | date | The time during which the plan definition is intended to be in use | PlanDefinition.effectivePeriod
identifier | token | External identifier for the plan definition | PlanDefinition.identifier
jurisdiction | token | Intended jurisdiction for the plan definition | PlanDefinition.jurisdiction
name | string | Computationally friendly name of the plan definition | PlanDefinition.name
predecessor | reference | What resource is being referenced | PlanDefinition.relatedArtifact.where(type='predecessor').resource (Any)
publisher | string | Name of the publisher of the plan definition | PlanDefinition.publisher
status | token | The current status of the plan definition | PlanDefinition.status
successor | reference | What resource is being referenced | PlanDefinition.relatedArtifact.where(type='successor').resource (Any)
title | string | The human-friendly name of the plan definition | PlanDefinition.title
topic | token | Topics associated with the module | PlanDefinition.topic
url | uri | The uri that identifies the plan definition | PlanDefinition.url
version | token | The business version of the plan definition | PlanDefinition.version

12.19.5 Resource PlanDefinition - Examples

URL: https://hl7.org/fhir/STU3/plandefinition-examples.html

This example illustrates the use of the PlanDefinition resource to represent a Suicide Risk Assessment order set.

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

12.19.5 Resource PlanDefinition - Examples

Clinical Decision Support Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Not linked to any defined compartments

Example List:

Low Suicide Risk Order Set | low-suicide-risk-order-set | XML | JSON | Turtle
Chlamydia Screening CDS example | chlamydia-screening-intervention | XML | JSON | Turtle
Exclusive Breastfeeding Intervention 01 | exclusive-breastfeeding-intervention-01 | XML | JSON | Turtle
Exclusive Breastfeeding Intervention 02 | exclusive-breastfeeding-intervention-02 | XML | JSON | Turtle
Exclusive Breastfeeding Intervention 03 | exclusive-breastfeeding-intervention-03 | XML | JSON | Turtle
Exclusive Breastfeeding Intervention 04 | exclusive-breastfeeding-intervention-04 | XML | JSON | Turtle
Obesity Assessment Protocol | protocol-example | XML | JSON | Turtle
Chemotherapy Regimen | KDN5 | XML | JSON | Turtle
Zika Virus Infection Management Process | zika-virus-intervention | XML | JSON | Turtle
Plan Definition Illustrating Related Actions | options-example | XML | JSON | Turtle
Plan Definition Illustrating Predecessor Relationship | zika-virus-intervention-initial | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

12.19.6 Resource PlanDefinition - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/plandefinition-definitions.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

12.19.6 Resource PlanDefinition - Detailed Descriptions

Clinical Decision Support Work Group | Maturity Level: 2 | Trial Use | Compartments: Not linked to any defined compartments

Detailed Descriptions for the elements in the PlanDefinition resource.

PlanDefinition
Definition | This resource allows for the definition of various types of plans as a sharable, consumable, and executable artifact. The resource is general enough to support the description of a broad range of clinical artifacts such as clinical decision support rules, order sets and protocols.
Control | 1..1
PlanDefinition.url
Definition | An absolute URI that is used to identify this plan definition when it is referenced in a specification, model, design or an instance. This SHALL be a URL, SHOULD be globally unique, and SHOULD be an address at which this plan definition is (or will be) published. The URL SHOULD include the major version of the plan definition. For more information see Technical and Business Versions.
Control | 0..1
Type | uri
Requirements | Allows the plan definition to be referenced by a single globally unique identifier.

Summary | true
Comments | Can be a urn:uuid: or a urn:oid:, but real http: addresses are preferred. Multiple instances may share the same url if they have a distinct version.
PlanDefinition.identifier
Definition | A formal identifier that is used to identify this plan definition when it is represented in other formats, or referenced in a specification, model, design or an instance.
Note | This is a business identifer, not a resource identifier (see discussion)
Control | 0..*
Type | Identifier
Requirements | Allows externally provided and/or usable business identifiers to be easily associated with the module.
Summary | true
Comments | Typically, this is used for identifiers that can go in an HL7 V3 II (instance identifier) data type, e.g., to identify this plan definition outside of FHIR, where it is not possible to use the logical URI.
PlanDefinition.version
Definition | The identifier that is used to identify this version of the plan definition when it is referenced in a specification, model, design or instance. This is an arbitrary value managed by the plan definition author and is not expected to be globally unique. For example, it might be a timestamp (e.g. yyyymmdd) if a managed version is not available. There is also no expectation that versions can be placed in a lexicographical sequence. To provide a version consistent with the Decision Support Service specification, use the format Major.Minor.Revision (e.g. 1.0.0). For more information on versioning knowledge assets, refer to the Decision Support Service specification. Note that a version is required for non-experimental active artifacts.
Note | This is a business versionId, not a resource version id (see discussion)
Control | 0..1
Type | string
Summary | true
Comments | There may be different plan definition instances that have the same identifier but different versions. The version can be appended to the url in a reference to allow a refrence to a particular business version of the plan definition with the format [url]|[version].
PlanDefinition.name
Definition | A natural language name identifying the plan definition. This name should be usable as an identifier for the module by machine processing applications such as code generation.
Control | 0..1
Type | string
Requirements | Support human navigation and code generation.
Summary | true
Comments | The name is not expected to be globally unique. The name should be a simple alpha-numeric type name to ensure that it is computable friendly.
PlanDefinition.title
Definition | A short, descriptive, user-friendly title for the plan definition.
Control | 0..1
Type | string
Summary | true
Comments | This name does not need to be machine-processing friendly and may contain punctuation, white-space, etc.
PlanDefinition.type
Definition | The type of asset the plan definition represents, e.g. an order set, protocol, or event-condition-action rule.
Control | 0..1
Terminology Binding | PlanDefinitionType (Extensible)
Type | CodeableConcept
Summary | true
PlanDefinition.status
Definition | The status of this plan definition. Enables tracking the life-cycle of the content.
Control | 1..1
Terminology Binding | PublicationStatus (Required)
Type | code
Is Modifier | true
Summary | true
Comments | Allows filtering of plan definitions that are appropriate for use vs. not. This is labeled as "Is Modifier" because applications should not use a retired plan definition without due consideration.
PlanDefinition.experimental
Definition | A boolean value to indicate that this plan definition is authored for testing purposes (or education/evaluation/marketing), and is not intended to be used for genuine usage.
Control | 0..1
Type | boolean
Is Modifier | true
Requirements | Enables experimental content to be developed following the same lifecycle that would be used for a production-level plan definition.
Summary | true
Comments | Allows filtering of plan definition that are appropriate for use vs. not. This is labeled as "Is Modifier" because applications should not use an experimental plan definition in production.
PlanDefinition.date
Definition | The date (and optionally time) when the plan definition was published. The date must change if and when the business version changes and it must change if the status code changes. In addition, it should change when the substantive content of the plan definition changes.
Control | 0..1
Type | dateTime
Alternate Names | Revision Date
Summary | true
Comments | Note that this is not the same as the resource last-modified-date, since the resource may be a secondary representation of the plan definition. Additional specific dates may be added as extensions or be found by consulting Provenances associated with past versions of the

resource.
PlanDefinition.publisher
Definition | The name of the individual or organization that published the plan definition.
Control | 0..1
Type | string
Requirements | Helps establish the "authority/credibility" of the plan definition. May also allow for contact.
Summary | true
Comments | Usually an organization, but may be an individual. The publisher (or steward) of the plan definition is the organization or individual primarily responsible for the maintenance and upkeep of the plan definition. This is not necessarily the same individual or organization that developed and initially authored the content. The publisher is the primary point of contact for questions or issues with the plan definition. This item SHOULD be populated unless the information is available from context.
PlanDefinition.description
Definition | A free text natural language description of the plan definition from a consumer's perspective.
Control | 0..1
Type | markdown
Summary | true
Comments | This description can be used to capture details such as why the plan definition was built, comments about misuse, instructions for clinical use and interpretation, literature references, examples from the paper world, etc. It is not a rendering of the plan definition as conveyed in the 'text' field of the resource itself. This item SHOULD be populated unless the information is available from context (e.g. the language of the profile is presumed to be the predominant language in the place the profile was created).
PlanDefinition.purpose
Definition | Explaination of why this plan definition is needed and why it has been designed as it has.
Control | 0..1
Type | markdown
Comments | This element does not describe the usage of the plan definition Instead it provides traceability of "why" the resource is either needed or "why" it is defined as it is. This may be used to point to source materials or specifications that drove the structure of this plan definition.
PlanDefinition.usage
Definition | A detailed description of how the asset is used from a clinical perspective.
Control | 0..1
Type | string
PlanDefinition.approvalDate
Definition | The date on which the resource content was approved by the publisher. Approval happens once when the content is officially approved for usage.
Control | 0..1
Type | date
Comments | The 'date' element may be more recent than the approval date because of minor changes / editorial corrections.
PlanDefinition.lastReviewDate
Definition | The date on which the resource content was last reviewed. Review happens periodically after approval, but doesn't change the original approval date.
Control | 0..1
Type | date
Requirements | Gives a sense of how "current" the content is. Resources that have not been reviewed in a long time may have a risk of being less appropriate/relevant.
Comments | If specified, this is usually after the approval date.
PlanDefinition.effectivePeriod
Definition | The period during which the plan definition content was or is planned to be in active use.
Control | 0..1
Type | Period
Requirements | Allows establishing a transition before a resource comes into effect and also allows for a sunsetting process when new versions of a the plan definition are or are expected to be used instead.
Summary | true
Comments | The effective period for a plan definition determines when the content is applicable for usage and is independent of publication and review dates. For example, a measure intended to be used for the year 2016 might be published in 2015.
PlanDefinition.useContext
Definition | The content was developed with a focus and intent of supporting the contexts that are listed. These terms may be used to assist with indexing and searching for appropriate plan definition instances.
Control | 0..*
Type | UsageContext
Requirements | Assist in searching for appropriate content.
Summary | true
Comments | When multiple useContexts are specified, there is no expectation whether all or any of the contexts apply.
PlanDefinition.jurisdiction
Definition | A legal or geographic region in which the plan definition is intended to be used.
Control | 0..*
Terminology Binding | Jurisdiction ValueSet (Extensible)
Type | CodeableConcept
Summary | true
Comments | It may be possible for the plan definition to be used in jurisdictions other than those for which it was originally designed or intended.
PlanDefinition.topic
Definition | Descriptive topics related to the content of the plan definition. Topics provide a high-level categorization of the definition that can be useful for filtering and searching.
Control | 0..*
Terminology Binding | DefinitionTopic (Example)

Type | CodeableConcept
Requirements | Repositories must be able to determine how to categorize the asset so that it can be found by topical searches.
PlanDefinition.contributor
Definition | A contributor to the content of the asset, including authors, editors, reviewers, and endorsers.
Control | 0..*
Type | Contributor
Requirements | Consumers of the content must be able to quickly determine who contributed to the content of the asset.
PlanDefinition.contact
Definition | Contact details to assist a user in finding and communicating with the publisher.
Control | 0..*
Type | ContactDetail
Summary | true
Comments | May be a web site, an email address, a telephone number, etc.
PlanDefinition.copyright
Definition | A copyright statement relating to the plan definition and/or its contents. Copyright statements are generally legal restrictions on the use and publishing of the plan definition.
Control | 0..1
Type | markdown
Requirements | Consumers must be able to determine any legal restrictions on the use of the plan definition and/or its content.
Alternate Names | License; Restrictions
PlanDefinition.relatedArtifact
Definition | Related artifacts such as additional documentation, justification, or bibliographic references.
Control | 0..*
Type | RelatedArtifact
Requirements | Assets must be able to provide enough information for consumers of the content (and/or interventions or results produced by the content) to be able to determine and understand the justification for and evidence in support of the content.
Comments | Each related artifact is either an attachment, or a reference to another resource, but not both.
PlanDefinition.library
Definition | A reference to a Library resource containing any formal logic used by the plan definition.
Control | 0..*
Type | Reference(Library)
PlanDefinition.goal
Definition | Goals that describe what the activities within the plan are intended to achieve. For example, weight loss, restoring an activity of daily living, obtaining herd immunity via immunization, meeting a process improvement objective, etc.
Control | 0..*
Requirements | Goal information needs to be captured for order sets, protocols, and care plan definitions to better describe the objectives of the protocol activities and to guide the creation of specific goals within the derived care plans and orders.
PlanDefinition.goal.category
Definition | Indicates a category the goal falls within.
Control | 0..1
Terminology Binding | GoalCategory (Example)
Type | CodeableConcept
PlanDefinition.goal.description
Definition | Human-readable and/or coded description of a specific desired objective of care, such as "control blood pressure" or "negotiate an obstacle course" or "dance with child at wedding".
Control | 1..1
Terminology Binding | SNOMED CT Clinical Findings (Example)
Type | CodeableConcept
Comments | If no code is available, use CodeableConcept.text.
PlanDefinition.goal.priority
Definition | Identifies the expected level of importance associated with reaching/sustaining the defined goal.
Control | 0..1
Terminology Binding | GoalPriority (Preferred)
Type | CodeableConcept
PlanDefinition.goal.start
Definition | The event after which the goal should begin being pursued.
Control | 0..1
Terminology Binding | GoalStartEvent (Example)
Type | CodeableConcept
PlanDefinition.goal.addresses
Definition | Identifies problems, conditions, issues, or concerns the goal is intended to address.
Control | 0..*
Terminology Binding | Condition/Problem/Diagnosis Codes (Example)
Type | CodeableConcept
PlanDefinition.goal.documentation
Definition | Didactic or other informational resources associated with the goal that provide further supporting information about the goal. Information resources can include inline text commentary and links to web resources.
Control | 0..*
Type | RelatedArtifact
PlanDefinition.goal.target
Definition | Indicates what should be done and within what timeframe.
Control | 0..*
PlanDefinition.goal.target.measure

Definition | The parameter whose value is to be tracked, e.g. body weigth, blood pressure, or hemoglobin A1c level.
Control | 0..1
Terminology Binding | LOINC Codes (Example)
Type | CodeableConcept
PlanDefinition.goal.target.detail[x]
Definition | The target value of the measure to be achieved to signify fulfillment of the goal, e.g. 150 pounds or 7.0%. Either the high or low or both values of the range can be specified. Whan a low value is missing, it indicates that the goal is achieved at any value at or below the high value. Similarly, if the high value is missing, it indicates that the goal is achieved at any value at or above the low value.
Control | 0..1
Type | Quantity|Range|CodeableConcept
[x] Note | See Choice of Data Types for further information about how to use [x]
PlanDefinition.goal.target.due
Definition | Indicates the timeframe after the start of the goal in which the goal should be met.
Control | 0..1
Type | Duration
PlanDefinition.action
Definition | An action to be taken as part of the plan.
Control | 0..*
PlanDefinition.action.label
Definition | A user-visible label for the action.
Control | 0..1
Type | string
PlanDefinition.action.title
Definition | The title of the action displayed to a user.
Control | 0..1
Type | string
PlanDefinition.action.description
Definition | A short description of the action used to provide a summary to display to the user.
Control | 0..1
Type | string
PlanDefinition.action.textEquivalent
Definition | A text equivalent of the action to be performed. This provides a human-interpretable description of the action when the definition is consumed by a system that may not be capable of interpreting it dynamically.
Control | 0..1
Type | string
PlanDefinition.action.code
Definition | A code that provides meaning for the action or action group. For example, a section may have a LOINC code for a the section of a documentation template.
Control | 0..*
Type | CodeableConcept
PlanDefinition.action.reason
Definition | A description of why this action is necessary or appropriate.
Control | 0..*
Type | CodeableConcept
Comments | This is different than the clinical evidence documentation, it's an actual business description of the reason for performing the action.
PlanDefinition.action.documentation
Definition | Didactic or other informational resources associated with the action that can be provided to the CDS recipient. Information resources can include inline text commentary and links to web resources.
Control | 0..*
Type | RelatedArtifact
PlanDefinition.action.goalId
Definition | Identifies goals that this action supports. The reference must be to a goal element defined within this plan definition.
Control | 0..*
Type | id
PlanDefinition.action.triggerDefinition
Definition | A description of when the action should be triggered.
Control | 0..*
Type | TriggerDefinition
PlanDefinition.action.condition
Definition | An expression that describes applicability criteria, or start/stop conditions for the action.
Control | 0..*
PlanDefinition.action.condition.kind
Definition | The kind of condition.
Control | 1..1
Terminology Binding | ActionConditionKind (Required)
Type | code
Comments | Applicability criteria are used to determine immediate applicability when a plan definition is applied to a given context. Start and stop criteria are carried through application and used to describe when enter/exit criteria for an action.
PlanDefinition.action.condition.description
Definition | A brief, natural language description of the condition that effectively communicates the intended semantics.
Control | 0..1
Type | string
PlanDefinition.action.condition.language

Definition | The media type of the language for the expression.
Control | 0..1
Type | string
PlanDefinition.action.condition.expression
Definition | An expression that returns true or false, indicating whether or not the condition is satisfied.
Control | 0..1
Type | string
Comments | The expression may be inlined, or may be a reference to a named expression within a logic library referenced by the library element.
PlanDefinition.action.input
Definition | Defines input data requirements for the action.
Control | 0..*
Type | DataRequirement
PlanDefinition.action.output
Definition | Defines the outputs of the action, if any.
Control | 0..*
Type | DataRequirement
PlanDefinition.action.relatedAction
Definition | A relationship to another action such as "before" or "30-60 minutes after start of".
Control | 0..*
Comments | When an action depends on multiple actions, the meaning is that all actions are dependencies, rather than that any of the actions are a dependency.
PlanDefinition.action.relatedAction.actionId
Definition | The element id of the related action.
Control | 1..1
Type | id
PlanDefinition.action.relatedAction.relationship
Definition | The relationship of this action to the related action.
Control | 1..1
Terminology Binding | ActionRelationshipType (Required)
Type | code
PlanDefinition.action.relatedAction.offset[x]
Definition | A duration or range of durations to apply to the relationship. For example, 30-60 minutes before.
Control | 0..1
Type | Duration|Range
[x] Note | See Choice of Data Types for further information about how to use [x]
PlanDefinition.action.timing[x]
Definition | An optional value describing when the action should be performed.
Control | 0..1
Type | dateTime|Period|Duration|Range|Timing
[x] Note | See Choice of Data Types for further information about how to use [x]
PlanDefinition.action.participant
Definition | Indicates who should participate in performing the action described.
Control | 0..*
PlanDefinition.action.participant.type
Definition | The type of participant in the action.
Control | 1..1
Terminology Binding | ActionParticipantType (Required)
Type | code
PlanDefinition.action.participant.role
Definition | The role the participant should play in performing the described action.
Control | 0..1
Terminology Binding | ActionParticipantRole (Example)
Type | CodeableConcept
PlanDefinition.action.type
Definition | The type of action to perform (create, update, remove).
Control | 0..1
Terminology Binding | ActionType (Required)
Type | Coding
PlanDefinition.action.groupingBehavior
Definition | Defines the grouping behavior for the action and its children.
Control | 0..1
Terminology Binding | ActionGroupingBehavior (Required)
Type | code
PlanDefinition.action.selectionBehavior
Definition | Defines the selection behavior for the action and its children.
Control | 0..1
Terminology Binding | ActionSelectionBehavior (Required)
Type | code
PlanDefinition.action.requiredBehavior
Definition | Defines the requiredness behavior for the action.
Control | 0..1
Terminology Binding | ActionRequiredBehavior (Required)

Type | code
PlanDefinition.action.precheckBehavior
Definition | Defines whether the action should usually be preselected.
Control | 0..1
Terminology Binding | ActionPrecheckBehavior (Required)
Type | code
PlanDefinition.action.cardinalityBehavior
Definition | Defines whether the action can be selected multiple times.
Control | 0..1
Terminology Binding | ActionCardinalityBehavior (Required)
Type | code
PlanDefinition.action.definition
Definition | A reference to an ActivityDefinition that describes the action to be taken in detail, or a PlanDefinition that describes a series of actions to be taken.
Control | 0..1
Type | Reference(ActivityDefinition | PlanDefinition)
Comments | Note that the definition is optional, and if no definition is specified, a dynamicValue with a root (~) path can be used to define the entire resource dynamically.
PlanDefinition.action.transform
Definition | A reference to a StructureMap resource that defines a transform that can be executed to produce the intent resource using the ActivityDefinition instance as the input.
Control | 0..1
Type | Reference(StructureMap)
PlanDefinition.action.dynamicValue
Definition | Customizations that should be applied to the statically defined resource. For example, if the dosage of a medication must be computed based on the patient's weight, a customization would be used to specify an expression that calculated the weight, and the path on the resource that would contain the result.
Control | 0..*
PlanDefinition.action.dynamicValue.description
Definition | A brief, natural language description of the intended semantics of the dynamic value.
Control | 0..1
Type | string
PlanDefinition.action.dynamicValue.path
Definition | The path to the element to be customized. This is the path on the resource that will hold the result of the calculation defined by the expression.
Control | 0..1
Type | string
PlanDefinition.action.dynamicValue.language
Definition | The media type of the language for the expression.
Control | 0..1
Type | string
PlanDefinition.action.dynamicValue.expression
Definition | An expression specifying the value of the customized element.
Control | 0..1
Type | string
Comments | The expression may be inlined, or may be a reference to a named expression within a logic library referenced by the library element.
PlanDefinition.action.action
Definition | Sub actions that are contained within the action. The behavior of this action determines the functionality of the sub-actions. For example, a selection behavior of at-most-one indicates that of the sub-actions, at most one may be chosen as part of realizing the action definition.
Control | 0..*
Type | See PlanDefinition.action

12.19.7 Resource PlanDefinition - Mappings

URL: https://hl7.org/fhir/STU3/plandefinition-mappings.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

12.19.7 Resource PlanDefinition - Mappings

Clinical Decision Support Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Not linked to any defined compartments

Mappings:

Workflow Mapping

Mappings for the plandefinition resource.

12.19.7.1 Mappings for Workflow Mapping (http://hl7.org/fhir/workflow)

PlanDefinition | Definition
url | Definition.url
identifier | Definition.identifier
version | Definition.version
name
title | Definition.title

type
status | Definition.status (different ValueSet)
experimental | Definition.experimental
date | Definition.date
publisher | Definition.publisher (as string instead of Reference)
description | Definition.description
purpose | Definition.purpose
usage
approvalDate | Definition.approvalDate
lastReviewDate | Definition.lastReviewDate
effectivePeriod | Definition.effectivePeriod
useContext | Definition.useContext
jurisdiction | Definition.jurisdiction
topic | Definition.subjectCodableConcept
contributor
contact | Definition.contact
copyright | Definition.copyright
relatedArtifact
library
goal
category
description
priority
start
addresses
documentation
target
measure
detail[x]
due
action | Is a contained Definition
label
title | Definition.title
description | Definition.description
textEquivalent | Definition.description
code
reason
documentation
goalId
triggerDefinition
condition
kind
description
language
expression
input
output
relatedAction
actionId
relationship
offset[x]
timing[x]
participant
type
role
type
groupingBehavior
selectionBehavior
requiredBehavior
precheckBehavior
cardinalityBehavior
definition | Definition.definition
transform
dynamicValue
description
path
language
expression
action | InverseRelationship of Definition.partOf

12.19.8 Resource PlanDefinition - Design Notes

URL: https://hl7.org/fhir/STU3/plandefinition-explanations.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

12.19.8 Resource PlanDefinition - Design Notes

to do

12.19.9 Resource PlanDefinition - Extensions & Profiles

URL: https://hl7.org/fhir/STU3/plandefinition-profiles.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

12.19.9 Resource PlanDefinition - Extensions & Profiles

Clinical Decision Support Work Group | Maturity Level: N/A | Ballot Status: Informative

This table lists profiles and extensions for the PlanDefinition resource. For background information, see Profiling Resources and Extensibility. Additional profiles and resources may be found in published Implementation Guides , or in the Conformance resource registry .

12.19.9.1 Profiles

No Profiles defined for this resource

12.19.9.2 Extensions

timing-daysOfCycle | daysOfCycle | for Common extensions for Timing data type

Extensions that reference this resource:

communicationrequest-definition | definition | for CommunicationRequest HL7 Extensions
condition-definition | definition | for Condition HL7 Extensions
event-definition | definition | for Event Pattern HL7 Extensions

12.19.9.3 Search Extensions

No Search Extensions defined for this resource

12.19.10 Resource PlanDefinition - Operations

URL: https://hl7.org/fhir/STU3/plandefinition-operations.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

12.19.10 Resource PlanDefinition - Operations

Clinical Decision Support Work Group | Maturity Level: 2 | Trial Use | Compartments: Not linked to any defined compartments

This resource has 2 operations associated with it:

$apply | Apply
$data-requirements | Data Requirements

For more information about operations, including how they are invoked, see Operations.

12.19.10.1 Apply

The apply operation applies a PlanDefinition to a given context

The official URL for this operation definition is

```
http://hl7.org/fhir/OperationDefinition/PlanDefinition-apply
```

Formal Definition (as a OperationDefinition).

URL: [base]/PlanDefinition/[id]/$apply

This is not an idempotent operation

In Parameters:

Name | Cardinality | Type | Binding | Profile | Documentation
patient | 1..1 | Reference(Patient) | | | The patient that is the target of the plan to be applied
encounter | 0..1 | Reference(Encounter) | | | The encounter in context, if any
practitioner | 0..1 | Reference(Practitioner) | | | The practitioner applying the plan definition
organization | 0..1 | Reference(Organization) | | | The organization applying the plan definition
userType | 0..1 | CodeableConcept | | | The type of user initiating the request, e.g. patient, healthcare provider, or specific type of healthcare provider (physician, nurse, etc.)
userLanguage | 0..1 | CodeableConcept | | | Preferred language of the person using the system
userTaskContext | 0..1 | CodeableConcept | | | The task the system user is performing, e.g. laboratory results review, medication list review, etc. This information can be used to tailor decision support outputs, such as recommended information resources
setting | 0..1 | CodeableConcept | | | The current setting of the request (inpatient, outpatient, etc)
settingContext | 0..1 | CodeableConcept | | | Additional detail about the setting of the request, if any

Out Parameters:

Name | Cardinality | Type | Binding | Profile | Documentation

return | 1..1 | CarePlan | | | The CarePlan that is the result of applying the plan definition Note: as this the only out parameter, it is a resource, and it has the name 'return', the result of this operation is returned directly as a resource

The result of this operation is a CarePlan resource, with activities for each of the applicable actions based on evaluating the applicability condition in context. For each applicable action, the activitydefinition is applied as described in the $apply operation of the ActivityDefinition resource, and the resulting resource is added as an activity to the CarePlan. If the ActivityDefinition includes library references, those libraries will be available to the evaluated expressions. If those libraries have parameters, those parameters will be bound by name to the parameters given to the operation

12.19.10.1.1 Examples

Apply the low suicide risk order set to patient 124 (Request):

```
GET [base]/PlanDefinition/example/$apply?patient=124
```

Returned on successful application of the order set (Response):

```
HTTP/1.1 200 OK

<?xml version="1.0" encoding="UTF-8"?>
<CarePlan xmlns="http://hl7.org/fhir">
  <!-- This example demonstrates the result of an $apply operation for the citalopramPrescription activity definition.
    -->
  <id value="apply-operation-response"/>
  <text>
  <status value="generated"/>
    <div xmlns="http://www.w3.org/1999/xhtml">
    Generated...
    </div>
  </text>
  <contained>
  <ReferralRequest>
    <id value="referralrequest"/>
    <status value="draft"/>
      <intent value="proposal"/>
      <serviceRequested>
    <coding>
      <system value="http://snomed.info/sct"/>
      <code value="306206005"/>
    </coding>
    <text value="Referral to service (procedure)"/>
      </serviceRequested>
      <subject>
        <reference value="Patient/124"/>
      </subject>
      <occurrenceDateTime value="2017-02-26T11:47:00.00"/>
      <requester>
    <agent>
      <reference value="Practitioner/123"/>
    </agent>
    </requester>
      <specialty>
      <coding>
        <system value="http://nucc.org/provider-taxonomy"/>
        <code value="261QM0850X"/>
        <display value="Adult Mental Health"/>
      </coding>
      </specialty>
    <reasonCode>
      <coding>
        <system value="http://example.org/CodeSystem/suicide-risk-assessment-score"/>
        <code value="Low"/>
        <display value="Low Suicide Risk"/>
      </coding>
      <text value="Low Suicide Risk"/>
    </reasonCode>
    <reasonReference>
      <reference value="RiskAssessment/suicide-risk-assessment"/>
    </reasonReference>
  </ReferralRequest>
  </contained>
  <contained>
    <Medication>
      <id value="citalopramMedication"/>
      <code>
        <coding>
          <system value="http://www.nlm.nih.gov/research/umls/rxnorm"/>
          <code value="200371"/>
        </coding>
        <text value="citalopram"/>
      </code>
        <form>
          <coding>
            <system value="http://snomed.info/sct"/>
            <code value="385055001"/>
            <display value="Tablet dose form"/>
          </coding>
```

```
        <text value="Tablet dose form"/>
      </form>
      <ingredient>
        <itemReference>
          <reference value="#citalopramSubstance"/>
        </itemReference>
        <amount>
          <numerator>
            <value value="20"/>
            <unit value="mg"/>
          </numerator>
          <denominator>
            <value value="1"/>
            <unit value="{tbl}"/>
          </denominator>
        </amount>
      </ingredient>
  </Medication>
</contained>
<contained>
  <Substance>
    <id value="citalopramSubstance"/>
    <code>
      <coding>
        <system value="http://www.nlm.nih.gov/research/umls/rxnorm"/>
        <code value="2556"/>
      </coding>
      <text value="citalopram"/>
    </code>
  </Substance>
</contained>
<contained>
<MedicationRequest>
  <id value="citalopramPrescription"/>
  <definition>
  <reference value="ActivityDefinition/citalopramPrescription"/>
  </definition>
  <status value="draft"/>
  <intent value="proposal"/>
  <medicationReference>
    <reference value="#citalopramMedication"/>
  </medicationReference>
  <subject>
    <reference value="Patient/124"/>
  </subject>
  <requester>
  <agent>
    <reference value="Practitioner/123"/>
  </agent>
  </requester>
  <reasonCode>
    <coding>
      <system value="http://example.org/CodeSystem/suicide-risk-assessment-score"/>
      <code value="Low"/>
      <display value="Low Suicide Risk"/>
    </coding>
    <text value="Low Suicide Risk"/>
  </reasonCode>
  <reasonReference>
    <reference value="RiskAssessment/suicide-risk-assessment"/>
  </reasonReference>
  <dosageInstruction>
  <text value="1 tablet oral 1 time daily"/>
  <timing>
    <repeat>
      <frequency value="1"/>
      <period value="1"/>
      <periodUnit value="d"/>
    </repeat>
  </timing>
  <route>
    <coding>
      <code value="26643006"/>
      <display value="Oral route (qualifier value)"/>
    </coding>
    <text value="Oral route (qualifier value)"/>
  </route>
  <doseQuantity>
    <value value="1"/>
    <unit value="{tbl}"/>
  </doseQuantity>
  </dosageInstruction>
  <dispenseRequest>
    <numberOfRepeatsAllowed value="3"/>
    <quantity>
      <value value="30"/>
      <unit value="{tbl}"/>
    </quantity>
  </dispenseRequest>
```

```
  </MedicationRequest>
  </contained>
  <definition>
    <reference value="PlanDefinition/example"/>
  </definition>
  <status value="draft"/>
  <intent value="proposal"/>
  <subject>
    <reference value="Patient/124"/>
  </subject>
  <activity>
  <reference>
    <reference value="#referralrequest"/>
  </reference>
  </activity>
  <activity>
    <reference>
      <reference value="#citalopramPrescription"/>
    </reference>
  </activity>
</CarePlan>
```

12.19.10.2 Data Requirements

The data-requirements operation aggregates and returns the parameters and data requirements for the plan definition and all its dependencies as a single module definition library

The official URL for this operation definition is

```
http://hl7.org/fhir/OperationDefinition/PlanDefinition-data-requirements
```

Formal Definition (as a OperationDefinition).

URL: [base]/PlanDefinition/[id]/$data-requirements

This is not an idempotent operation

Out Parameters:

Name | Cardinality | Type | Binding | Profile | Documentation

return | 1..1 | Library | | | The result of the requirements gathering is a module-definition Library that describes the aggregate parameters, data requirements, and dependencies of the plan definition Note: as this the only out parameter, it is a resource, and it has the name 'return', the result of this operation is returned directly as a resource

The effect of invoking this operation is to determine the aggregate set of data requirements and dependencies for the plan definition. The result is a Library resource with a type of module-definition that contains all the parameter definitions and data requirements of the plan definition and any libraries referenced by it. Implementations SHOULD aggregate data requirements intelligently (i.e. by collapsing overlapping data requirements)

12.19.10.2.1 Examples

Request data requirements for the Zika Virus intervention plan definition (Request):

```
GET [base]/PlanDefinition/zika-virus-intervention/$data-requirements
```

Returned on successful evaluation (Response):

```
HTTP/1.1 200 OK

{
  "resourceType": "Library",
  "type": {
    "coding": [
      {
        "code": "module-definition"
      }
    ]
  },
  "relatedArtifact": [
    {
      "type": "depends-on",
      "display": "Zika Affected Areas",
      "resource": {
        "reference": "ValueSet/zika-affected-areas"
      }
    },
    {
      "type": "depends-on",
      "display": "Pregnancy",
      "resource": {
        "identifier": "urn:oid:X.Y.Z",
        "display": "Pregnancy value set"
      }
    },
    {
      "type": "depends-on",
      "display": "Zika Exposure Assessment",
      "resource": {
        "identifier": "urn:oid:X.Y.Z",
        "display": "Zika Virus Exposure Assessment value set"
      }
    },
```

```
    {
      "type": "depends-on",
      "display": "Zika Virus Signs and Symptoms",
      "url": "https://phinvads.cdc.gov/vads/ViewValueSet.action?oid=2.16.840.1.114222.4.11.7459",
      "resource": {
        "identifier": "urn:oid:2.16.840.1.114222.4.11.7459",
        "display": "Zika Virus Signs and Symptoms value set"
      }
    },
  ]
  "dataRequirement": [
    {
      "type": "Condition",
      "codeFilter": [
        {
          "path": "code",
          "valueSetString": "Pregnancy"
        }
      ]
    },
    {
      "type": "Patient"
    },
    {
      "type": "Observation",
      "codeFilter": [
        {
          "path": "code",
          "valueSetString": [
            "Zika Exposure Assessment"
          ]
        },
        {
          "path": "status",
          "valueCode": [
            "final"
          ]
        }
      ]
    },
    {
      "type": "Condition",
      "codeFilter": [
        {
          "path": "category",
          "valueCode": [
            "diagnosis"
          ]
        },
        {
          "path": "clinicalStatus",
          "valueCode": [
            "confirmed"
          ]
        },
        {
          "path": "code",
          "valueSetString": "Zika Virus Signs and Symptoms"
        }
      ]
    }
  ]
}
```

12.20 Resource ProcessRequest - Content

URL: https://hl7.org/fhir/STU3/processrequest.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

Financial Management Work Group | Maturity Level: 2 | Trial Use | Compartments: Practitioner

This resource provides the target, request and response, and action details for an action to be performed by the target on or about existing resources.

12.20.1 Scope and Usage

NOTE: This resource is slated to be refactored and reduced in scope as some of its functionality is taken over by the Task resource as part of the FHIR Workflow refactoring. Readers of this resource are encouraged to review the Task resource and consider using it instead and/or to provide feedback if they feel it will be inadequate to satisfy the use-cases associated with this resource and to express opinions about which purposes they feel ProcessRequest and ProcessResponse should be retained for.

The ProcessRequest resource allows for the specification of an action to be performed on an existing resource or resources and then provides the additional supporting information to support that action. The actions currently defined are: cancel, poll,

reprocess, and status.

Cancel indicates the resource which is to be reversed and provides both supporting information for the reversal and whether the receiving system is permitted to retain a copy of the reversed resource.

The Cancel is the formal request to cease processing an incomplete prior request or to reverse and/or nullify a complete prior request or information submission. When nullify=true then all copies of the original submission are to be purged, although audit logs may be retained. When Nullify=false a copy of the original request may be retained. A ProcessResponse may be craeated or returned to indicate whether the requestion action was accepted and successful.

Poll provides supporting information for the poll request. The response to this is a previously undelivered response or a ProcessResponse which may contain errors.

This is a formal request for payors or systems which require such and/or transports which don't support a 'Get Operation', for the retrieval of pended, held, resources.

A simple Poll request, one which doesn't specify: request, include, exclude or period; would return any pended resource. Specific types of business behaviors may be supported by providing values for the filtering elements, for example:

* Get any pended resource - no filters specified

* Get deferred response to a Claim - specify the Claim in the 'request'

* Get all supporting information - specify 'Communication' as an 'include'

* Get an Explanation of Benefit - specify 'ExplanationOfBenefit' as an 'include'

* Get a payment reconciliation - specify a 'period' which contains the expected reconciliation creation date, and specify 'PaymentReconciliation' as an 'include'

Reprocess indicates the resource which is to be reprocessed, for example a claim to be readjudicated or a specimen or diagnostic image to be re-examined, and provides both supporting information for the reprocessing and the line items which are to be reprocessed.

This is necessary for the limited supporters who require the ability to formally request the reprocessing of specified service sub-trees from an already processed resource such as a previously adjudicated Claim.

Status indicates the resource for which the processing status is requested and provides supporting information for the status request.

This is a formal request for payors which require such and/or transports which don't support a 'Get Operation', for the processing status of a previously submitted processing request.

This resource is referenced by paymentreconciliation

12.20.2 Resource Content

* Structure

* UML

* XML

* JSON

* Turtle

* R2 Diff

* All

Structure

Name | Flags | Card. | Type | Description & Constraints
ProcessRequest | | | DomainResource | Request to perform some action on or in regards to an existing resource Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | | 0..* | Identifier | Business Identifier
status | ?!? | 0..1 | code | active | cancelled | draft | entered-in-error Financial Resource Status Codes (Required)
action | | 0..1 | code | cancel | poll | reprocess | status ActionList (Required)
target | | 0..1 | Reference(Organization) | Party which is the target of the request
created | | 0..1 | dateTime | Creation date
provider | | 0..1 | Reference(Practitioner) | Responsible practitioner
organization | | 0..1 | Reference(Organization) | Responsible organization
request | | 0..1 | Reference(Any) | Reference to the Request resource
response | | 0..1 | Reference(Any) | Reference to the Response resource
nullify | | 0..1 | boolean | Remove history
reference | | 0..1 | string | Reference number/string
item | | 0..* | BackboneElement | Items to re-adjudicate
sequenceLinkId | | 1..1 | integer | Service instance
include | | 0..* | string | Resource type(s) to include
exclude | | 0..* | string | Resource type(s) to exclude
period | | 0..1 | Period | Selection period
Documentation for this format

UML Diagram (Legend)

XML Template

```
<ProcessRequest xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Business Identifier --></identifier>
 <status value="[code]"/><!-- 0..1 active | cancelled | draft | entered-in-error -->
 <action value="[code]"/><!-- 0..1 cancel | poll | reprocess | status -->
 <target><!-- 0..1 Reference(Organization) Party which is the target of the request --></target>
 <created value="[dateTime]"/><!-- 0..1 Creation date -->
 <provider><!-- 0..1 Reference(Practitioner) Responsible practitioner --></provider>
 <organization><!-- 0..1 Reference(Organization) Responsible organization --></organization>
 <request><!-- 0..1 Reference(Any) Reference to the Request resource --></request>
 <response><!-- 0..1 Reference(Any) Reference to the Response resource --></response>
 <nullify value="[boolean]"/><!-- 0..1 Remove history -->
 <reference value="[string]"/><!-- 0..1 Reference number/string -->
 <item>  <!-- 0..* Items to re-adjudicate -->
  <sequenceLinkId value="[integer]"/><!-- 1..1 Service instance -->
 </item>
 <include value="[string]"/><!-- 0..* Resource type(s) to include -->
 <exclude value="[string]"/><!-- 0..* Resource type(s) to exclude -->
 <period><!-- 0..1 Period Selection period --></period>
</ProcessRequest>
```

JSON Template

```
{
  "resourceType" : "ProcessRequest",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Business Identifier
  "status" : "<code>", // active | cancelled | draft | entered-in-error
  "action" : "<code>", // cancel | poll | reprocess | status
  "target" : { Reference(Organization) }, // Party which is the target of the request
  "created" : "<dateTime>", // Creation date
  "provider" : { Reference(Practitioner) }, // Responsible practitioner
  "organization" : { Reference(Organization) }, // Responsible organization
  "request" : { Reference(Any) }, // Reference to the Request resource
  "response" : { Reference(Any) }, // Reference to the Response resource
  "nullify" : <boolean>, // Remove history
  "reference" : "<string>", // Reference number/string
  "item" : [{ // Items to re-adjudicate
    "sequenceLinkId" : <integer> // R!  Service instance
  }],
  "include" : ["<string>"], // Resource type(s) to include
  "exclude" : ["<string>"], // Resource type(s) to exclude
  "period" : { Period } // Selection period
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:ProcessRequest;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:ProcessRequest.identifier [ Identifier ], ... ; # 0..* Business Identifier
  fhir:ProcessRequest.status [ code ]; # 0..1 active | cancelled | draft | entered-in-error
  fhir:ProcessRequest.action [ code ]; # 0..1 cancel | poll | reprocess | status
  fhir:ProcessRequest.target [ Reference(Organization) ]; # 0..1 Party which is the target of the request
  fhir:ProcessRequest.created [ dateTime ]; # 0..1 Creation date
  fhir:ProcessRequest.provider [ Reference(Practitioner) ]; # 0..1 Responsible practitioner
  fhir:ProcessRequest.organization [ Reference(Organization) ]; # 0..1 Responsible organization
  fhir:ProcessRequest.request [ Reference(Any) ]; # 0..1 Reference to the Request resource
  fhir:ProcessRequest.response [ Reference(Any) ]; # 0..1 Reference to the Response resource
  fhir:ProcessRequest.nullify [ boolean ]; # 0..1 Remove history
  fhir:ProcessRequest.reference [ string ]; # 0..1 Reference number/string
  fhir:ProcessRequest.item [ # 0..* Items to re-adjudicate
    fhir:ProcessRequest.item.sequenceLinkId [ integer ]; # 1..1 Service instance
  ], ...;
  fhir:ProcessRequest.include [ string ], ... ; # 0..* Resource type(s) to include
  fhir:ProcessRequest.exclude [ string ], ... ; # 0..* Resource type(s) to exclude
  fhir:ProcessRequest.period [ Period ]; # 0..1 Selection period
]
```

Changes since DSTU2

ProcessRequest

ProcessRequest.status | Added Element

ProcessRequest.action | Min Cardinality changed from 1 to 0

ProcessRequest.ruleset | deleted

ProcessRequest.originalRuleset | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

Structure

Name | Flags | Card. | Type | Description & Constraints

ProcessRequest | | | DomainResource | Request to perform some action on or in regards to an existing resource Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | | 0..* | Identifier | Business Identifier
status | ?!? | 0..1 | code | active | cancelled | draft | entered-in-error Financial Resource Status Codes (Required)
action | | 0..1 | code | cancel | poll | reprocess | status ActionList (Required)
target | | 0..1 | Reference(Organization) | Party which is the target of the request
created | | 0..1 | dateTime | Creation date
provider | | 0..1 | Reference(Practitioner) | Responsible practitioner
organization | | 0..1 | Reference(Organization) | Responsible organization
request | | 0..1 | Reference(Any) | Reference to the Request resource
response | | 0..1 | Reference(Any) | Reference to the Response resource
nullify | | 0..1 | boolean | Remove history
reference | | 0..1 | string | Reference number/string
item | | 0..* | BackboneElement | Items to re-adjudicate
sequenceLinkId | | 1..1 | integer | Service instance
include | | 0..* | string | Resource type(s) to include
exclude | | 0..* | string | Resource type(s) to exclude
period | | 0..1 | Period | Selection period
Documentation for this format

UML Diagram (Legend)

XML Template

```
<ProcessRequest xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Business Identifier --></identifier>
 <status value="[code]"/><!-- 0..1 active | cancelled | draft | entered-in-error -->
 <action value="[code]"/><!-- 0..1 cancel | poll | reprocess | status -->
 <target><!-- 0..1 Reference(Organization) Party which is the target of the request --></target>
 <created value="[dateTime]"/><!-- 0..1 Creation date -->
 <provider><!-- 0..1 Reference(Practitioner) Responsible practitioner --></provider>
 <organization><!-- 0..1 Reference(Organization) Responsible organization --></organization>
 <request><!-- 0..1 Reference(Any) Reference to the Request resource --></request>
 <response><!-- 0..1 Reference(Any) Reference to the Response resource --></response>
 <nullify value="[boolean]"/><!-- 0..1 Remove history -->
 <reference value="[string]"/><!-- 0..1 Reference number/string -->
 <item>  <!-- 0..* Items to re-adjudicate -->
  <sequenceLinkId value="[integer]"/><!-- 1..1 Service instance -->
 </item>
 <include value="[string]"/><!-- 0..* Resource type(s) to include -->
 <exclude value="[string]"/><!-- 0..* Resource type(s) to exclude -->
 <period><!-- 0..1 Period Selection period --></period>
</ProcessRequest>
```

JSON Template

```
{
  "resourceType" : "ProcessRequest",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Business Identifier
  "status" : "<code>", // active | cancelled | draft | entered-in-error
  "action" : "<code>", // cancel | poll | reprocess | status
  "target" : { Reference(Organization) }, // Party which is the target of the request
  "created" : "<dateTime>", // Creation date
  "provider" : { Reference(Practitioner) }, // Responsible practitioner
  "organization" : { Reference(Organization) }, // Responsible organization
  "request" : { Reference(Any) }, // Reference to the Request resource
  "response" : { Reference(Any) }, // Reference to the Response resource
  "nullify" : <boolean>, // Remove history
  "reference" : "<string>", // Reference number/string
  "item" : [{ // Items to re-adjudicate
    "sequenceLinkId" : <integer> // R!  Service instance
  }],
  "include" : ["<string>"], // Resource type(s) to include
  "exclude" : ["<string>"], // Resource type(s) to exclude
  "period" : { Period } // Selection period
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:ProcessRequest;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:ProcessRequest.identifier [ Identifier ], ... ; # 0..* Business Identifier
  fhir:ProcessRequest.status [ code ]; # 0..1 active | cancelled | draft | entered-in-error
  fhir:ProcessRequest.action [ code ]; # 0..1 cancel | poll | reprocess | status
  fhir:ProcessRequest.target [ Reference(Organization) ]; # 0..1 Party which is the target of the request
  fhir:ProcessRequest.created [ dateTime ]; # 0..1 Creation date
  fhir:ProcessRequest.provider [ Reference(Practitioner) ]; # 0..1 Responsible practitioner
  fhir:ProcessRequest.organization [ Reference(Organization) ]; # 0..1 Responsible organization
```

```
  fhir:ProcessRequest.request [ Reference(Any) ]; # 0..1 Reference to the Request resource
  fhir:ProcessRequest.response [ Reference(Any) ]; # 0..1 Reference to the Response resource
  fhir:ProcessRequest.nullify [ boolean ]; # 0..1 Remove history
  fhir:ProcessRequest.reference [ string ]; # 0..1 Reference number/string
  fhir:ProcessRequest.item [ # 0..* Items to re-adjudicate
    fhir:ProcessRequest.item.sequenceLinkId [ integer ]; # 1..1 Service instance
  ], ...;
  fhir:ProcessRequest.include [ string ], ... ; # 0..* Resource type(s) to include
  fhir:ProcessRequest.exclude [ string ], ... ; # 0..* Resource type(s) to exclude
  fhir:ProcessRequest.period [ Period ]; # 0..1 Selection period
]
```

Changes since DSTU2

ProcessRequest
ProcessRequest.status | Added Element
ProcessRequest.action | Min Cardinality changed from 1 to 0
ProcessRequest.ruleset | deleted
ProcessRequest.originalRuleset | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

Alternate definitions: Master Definition (XML, JSON), XML Schema/Schematron (for) + JSON Schema, ShEx (for Turtle)

12.20.2.1 Terminology Bindings

Path | Definition | Type | Reference
ProcessRequest.status | A code specifying the state of the resource instance. | Required | Financial Resource Status Codes
ProcessRequest.action | List of allowable action which this resource can request. | Required | ActionList

12.20.3 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
action | token | The action requested by this resource | ProcessRequest.action
identifier | token | The business identifier of the ProcessRequest | ProcessRequest.identifier
organization | reference | The organization who generated this request | ProcessRequest.organization (Organization)
provider | reference | The provider who regenerated this request | ProcessRequest.provider (Practitioner)

12.20.4 Resource ProcessRequest - Examples

URL: https://hl7.org/fhir/STU3/processrequest-examples.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

12.20.4 Resource ProcessRequest - Examples

Financial Management Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Practitioner

Example List:

All Pended Checking Example | 1110 | XML | JSON | Turtle
Check for responses for a specific Request | 1111 | XML | JSON | Turtle
Check for responses for types of resources | 1112 | XML | JSON | Turtle
Check for responses for excluding types of resources | 1113 | XML | JSON | Turtle
Check for Payment Reconciliation | 1114 | XML | JSON | Turtle
Check for Explanation of Benefit | 1115 | XML | JSON | Turtle
Claim Readjudicate Example | 44654 | XML | JSON | Turtle
Claim Reversal Example | 87654 | XML | JSON | Turtle
Claim Status Checking Example | 87655 | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

12.20.5 Resource ProcessRequest - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/processrequest-definitions.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

12.20.5 Resource ProcessRequest - Detailed Descriptions

Financial Management Work Group | Maturity Level: 2 | Trial Use | Compartments: Practitioner

Detailed Descriptions for the elements in the ProcessRequest resource.

ProcessRequest

Definition | This resource provides the target, request and response, and action details for an action to be performed by the target on or about existing resources.
Control | 1..1
ProcessRequest.identifier
Definition | The ProcessRequest business identifier.
Note | This is a business identifer, not a resource identifier (see discussion)
Control | 0..*
Type | Identifier
ProcessRequest.status
Definition | The status of the resource instance.
Control | 0..1
Terminology Binding | Financial Resource Status Codes (Required)
Type | code
Is Modifier | true
Summary | true
Comments | This element is labeled as a modifier because the status contains codes that mark the resource as not currently valid.
ProcessRequest.action
Definition | The type of processing action being requested, for example Reversal, Readjudication, StatusRequest,PendedRequest.
Control | 0..1
Terminology Binding | ActionList (Required)
Type | code
ProcessRequest.target
Definition | The organization which is the target of the request.
Control | 0..1
Type | Reference(Organization)
ProcessRequest.created
Definition | The date when this resource was created.
Control | 0..1
Type | dateTime
ProcessRequest.provider
Definition | The practitioner who is responsible for the action specified in this request.
Control | 0..1
Type | Reference(Practitioner)
ProcessRequest.organization
Definition | The organization which is responsible for the action speccified in this request.
Control | 0..1
Type | Reference(Organization)
ProcessRequest.request
Definition | Reference of resource which is the target or subject of this action.
Control | 0..1
Type | Reference(Any)
ProcessRequest.response
Definition | Reference of a prior response to resource which is the target or subject of this action.
Control | 0..1
Type | Reference(Any)
ProcessRequest.nullify
Definition | If true remove all history excluding audit.
Control | 0..1
Type | boolean
Requirements | Some resources must not simply be reversed in a processing or accounting sense but rather must have all history removed, such as the accidental submission of sensitive and/or wrong information. If the receiver cannot comply with a Nullify request then they must reject the request.
ProcessRequest.reference
Definition | A reference to supply which authenticates the process.
Control | 0..1
Type | string
ProcessRequest.item
Definition | List of top level items to be re-adjudicated, if none specified then the entire submission is re-adjudicated.
Control | 0..*
ProcessRequest.item.sequenceLinkId
Definition | A service line number.
Control | 1..1
Type | integer
ProcessRequest.include
Definition | Names of resource types to include.
Control | 0..*
Type | string
ProcessRequest.exclude
Definition | Names of resource types to exclude.
Control | 0..*
Type | string
ProcessRequest.period

Definition | A period of time during which the fulfilling resources would have been created.
Control | 0..1
Type | Period

12.20.6 Resource ProcessRequest - Mappings

URL: https://hl7.org/fhir/STU3/processrequest-mappings.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

12.20.6 Resource ProcessRequest - Mappings

Financial Management Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Practitioner

Mappings:

Canadian Dental Association eclaims standard

Workflow Mapping

Mappings for the processrequest resource.

12.20.6.1 Mappings for Canadian Dental Association eclaims standard (http://www.cda-adc.ca/en/services/cdanet/)

ProcessRequest
identifier
status
action
target
created
provider | B01
organization
request
response
nullify
reference
item
sequenceLinkId
include
exclude
period

12.20.6.2 Mappings for Workflow Mapping (http://hl7.org/fhir/workflow)

ProcessRequest | Request
identifier | Request.identifier
status | Request.status
action
target
created | Request.authoredOn
provider | Request.requester.agent
organization | Request.requester.onBehalfOf
request | Request.basedOn
response
nullify
reference
item
sequenceLinkId
include
exclude
period

12.20.7 Resource ProcessRequest - Design Notes

URL: https://hl7.org/fhir/STU3/processrequest-explanations.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

12.20.7 Resource ProcessRequest - Design Notes

to do

12.20.8 Resource ProcessRequest - Extensions & Profiles

URL: https://hl7.org/fhir/STU3/processrequest-profiles.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

12.20.8 Resource ProcessRequest - Extensions & Profiles

Financial Management Work Group | Maturity Level: N/A | Ballot Status: Informative

This table lists profiles and extensions for the ProcessRequest resource. For background information, see Profiling Resources and Extensibility. Additional profiles and resources may be found in published Implementation Guides , or in the Conformance resource registry .

12.20.8.1 Profiles

No Profiles defined for this resource

12.20.8.2 Extensions

No Extensions defined for this resource

Extensions that reference this resource:

No Extensions refer to this resource

12.20.8.3 Search Extensions

No Search Extensions defined for this resource

12.21 Resource ProcessResponse - Content

URL: https://hl7.org/fhir/STU3/processresponse.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

Financial Management Work Group | Maturity Level: 2 | Trial Use | Compartments: Practitioner

This resource provides processing status, errors and notes from the processing of a resource.

12.21.1 Scope and Usage

NOTE: This resource is slated to be refactored and reduced in scope as some of its functionality is taken over by the Task resource as part of the FHIR Workflow refactoring. Readers of this resource are encouraged to review the Task resource and consider using it instead and/or to provide feedback if they feel it will be inadequate to satisfy the use-cases associated with this resource and to express opinions about which purposes for they feel ProcessRequest and ProcessResponse should be retained.

The ProcessResponse resource indicates the resource for which the processing status is requested and provides simple acknowledgement and status information of application level errors. It may also be used to convey additional processing requirements in a text form.

This is the formal response to a ProcessRequest and may be used as a application level response to the PaymentNotice resource.

12.21.2 Resource Content

* Structure
* UML
* XML
* JSON
* Turtle
* R2 Diff
* All

Structure

Name | Flags | Card. | Type | Description & Constraints
ProcessResponse | | | DomainResource | ProcessResponse resource Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | | 0..* | Identifier | Business Identifier
status | ?!? | 0..1 | code | active | cancelled | draft | entered-in-error Financial Resource Status Codes (Required)
created | | 0..1 | dateTime | Creation date
organization | | 0..1 | Reference(Organization) | Authoring Organization
request | | 0..1 | Reference(Any) | Request reference
outcome | | 0..1 | CodeableConcept | Processing outcome Process Outcome Codes (Example)
disposition | | 0..1 | string | Disposition Message
requestProvider | | 0..1 | Reference(Practitioner) | Responsible Practitioner
requestOrganization | | 0..1 | Reference(Organization) | Responsible organization
form | | 0..1 | CodeableConcept | Printed Form Identifier Form Codes (Example)
processNote | | 0..* | BackboneElement | Processing comments or additional requirements

type | | 0..1 | CodeableConcept | display | print | printoper NoteType (Required)
text | | 0..1 | string | Comment on the processing
error | | 0..* | CodeableConcept | Error code Adjudication Error Codes (Example)
communicationRequest | | 0..* | Reference(CommunicationRequest) | Request for additional information
Documentation for this format

UML Diagram (Legend)

XML Template

```
<ProcessResponse xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Business Identifier --></identifier>
 <status value="[code]"/><!-- 0..1 active | cancelled | draft | entered-in-error -->
 <created value="[dateTime]"/><!-- 0..1 Creation date -->
 <organization><!-- 0..1 Reference(Organization) Authoring Organization --></organization>
 <request><!-- 0..1 Reference(Any) Request reference --></request>
 <outcome><!-- 0..1 CodeableConcept Processing outcome --></outcome>
 <disposition value="[string]"/><!-- 0..1 Disposition Message -->
 <requestProvider><!-- 0..1 Reference(Practitioner) Responsible Practitioner --></requestProvider>
 <requestOrganization><!-- 0..1 Reference(Organization) Responsible organization --></requestOrganization>
 <form><!-- 0..1 CodeableConcept Printed Form Identifier --></form>
 <processNote>  <!-- 0..* Processing comments or additional requirements -->
  <type><!-- 0..1 CodeableConcept display | print | printoper --></type>
  <text value="[string]"/><!-- 0..1 Comment on the processing -->
 </processNote>
 <error><!-- 0..* CodeableConcept Error code --></error>
 <communicationRequest><!-- 0..* Reference(CommunicationRequest) Request for additional information
    --></communicationRequest>
</ProcessResponse>
```

JSON Template

```
{
  "resourceType" : "ProcessResponse",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Business Identifier
  "status" : "<code>", // active | cancelled | draft | entered-in-error
  "created" : "<dateTime>", // Creation date
  "organization" : { Reference(Organization) }, // Authoring Organization
  "request" : { Reference(Any) }, // Request reference
  "outcome" : { CodeableConcept }, // Processing outcome
  "disposition" : "<string>", // Disposition Message
  "requestProvider" : { Reference(Practitioner) }, // Responsible Practitioner
  "requestOrganization" : { Reference(Organization) }, // Responsible organization
  "form" : { CodeableConcept }, // Printed Form Identifier
  "processNote" : [{ // Processing comments or additional requirements
    "type" : { CodeableConcept }, // display | print | printoper
    "text" : "<string>" // Comment on the processing
  }],
  "error" : [{ CodeableConcept }], // Error code
  "communicationRequest" : [{ Reference(CommunicationRequest) }] // Request for additional information
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:ProcessResponse;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:ProcessResponse.identifier [ Identifier ], ... ; # 0..* Business Identifier
  fhir:ProcessResponse.status [ code ]; # 0..1 active | cancelled | draft | entered-in-error
  fhir:ProcessResponse.created [ dateTime ]; # 0..1 Creation date
  fhir:ProcessResponse.organization [ Reference(Organization) ]; # 0..1 Authoring Organization
  fhir:ProcessResponse.request [ Reference(Any) ]; # 0..1 Request reference
  fhir:ProcessResponse.outcome [ CodeableConcept ]; # 0..1 Processing outcome
  fhir:ProcessResponse.disposition [ string ]; # 0..1 Disposition Message
  fhir:ProcessResponse.requestProvider [ Reference(Practitioner) ]; # 0..1 Responsible Practitioner
  fhir:ProcessResponse.requestOrganization [ Reference(Organization) ]; # 0..1 Responsible organization
  fhir:ProcessResponse.form [ CodeableConcept ]; # 0..1 Printed Form Identifier
  fhir:ProcessResponse.processNote [ # 0..* Processing comments or additional requirements
    fhir:ProcessResponse.processNote.type [ CodeableConcept ]; # 0..1 display | print | printoper
    fhir:ProcessResponse.processNote.text [ string ]; # 0..1 Comment on the processing
  ], ...;
  fhir:ProcessResponse.error [ CodeableConcept ], ... ; # 0..* Error code
  fhir:ProcessResponse.communicationRequest [ Reference(CommunicationRequest) ], ... ; # 0..* Request for additional
    information
]
```

Changes since DSTU2

ProcessResponse
ProcessResponse.status | Added Element
ProcessResponse.outcome | Type changed from Coding to CodeableConcept
ProcessResponse.form | Type changed from Coding to CodeableConcept Remove Binding http://hl7.org/fhir/ValueSet/forms (required)

ProcessResponse.processNote | Added Element
ProcessResponse.processNote.type | Added Element
ProcessResponse.processNote.text | Added Element
ProcessResponse.error | Type changed from Coding to CodeableConcept Remove Binding http://hl7.org/fhir/ValueSet/adjudication-error (required)
ProcessResponse.communicationRequest | Added Element
ProcessResponse.ruleset | deleted
ProcessResponse.originalRuleset | deleted
ProcessResponse.notes | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

Structure

Name | Flags | Card. | Type | Description & Constraints
ProcessResponse | | | DomainResource | ProcessResponse resource Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | | 0..* | Identifier | Business Identifier
status | ?!? | 0..1 | code | active | cancelled | draft | entered-in-error Financial Resource Status Codes (Required)
created | | 0..1 | dateTime | Creation date
organization | | 0..1 | Reference(Organization) | Authoring Organization
request | | 0..1 | Reference(Any) | Request reference
outcome | | 0..1 | CodeableConcept | Processing outcome Process Outcome Codes (Example)
disposition | | 0..1 | string | Disposition Message
requestProvider | | 0..1 | Reference(Practitioner) | Responsible Practitioner
requestOrganization | | 0..1 | Reference(Organization) | Responsible organization
form | | 0..1 | CodeableConcept | Printed Form Identifier Form Codes (Example)
processNote | | 0..* | BackboneElement | Processing comments or additional requirements
type | | 0..1 | CodeableConcept | display | print | printoper NoteType (Required)
text | | 0..1 | string | Comment on the processing
error | | 0..* | CodeableConcept | Error code Adjudication Error Codes (Example)
communicationRequest | | 0..* | Reference(CommunicationRequest) | Request for additional information
Documentation for this format

UML Diagram (Legend)

XML Template

```
<ProcessResponse xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Business Identifier --></identifier>
 <status value="[code]"/><!-- 0..1 active | cancelled | draft | entered-in-error -->
 <created value="[dateTime]"/><!-- 0..1 Creation date -->
 <organization><!-- 0..1 Reference(Organization) Authoring Organization --></organization>
 <request><!-- 0..1 Reference(Any) Request reference --></request>
 <outcome><!-- 0..1 CodeableConcept Processing outcome --></outcome>
 <disposition value="[string]"/><!-- 0..1 Disposition Message -->
 <requestProvider><!-- 0..1 Reference(Practitioner) Responsible Practitioner --></requestProvider>
 <requestOrganization><!-- 0..1 Reference(Organization) Responsible organization --></requestOrganization>
 <form><!-- 0..1 CodeableConcept Printed Form Identifier --></form>
 <processNote>  <!-- 0..* Processing comments or additional requirements -->
  <type><!-- 0..1 CodeableConcept display | print | printoper --></type>
  <text value="[string]"/><!-- 0..1 Comment on the processing -->
 </processNote>
 <error><!-- 0..* CodeableConcept Error code --></error>
 <communicationRequest><!-- 0..* Reference(CommunicationRequest) Request for additional information
    --></communicationRequest>
</ProcessResponse>
```

JSON Template

```
{
  "resourceType" : "ProcessResponse",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Business Identifier
  "status" : "<code>", // active | cancelled | draft | entered-in-error
  "created" : "<dateTime>", // Creation date
  "organization" : { Reference(Organization) }, // Authoring Organization
  "request" : { Reference(Any) }, // Request reference
  "outcome" : { CodeableConcept }, // Processing outcome
  "disposition" : "<string>", // Disposition Message
  "requestProvider" : { Reference(Practitioner) }, // Responsible Practitioner
  "requestOrganization" : { Reference(Organization) }, // Responsible organization
  "form" : { CodeableConcept }, // Printed Form Identifier
  "processNote" : [{ // Processing comments or additional requirements
    "type" : { CodeableConcept }, // display | print | printoper
    "text" : "<string>" // Comment on the processing
  }],
  "error" : [{ CodeableConcept }], // Error code
  "communicationRequest" : [{ Reference(CommunicationRequest) }] // Request for additional information
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:ProcessResponse;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:ProcessResponse.identifier [ Identifier ], ... ; # 0..* Business Identifier
  fhir:ProcessResponse.status [ code ]; # 0..1 active | cancelled | draft | entered-in-error
  fhir:ProcessResponse.created [ dateTime ]; # 0..1 Creation date
  fhir:ProcessResponse.organization [ Reference(Organization) ]; # 0..1 Authoring Organization
  fhir:ProcessResponse.request [ Reference(Any) ]; # 0..1 Request reference
  fhir:ProcessResponse.outcome [ CodeableConcept ]; # 0..1 Processing outcome
  fhir:ProcessResponse.disposition [ string ]; # 0..1 Disposition Message
  fhir:ProcessResponse.requestProvider [ Reference(Practitioner) ]; # 0..1 Responsible Practitioner
  fhir:ProcessResponse.requestOrganization [ Reference(Organization) ]; # 0..1 Responsible organization
  fhir:ProcessResponse.form [ CodeableConcept ]; # 0..1 Printed Form Identifier
  fhir:ProcessResponse.processNote [ # 0..* Processing comments or additional requirements
    fhir:ProcessResponse.processNote.type [ CodeableConcept ]; # 0..1 display | print | printoper
    fhir:ProcessResponse.processNote.text [ string ]; # 0..1 Comment on the processing
  ], ...;
  fhir:ProcessResponse.error [ CodeableConcept ], ... ; # 0..* Error code
  fhir:ProcessResponse.communicationRequest [ Reference(CommunicationRequest) ], ... ; # 0..* Request for additional
    information
]
```

Changes since DSTU2

ProcessResponse
ProcessResponse.status | Added Element
ProcessResponse.outcome | Type changed from Coding to CodeableConcept
ProcessResponse.form | Type changed from Coding to CodeableConcept Remove Binding http://hl7.org/fhir/ValueSet/forms (required)
ProcessResponse.processNote | Added Element
ProcessResponse.processNote.type | Added Element
ProcessResponse.processNote.text | Added Element
ProcessResponse.error | Type changed from Coding to CodeableConcept Remove Binding http://hl7.org/fhir/ValueSet/adjudication-error (required)
ProcessResponse.communicationRequest | Added Element
ProcessResponse.ruleset | deleted
ProcessResponse.originalRuleset | deleted
ProcessResponse.notes | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

Alternate definitions: Master Definition (XML, JSON), XML Schema/Schematron (for) + JSON Schema, ShEx (for Turtle)

12.21.2.1 Terminology Bindings

Path | Definition | Type | Reference
ProcessResponse.status | A code specifying the state of the resource instance. | Required | Financial Resource Status Codes
ProcessResponse.outcome | Local status of outcome codes | Example | Process Outcome Codes
ProcessResponse.form | The forms codes. | Example | Form Codes
ProcessResponse.processNote.type | The presentation types of notes. | Required | NoteType
ProcessResponse.error | The error codes for adjudication processing. | Example | Adjudication Error Codes

12.21.3 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
identifier | token | The business identifier of the Explanation of Benefit | ProcessResponse.identifier
organization | reference | The organization who generated this resource | ProcessResponse.organization (Organization)
request | reference | The reference to the claim | ProcessResponse.request (Any)
request-organization | reference | The Organization who is responsible the request transaction | ProcessResponse.requestOrganization (Organization)
request-provider | reference | The Provider who is responsible the request transaction | ProcessResponse.requestProvider (Practitioner)

12.21.4 Resource ProcessResponse - Examples

URL: https://hl7.org/fhir/STU3/processresponse-examples.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

12.21.4 Resource ProcessResponse - Examples

Financial Management Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Practitioner

Example List:

General Eligibility Status Check Example | SR2500 | XML | JSON | Turtle

Sample ProcessResponse with error | SR2349 | XML | JSON | Turtle
ProcessResponse with Pended status and request for additional information | SR2499 | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

12.21.5 Resource ProcessResponse - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/processresponse-definitions.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

12.21.5 Resource ProcessResponse - Detailed Descriptions

Financial Management Work Group | Maturity Level: 2 | Trial Use | Compartments: Practitioner

Detailed Descriptions for the elements in the ProcessResponse resource.

ProcessResponse
Definition | This resource provides processing status, errors and notes from the processing of a resource.
Control | 1..1
ProcessResponse.identifier
Definition | The Response business identifier.
Note | This is a business identifer, not a resource identifier (see discussion)
Control | 0..*
Type | Identifier
ProcessResponse.status
Definition | The status of the resource instance.
Control | 0..1
Terminology Binding | Financial Resource Status Codes (Required)
Type | code
Is Modifier | true
Summary | true
Comments | This element is labeled as a modifier because the status contains codes that mark the resource as not currently valid.
ProcessResponse.created
Definition | The date when the enclosed suite of services were performed or completed.
Control | 0..1
Type | dateTime
ProcessResponse.organization
Definition | The organization who produced this adjudicated response.
Control | 0..1
Type | Reference(Organization)
ProcessResponse.request
Definition | Original request resource reference.
Control | 0..1
Type | Reference(Any)
ProcessResponse.outcome
Definition | Transaction status: error, complete, held.
Control | 0..1
Terminology Binding | Process Outcome Codes (Example)
Type | CodeableConcept
ProcessResponse.disposition
Definition | A description of the status of the adjudication or processing.
Control | 0..1
Type | string
ProcessResponse.requestProvider
Definition | The practitioner who is responsible for the services rendered to the patient.
Control | 0..1
Type | Reference(Practitioner)
ProcessResponse.requestOrganization
Definition | The organization which is responsible for the services rendered to the patient.
Control | 0..1
Type | Reference(Organization)
ProcessResponse.form
Definition | The form to be used for printing the content.
Control | 0..1
Terminology Binding | Form Codes (Example)
Type | CodeableConcept
ProcessResponse.processNote
Definition | Suite of processing notes or additional requirements if the processing has been held.
Control | 0..*
ProcessResponse.processNote.type
Definition | The note purpose: Print/Display.
Control | 0..1

Terminology Binding | NoteType (Required)
Type | CodeableConcept
ProcessResponse.processNote.text
Definition | The note text.
Control | 0..1
Type | string
ProcessResponse.error
Definition | Processing errors.
Control | 0..*
Terminology Binding | Adjudication Error Codes (Example)
Type | CodeableConcept
ProcessResponse.communicationRequest
Definition | Request for additional supporting or authorizing information, such as: documents, images or resources.
Control | 0..*
Type | Reference(CommunicationRequest)

12.21.6 Resource ProcessResponse - Mappings

URL: https://hl7.org/fhir/STU3/processresponse-mappings.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

12.21.6 Resource ProcessResponse - Mappings

Financial Management Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Practitioner

Mappings:

Canadian Dental Association eclaims standard

Workflow Mapping

Mappings for the processresponse resource.

12.21.6.1 Mappings for Canadian Dental Association eclaims standard (http://www.cda-adc.ca/en/services/cdanet/)

ProcessResponse
identifier
status
created
organization
request | A02|G01
outcome
disposition
requestProvider | B01
requestOrganization
form | G42
processNote
type
text | G32
error
communicationRequest

12.21.6.2 Mappings for Workflow Mapping (http://hl7.org/fhir/workflow)

ProcessResponse | Event
identifier | Event.identifier
status | Event.status
created | Event.occurrence
organization | Event.performer.onBehalfOf
request
outcome
disposition | Event.note
requestProvider
requestOrganization
form
processNote
type
text
error
communicationRequest

12.21.7 Resource ProcessResponse - Design Notes

URL: https://hl7.org/fhir/STU3/processresponse-explanations.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

12.21.7 Resource ProcessResponse - Design Notes

to do

12.21.8 Resource ProcessResponse - Extensions & Profiles

URL: https://hl7.org/fhir/STU3/processresponse-profiles.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

12.21.8 Resource ProcessResponse - Extensions & Profiles

Financial Management Work Group | Maturity Level: N/A | Ballot Status: Informative

This table lists profiles and extensions for the ProcessResponse resource. For background information, see Profiling Resources and Extensibility. Additional profiles and resources may be found in published Implementation Guides , or in the Conformance resource registry .

12.21.8.1 Profiles

No Profiles defined for this resource

12.21.8.2 Extensions

No Extensions defined for this resource

Extensions that reference this resource:

No Extensions refer to this resource

12.21.8.3 Search Extensions

No Search Extensions defined for this resource

12.22 Resource DeviceRequest - Content

URL: https://hl7.org/fhir/STU3/devicerequest.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

Orders and Observations Work Group | Maturity Level: 0 | Draft | Compartments: Device, Encounter, Patient, Practitioner

Represents a request for a patient to employ a medical device. The device may be an implantable device, or an external assistive device, such as a walker.

12.22.1 Scope and Usage

This resource is a request resource from a FHIR workflow perspective - see Workflow.

This resource describes the request for the use of a device by a patient. The device may be any pertinent device specified in the Device resource. Examples of devices that may be requested include wheelchair, hearing aids, or an insulin pump. The request may lead to the dispensing of the device to the patient or for use by the patient.

The device use request may represent an order or a prescription entered by a practitioner in a CPOE system or a proposal made by a clinical decision support (CDS) system based on a patient's clinical record and context of care.

12.22.2 Boundaries and Relationships

This resource deals with the allocation of a device to a patient and while it may contain instructions on how to use the device, the data about getting the device to the patient is addressed in other resources. For example, Certain devices must be implanted via a surgical or other procedure and the implantation or explantation is represented in the [Procedure](procedure.html) or [ProcedureRequest](procedurerequest.html) resource.

The SupplyRequest resource is similar in that it deals with requesting a particular item for a specific patient. However, DeviceRequest is concerned with items where there is a patient focus, or instructions regarding their use, whereas SupplyRequest is concerned with inventory management.

This resource is referenced by CarePlan and Observation

12.22.3 Resource Content

* Structure

* UML

* XML

* JSON

* Turtle

* R2 Diff

* All

Structure

Name | Flags | Card. | Type | Description & Constraints
DeviceRequest | | | DomainResource | Medical device request Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | ? | 0..* | Identifier | External Request identifier
definition | ? | 0..* | Reference(ActivityDefinition | PlanDefinition) | Protocol or definition
basedOn | ? | 0..* | Reference(Any) | What request fulfills
priorRequest | ? | 0..* | Reference(Any) | What request replaces
groupIdentifier | ? | 0..1 | Identifier | Identifier of composite request
status | ?!? | 0..1 | code | draft | active | suspended | completed | entered-in-error | cancelled RequestStatus (Required)
intent | ?!? | 1..1 | CodeableConcept | proposal | plan | original-order | encoded | reflex-order RequestIntent (Required)
priority | ? | 0..1 | code | Indicates how quickly the {{title}} should be addressed with respect to other requests RequestPriority (Required)
code[x] | ? | 1..1 | | Device requested FHIR Device Types (Example)
codeReference | | | Reference(Device)
codeCodeableConcept | | | CodeableConcept
subject | ? | 1..1 | Reference(Patient | Group | Location | Device) | Focus of request
context | ? | 0..1 | Reference(Encounter | EpisodeOfCare) | Encounter or Episode motivating request
occurrence[x] | ? | 0..1 | | Desired time or schedule for use
occurrenceDateTime | | | dateTime
occurrencePeriod | | | Period
occurrenceTiming | | | Timing
authoredOn | ? | 0..1 | dateTime | When recorded
requester | ? | 0..1 | BackboneElement | Who/what is requesting diagnostics
agent | ? | 1..1 | Reference(Device | Practitioner | Organization) | Individual making the request
onBehalfOf | ? | 0..1 | Reference(Organization) | Organization agent is acting for
performerType | ? | 0..1 | CodeableConcept | Fille role Participant Roles (Example)
performer | ? | 0..1 | Reference(Practitioner | Organization | Patient | Device | RelatedPerson | HealthcareService) | Requested Filler
reasonCode | ? | 0..* | CodeableConcept | Coded Reason for request Condition/Problem/Diagnosis Codes (Example)
reasonReference | ? | 0..* | Reference(Any) | Linked Reason for request
supportingInfo | | 0..* | Reference(Any) | Additional clinical information
note | | 0..* | Annotation | Notes or comments
relevantHistory | | 0..* | Reference(Provenance) | Request provenance
Documentation for this format

UML Diagram (Legend)

XML Template

```
<DeviceRequest xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier External Request identifier --></identifier>
 <definition><!-- 0..* Reference(ActivityDefinition|PlanDefinition) Protocol or definition --></definition>
 <basedOn><!-- 0..* Reference(Any) What request fulfills --></basedOn>
 <priorRequest><!-- 0..* Reference(Any) What request replaces --></priorRequest>
 <groupIdentifier><!-- 0..1 Identifier Identifier of composite request --></groupIdentifier>
 <status value="[code]"/><!-- 0..1 draft | active | suspended | completed | entered-in-error | cancelled -->
 <intent><!-- 1..1 CodeableConcept proposal | plan | original-order | encoded | reflex-order  --></intent>
 <priority value="[code]"/><!-- 0..1 Indicates how quickly the {{title}} should be addressed with respect to other
    requests -->
 <code[x]><!-- 1..1 Reference(Device)|CodeableConcept Device requested --></code[x]>
 <subject><!-- 1..1 Reference(Patient|Group|Location|Device) Focus of request --></subject>
 <context><!-- 0..1 Reference(Encounter|EpisodeOfCare) Encounter or Episode motivating request --></context>
 <occurrence[x]><!-- 0..1 dateTime|Period|Timing Desired time or schedule for use --></occurrence[x]>
 <authoredOn value="[dateTime]"/><!-- 0..1 When recorded -->
 <requester>  <!-- 0..1 Who/what is requesting diagnostics -->
  <agent><!-- 1..1 Reference(Device|Practitioner|Organization) Individual making the request --></agent>
  <onBehalfOf><!-- 0..1 Reference(Organization) Organization agent is acting for --></onBehalfOf>
 </requester>
 <performerType><!-- 0..1 CodeableConcept Fille role --></performerType>
 <performer><!-- 0..1 Reference(Practitioner|Organization|Patient|Device|
   RelatedPerson|HealthcareService) Requested Filler --></performer>
 <reasonCode><!-- 0..* CodeableConcept Coded Reason for request --></reasonCode>
 <reasonReference><!-- 0..* Reference(Any) Linked Reason for request --></reasonReference>
 <supportingInfo><!-- 0..* Reference(Any) Additional clinical information --></supportingInfo>
 <note><!-- 0..* Annotation Notes or comments --></note>
 <relevantHistory><!-- 0..* Reference(Provenance) Request provenance --></relevantHistory>
</DeviceRequest>
```

JSON Template

```
{
  "resourceType" : "DeviceRequest",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // External Request identifier
  "definition" : [{ Reference(ActivityDefinition|PlanDefinition) }], // Protocol or definition
  "basedOn" : [{ Reference(Any) }], // What request fulfills
  "priorRequest" : [{ Reference(Any) }], // What request replaces
  "groupIdentifier" : { Identifier }, // Identifier of composite request
```

```
  "status" : "<code>", // draft | active | suspended | completed | entered-in-error | cancelled
  "intent" : { CodeableConcept }, // R!  proposal | plan | original-order | encoded | reflex-order
  "priority" : "<code>", // Indicates how quickly the {{title}} should be addressed with respect to other requests
  // code[x]: Device requested. One of these 2:
  "codeReference" : { Reference(Device) },
  "codeCodeableConcept" : { CodeableConcept },
  "subject" : { Reference(Patient|Group|Location|Device) }, // R!  Focus of request
  "context" : { Reference(Encounter|EpisodeOfCare) }, // Encounter or Episode motivating request
  // occurrence[x]: Desired time or schedule for use. One of these 3:
  "occurrenceDateTime" : "<dateTime>",
  "occurrencePeriod" : { Period },
  "occurrenceTiming" : { Timing },
  "authoredOn" : "<dateTime>", // When recorded
  "requester" : { // Who/what is requesting diagnostics
    "agent" : { Reference(Device|Practitioner|Organization) }, // R!  Individual making the request
    "onBehalfOf" : { Reference(Organization) } // Organization agent is acting for
  },
  "performerType" : { CodeableConcept }, // Fille role
  "performer" : { Reference(Practitioner|Organization|Patient|Device|
   RelatedPerson|HealthcareService) }, // Requested Filler
  "reasonCode" : [{ CodeableConcept }], // Coded Reason for request
  "reasonReference" : [{ Reference(Any) }], // Linked Reason for request
  "supportingInfo" : [{ Reference(Any) }], // Additional clinical information
  "note" : [{ Annotation }], // Notes or comments
  "relevantHistory" : [{ Reference(Provenance) }] // Request provenance
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:DeviceRequest;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:DeviceRequest.identifier [ Identifier ], ... ; # 0..* External Request identifier
  fhir:DeviceRequest.definition [ Reference(ActivityDefinition|PlanDefinition) ], ... ; # 0..* Protocol or definition
  fhir:DeviceRequest.basedOn [ Reference(Any) ], ... ; # 0..* What request fulfills
  fhir:DeviceRequest.priorRequest [ Reference(Any) ], ... ; # 0..* What request replaces
  fhir:DeviceRequest.groupIdentifier [ Identifier ]; # 0..1 Identifier of composite request
  fhir:DeviceRequest.status [ code ]; # 0..1 draft | active | suspended | completed | entered-in-error | cancelled
  fhir:DeviceRequest.intent [ CodeableConcept ]; # 1..1 proposal | plan | original-order | encoded | reflex-order
  fhir:DeviceRequest.priority [ code ]; # 0..1 Indicates how quickly the {{title}} should be addressed with respect to
    other requests
  # DeviceRequest.code[x] : 1..1 Device requested. One of these 2
    fhir:DeviceRequest.codeReference [ Reference(Device) ]
    fhir:DeviceRequest.codeCodeableConcept [ CodeableConcept ]
  fhir:DeviceRequest.subject [ Reference(Patient|Group|Location|Device) ]; # 1..1 Focus of request
  fhir:DeviceRequest.context [ Reference(Encounter|EpisodeOfCare) ]; # 0..1 Encounter or Episode motivating request
  # DeviceRequest.occurrence[x] : 0..1 Desired time or schedule for use. One of these 3
    fhir:DeviceRequest.occurrenceDateTime [ dateTime ]
    fhir:DeviceRequest.occurrencePeriod [ Period ]
    fhir:DeviceRequest.occurrenceTiming [ Timing ]
  fhir:DeviceRequest.authoredOn [ dateTime ]; # 0..1 When recorded
  fhir:DeviceRequest.requester [ # 0..1 Who/what is requesting diagnostics
    fhir:DeviceRequest.requester.agent [ Reference(Device|Practitioner|Organization) ]; # 1..1 Individual making the
    request
    fhir:DeviceRequest.requester.onBehalfOf [ Reference(Organization) ]; # 0..1 Organization agent is acting for
  ];
  fhir:DeviceRequest.performerType [ CodeableConcept ]; # 0..1 Fille role
  fhir:DeviceRequest.performer [ Reference(Practitioner|Organization|Patient|Device|RelatedPerson|HealthcareService) ];
    # 0..1 Requested Filler
  fhir:DeviceRequest.reasonCode [ CodeableConcept ], ... ; # 0..* Coded Reason for request
  fhir:DeviceRequest.reasonReference [ Reference(Any) ], ... ; # 0..* Linked Reason for request
  fhir:DeviceRequest.supportingInfo [ Reference(Any) ], ... ; # 0..* Additional clinical information
  fhir:DeviceRequest.note [ Annotation ], ... ; # 0..* Notes or comments
  fhir:DeviceRequest.relevantHistory [ Reference(Provenance) ], ... ; # 0..* Request provenance
]
```

Changes since DSTU2

DeviceRequest

DeviceRequest.definition | Added Element

DeviceRequest.basedOn | Added Element

DeviceRequest.priorRequest | Added Element

DeviceRequest.groupIdentifier | Added Element

DeviceRequest.status | Change value set from http://hl7.org/fhir/ValueSet/device-use-request-status to http://hl7.org/fhir/ValueSet/request-status

DeviceRequest.intent | Added Element

DeviceRequest.priority | Change value set from http://hl7.org/fhir/ValueSet/device-use-request-priority to http://hl7.org/fhir/ValueSet/request-priority
Default Value "If missing, normal priority" added

DeviceRequest.code[x] | Added Element

DeviceRequest.subject | Add Reference(Group), Add Reference(Location), Add Reference(Device)

DeviceRequest.context | Renamed from encounter to context Add Reference(EpisodeOfCare)

DeviceRequest.occurrence[x] | Added Element

DeviceRequest.authoredOn | Added Element

DeviceRequest.requester | Added Element
DeviceRequest.requester.agent | Added Element
DeviceRequest.requester.onBehalfOf | Added Element
DeviceRequest.performerType | Added Element
DeviceRequest.performer | Added Element
DeviceRequest.reasonCode | Renamed from indication to reasonCode
DeviceRequest.reasonReference | Added Element
DeviceRequest.supportingInfo | Added Element
DeviceRequest.note | Renamed from notes to note Type changed from string to Annotation
DeviceRequest.relevantHistory | Added Element
DeviceRequest.bodySite[x] | deleted
DeviceRequest.device | deleted
DeviceRequest.prnReason | deleted
DeviceRequest.orderedOn | deleted
DeviceRequest.recordedOn | deleted
DeviceRequest.timing[x] | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

Structure

Name | Flags | Card. | Type | Description & Constraints
DeviceRequest | | | DomainResource | Medical device request Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | ? | 0..* | Identifier | External Request identifier
definition | ? | 0..* | Reference(ActivityDefinition | PlanDefinition) | Protocol or definition
basedOn | ? | 0..* | Reference(Any) | What request fulfills
priorRequest | ? | 0..* | Reference(Any) | What request replaces
groupIdentifier | ? | 0..1 | Identifier | Identifier of composite request
status | ?!? | 0..1 | code | draft | active | suspended | completed | entered-in-error | cancelled RequestStatus (Required)
intent | ?!? | 1..1 | CodeableConcept | proposal | plan | original-order | encoded | reflex-order RequestIntent (Required)
priority | ? | 0..1 | code | Indicates how quickly the {{title}} should be addressed with respect to other requests RequestPriority (Required)
code[x] | ? | 1..1 | | Device requested FHIR Device Types (Example)
codeReference | | | Reference(Device)
codeCodeableConcept | | | CodeableConcept
subject | ? | 1..1 | Reference(Patient | Group | Location | Device) | Focus of request
context | ? | 0..1 | Reference(Encounter | EpisodeOfCare) | Encounter or Episode motivating request
occurrence[x] | ? | 0..1 | | Desired time or schedule for use
occurrenceDateTime | | | dateTime
occurrencePeriod | | | Period
occurrenceTiming | | | Timing
authoredOn | ? | 0..1 | dateTime | When recorded
requester | ? | 0..1 | BackboneElement | Who/what is requesting diagnostics
agent | ? | 1..1 | Reference(Device | Practitioner | Organization) | Individual making the request
onBehalfOf | ? | 0..1 | Reference(Organization) | Organization agent is acting for
performerType | ? | 0..1 | CodeableConcept | Fille role Participant Roles (Example)
performer | ? | 0..1 | Reference(Practitioner | Organization | Patient | Device | RelatedPerson | HealthcareService) | Requested Filler
reasonCode | ? | 0..* | CodeableConcept | Coded Reason for request Condition/Problem/Diagnosis Codes (Example)
reasonReference | ? | 0..* | Reference(Any) | Linked Reason for request
supportingInfo | | 0..* | Reference(Any) | Additional clinical information
note | | 0..* | Annotation | Notes or comments
relevantHistory | | 0..* | Reference(Provenance) | Request provenance
Documentation for this format

UML Diagram (Legend)

XML Template

```
<DeviceRequest xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier External Request identifier --></identifier>
 <definition><!-- 0..* Reference(ActivityDefinition|PlanDefinition) Protocol or definition --></definition>
 <basedOn><!-- 0..* Reference(Any) What request fulfills --></basedOn>
 <priorRequest><!-- 0..* Reference(Any) What request replaces --></priorRequest>
 <groupIdentifier><!-- 0..1 Identifier Identifier of composite request --></groupIdentifier>
 <status value="[code]"/><!-- 0..1 draft | active | suspended | completed | entered-in-error | cancelled -->
 <intent><!-- 1..1 CodeableConcept proposal | plan | original-order | encoded | reflex-order  --></intent>
 <priority value="[code]"/><!-- 0..1 Indicates how quickly the {{title}} should be addressed with respect to other
    requests -->
 <code[x]><!-- 1..1 Reference(Device)|CodeableConcept Device requested --></code[x]>
 <subject><!-- 1..1 Reference(Patient|Group|Location|Device) Focus of request --></subject>
 <context><!-- 0..1 Reference(Encounter|EpisodeOfCare) Encounter or Episode motivating request --></context>
 <occurrence[x]><!-- 0..1 dateTime|Period|Timing Desired time or schedule for use --></occurrence[x]>
 <authoredOn value="[dateTime]"/><!-- 0..1 When recorded -->
 <requester>  <!-- 0..1 Who/what is requesting diagnostics -->
  <agent><!-- 1..1 Reference(Device|Practitioner|Organization) Individual making the request --></agent>
  <onBehalfOf><!-- 0..1 Reference(Organization) Organization agent is acting for --></onBehalfOf>
 </requester>
```

```
 <performerType><!-- 0..1 CodeableConcept Fille role --></performerType>
 <performer><!-- 0..1 Reference(Practitioner|Organization|Patient|Device|
   RelatedPerson|HealthcareService) Requested Filler --></performer>
 <reasonCode><!-- 0..* CodeableConcept Coded Reason for request --></reasonCode>
 <reasonReference><!-- 0..* Reference(Any) Linked Reason for request --></reasonReference>
 <supportingInfo><!-- 0..* Reference(Any) Additional clinical information --></supportingInfo>
 <note><!-- 0..* Annotation Notes or comments --></note>
 <relevantHistory><!-- 0..* Reference(Provenance) Request provenance --></relevantHistory>
</DeviceRequest>
```

JSON Template

```
{
  "resourceType" : "DeviceRequest",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // External Request identifier
  "definition" : [{ Reference(ActivityDefinition|PlanDefinition) }], // Protocol or definition
  "basedOn" : [{ Reference(Any) }], // What request fulfills
  "priorRequest" : [{ Reference(Any) }], // What request replaces
  "groupIdentifier" : { Identifier }, // Identifier of composite request
  "status" : "<code>", // draft | active | suspended | completed | entered-in-error | cancelled
  "intent" : { CodeableConcept }, // R!  proposal | plan | original-order | encoded | reflex-order
  "priority" : "<code>", // Indicates how quickly the {{title}} should be addressed with respect to other requests
  // code[x]: Device requested. One of these 2:
  "codeReference" : { Reference(Device) },
  "codeCodeableConcept" : { CodeableConcept },
  "subject" : { Reference(Patient|Group|Location|Device) }, // R!  Focus of request
  "context" : { Reference(Encounter|EpisodeOfCare) }, // Encounter or Episode motivating request
  // occurrence[x]: Desired time or schedule for use. One of these 3:
  "occurrenceDateTime" : "<dateTime>",
  "occurrencePeriod" : { Period },
  "occurrenceTiming" : { Timing },
  "authoredOn" : "<dateTime>", // When recorded
  "requester" : { // Who/what is requesting diagnostics
    "agent" : { Reference(Device|Practitioner|Organization) }, // R!  Individual making the request
    "onBehalfOf" : { Reference(Organization) } // Organization agent is acting for
  },
  "performerType" : { CodeableConcept }, // Fille role
  "performer" : { Reference(Practitioner|Organization|Patient|Device|
   RelatedPerson|HealthcareService) }, // Requested Filler
  "reasonCode" : [{ CodeableConcept }], // Coded Reason for request
  "reasonReference" : [{ Reference(Any) }], // Linked Reason for request
  "supportingInfo" : [{ Reference(Any) }], // Additional clinical information
  "note" : [{ Annotation }], // Notes or comments
  "relevantHistory" : [{ Reference(Provenance) }] // Request provenance
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:DeviceRequest;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:DeviceRequest.identifier [ Identifier ], ... ; # 0..* External Request identifier
  fhir:DeviceRequest.definition [ Reference(ActivityDefinition|PlanDefinition) ], ... ; # 0..* Protocol or definition
  fhir:DeviceRequest.basedOn [ Reference(Any) ], ... ; # 0..* What request fulfills
  fhir:DeviceRequest.priorRequest [ Reference(Any) ], ... ; # 0..* What request replaces
  fhir:DeviceRequest.groupIdentifier [ Identifier ]; # 0..1 Identifier of composite request
  fhir:DeviceRequest.status [ code ]; # 0..1 draft | active | suspended | completed | entered-in-error | cancelled
  fhir:DeviceRequest.intent [ CodeableConcept ]; # 1..1 proposal | plan | original-order | encoded | reflex-order
  fhir:DeviceRequest.priority [ code ]; # 0..1 Indicates how quickly the {{title}} should be addressed with respect to
    other requests
  # DeviceRequest.code[x] : 1..1 Device requested. One of these 2
    fhir:DeviceRequest.codeReference [ Reference(Device) ]
    fhir:DeviceRequest.codeCodeableConcept [ CodeableConcept ]
  fhir:DeviceRequest.subject [ Reference(Patient|Group|Location|Device) ]; # 1..1 Focus of request
  fhir:DeviceRequest.context [ Reference(Encounter|EpisodeOfCare) ]; # 0..1 Encounter or Episode motivating request
  # DeviceRequest.occurrence[x] : 0..1 Desired time or schedule for use. One of these 3
    fhir:DeviceRequest.occurrenceDateTime [ dateTime ]
    fhir:DeviceRequest.occurrencePeriod [ Period ]
    fhir:DeviceRequest.occurrenceTiming [ Timing ]
  fhir:DeviceRequest.authoredOn [ dateTime ]; # 0..1 When recorded
  fhir:DeviceRequest.requester [ # 0..1 Who/what is requesting diagnostics
    fhir:DeviceRequest.requester.agent [ Reference(Device|Practitioner|Organization) ]; # 1..1 Individual making the
    request
    fhir:DeviceRequest.requester.onBehalfOf [ Reference(Organization) ]; # 0..1 Organization agent is acting for
  ];
  fhir:DeviceRequest.performerType [ CodeableConcept ]; # 0..1 Fille role
  fhir:DeviceRequest.performer [ Reference(Practitioner|Organization|Patient|Device|RelatedPerson|HealthcareService) ];
    # 0..1 Requested Filler
  fhir:DeviceRequest.reasonCode [ CodeableConcept ], ... ; # 0..* Coded Reason for request
  fhir:DeviceRequest.reasonReference [ Reference(Any) ], ... ; # 0..* Linked Reason for request
  fhir:DeviceRequest.supportingInfo [ Reference(Any) ], ... ; # 0..* Additional clinical information
  fhir:DeviceRequest.note [ Annotation ], ... ; # 0..* Notes or comments
  fhir:DeviceRequest.relevantHistory [ Reference(Provenance) ], ... ; # 0..* Request provenance
]
```

Changes since DSTU2

DeviceRequest
DeviceRequest.definition | Added Element
DeviceRequest.basedOn | Added Element
DeviceRequest.priorRequest | Added Element
DeviceRequest.groupIdentifier | Added Element
DeviceRequest.status | Change value set from http://hl7.org/fhir/ValueSet/device-use-request-status to http://hl7.org/fhir/ValueSet/request-status
DeviceRequest.intent | Added Element
DeviceRequest.priority | Change value set from http://hl7.org/fhir/ValueSet/device-use-request-priority to http://hl7.org/fhir/ValueSet/request-priority Default Value "If missing, normal priority" added
DeviceRequest.code[x] | Added Element
DeviceRequest.subject | Add Reference(Group), Add Reference(Location), Add Reference(Device)
DeviceRequest.context | Renamed from encounter to context Add Reference(EpisodeOfCare)
DeviceRequest.occurrence[x] | Added Element
DeviceRequest.authoredOn | Added Element
DeviceRequest.requester | Added Element
DeviceRequest.requester.agent | Added Element
DeviceRequest.requester.onBehalfOf | Added Element
DeviceRequest.performerType | Added Element
DeviceRequest.performer | Added Element
DeviceRequest.reasonCode | Renamed from indication to reasonCode
DeviceRequest.reasonReference | Added Element
DeviceRequest.supportingInfo | Added Element
DeviceRequest.note | Renamed from notes to note Type changed from string to Annotation
DeviceRequest.relevantHistory | Added Element
DeviceRequest.bodySite[x] | deleted
DeviceRequest.device | deleted
DeviceRequest.prnReason | deleted
DeviceRequest.orderedOn | deleted
DeviceRequest.recordedOn | deleted
DeviceRequest.timing[x] | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

Alternate definitions: Master Definition (XML, JSON), XML Schema/Schematron (for) + JSON Schema, ShEx (for Turtle)

12.22.3.1 Terminology Bindings

Path | Definition | Type | Reference
DeviceRequest.status | Codes representing the status of the request. | Required | RequestStatus
DeviceRequest.intent | The kind of diagnostic request | Required | valueset-request-intent.html
DeviceRequest.priority | Identifies the level of importance to be assigned to actioning the request | Required | RequestPriority
DeviceRequest.code[x] | Codes for devices that can be requested | Example | FHIR Device Types
DeviceRequest.performerType | Indicates specific responsibility of an individual within the care team, such as "Primary physician", "Team coordinator", "Caregiver", etc. | Example | Participant Roles
DeviceRequest.reasonCode | Diagnosis or problem codes justifying the reason for requesting the device. | Example | Condition/Problem/Diagnosis Codes

Notes to reviewers:

At this time, the code bindings are placeholders to be fleshed out upon further review by the community.

12.22.4 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
authored-on | date | When the request transitioned to being actionable | DeviceRequest.authoredOn
based-on | reference | Plan/proposal/order fulfilled by this request | DeviceRequest.basedOn (Any)
code | token | Code for what is being requested/ordered | DeviceRequest.code.as(CodeableConcept) | 8 Resources
definition | reference | Protocol or definition followed by this request | DeviceRequest.definition (PlanDefinition, ActivityDefinition)
device | reference | Reference to resource that is being requested/ordered | DeviceRequest.code.as(Reference) (Device)
encounter | reference | Encounter or Episode during which request was created | DeviceRequest.context (EpisodeOfCare, Encounter) | 12 Resources
event-date | date | When service should occur | DeviceRequest.occurrence.as(DateTime) | DeviceRequest.occurrence.as(Period)
group-identifier | token | Composite request this is part of | DeviceRequest.groupIdentifier
identifier | token | Business identifier for request/order | DeviceRequest.identifier | 26 Resources
intent | token | proposal | plan | original-order |reflex-order | DeviceRequest.intent
patient | reference | Individual the service is ordered for | DeviceRequest.subject (Patient) | 31 Resources
performer | reference | Desired performer for service | DeviceRequest.performer (Practitioner, Organization, Device, Patient, HealthcareService, RelatedPerson)
priorrequest | reference | Request takes the place of referenced completed or terminated requests | DeviceRequest.priorRequest (Any)
requester | reference | Who/what is requesting service | DeviceRequest.requester.agent (Practitioner, Organization, Device)
status | token | entered-in-error | draft | active |suspended | completed | DeviceRequest.status

subject | reference | Individual the service is ordered for | DeviceRequest.subject (Group, Device, Patient, Location)

12.22.5 Resource DeviceRequest - Examples

URL: https://hl7.org/fhir/STU3/devicerequest-examples.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

12.22.5 Resource DeviceRequest - Examples

Orders and Observations Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Device, Encounter, Patient, Practitioner

An introduction to this page as well as proper examples are to be submitted shortly.

Example List:

Wheelchair assignment | example | XML | JSON | Turtle
Insulin Pump request | insulinpump | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

12.22.6 Resource DeviceRequest - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/devicerequest-definitions.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

12.22.6 Resource DeviceRequest - Detailed Descriptions

Orders and Observations Work Group | Maturity Level: 0 | Draft | Compartments: Device, Encounter, Patient, Practitioner

Detailed Descriptions for the elements in the DeviceRequest resource.

DeviceRequest
Definition | Represents a request for a patient to employ a medical device. The device may be an implantable device, or an external assistive device, such as a walker.
Control | 1..1
DeviceRequest.identifier
Definition | Identifiers assigned to this order by the orderer or by the receiver.
Note | This is a business identifer, not a resource identifier (see discussion)
Control | 0..*
Type | Identifier
Summary | true
DeviceRequest.definition
Definition | Protocol or definition followed by this request. For example: The proposed act must be performed if the indicated conditions occur, e.g.., shortness of breath, SpO2 less than x%.
Control | 0..*
Type | Reference(ActivityDefinition | PlanDefinition)
Summary | true
DeviceRequest.basedOn
Definition | Plan/proposal/order fulfilled by this request.
Control | 0..*
Type | Reference(Any)
Summary | true
DeviceRequest.priorRequest
Definition | The request takes the place of the referenced completed or terminated request(s).
Control | 0..*
Type | Reference(Any)
Summary | true
DeviceRequest.groupIdentifier
Definition | Composite request this is part of.
Control | 0..1
Type | Identifier
Summary | true
DeviceRequest.status
Definition | The status of the request.
Control | 0..1
Terminology Binding | RequestStatus (Required)
Type | code
Is Modifier | true
Summary | true
Comments | This element is labeled as a modifier because the status contains the codes cancelled and entered-in-error that mark the request as not currently valid.
DeviceRequest.intent
Definition | Whether the request is a proposal, plan, an original order or a reflex order.

Control | 1..1
Terminology Binding | RequestIntent :
Type | CodeableConcept
Is Modifier | true
Summary | true
DeviceRequest.priority
Definition | Indicates how quickly the {{title}} should be addressed with respect to other requests.
Control | 0..1
Terminology Binding | RequestPriority (Required)
Type | code
Default Value | If missing, normal priority
Summary | true
DeviceRequest.code[x]
Definition | The details of the device to be used.
Control | 1..1
Terminology Binding | FHIR Device Types (Example)
Type | Reference(Device)|CodeableConcept
[x] Note | See Choice of Data Types for further information about how to use [x]
Summary | true
DeviceRequest.subject
Definition | The patient who will use the device.
Control | 1..1
Type | Reference(Patient | Group | Location | Device)
Summary | true
DeviceRequest.context
Definition | An encounter that provides additional context in which this request is made.
Control | 0..1
Type | Reference(Encounter | EpisodeOfCare)
Summary | true
DeviceRequest.occurrence[x]
Definition | The timing schedule for the use of the device. The Schedule data type allows many different expressions, for example. "Every 8 hours"; "Three times a day"; "1/2 an hour before breakfast for 10 days from 23-Dec 2011:"; "15 Oct 2013, 17 Oct 2013 and 1 Nov 2013".
Control | 0..1
Type | dateTime|Period|Timing
[x] Note | See Choice of Data Types for further information about how to use [x]
Summary | true
DeviceRequest.authoredOn
Definition | When the request transitioned to being actionable.
Control | 0..1
Type | dateTime
Summary | true
DeviceRequest.requester
Definition | The individual who initiated the request and has responsibility for its activation.
Control | 0..1
Summary | true
DeviceRequest.requester.agent
Definition | The device, practitioner, etc. who initiated the request.
Control | 1..1
Type | Reference(Device | Practitioner | Organization)
Summary | true
DeviceRequest.requester.onBehalfOf
Definition | The organization the device or practitioner was acting on behalf of.
Control | 0..1
Type | Reference(Organization)
Requirements | Practitioners and Devices can be associated with multiple organizations. This element indicates which organization they were acting on behalf of when authoring the request.
Summary | true
DeviceRequest.performerType
Definition | Desired type of performer for doing the diagnostic testing.
Control | 0..1
Terminology Binding | Participant Roles (Example)
Type | CodeableConcept
Summary | true
DeviceRequest.performer
Definition | The desired perfomer for doing the diagnostic testing.
Control | 0..1
Type | Reference(Practitioner | Organization | Patient | Device | RelatedPerson | HealthcareService)
Summary | true
DeviceRequest.reasonCode
Definition | Reason or justification for the use of this device.
Control | 0..*
Terminology Binding | Condition/Problem/Diagnosis Codes (Example)

Type | CodeableConcept
Summary | true
DeviceRequest.reasonReference
Definition | Reason or justification for the use of this device.
Control | 0..*
Type | Reference(Any)
Summary | true
DeviceRequest.supportingInfo
Definition | Additional clinical information about the patient that may influence the request fulfilment. For example, this may includes body where on the subject's the device will be used (i.e. the target site).
Control | 0..*
Type | Reference(Any)
Requirements | Knowing where the device is targeted is important for tracking if multiple sites are possible.
DeviceRequest.note
Definition | Details about this request that were not represented at all or sufficiently in one of the attributes provided in a class. These may include for example a comment, an instruction, or a note associated with the statement.
Control | 0..*
Type | Annotation
DeviceRequest.relevantHistory
Definition | Key events in the history of the request.
Control | 0..*
Type | Reference(Provenance)
Comments | This may not include provenances for all versions of the request – only those deemed “relevant” or important. This SHALL NOT include the Provenance associated with this current version of the resource. (If that provenance is deemed to be a “relevant” change, it will need to be added as part of a later update. Until then, it can be queried directly as the Provenance that points to this version using _revinclude All Provenances should have some historical version of this Request as their subject.

12.22.7 Resource DeviceRequest - Mappings

URL: https://hl7.org/fhir/STU3/devicerequest-mappings.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

12.22.7 Resource DeviceRequest - Mappings

Orders and Observations Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Device, Encounter, Patient, Practitioner

Mappings:

RIM Mapping

HL7 v2 Mapping

Quality Improvement and Clinical Knowledge (QUICK)

Mappings for the devicerequest resource.

12.22.7.1 Mappings for RIM Mapping (http://hl7.org/v3)

DeviceRequest | Act[moodCode<=INT]
identifier | .identifier
definition | .outboundRelationship[typeCode=DEFN].target
basedOn | .outboundRelationship[typeCode=FLFS].target
priorRequest | .outboundRelationship[typeCode=RPLC].target
groupIdentifier | .inboundRelationship(typeCode=COMP].source[moodCode=INT].identifier
status | .status
intent | .moodCode (nuances beyond PRP/PLAN/RQO would need to be elsewhere)
priority | .priorityCode
code[x] | .code
subject | .participation[typeCode=SBJ].role
context | .inboundRelationship(typeCode=COMP].source[classCode<=PCPR, moodCode=EVN]
occurrence[x] | .effectiveTime
authoredOn | .participation[typeCode=AUT].time
requester | .participation[typeCode=AUT].role
agent | .player
onBehalfOf | .scoper
performerType | .participation[typeCode=PRF].role[scoper.determinerCode=KIND].code
performer | .participation[typeCode=PRF].role[scoper.determinerCode=INSTANCE]
reasonCode | .reasonCode
reasonReference | .outboundRelationship[typeCode=RSON].target
supportingInfo | .outboundRelationship[typeCode=PERT].target
note | .inboundRelationship(typeCode=SUBJ].source[classCode=ANNGEN, moodCode=EVN].value[xsi:type=ST]
relevantHistory | .inboundRelationship(typeCode=SUBJ].source[classCode=CACT, moodCode=EVN]

12.22.7.2 Mappings for HL7 v2 Mapping (http://hl7.org/v2)

DeviceRequest | ORC

identifier | ORC.2, ORC.3
definition | Varies by domain
basedOn | ORC.8 (plus others)
priorRequest | Handled by message location of ORC (ORC.1=RO or RU)
groupIdentifier | ORC.4
status | ORC.5
intent | N/A
priority | TQ1.9
code[x] | Varies by domain
subject | Accompanying PID segment
context | Accompanying PV1
occurrence[x] | Accompanying TQ1/TQ2 segments
authoredOn | ORC.9
requester | ORC.12
agent | ORC.12
onBehalfOf | N/A
performerType | PRT
performer | PRT
reasonCode | ORC.16
reasonReference | ORC.16
supportingInfo | Accompanying segments
note | NTE
relevantHistory | N/A

12.22.7.3 Mappings for Quality Improvement and Clinical Knowledge (QUICK) (http://wiki.siframework.org/CQF)

DeviceRequest
identifier | ClinicalStatement.identifier
definition
basedOn | Proposal.prnReason.reason
priorRequest
groupIdentifier
status | Action.currentStatus
intent
priority
code[x] | DeviceUse.device
subject | ClinicalStatement.subject
context | ClinicalStatement.encounter
occurrence[x] | DeviceUse.applicationSchedule
authoredOn | ClinicalStatement.statementDateTime
requester
agent
onBehalfOf
performerType
performer
reasonCode | Action.indication.reason
reasonReference
supportingInfo
note | ClinicalStatement.additionalText
relevantHistory

12.22.8 Resource DeviceRequest - Design Notes

URL: https://hl7.org/fhir/STU3/devicerequest-explanations.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

12.22.8 Resource DeviceRequest - Design Notes

to do

12.22.9 Resource DeviceRequest - Extensions & Profiles

URL: https://hl7.org/fhir/STU3/devicerequest-profiles.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

12.22.9 Resource DeviceRequest - Extensions & Profiles

Orders and Observations Work Group | Maturity Level: N/A | Ballot Status: Informative

This table lists profiles and extensions for the DeviceRequest resource. For background information, see Profiling Resources and Extensibility. Additional profiles and resources may be found in published Implementation Guides , or in the Conformance resource registry .

12.22.9.1 Profiles

No Profiles defined for this resource

12.22.9.2 Extensions

devicerequest-patientInstruction | patientInstruction | for DeviceRequest HL7 Extensions
devicerequest-reasonRejected | reasonRejected | for DeviceRequest HL7 Extensions

Extensions that reference this resource:

No Extensions refer to this resource

12.22.9.3 Search Extensions

No Search Extensions defined for this resource

12.22.10 DeviceRequest HL7 Extensions

URL: https://hl7.org/fhir/STU3/devicerequest-extensions.html

Defines common extensions used with or related to the DeviceRequest resource

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

12.22.10 DeviceRequest HL7 Extensions

(No assigned work group) | Maturity Level: N/A | Ballot Status: Informative

12.22.10.1 Content

Extensions:
devicerequest-reasonRejected | reasonRejected : The reason associated with the rejection by the receiver of this device use request. Applicable only if the status is rejected.
devicerequest-patientInstruction | patientInstruction : Simple concise instructions to be read by the patient. For example “twice a day” rather than “BID.”.

12.22.11 Resource DeviceRequest - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/devicerequest-version-maps.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

12.22.11 Resource DeviceRequest - Detailed Descriptions

Orders and Observations Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Device, Encounter, Patient, Practitioner

R2 : R3 Converson maps for DeviceRequest.

Functional status for this map: Not Mapped (see documentation)

12.22.11.1 R2 to R3

```
map "http://hl7.org/fhir/StructureMap/DeviceUseRequest2to3" = "R2 to R3 Conversions for DeviceUseRequest"

conceptmap "RequestStatus" {
  prefix s = "http://hl7.org/fhir/device-use-request-status"
  prefix t = "http://hl7.org/fhir/request-status"

  s:proposed = t:draft
  s:planned = t:draft
  s:requested = t:active
  s:received = t:active
  s:accepted = t:active
  s:"in-progress" = t:active
  s:completed = t:completed
  s:suspended = t:suspended
  s:rejected = t:cancelled
  s:aborted = t:cancelled
}

uses "http://hl7.org/fhir/DSTU2/StructureDefinition/DeviceUseRequest" alias DeviceUseRequestR2 as source
uses "http://hl7.org/fhir/StructureDefinition/DeviceRequest" alias DeviceRequest as target

imports "http://hl7.org/fhir/StructureMap/*2to3"

group for type+types  DeviceUseRequest extends DomainResource
  input src : DeviceUseRequestR2 as source
  input tgt : DeviceRequest as target

  // drop bodySite
```

```
  "DeviceUseRequest-status" : for src.status as v make tgt.status = translate(v, "#RequestStatus", "code")
  "DeviceUseRequest-device" : for src.device make tgt.device
  "DeviceUseRequest-context" : for src.encounter make tgt.context
  "DeviceUseRequest-identifier" : for src.identifier make tgt.identifier
  "DeviceUseRequest-reasonCode" : for src.indication make tgt.reasonCode
  "DeviceUseRequest-notes" : for src.notes make tgt.note
  // drop prnReason
  "DeviceUseRequest-authored" : for src.orderedOn make tgt.authored
  "DeviceUseRequest-subject" : for src.subject make tgt.subject
  "DeviceUseRequest-Timing" : for src.timing make tgt.occurrence
  // drop priority

endgroup
```

12.22.11.2 R3 to R2

```
map "http://hl7.org/fhir/StructureMap/DeviceUseRequest3to2" = "R3 to R2 Conversion for DeviceUseRequest"

conceptmap "RequestStatus" {
  prefix s = "http://hl7.org/fhir/device-use-request-status"
  prefix t = "http://hl7.org/fhir/request-status"

  s:draft = t:proposed
  s:active = t:"in-progress"
  s:suspended = t:suspended
  s:completed = t:completed
  s:cancelled = t:rejected
}

uses "http://hl7.org/fhir/StructureDefinition/DeviceRequest" alias DeviceRequest as source
uses "http://hl7.org/fhir/DSTU2/StructureDefinition/DeviceUseRequest" alias DeviceUseRequestR2 as target

imports "http://hl7.org/fhir/StructureMap/*3to2"

group for type+types  DeviceRequest extends DomainResource
  input src : DeviceRequest as source
  input tgt : DeviceUseRequestR2 as target

  "DeviceRequest-status" : for src.status as v check src.status != 'entered-in-error' make tgt.status = translate(v,
    "#RequestStatus", "code")
  "DeviceRequest-device" : for src.device make tgt.device
  "DeviceRequest-context" : for src.context make tgt.encounter
  "DeviceRequest-identifier" : for src.identifier make tgt.identifier
  "DeviceRequest-reasonCode" : for src.reasonCode make tgt.indication
  "DeviceRequest-notes" : for src.note make tgt.notes
  "DeviceRequest-authored" : for src.authored make tgt.orderedOn
  "DeviceRequest-subject" : for src.subject make tgt.subject
  "DeviceRequest-Timing" : for src.occurrence make tgt.timing
endgroup
```

12.22.11.3 R3 Validation Errors

n/a

12.23 Resource DeviceUseStatement - Content

URL: https://hl7.org/fhir/STU3/deviceusestatement.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

Orders and Observations Work Group | Maturity Level: 0 | Draft | Compartments: Device, Patient

A record of a device being used by a patient where the record is the result of a report from the patient or another clinician.

12.23.1 Scope and Usage

These resources have not yet undergone proper review by PC, CQI, CDS, and OO. At this time, they are to be considered only as draft resource proposals for potential submission.

This resource is an event resource from a FHIR workflow perspective - see Workflow. It is the intent of the Orders and Observation Workgroup to align this resource with the workflow pattern for event resources.

This resource records the use of a healthcare-related device by a patient. The record is the result of a report of use by the patient, another provider or a related person. The resource can be used to note the use of an assistive device such as a wheelchair or hearing aid, a contraceptive such an intra-uterine device, or other implanted devices such as a pacemaker.

12.23.2 Boundaries and Relationships

This resource is different from DeviceRequest which records a request to use the device. This also is distinct from the Procedure resource which may describe the implantation or explantation of a device.

12.23.3 Resource Content

* Structure

* UML

* XML
* JSON
* Turtle
* R2 Diff
* All

Structure

Name | Flags | Card. | Type | Description & Constraints
DeviceUseStatement | | | DomainResource | Record of use of a device Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | | 0..* | Identifier | External identifier for this record
status | ?!? | 1..1 | code | active | completed | entered-in-error + DeviceUseStatementStatus (Required)
subject | | 1..1 | Reference(Patient | Group) | Patient using device
whenUsed | | 0..1 | Period | Period device was used
timing[x] | | 0..1 | | How often the device was used
timingTiming | | | Timing
timingPeriod | | | Period
timingDateTime | | | dateTime
recordedOn | | 0..1 | dateTime | When statement was recorded
source | | 0..1 | Reference(Patient | Practitioner | RelatedPerson) | Who made the statement
device | | 1..1 | Reference(Device) | Reference to device used
indication | | 0..* | CodeableConcept | Why device was used
bodySite | | 0..1 | CodeableConcept | Target body site SNOMED CT Body Structures (Example)
note | | 0..* | Annotation | Addition details (comments, instructions)
Documentation for this format

UML Diagram (Legend)

XML Template

```
<DeviceUseStatement xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier External identifier for this record --></identifier>
 <status value="[code]"/><!-- 1..1 active | completed | entered-in-error + -->
 <subject><!-- 1..1 Reference(Patient|Group) Patient using device --></subject>
 <whenUsed><!-- 0..1 Period Period device was used --></whenUsed>
 <timing[x]><!-- 0..1 Timing|Period|dateTime How often  the device was used --></timing[x]>
 <recordedOn value="[dateTime]"/><!-- 0..1 When statement was recorded -->
 <source><!-- 0..1 Reference(Patient|Practitioner|RelatedPerson) Who made the statement --></source>
 <device><!-- 1..1 Reference(Device) Reference to device used --></device>
 <indication><!-- 0..* CodeableConcept Why device was used --></indication>
 <bodySite><!-- 0..1 CodeableConcept Target body site --></bodySite>
 <note><!-- 0..* Annotation Addition details (comments, instructions) --></note>
</DeviceUseStatement>
```

JSON Template

```
{
  "resourceType" : "DeviceUseStatement",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // External identifier for this record
  "status" : "<code>", // R!  active | completed | entered-in-error +
  "subject" : { Reference(Patient|Group) }, // R!  Patient using device
  "whenUsed" : { Period }, // Period device was used
  // timing[x]: How often  the device was used. One of these 3:
  "timingTiming" : { Timing },
  "timingPeriod" : { Period },
  "timingDateTime" : "<dateTime>",
  "recordedOn" : "<dateTime>", // When statement was recorded
  "source" : { Reference(Patient|Practitioner|RelatedPerson) }, // Who made the statement
  "device" : { Reference(Device) }, // R!  Reference to device used
  "indication" : [{ CodeableConcept }], // Why device was used
  "bodySite" : { CodeableConcept }, // Target body site
  "note" : [{ Annotation }] // Addition details (comments, instructions)
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:DeviceUseStatement;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:DeviceUseStatement.identifier [ Identifier ], ... ; # 0..* External identifier for this record
  fhir:DeviceUseStatement.status [ code ]; # 1..1 active | completed | entered-in-error +
  fhir:DeviceUseStatement.subject [ Reference(Patient|Group) ]; # 1..1 Patient using device
  fhir:DeviceUseStatement.whenUsed [ Period ]; # 0..1 Period device was used
  # DeviceUseStatement.timing[x] : 0..1 How often  the device was used. One of these 3
    fhir:DeviceUseStatement.timingTiming [ Timing ]
    fhir:DeviceUseStatement.timingPeriod [ Period ]
```

```
    fhir:DeviceUseStatement.timingDateTime [ dateTime ]
  fhir:DeviceUseStatement.recordedOn [ dateTime ]; # 0..1 When statement was recorded
  fhir:DeviceUseStatement.source [ Reference(Patient|Practitioner|RelatedPerson) ]; # 0..1 Who made the statement
  fhir:DeviceUseStatement.device [ Reference(Device) ]; # 1..1 Reference to device used
  fhir:DeviceUseStatement.indication [ CodeableConcept ], ... ; # 0..* Why device was used
  fhir:DeviceUseStatement.bodySite [ CodeableConcept ]; # 0..1 Target body site
  fhir:DeviceUseStatement.note [ Annotation ], ... ; # 0..* Addition details (comments, instructions)
]
```

Changes since DSTU2

DeviceUseStatement

DeviceUseStatement.status | Added Element

DeviceUseStatement.subject | Add Reference(Group)

DeviceUseStatement.source | Added Element

DeviceUseStatement.bodySite | Renamed from bodySite[x] to bodySite Remove Reference(BodySite)

DeviceUseStatement.note | Added Element

DeviceUseStatement.notes | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R2 <--> R3 Conversion Maps (status = 1 test that all execute ok. All tests pass round-trip testing and 1 r3 resources are invalid (1 errors).).

Structure

Name | Flags | Card. | Type | Description & Constraints

DeviceUseStatement | | | DomainResource | Record of use of a device Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension

identifier | | 0..* | Identifier | External identifier for this record

status | ?!? | 1..1 | code | active | completed | entered-in-error + DeviceUseStatementStatus (Required)

subject | | 1..1 | Reference(Patient | Group) | Patient using device

whenUsed | | 0..1 | Period | Period device was used

timing[x] | | 0..1 | | How often the device was used

timingTiming | | | Timing

timingPeriod | | | Period

timingDateTime | | | dateTime

recordedOn | | 0..1 | dateTime | When statement was recorded

source | | 0..1 | Reference(Patient | Practitioner | RelatedPerson) | Who made the statement

device | | 1..1 | Reference(Device) | Reference to device used

indication | | 0..* | CodeableConcept | Why device was used

bodySite | | 0..1 | CodeableConcept | Target body site SNOMED CT Body Structures (Example)

note | | 0..* | Annotation | Addition details (comments, instructions)

Documentation for this format

UML Diagram (Legend)

XML Template

```
<DeviceUseStatement xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier External identifier for this record --></identifier>
 <status value="[code]"/><!-- 1..1 active | completed | entered-in-error + -->
 <subject><!-- 1..1 Reference(Patient|Group) Patient using device --></subject>
 <whenUsed><!-- 0..1 Period Period device was used --></whenUsed>
 <timing[x]><!-- 0..1 Timing|Period|dateTime How often  the device was used --></timing[x]>
 <recordedOn value="[dateTime]"/><!-- 0..1 When statement was recorded -->
 <source><!-- 0..1 Reference(Patient|Practitioner|RelatedPerson) Who made the statement --></source>
 <device><!-- 1..1 Reference(Device) Reference to device used --></device>
 <indication><!-- 0..* CodeableConcept Why device was used --></indication>
 <bodySite><!-- 0..1 CodeableConcept Target body site --></bodySite>
 <note><!-- 0..* Annotation Addition details (comments, instructions) --></note>
</DeviceUseStatement>
```

JSON Template

```
{
  "resourceType" : "DeviceUseStatement",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // External identifier for this record
  "status" : "<code>", // R!  active | completed | entered-in-error +
  "subject" : { Reference(Patient|Group) }, // R!  Patient using device
  "whenUsed" : { Period }, // Period device was used
  // timing[x]: How often  the device was used. One of these 3:
  "timingTiming" : { Timing },
  "timingPeriod" : { Period },
  "timingDateTime" : "<dateTime>",
  "recordedOn" : "<dateTime>", // When statement was recorded
  "source" : { Reference(Patient|Practitioner|RelatedPerson) }, // Who made the statement
  "device" : { Reference(Device) }, // R!  Reference to device used
  "indication" : [{ CodeableConcept }], // Why device was used
  "bodySite" : { CodeableConcept }, // Target body site
  "note" : [{ Annotation }] // Addition details (comments, instructions)
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:DeviceUseStatement;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:DeviceUseStatement.identifier [ Identifier ], ... ; # 0..* External identifier for this record
  fhir:DeviceUseStatement.status [ code ]; # 1..1 active | completed | entered-in-error +
  fhir:DeviceUseStatement.subject [ Reference(Patient|Group) ]; # 1..1 Patient using device
  fhir:DeviceUseStatement.whenUsed [ Period ]; # 0..1 Period device was used
  # DeviceUseStatement.timing[x] : 0..1 How often  the device was used. One of these 3
    fhir:DeviceUseStatement.timingTiming [ Timing ]
    fhir:DeviceUseStatement.timingPeriod [ Period ]
    fhir:DeviceUseStatement.timingDateTime [ dateTime ]
  fhir:DeviceUseStatement.recordedOn [ dateTime ]; # 0..1 When statement was recorded
  fhir:DeviceUseStatement.source [ Reference(Patient|Practitioner|RelatedPerson) ]; # 0..1 Who made the statement
  fhir:DeviceUseStatement.device [ Reference(Device) ]; # 1..1 Reference to device used
  fhir:DeviceUseStatement.indication [ CodeableConcept ], ... ; # 0..* Why device was used
  fhir:DeviceUseStatement.bodySite [ CodeableConcept ]; # 0..1 Target body site
  fhir:DeviceUseStatement.note [ Annotation ], ... ; # 0..* Addition details (comments, instructions)
]
```

Changes since DSTU2

DeviceUseStatement
DeviceUseStatement.status | Added Element
DeviceUseStatement.subject | Add Reference(Group)
DeviceUseStatement.source | Added Element
DeviceUseStatement.bodySite | Renamed from bodySite[x] to bodySite Remove Reference(BodySite)
DeviceUseStatement.note | Added Element
DeviceUseStatement.notes | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R2 <--> R3 Conversion Maps (status = 1 test that all execute ok. All tests pass round-trip testing and 1 r3 resources are invalid (1 errors).).

Alternate definitions: Master Definition (XML, JSON), XML Schema/Schematron (for) + JSON Schema, ShEx (for Turtle)

12.23.3.1 Terminology Bindings

Path | Definition | Type | Reference
DeviceUseStatement.status | A coded concept indicating the current status of a the Device Usage | Required | DeviceUseStatementStatus
DeviceUseStatement.bodySite | Codes describing anatomical locations. May include laterality. | Example | SNOMED CT Body Structures

Notes to reviewers:

At this time, the code bindings are placeholders to be fleshed out upon further review by the community.

12.23.4 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
device | reference | Search by device | DeviceUseStatement.device (Device)
identifier | token | Search by identifier | DeviceUseStatement.identifier
patient | reference | Search by subject - a patient | DeviceUseStatement.subject (Group, Patient) | 31 Resources
subject | reference | Search by subject | DeviceUseStatement.subject (Group, Patient)

12.23.5 Resource DeviceUseStatement - Examples

URL: https://hl7.org/fhir/STU3/deviceusestatement-examples.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

12.23.5 Resource DeviceUseStatement - Examples

Orders and Observations Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Device, Patient

An introduction to this page as well as proper examples are to be submitted shortly.

Example List:

The provision of a wheelchair to a patient | example | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

12.23.6 Resource DeviceUseStatement - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/deviceusestatement-definitions.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

12.23.6 Resource DeviceUseStatement - Detailed Descriptions

Orders and Observations Work Group | Maturity Level: 0 | Draft | Compartments: Device, Patient

Detailed Descriptions for the elements in the DeviceUseStatement resource.

DeviceUseStatement
Definition | A record of a device being used by a patient where the record is the result of a report from the patient or another clinician.
Control | 1..1
DeviceUseStatement.identifier
Definition | An external identifier for this statement such as an IRI.
Note | This is a business identifer, not a resource identifier (see discussion)
Control | 0..*
Type | Identifier
DeviceUseStatement.status
Definition | A code representing the patient or other source's judgment about the state of the device used that this statement is about. Generally this will be active or completed.
Control | 1..1
Terminology Binding | DeviceUseStatementStatus (Required)
Type | code
Is Modifier | true
Summary | true
Comments | DeviceUseStatment is a statement at a point in time. The status is only representative at the point when it was asserted. The value set for contains codes that assert the status of the use by the patient (for example, stopped or on hold) as well as codes that assert the status of the resource itself (for example, entered in error). This element is labeled as a modifier because the status contains the codes that mark the statement as not currently valid.
DeviceUseStatement.subject
Definition | The patient who used the device.
Control | 1..1
Type | Reference(Patient | Group)
DeviceUseStatement.whenUsed
Definition | The time period over which the device was used.
Control | 0..1
Type | Period
DeviceUseStatement.timing[x]
Definition | How often the device was used.
Control | 0..1
Type | Timing|Period|dateTime
[x] Note | See Choice of Data Types for further information about how to use [x]
DeviceUseStatement.recordedOn
Definition | The time at which the statement was made/recorded.
Control | 0..1
Type | dateTime
DeviceUseStatement.source
Definition | Who reported the device was being used by the patient.
Control | 0..1
Type | Reference(Patient | Practitioner | RelatedPerson)
DeviceUseStatement.device
Definition | The details of the device used.
Control | 1..1
Type | Reference(Device)
DeviceUseStatement.indication
Definition | Reason or justification for the use of the device.
Control | 0..*
Type | CodeableConcept
DeviceUseStatement.bodySite
Definition | Indicates the site on the subject's body where the device was used (i.e. the target site).
Control | 0..1
Terminology Binding | SNOMED CT Body Structures (Example)
Type | CodeableConcept
Requirements | Knowing where the device is targeted is important for tracking if multiple sites are possible. If more information than just a code is required, use the extension http://hl7.org/fhir/StructureDefinition/body-site-instance.
DeviceUseStatement.note
Definition | Details about the device statement that were not represented at all or sufficiently in one of the attributes provided in a class. These may include for example a comment, an instruction, or a note associated with the statement.
Control | 0..*
Type | Annotation

12.23.7 Resource DeviceUseStatement - Mappings

URL: https://hl7.org/fhir/STU3/deviceusestatement-mappings.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

12.23.7 Resource DeviceUseStatement - Mappings

Orders and Observations Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Device, Patient

Mappings:

RIM Mapping

Quality Improvement and Clinical Knowledge (QUICK)

Mappings for the deviceusestatement resource.

12.23.7.1 Mappings for RIM Mapping (http://hl7.org/v3)

DeviceUseStatement
identifier
status | .statusCode
subject
whenUsed
timing[x]
recordedOn
source
device
indication
bodySite | targetSiteCode
note

12.23.7.2 Mappings for Quality Improvement and Clinical Knowledge (QUICK) (http://wiki.siframework.org/CQF)

DeviceUseStatement | DeviceUsePerformanceOccurrence
identifier | ClinicalStatement.identifier
status | …status
subject | ClinicalStatement.subject
whenUsed | Performance.performanceTime
timing[x] | DeviceUse.applicationSchedule
recordedOn | ClinicalStatement.statementDateTime
source
device | DeviceUse.device
indication | Action.indication.reason
bodySite
note | ClinicalStatement.additionalText

12.23.8 Resource DeviceUseStatement - Design Notes

URL: https://hl7.org/fhir/STU3/deviceusestatement-explanations.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

12.23.8 Resource DeviceUseStatement - Design Notes

to do

12.23.9 Resource DeviceUseStatement - Extensions & Profiles

URL: https://hl7.org/fhir/STU3/deviceusestatement-profiles.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

12.23.9 Resource DeviceUseStatement - Extensions & Profiles

Orders and Observations Work Group | Maturity Level: N/A | Ballot Status: Informative

This table lists profiles and extensions for the DeviceUseStatement resource. For background information, see Profiling Resources and Extensibility. Additional profiles and resources may be found in published Implementation Guides , or in the Conformance resource registry .

12.23.9.1 Profiles

No Profiles defined for this resource

12.23.9.2 Extensions

No Extensions defined for this resource

Extensions that reference this resource:

No Extensions refer to this resource

12.23.9.3 Search Extensions

No Search Extensions defined for this resource

12.23.10 DeviceUseStatement HL7 Extensions

URL: https://hl7.org/fhir/STU3/deviceusestatement-extensions.html

Defines common extensions used with or related to the DeviceUseStatement resource

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

12.23.10 DeviceUseStatement HL7 Extensions

(No assigned work group) | Maturity Level: N/A | Ballot Status: Informative

12.23.10.1 Content

12.23.11 Resource DeviceUseStatement - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/deviceusestatement-version-maps.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

12.23.11 Resource DeviceUseStatement - Detailed Descriptions

Orders and Observations Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Device, Patient

R2 : R3 Converson maps for DeviceUseStatement.

Functional status for this map: 1 test that all execute ok. All tests pass round-trip testing and 1 r3 resources are invalid (1 errors). (see documentation)

12.23.11.1 R2 to R3

```
map "http://hl7.org/fhir/StructureMap/DeviceUseStatement2to3" = "R2 to R3 Conversions for DeviceUseStatement"

uses "http://hl7.org/fhir/DSTU2/StructureDefinition/DeviceUseStatement" alias DeviceUseStatementR2 as source
uses "http://hl7.org/fhir/StructureDefinition/DeviceUseStatement" alias DeviceUseStatement as target

imports "http://hl7.org/fhir/StructureMap/*2to3"

group for type+types  DeviceUseStatement extends DomainResource
  input src : DeviceUseStatementR2 as source
  input tgt : DeviceUseStatement as target

  "DeviceUseStatement-bodySite" : for src.bodySiteCodeableConcept make tgt.bodySite
  "DeviceUseStatement-whenUsed" : for src.whenUsed make tgt.whenUsed
  "DeviceUseStatement-device" : for src.device make tgt.device
  "DeviceUseStatement-identifier" : for src.identifier make tgt.identifier
  "DeviceUseStatement-indication" : for src.indication make tgt.indication
  "DeviceUseStatement-notes" : for src.notes as v make tgt.note as n, n.text = v
  "DeviceUseStatement-recordedOn" : for src.recordedOn make tgt.recordedOn
  "DeviceUseStatement-subject" : for src.subject make tgt.subject
  "DeviceUseStatement-timing" : for src.timing make tgt.timing
endgroup
```

12.23.11.2 R3 to R2

```
map "http://hl7.org/fhir/StructureMap/DeviceUseStatement3to2" = "R3 to R2 Conversion for DeviceUseStatement"

uses "http://hl7.org/fhir/StructureDefinition/DeviceUseStatement" alias DeviceUseStatement as source
uses "http://hl7.org/fhir/DSTU2/StructureDefinition/DeviceUseStatement" alias DeviceUseStatementR2 as target

imports "http://hl7.org/fhir/StructureMap/*3to2"

group for type+types  DeviceUseStatement extends DomainResource
  input src : DeviceUseStatement as source
  input tgt : DeviceUseStatementR2 as target

  "DeviceUseStatement-bodySite-CodeableConcept" : for src.bodySite : CodeableConcept as vs make tgt.bodySite =
    create("CodeableConcept") as vt then CodeableConcept(vs, vt)
  "DeviceUseStatement-whenUsed" : for src.whenUsed make tgt.whenUsed
  "DeviceUseStatement-device" : for src.device make tgt.device
  "DeviceUseStatement-identifier" : for src.identifier make tgt.identifier
  "DeviceUseStatement-indication" : for src.indication make tgt.indication
  "DeviceUseStatement-notes" : for src.note as n then dusNote(n, tgt)
  "DeviceUseStatement-recordedOn" : for src.recordedOn make tgt.recordedOn
  "DeviceUseStatement-subject" : for src.subject make tgt.subject
  "DeviceUseStatement-timing" : for src.timing make tgt.timing
endgroup
```

```
group dusNote extends BackboneElement
  input src as source
  input tgt as target

  "DeviceUseStatement.note-notes" : for src.text make tgt.notes
endgroup
```

12.23.11.3 R3 Validation Errors

example | ERROR @ DeviceUseStatement: Profile http://hl7.org/fhir/StructureDefinition/DeviceUseStatement, Element 'DeviceUseStatement.status': minimum required = 1, but only found 0 (src = InstanceValidator)

12.24 Resource SupplyRequest - Content

URL: https://hl7.org/fhir/STU3/supplyrequest.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

Orders and Observations Work Group | Maturity Level: 1 | Trial Use | Compartments: Device, Patient, Practitioner, RelatedPerson

A record of a request for a medication, substance or device used in the healthcare setting.

12.24.1 Scope and Usage

This resource is a request resource from a FHIR workflow perspective - see Workflow. It is the intent of the Orders and Observation Workgroup to align this resource with the workflow pattern for request resources.

The scope of the SupplyRequest resource is for recording the request of supplies used in the healthcare process. This includes supplies specifically used in the treatment of patients as well as supply movement within an institution (transport a set of supplies from materials management to a service unit (nurse station). This resource does not include the provisioning of transportation services.

The SupplyRequest resource allows requesting only a single item. If a workflow requires requesting multiple items simultaneously, this is done using multiple instances of this resource. These instances can be linked in different ways, depending on the needs of the workflow. For guidance, refer to the Request pattern

12.24.2 Boundaries and Relationships

Note that the SupplyRequest records the fact that a request was made. To actually act on that request, additional workflow beyond simply the existence of a SupplyRequest is required. This can be achieved by using an Task resource, with the SupplyRequest referenced from the Task.focus, or by using the SupplyRequest resource in the context of an messaging or service workflow where the request is explicit or implicit. The SupplyDelivery resource represents the fulfillment as a result of SupplyRequest being acted upon.

The SupplyRequest resource is used for inventory management. When requesting medication, substances and devices when there is a patient focus or instructions regarding their use, DeviceRequest or MedicationRequest should be used instead

This resource is referenced by supplydelivery

12.24.3 Resource Content

* Structure

* UML

* XML

* JSON

* Turtle

* R2 Diff

* All

Structure

Name | Flags | Card. | Type | Description & Constraints
SupplyRequest | | | DomainResource | Request for a medication, substance or device Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | ? | 0..1 | Identifier | Unique identifier
status | ?!? | 0..1 | code | draft | active | suspended + SupplyRequestStatus (Required)
category | ? | 0..1 | CodeableConcept | The kind of supply (central, non-stock, etc.) Supply Type (Example)
priority | ? | 0..1 | code | routine | urgent | asap | stat RequestPriority (Required)
orderedItem | ? | 0..1 | BackboneElement | The item being requested
quantity | ? | 1..1 | Quantity | The requested amount of the item indicated
item[x] | ? | 0..1 | | Medication, Substance, or Device requested to be supplied SNOMED CT Supply Item (Example)
itemCodeableConcept | | | CodeableConcept
itemReference | | | Reference(Medication | Substance | Device)
occurrence[x] | ? | 0..1 | | When the request should be fulfilled
occurrenceDateTime | | | dateTime
occurrencePeriod | | | Period
occurrenceTiming | | | Timing

authoredOn | ? | 0..1 | dateTime | When the request was made
requester | ? | 0..1 | BackboneElement | Who/what is requesting service
agent | ? | 1..1 | Reference(Practitioner | Organization | Patient | RelatedPerson | Device) | Individual making the request
onBehalfOf | | 0..1 | Reference(Organization) | Organization agent is acting for
supplier | ? | 0..* | Reference(Organization) | Who is intended to fulfill the request
reason[x] | | 0..1 | | Why the supply item was requested SupplyRequestReason (Example)
reasonCodeableConcept | | | CodeableConcept
reasonReference | | | Reference(Any)
deliverFrom | | 0..1 | Reference(Organization | Location) | The origin of the supply
deliverTo | | 0..1 | Reference(Organization | Location | Patient) | The destination of the supply
Documentation for this format

UML Diagram (Legend)

XML Template

```
<SupplyRequest xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..1 Identifier Unique identifier --></identifier>
 <status value="[code]"/><!-- 0..1 draft | active | suspended + -->
 <category><!-- 0..1 CodeableConcept The kind of supply (central, non-stock, etc.) --></category>
 <priority value="[code]"/><!-- 0..1 routine | urgent | asap | stat -->
 <orderedItem>  <!-- 0..1 The item being requested -->
  <quantity><!-- 1..1 Quantity The requested amount of the item indicated --></quantity>
  <item[x]><!-- 0..1 CodeableConcept|Reference(Medication|Substance|Device) Medication, Substance, or Device requested
    to be supplied --></item[x]>
 </orderedItem>
 <occurrence[x]><!-- 0..1 dateTime|Period|Timing When the request should be fulfilled --></occurrence[x]>
 <authoredOn value="[dateTime]"/><!-- 0..1 When the request was made -->
 <requester>  <!-- 0..1 Who/what is requesting service -->
  <agent><!-- 1..1 Reference(Practitioner|Organization|Patient|RelatedPerson|
    Device) Individual making the request --></agent>
  <onBehalfOf><!-- 0..1 Reference(Organization) Organization agent is acting for --></onBehalfOf>
 </requester>
 <supplier><!-- 0..* Reference(Organization) Who is intended to fulfill the request --></supplier>
 <reason[x]><!-- 0..1 CodeableConcept|Reference(Any) Why the supply item was requested --></reason[x]>
 <deliverFrom><!-- 0..1 Reference(Organization|Location) The origin of the supply --></deliverFrom>
 <deliverTo><!-- 0..1 Reference(Organization|Location|Patient) The destination of the supply --></deliverTo>
</SupplyRequest>
```

JSON Template

```
{
  "resourceType" : "SupplyRequest",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : { Identifier }, // Unique identifier
  "status" : "<code>", // draft | active | suspended +
  "category" : { CodeableConcept }, // The kind of supply (central, non-stock, etc.)
  "priority" : "<code>", // routine | urgent | asap | stat
  "orderedItem" : { // The item being requested
    "quantity" : { Quantity }, // R!  The requested amount of the item indicated
    // item[x]: Medication, Substance, or Device requested to be supplied. One of these 2:
    "itemCodeableConcept" : { CodeableConcept }
    "itemReference" : { Reference(Medication|Substance|Device) }
  },
  // occurrence[x]: When the request should be fulfilled. One of these 3:
  "occurrenceDateTime" : "<dateTime>",
  "occurrencePeriod" : { Period },
  "occurrenceTiming" : { Timing },
  "authoredOn" : "<dateTime>", // When the request was made
  "requester" : { // Who/what is requesting service
    "agent" : { Reference(Practitioner|Organization|Patient|RelatedPerson|
    Device) }, // R!  Individual making the request
    "onBehalfOf" : { Reference(Organization) } // Organization agent is acting for
  },
  "supplier" : [{ Reference(Organization) }], // Who is intended to fulfill the request
  // reason[x]: Why the supply item was requested. One of these 2:
  "reasonCodeableConcept" : { CodeableConcept },
  "reasonReference" : { Reference(Any) },
  "deliverFrom" : { Reference(Organization|Location) }, // The origin of the supply
  "deliverTo" : { Reference(Organization|Location|Patient) } // The destination of the supply
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:SupplyRequest;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:SupplyRequest.identifier [ Identifier ]; # 0..1 Unique identifier
  fhir:SupplyRequest.status [ code ]; # 0..1 draft | active | suspended +
  fhir:SupplyRequest.category [ CodeableConcept ]; # 0..1 The kind of supply (central, non-stock, etc.)
  fhir:SupplyRequest.priority [ code ]; # 0..1 routine | urgent | asap | stat
```

```
  fhir:SupplyRequest.orderedItem [ # 0..1 The item being requested
    fhir:SupplyRequest.orderedItem.quantity [ Quantity ]; # 1..1 The requested amount of the item indicated
    # SupplyRequest.orderedItem.item[x] : 0..1 Medication, Substance, or Device requested to be supplied. One of these 2
      fhir:SupplyRequest.orderedItem.itemCodeableConcept [ CodeableConcept ]
      fhir:SupplyRequest.orderedItem.itemReference [ Reference(Medication|Substance|Device) ]
  ];
  # SupplyRequest.occurrence[x] : 0..1 When the request should be fulfilled. One of these 3
    fhir:SupplyRequest.occurrenceDateTime [ dateTime ]
    fhir:SupplyRequest.occurrencePeriod [ Period ]
    fhir:SupplyRequest.occurrenceTiming [ Timing ]
  fhir:SupplyRequest.authoredOn [ dateTime ]; # 0..1 When the request was made
  fhir:SupplyRequest.requester [ # 0..1 Who/what is requesting service
    fhir:SupplyRequest.requester.agent [ Reference(Practitioner|Organization|Patient|RelatedPerson|Device) ]; # 1..1
    Individual making the request
    fhir:SupplyRequest.requester.onBehalfOf [ Reference(Organization) ]; # 0..1 Organization agent is acting for
  ];
  fhir:SupplyRequest.supplier [ Reference(Organization) ], ... ; # 0..* Who is intended to fulfill the request
  # SupplyRequest.reason[x] : 0..1 Why the supply item was requested. One of these 2
    fhir:SupplyRequest.reasonCodeableConcept [ CodeableConcept ]
    fhir:SupplyRequest.reasonReference [ Reference(Any) ]
  fhir:SupplyRequest.deliverFrom [ Reference(Organization|Location) ]; # 0..1 The origin of the supply
  fhir:SupplyRequest.deliverTo [ Reference(Organization|Location|Patient) ]; # 0..1 The destination of the supply
]
```

Changes since DSTU2

SupplyRequest
SupplyRequest.category | Renamed from kind to category Remove Binding http://hl7.org/fhir/ValueSet/supplyrequest-kind (required)
SupplyRequest.priority | Added Element
SupplyRequest.orderedItem | Remove Reference(Medication), Remove Reference(Substance), Remove Reference(Device)
SupplyRequest.orderedItem.quantity | Added Element
SupplyRequest.orderedItem.item[x] | Added Element
SupplyRequest.occurrence[x] | Added Element
SupplyRequest.authoredOn | Renamed from date to authoredOn
SupplyRequest.requester | Added Element
SupplyRequest.requester.agent | Added Element
SupplyRequest.requester.onBehalfOf | Added Element
SupplyRequest.deliverFrom | Added Element
SupplyRequest.deliverTo | Added Element
SupplyRequest.patient | deleted
SupplyRequest.source | deleted
SupplyRequest.when | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R2 <--> R3 Conversion Maps (status = 1 test that all execute ok. All tests pass round-trip testing and all r3 resources are valid.).

Structure

Name | Flags | Card. | Type | Description & Constraints
SupplyRequest | | | DomainResource | Request for a medication, substance or device Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | ? | 0..1 | Identifier | Unique identifier
status | ?!? | 0..1 | code | draft | active | suspended + SupplyRequestStatus (Required)
category | ? | 0..1 | CodeableConcept | The kind of supply (central, non-stock, etc.) Supply Type (Example)
priority | ? | 0..1 | code | routine | urgent | asap | stat RequestPriority (Required)
orderedItem | ? | 0..1 | BackboneElement | The item being requested
quantity | ? | 1..1 | Quantity | The requested amount of the item indicated
item[x] | ? | 0..1 | | Medication, Substance, or Device requested to be supplied SNOMED CT Supply Item (Example)
itemCodeableConcept | | | CodeableConcept
itemReference | | | Reference(Medication | Substance | Device)
occurrence[x] | ? | 0..1 | | When the request should be fulfilled
occurrenceDateTime | | | dateTime
occurrencePeriod | | | Period
occurrenceTiming | | | Timing
authoredOn | ? | 0..1 | dateTime | When the request was made
requester | ? | 0..1 | BackboneElement | Who/what is requesting service
agent | ? | 1..1 | Reference(Practitioner | Organization | Patient | RelatedPerson | Device) | Individual making the request
onBehalfOf | | 0..1 | Reference(Organization) | Organization agent is acting for
supplier | ? | 0..* | Reference(Organization) | Who is intended to fulfill the request
reason[x] | | 0..1 | | Why the supply item was requested SupplyRequestReason (Example)
reasonCodeableConcept | | | CodeableConcept
reasonReference | | | Reference(Any)
deliverFrom | | 0..1 | Reference(Organization | Location) | The origin of the supply
deliverTo | | 0..1 | Reference(Organization | Location | Patient) | The destination of the supply
Documentation for this format

UML Diagram (Legend)

XML Template

```
<SupplyRequest xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..1 Identifier Unique identifier --></identifier>
 <status value="[code]"/><!-- 0..1 draft | active | suspended + -->
 <category><!-- 0..1 CodeableConcept The kind of supply (central, non-stock, etc.) --></category>
 <priority value="[code]"/><!-- 0..1 routine | urgent | asap | stat -->
 <orderedItem>  <!-- 0..1 The item being requested -->
  <quantity><!-- 1..1 Quantity The requested amount of the item indicated --></quantity>
  <item[x]><!-- 0..1 CodeableConcept|Reference(Medication|Substance|Device) Medication, Substance, or Device requested
    to be supplied --></item[x]>
 </orderedItem>
 <occurrence[x]><!-- 0..1 dateTime|Period|Timing When the request should be fulfilled --></occurrence[x]>
 <authoredOn value="[dateTime]"/><!-- 0..1 When the request was made -->
 <requester>  <!-- 0..1 Who/what is requesting service -->
  <agent><!-- 1..1 Reference(Practitioner|Organization|Patient|RelatedPerson|
    Device) Individual making the request --></agent>
  <onBehalfOf><!-- 0..1 Reference(Organization) Organization agent is acting for --></onBehalfOf>
 </requester>
 <supplier><!-- 0..* Reference(Organization) Who is intended to fulfill the request --></supplier>
 <reason[x]><!-- 0..1 CodeableConcept|Reference(Any) Why the supply item was requested --></reason[x]>
 <deliverFrom><!-- 0..1 Reference(Organization|Location) The origin of the supply --></deliverFrom>
 <deliverTo><!-- 0..1 Reference(Organization|Location|Patient) The destination of the supply --></deliverTo>
</SupplyRequest>
```

JSON Template

```
{
  "resourceType" : "SupplyRequest",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : { Identifier }, // Unique identifier
  "status" : "<code>", // draft | active | suspended +
  "category" : { CodeableConcept }, // The kind of supply (central, non-stock, etc.)
  "priority" : "<code>", // routine | urgent | asap | stat
  "orderedItem" : { // The item being requested
    "quantity" : { Quantity }, // R!  The requested amount of the item indicated
    // item[x]: Medication, Substance, or Device requested to be supplied. One of these 2:
    "itemCodeableConcept" : { CodeableConcept }
    "itemReference" : { Reference(Medication|Substance|Device) }
  },
  // occurrence[x]: When the request should be fulfilled. One of these 3:
  "occurrenceDateTime" : "<dateTime>",
  "occurrencePeriod" : { Period },
  "occurrenceTiming" : { Timing },
  "authoredOn" : "<dateTime>", // When the request was made
  "requester" : { // Who/what is requesting service
    "agent" : { Reference(Practitioner|Organization|Patient|RelatedPerson|
    Device) }, // R!  Individual making the request
    "onBehalfOf" : { Reference(Organization) } // Organization agent is acting for
  },
  "supplier" : [{ Reference(Organization) }], // Who is intended to fulfill the request
  // reason[x]: Why the supply item was requested. One of these 2:
  "reasonCodeableConcept" : { CodeableConcept },
  "reasonReference" : { Reference(Any) },
  "deliverFrom" : { Reference(Organization|Location) }, // The origin of the supply
  "deliverTo" : { Reference(Organization|Location|Patient) } // The destination of the supply
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:SupplyRequest;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:SupplyRequest.identifier [ Identifier ]; # 0..1 Unique identifier
  fhir:SupplyRequest.status [ code ]; # 0..1 draft | active | suspended +
  fhir:SupplyRequest.category [ CodeableConcept ]; # 0..1 The kind of supply (central, non-stock, etc.)
  fhir:SupplyRequest.priority [ code ]; # 0..1 routine | urgent | asap | stat
  fhir:SupplyRequest.orderedItem [ # 0..1 The item being requested
    fhir:SupplyRequest.orderedItem.quantity [ Quantity ]; # 1..1 The requested amount of the item indicated
    # SupplyRequest.orderedItem.item[x] : 0..1 Medication, Substance, or Device requested to be supplied. One of these 2
      fhir:SupplyRequest.orderedItem.itemCodeableConcept [ CodeableConcept ]
      fhir:SupplyRequest.orderedItem.itemReference [ Reference(Medication|Substance|Device) ]
  ];
  # SupplyRequest.occurrence[x] : 0..1 When the request should be fulfilled. One of these 3
    fhir:SupplyRequest.occurrenceDateTime [ dateTime ]
    fhir:SupplyRequest.occurrencePeriod [ Period ]
    fhir:SupplyRequest.occurrenceTiming [ Timing ]
  fhir:SupplyRequest.authoredOn [ dateTime ]; # 0..1 When the request was made
  fhir:SupplyRequest.requester [ # 0..1 Who/what is requesting service
    fhir:SupplyRequest.requester.agent [ Reference(Practitioner|Organization|Patient|RelatedPerson|Device) ]; # 1..1
    Individual making the request
    fhir:SupplyRequest.requester.onBehalfOf [ Reference(Organization) ]; # 0..1 Organization agent is acting for
  ];
```

```
  fhir:SupplyRequest.supplier [ Reference(Organization) ], ... ; # 0..* Who is intended to fulfill the request
  # SupplyRequest.reason[x] : 0..1 Why the supply item was requested. One of these 2
    fhir:SupplyRequest.reasonCodeableConcept [ CodeableConcept ]
    fhir:SupplyRequest.reasonReference [ Reference(Any) ]
  fhir:SupplyRequest.deliverFrom [ Reference(Organization|Location) ]; # 0..1 The origin of the supply
  fhir:SupplyRequest.deliverTo [ Reference(Organization|Location|Patient) ]; # 0..1 The destination of the supply
]
```

Changes since DSTU2

SupplyRequest
SupplyRequest.category | Renamed from kind to category Remove Binding http://hl7.org/fhir/ValueSet/supplyrequest-kind (required)
SupplyRequest.priority | Added Element
SupplyRequest.orderedItem | Remove Reference(Medication), Remove Reference(Substance), Remove Reference(Device)
SupplyRequest.orderedItem.quantity | Added Element
SupplyRequest.orderedItem.item[x] | Added Element
SupplyRequest.occurrence[x] | Added Element
SupplyRequest.authoredOn | Renamed from date to authoredOn
SupplyRequest.requester | Added Element
SupplyRequest.requester.agent | Added Element
SupplyRequest.requester.onBehalfOf | Added Element
SupplyRequest.deliverFrom | Added Element
SupplyRequest.deliverTo | Added Element
SupplyRequest.patient | deleted
SupplyRequest.source | deleted
SupplyRequest.when | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R2 <--> R3 Conversion Maps (status = 1 test that all execute ok. All tests pass round-trip testing and all r3 resources are valid.).

Alternate definitions: Master Definition (XML, JSON), XML Schema/Schematron (for) + JSON Schema, ShEx (for Turtle)

12.24.3.1 Terminology Bindings

Path | Definition | Type | Reference
SupplyRequest.status | Status of the supply request | Required | SupplyRequestStatus
SupplyRequest.category | Category of supply request | Example | Supply Type
SupplyRequest.priority | Identifies the level of importance to be assigned to actioning the request | Required | RequestPriority
SupplyRequest.orderedItem.item[x] | The item that was requested | Example | SNOMED CT Supply Item
SupplyRequest.reason[x] | Why the supply item was requested | Example | SupplyRequestReason

12.24.4 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
category | token | The kind of supply (central, non-stock, etc.) | SupplyRequest.category
date | date | When the request was made | SupplyRequest.authoredOn | 18 Resources
identifier | token | Unique identifier | SupplyRequest.identifier | 26 Resources
requester | reference | Individual making the request | SupplyRequest.requester.agent (Practitioner, Organization, Device, Patient, RelatedPerson)
status | token | draft | active | suspended + | SupplyRequest.status
supplier | reference | Who is intended to fulfill the request | SupplyRequest.supplier (Organization)

12.24.5 Resource SupplyRequest - Examples

URL: https://hl7.org/fhir/STU3/supplyrequest-examples.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

12.24.5 Resource SupplyRequest - Examples

Orders and Observations Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Device, Patient, Practitioner, RelatedPerson

Example List:

Simple order for resupply | simpleorder | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

12.24.6 Resource SupplyRequest - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/supplyrequest-definitions.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

12.24.6 Resource SupplyRequest - Detailed Descriptions

Orders and Observations Work Group | Maturity Level: 1 | Trial Use | Compartments: Device, Patient, Practitioner, RelatedPerson

Detailed Descriptions for the elements in the SupplyRequest resource.

SupplyRequest
Definition | A record of a request for a medication, substance or device used in the healthcare setting.
Control | 1..1
SupplyRequest.identifier
Definition | Unique identifier for this supply request.
Note | This is a business identifer, not a resource identifier (see discussion)
Control | 0..1
Type | Identifier
Summary | true
Comments | This is assigned by the requester, and used to refer to this order in other external standards.
SupplyRequest.status
Definition | Status of the supply request.
Control | 0..1
Terminology Binding | SupplyRequestStatus (Required)
Type | code
Is Modifier | true
Summary | true
Comments | This element is labeled as a modifier because the status contains codes that mark the resource as not currently valid.
SupplyRequest.category
Definition | Category of supply, e.g. central, non-stock, etc. This is used to support work flows associated with the supply process.
Control | 0..1
Terminology Binding | Supply Type (Example)
Type | CodeableConcept
Alternate Names | kind
Summary | true
SupplyRequest.priority
Definition | Indicates how quickly this SupplyRequest should be addressed with respect to other requests.
Control | 0..1
Terminology Binding | RequestPriority (Required)
Type | code
Meaning if Missing | If missing, this task should be performed with normal priority
Summary | true
SupplyRequest.orderedItem
Definition | The item being requested.
Control | 0..1
Summary | true
SupplyRequest.orderedItem.quantity
Definition | The amount that is being ordered of the indicated item.
Control | 1..1
Type | Quantity
Summary | true
SupplyRequest.orderedItem.item[x]
Definition | The item that is requested to be supplied. This is either a link to a resource representing the details of the item or a code that identifies the item from a known list.
Control | 0..1
Terminology Binding | SNOMED CT Supply Item (Example)
Type | CodeableConcept|Reference(Medication | Substance | Device)
[x] Note | See Choice of Data Types for further information about how to use [x]
Summary | true
Comments | Note that there's a difference between a prescription - an instruction to take a medication, along with a (sometimes) implicit supply, and an explicit request to supply, with no explicit instructions.
SupplyRequest.occurrence[x]
Definition | When the request should be fulfilled.
Control | 0..1
Type | dateTime|Period|Timing
[x] Note | See Choice of Data Types for further information about how to use [x]
Alternate Names | when; timing
Summary | true
SupplyRequest.authoredOn
Definition | When the request was made.
Control | 0..1
Type | dateTime
Alternate Names | date; createdOn; signedOn
Summary | true
SupplyRequest.requester
Definition | The individual who initiated the request and has responsibility for its activation.
Control | 0..1
Alternate Names | author; source

Summary | true
SupplyRequest.requester.agent
Definition | The device, practitioner, etc. who initiated the request.
Control | 1..1
Type | Reference(Practitioner | Organization | Patient | RelatedPerson | Device)
Summary | true
SupplyRequest.requester.onBehalfOf
Definition | The organization the device or practitioner was acting on behalf of.
Control | 0..1
Type | Reference(Organization)
Requirements | Practitioners and Devices can be associated with multiple organizations. This element indicates which organization they were acting on behalf of when authoring the request.
SupplyRequest.supplier
Definition | Who is intended to fulfill the request.
Control | 0..*
Type | Reference(Organization)
Alternate Names | performer
Summary | true
SupplyRequest.reason[x]
Definition | Why the supply item was requested.
Control | 0..1
Terminology Binding | SupplyRequestReason (Example)
Type | CodeableConcept|Reference(Any)
[x] Note | See Choice of Data Types for further information about how to use [x]
SupplyRequest.deliverFrom
Definition | Where the supply is expected to come from.
Control | 0..1
Type | Reference(Organization | Location)
SupplyRequest.deliverTo
Definition | Where the supply is destined to go.
Control | 0..1
Type | Reference(Organization | Location | Patient)

12.24.7 Resource SupplyRequest - Mappings

URL: https://hl7.org/fhir/STU3/supplyrequest-mappings.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

12.24.7 Resource SupplyRequest - Mappings

Orders and Observations Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Device, Patient, Practitioner, RelatedPerson

Mappings:

RIM Mapping

HL7 v2 Mapping

Workflow Mapping

Mappings for the supplyrequest resource.

12.24.7.1 Mappings for RIM Mapping (http://hl7.org/v3)

SupplyRequest | Act[moodCode<=INT]
identifier | .identifier
status | .status
category | .code
priority | .priorityCode
orderedItem | .code
quantity | .quantity
item[x] | .code
occurrence[x] | .effectiveTime
authoredOn | .participation[typeCode=AUT].time
requester | .participation[typeCode=AUT].role
agent | .player
onBehalfOf | .scoper
supplier | .participation[typeCode=PRF].role[scoper.determinerCode=INSTANCE]
reason[x] | .reasonCode, .outboundRelationship[typeCode=RSON].target
deliverFrom | .participation
deliverTo | .participation

12.24.7.2 Mappings for HL7 v2 Mapping (http://hl7.org/v2)

SupplyRequest | ORC
identifier | ORC.2, ORC.3

status | ORC.5
category | N/A
priority | TQ1.9
orderedItem | Varies by domain
quantity
item[x]
occurrence[x] | Accompanying TQ1/TQ2 segments
authoredOn | ORC.9
requester | ORC.12
agent | ORC.12
onBehalfOf | N/A
supplier | PRT
reason[x] | ORC.16
deliverFrom | PRT
deliverTo | PRT

12.24.7.3 Mappings for Workflow Mapping (http://hl7.org/fhir/workflow)

SupplyRequest | Request
identifier | Request.identifier
status | Request.status
category
priority | Request.priority
orderedItem | Request.code
quantity
item[x]
occurrence[x] | Request.occurrence.occurrenceTiming
authoredOn | Request.authoredOn
requester | Request.requester
agent | Request.requester.agent
onBehalfOf | Request.requester.onBehalfOf
supplier | Request.performer
reason[x] | Request.reasonCode, Request.reasonReference
deliverFrom
deliverTo

12.24.8 Resource SupplyRequest - Design Notes

URL: https://hl7.org/fhir/STU3/supplyrequest-explanations.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

12.24.8 Resource SupplyRequest - Design Notes

to do

12.24.9 Resource SupplyRequest - Extensions & Profiles

URL: https://hl7.org/fhir/STU3/supplyrequest-profiles.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

12.24.9 Resource SupplyRequest - Extensions & Profiles

Orders and Observations Work Group | Maturity Level: N/A | Ballot Status: Informative

This table lists profiles and extensions for the SupplyRequest resource. For background information, see Profiling Resources and Extensibility. Additional profiles and resources may be found in published Implementation Guides , or in the Conformance resource registry .

12.24.9.1 Profiles

No Profiles defined for this resource

12.24.9.2 Extensions

No Extensions defined for this resource

Extensions that reference this resource:

No Extensions refer to this resource

12.24.9.3 Search Extensions

No Search Extensions defined for this resource

12.24.10 Resource SupplyRequest - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/supplyrequest-version-maps.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

12.24.10 Resource SupplyRequest - Detailed Descriptions

Orders and Observations Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Device, Patient, Practitioner, RelatedPerson

R2 : R3 Converson maps for SupplyRequest.

Functional status for this map: 1 test that all execute ok. All tests pass round-trip testing and all r3 resources are valid. (see documentation)

12.24.10.1 R2 to R3

```
map "http://hl7.org/fhir/StructureMap/SupplyRequest2to3" = "R2 to R3 Conversions for SupplyRequest"

uses "http://hl7.org/fhir/DSTU2/StructureDefinition/SupplyRequest" alias SupplyRequestR2 as source
uses "http://hl7.org/fhir/StructureDefinition/SupplyRequest" alias SupplyRequest as target

imports "http://hl7.org/fhir/StructureMap/*2to3"

group for type+types  SupplyRequest extends DomainResource
  input src : SupplyRequestR2 as source
  input tgt : SupplyRequest as target

  "SupplyRequest-patient" : for src.patient as v make tgt.extension as t, t.url =
    'http://hl7.org/fhir/SupplyRequest-r2-patient', t.value = v
  "SupplyRequest-source" : for src.source as v make tgt.requester as r, r.agent as v
  "SupplyRequest-date" : for src.date make tgt.authoredOn
  "SupplyRequest-identifier" : for src.identifier make tgt.identifier
  "SupplyRequest-status" : for src.status make tgt.status
  "SupplyRequest-kind" : for src.kind make tgt.category
  "SupplyRequest-orderedItem" : for src.orderedItem make tgt.orderedItem
  "SupplyRequest-supplier" : for src.supplier make tgt.supplier
  "SupplyRequest-reason" : for src.reason make tgt.reason
  "SupplyRequest-when" : for src.when as vs0 then srWhen(vs0, tgt)
endgroup

group srWhen extends BackboneElement
  input src as source
  input tgt as target

  "SupplyRequest.when-schedule" : for src.schedule make tgt.occurrence
endgroup
```

12.24.10.2 R3 to R2

```
map "http://hl7.org/fhir/StructureMap/SupplyRequest3to2" = "R3 to R2 Conversion for SupplyRequest"

uses "http://hl7.org/fhir/StructureDefinition/SupplyRequest" alias SupplyRequest as source
uses "http://hl7.org/fhir/DSTU2/StructureDefinition/SupplyRequest" alias SupplyRequestR2 as target

imports "http://hl7.org/fhir/StructureMap/*3to2"

group for type+types  SupplyRequest extends DomainResource
  input src : SupplyRequest as source
  input tgt : SupplyRequestR2 as target

  "SupplyRequest-patient" : for src.extension as e where e.url = 'http://hl7.org/fhir/SupplyRequest-r2-patient' make
    tgt.patient = (e.value)
  "SupplyRequest-source" : for src.requester as v then sprRequest(v, tgt)
  "SupplyRequest-date" : for src.authoredOn make tgt.date
  "SupplyRequest-identifier" : for src.identifier make tgt.identifier
  "SupplyRequest-status" : for src.status make tgt.status
  "SupplyRequest-kind" : for src.category make tgt.kind
  "SupplyRequest-orderedItem" : for src.orderedItem make tgt.orderedItem
  "SupplyRequest-supplier" : for src.supplier make tgt.supplier
  "SupplyRequest-reason" : for src.reason make tgt.reason
  "SupplyRequest-when" : for src.occurrence as vs0 make tgt.when as vt0, vt0.schedule = vs0
endgroup

group sprRequest extends BackboneElement
  input src as source
  input tgt as target

  "SupplyRequest.source-agent" : for src.avent make tgt.source
endgroup
```

12.24.10.3 R3 Validation Errors

12.25 Resource SupplyDelivery - Content

URL: https://hl7.org/fhir/STU3/supplydelivery.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

Orders and Observations Work Group | Maturity Level: 1 | Trial Use | Compartments: Patient, Practitioner

Record of delivery of what is supplied.

12.25.1 Scope and Usage

This resource is an event resource from a FHIR workflow perspective - see Workflow. It is the intent of the Orders and Observation Workgroup to align this resource with the workflow pattern for event resources.

The scope of the supply resource is for supplies used in the healthcare process. This includes supplies specifically used in the treatment of patients as well as supply movement within an institution (transport a set of supplies from materials management to a service unit (nurse station). This resource does not include the provisioning of transportation services.

12.25.2 Boundaries and Relationships

This resource overlaps with others such as Device and Medication. The Supply resource may be used to describe medications and devices when handling them generically (as any other supply). For example, when processing bulk orders, etc. However, when the medication, device or other aspects of the resource are important, (e.g. in a MedicationRequest or when identifying a device as a performer), the more detailed resource must be used.

This resource is referenced by chargeitem

12.25.3 Resource Content

* Structure

* UML

* XML

* JSON

* Turtle

* R2 Diff

* All

Structure

Name | Flags | Card. | Type | Description & Constraints
SupplyDelivery | | | DomainResource | Delivery of bulk Supplies Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | | 0..1 | Identifier | External identifier
basedOn | ? | 0..* | Reference(SupplyRequest) | Fulfills plan, proposal or order
partOf | ? | 0..* | Reference(SupplyDelivery | Contract) | Part of referenced event
status | ?!? | 0..1 | code | in-progress | completed | abandoned | entered-in-error SupplyDeliveryStatus (Required)
patient | | 0..1 | Reference(Patient) | Patient for whom the item is supplied
type | | 0..1 | CodeableConcept | Category of dispense event Supply Item Type (Required)
suppliedItem | | 0..1 | BackboneElement | The item that is delivered or supplied
quantity | | 0..1 | SimpleQuantity | Amount dispensed
item[x] | | 0..1 | | Medication, Substance, or Device supplied SNOMED CT Supply Item (Example)
itemCodeableConcept | | | CodeableConcept
itemReference | | | Reference(Medication | Substance | Device)
occurrence[x] | ? | 0..1 | | When event occurred
occurrenceDateTime | | | dateTime
occurrencePeriod | | | Period
occurrenceTiming | | | Timing
supplier | | 0..1 | Reference(Practitioner | Organization) | Dispenser
destination | | 0..1 | Reference(Location) | Where the Supply was sent
receiver | | 0..* | Reference(Practitioner) | Who collected the Supply
Documentation for this format

UML Diagram (Legend)

XML Template

```
<SupplyDelivery xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..1 Identifier External identifier --></identifier>
 <basedOn><!-- 0..* Reference(SupplyRequest) Fulfills plan, proposal or order --></basedOn>
 <partOf><!-- 0..* Reference(SupplyDelivery|Contract) Part of referenced event --></partOf>
 <status value="[code]"/><!-- 0..1 in-progress | completed | abandoned | entered-in-error -->
 <patient><!-- 0..1 Reference(Patient) Patient for whom the item is supplied --></patient>
 <type><!-- 0..1 CodeableConcept Category of dispense event --></type>
 <suppliedItem>  <!-- 0..1 The item that is delivered or supplied -->
  <quantity><!-- 0..1 Quantity(SimpleQuantity) Amount dispensed --></quantity>
  <item[x]><!-- 0..1 CodeableConcept|Reference(Medication|Substance|Device) Medication, Substance, or Device supplied
    --></item[x]>
 </suppliedItem>
 <occurrence[x]><!-- 0..1 dateTime|Period|Timing When event occurred --></occurrence[x]>
```

```
 <supplier><!-- 0..1 Reference(Practitioner|Organization) Dispenser --></supplier>
 <destination><!-- 0..1 Reference(Location) Where the Supply was sent --></destination>
 <receiver><!-- 0..* Reference(Practitioner) Who collected the Supply --></receiver>
</SupplyDelivery>
```

JSON Template

```
{
  "resourceType" : "SupplyDelivery",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : { Identifier }, // External identifier
  "basedOn" : [{ Reference(SupplyRequest) }], // Fulfills plan, proposal or order
  "partOf" : [{ Reference(SupplyDelivery|Contract) }], // Part of referenced event
  "status" : "<code>", // in-progress | completed | abandoned | entered-in-error
  "patient" : { Reference(Patient) }, // Patient for whom the item is supplied
  "type" : { CodeableConcept }, // Category of dispense event
  "suppliedItem" : { // The item that is delivered or supplied
    "quantity" : { Quantity(SimpleQuantity) }, // Amount dispensed
    // item[x]: Medication, Substance, or Device supplied. One of these 2:
    "itemCodeableConcept" : { CodeableConcept }
    "itemReference" : { Reference(Medication|Substance|Device) }
  },
  // occurrence[x]: When event occurred. One of these 3:
  "occurrenceDateTime" : "<dateTime>",
  "occurrencePeriod" : { Period },
  "occurrenceTiming" : { Timing },
  "supplier" : { Reference(Practitioner|Organization) }, // Dispenser
  "destination" : { Reference(Location) }, // Where the Supply was sent
  "receiver" : [{ Reference(Practitioner) }] // Who collected the Supply
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:SupplyDelivery;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:SupplyDelivery.identifier [ Identifier ]; # 0..1 External identifier
  fhir:SupplyDelivery.basedOn [ Reference(SupplyRequest) ], ... ; # 0..* Fulfills plan, proposal or order
  fhir:SupplyDelivery.partOf [ Reference(SupplyDelivery|Contract) ], ... ; # 0..* Part of referenced event
  fhir:SupplyDelivery.status [ code ]; # 0..1 in-progress | completed | abandoned | entered-in-error
  fhir:SupplyDelivery.patient [ Reference(Patient) ]; # 0..1 Patient for whom the item is supplied
  fhir:SupplyDelivery.type [ CodeableConcept ]; # 0..1 Category of dispense event
  fhir:SupplyDelivery.suppliedItem [ # 0..1 The item that is delivered or supplied
    fhir:SupplyDelivery.suppliedItem.quantity [ Quantity(SimpleQuantity) ]; # 0..1 Amount dispensed
    # SupplyDelivery.suppliedItem.item[x] : 0..1 Medication, Substance, or Device supplied. One of these 2
      fhir:SupplyDelivery.suppliedItem.itemCodeableConcept [ CodeableConcept ]
      fhir:SupplyDelivery.suppliedItem.itemReference [ Reference(Medication|Substance|Device) ]
  ];
  # SupplyDelivery.occurrence[x] : 0..1 When event occurred. One of these 3
    fhir:SupplyDelivery.occurrenceDateTime [ dateTime ]
    fhir:SupplyDelivery.occurrencePeriod [ Period ]
    fhir:SupplyDelivery.occurrenceTiming [ Timing ]
  fhir:SupplyDelivery.supplier [ Reference(Practitioner|Organization) ]; # 0..1 Dispenser
  fhir:SupplyDelivery.destination [ Reference(Location) ]; # 0..1 Where the Supply was sent
  fhir:SupplyDelivery.receiver [ Reference(Practitioner) ], ... ; # 0..* Who collected the Supply
]
```

Changes since DSTU2

SupplyDelivery
SupplyDelivery.basedOn | Added Element
SupplyDelivery.partOf | Added Element
SupplyDelivery.suppliedItem | Remove Reference(Medication), Remove Reference(Substance), Remove Reference(Device)
SupplyDelivery.suppliedItem.quantity | Added Element
SupplyDelivery.suppliedItem.item[x] | Added Element
SupplyDelivery.occurrence[x] | Added Element
SupplyDelivery.supplier | Add Reference(Organization)
SupplyDelivery.quantity | deleted
SupplyDelivery.whenPrepared | deleted
SupplyDelivery.time | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R2 <--> R3 Conversion Maps (status = 1 test that all execute ok. All tests pass round-trip testing and all r3 resources are valid.).

Structure

Name | Flags | Card. | Type | Description & Constraints
SupplyDelivery | | | DomainResource | Delivery of bulk Supplies Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | | 0..1 | Identifier | External identifier

basedOn | ? | 0..* | Reference(SupplyRequest) | Fulfills plan, proposal or order
partOf | ? | 0..* | Reference(SupplyDelivery | Contract) | Part of referenced event
status | ?!? | 0..1 | code | in-progress | completed | abandoned | entered-in-error SupplyDeliveryStatus (Required)
patient | | 0..1 | Reference(Patient) | Patient for whom the item is supplied
type | | 0..1 | CodeableConcept | Category of dispense event Supply Item Type (Required)
suppliedItem | | 0..1 | BackboneElement | The item that is delivered or supplied
quantity | | 0..1 | SimpleQuantity | Amount dispensed
item[x] | | 0..1 | | Medication, Substance, or Device supplied SNOMED CT Supply Item (Example)
itemCodeableConcept | | | CodeableConcept
itemReference | | | Reference(Medication | Substance | Device)
occurrence[x] | ? | 0..1 | | When event occurred
occurrenceDateTime | | | dateTime
occurrencePeriod | | | Period
occurrenceTiming | | | Timing
supplier | | 0..1 | Reference(Practitioner | Organization) | Dispenser
destination | | 0..1 | Reference(Location) | Where the Supply was sent
receiver | | 0..* | Reference(Practitioner) | Who collected the Supply
Documentation for this format

UML Diagram (Legend)

XML Template

```
<SupplyDelivery xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..1 Identifier External identifier --></identifier>
 <basedOn><!-- 0..* Reference(SupplyRequest) Fulfills plan, proposal or order --></basedOn>
 <partOf><!-- 0..* Reference(SupplyDelivery|Contract) Part of referenced event --></partOf>
 <status value="[code]"/><!-- 0..1 in-progress | completed | abandoned | entered-in-error -->
 <patient><!-- 0..1 Reference(Patient) Patient for whom the item is supplied --></patient>
 <type><!-- 0..1 CodeableConcept Category of dispense event --></type>
 <suppliedItem>  <!-- 0..1 The item that is delivered or supplied -->
  <quantity><!-- 0..1 Quantity(SimpleQuantity) Amount dispensed --></quantity>
  <item[x]><!-- 0..1 CodeableConcept|Reference(Medication|Substance|Device) Medication, Substance, or Device supplied
    --></item[x]>
 </suppliedItem>
 <occurrence[x]><!-- 0..1 dateTime|Period|Timing When event occurred --></occurrence[x]>
 <supplier><!-- 0..1 Reference(Practitioner|Organization) Dispenser --></supplier>
 <destination><!-- 0..1 Reference(Location) Where the Supply was sent --></destination>
 <receiver><!-- 0..* Reference(Practitioner) Who collected the Supply --></receiver>
</SupplyDelivery>
```

JSON Template

```
{
  "resourceType" : "SupplyDelivery",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : { Identifier }, // External identifier
  "basedOn" : [{ Reference(SupplyRequest) }], // Fulfills plan, proposal or order
  "partOf" : [{ Reference(SupplyDelivery|Contract) }], // Part of referenced event
  "status" : "<code>", // in-progress | completed | abandoned | entered-in-error
  "patient" : { Reference(Patient) }, // Patient for whom the item is supplied
  "type" : { CodeableConcept }, // Category of dispense event
  "suppliedItem" : { // The item that is delivered or supplied
    "quantity" : { Quantity(SimpleQuantity) }, // Amount dispensed
    // item[x]: Medication, Substance, or Device supplied. One of these 2:
    "itemCodeableConcept" : { CodeableConcept }
    "itemReference" : { Reference(Medication|Substance|Device) }
  },
  // occurrence[x]: When event occurred. One of these 3:
  "occurrenceDateTime" : "<dateTime>",
  "occurrencePeriod" : { Period },
  "occurrenceTiming" : { Timing },
  "supplier" : { Reference(Practitioner|Organization) }, // Dispenser
  "destination" : { Reference(Location) }, // Where the Supply was sent
  "receiver" : [{ Reference(Practitioner) }] // Who collected the Supply
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:SupplyDelivery;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:SupplyDelivery.identifier [ Identifier ]; # 0..1 External identifier
  fhir:SupplyDelivery.basedOn [ Reference(SupplyRequest) ], ... ; # 0..* Fulfills plan, proposal or order
  fhir:SupplyDelivery.partOf [ Reference(SupplyDelivery|Contract) ], ... ; # 0..* Part of referenced event
  fhir:SupplyDelivery.status [ code ]; # 0..1 in-progress | completed | abandoned | entered-in-error
  fhir:SupplyDelivery.patient [ Reference(Patient) ]; # 0..1 Patient for whom the item is supplied
  fhir:SupplyDelivery.type [ CodeableConcept ]; # 0..1 Category of dispense event
  fhir:SupplyDelivery.suppliedItem [ # 0..1 The item that is delivered or supplied
    fhir:SupplyDelivery.suppliedItem.quantity [ Quantity(SimpleQuantity) ]; # 0..1 Amount dispensed
```

```
    # SupplyDelivery.suppliedItem.item[x] : 0..1 Medication, Substance, or Device supplied. One of these 2
      fhir:SupplyDelivery.suppliedItem.itemCodeableConcept [ CodeableConcept ]
      fhir:SupplyDelivery.suppliedItem.itemReference [ Reference(Medication|Substance|Device) ]
  ];
  # SupplyDelivery.occurrence[x] : 0..1 When event occurred. One of these 3
    fhir:SupplyDelivery.occurrenceDateTime [ dateTime ]
    fhir:SupplyDelivery.occurrencePeriod [ Period ]
    fhir:SupplyDelivery.occurrenceTiming [ Timing ]
  fhir:SupplyDelivery.supplier [ Reference(Practitioner|Organization) ]; # 0..1 Dispenser
  fhir:SupplyDelivery.destination [ Reference(Location) ]; # 0..1 Where the Supply was sent
  fhir:SupplyDelivery.receiver [ Reference(Practitioner) ], ... ; # 0..* Who collected the Supply
]
```

Changes since DSTU2

SupplyDelivery
SupplyDelivery.basedOn | Added Element
SupplyDelivery.partOf | Added Element
SupplyDelivery.suppliedItem | Remove Reference(Medication), Remove Reference(Substance), Remove Reference(Device)
SupplyDelivery.suppliedItem.quantity | Added Element
SupplyDelivery.suppliedItem.item[x] | Added Element
SupplyDelivery.occurrence[x] | Added Element
SupplyDelivery.supplier | Add Reference(Organization)
SupplyDelivery.quantity | deleted
SupplyDelivery.whenPrepared | deleted
SupplyDelivery.time | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R2 <--> R3 Conversion Maps (status = 1 test that all execute ok. All tests pass round-trip testing and all r3 resources are valid.).

Alternate definitions: Master Definition (XML, JSON), XML Schema/Schematron (for) + JSON Schema, ShEx (for Turtle)

12.25.3.1 Terminology Bindings

Path | Definition | Type | Reference
SupplyDelivery.status | Status of the supply delivery. | Required | SupplyDeliveryStatus
SupplyDelivery.type | The type of supply dispense. | Required | Supply Item Type
SupplyDelivery.suppliedItem.item[x] | The item that was delivered. | Example | SNOMED CT Supply Item

12.25.4 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
identifier | token | External identifier | SupplyDelivery.identifier | 26 Resources
patient | reference | Patient for whom the item is supplied | SupplyDelivery.patient (Patient) | 31 Resources
receiver | reference | Who collected the Supply | SupplyDelivery.receiver (Practitioner)
status | token | in-progress | completed | abandoned | entered-in-error | SupplyDelivery.status
supplier | reference | Dispenser | SupplyDelivery.supplier (Practitioner, Organization)

12.25.5 Resource SupplyDelivery - Examples

URL: https://hl7.org/fhir/STU3/supplydelivery-examples.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

12.25.5 Resource SupplyDelivery - Examples

Orders and Observations Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Patient, Practitioner

Example List:

Simple delivery for resupply | simpledelivery | XML | JSON | Turtle
Delivery of an insulin pump | pumpdelivery | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

12.25.6 Resource SupplyDelivery - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/supplydelivery-definitions.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

12.25.6 Resource SupplyDelivery - Detailed Descriptions

Orders and Observations Work Group | Maturity Level: 1 | Trial Use | Compartments: Patient, Practitioner

Detailed Descriptions for the elements in the SupplyDelivery resource.

SupplyDelivery
Definition | Record of delivery of what is supplied.
Control | 1..1
SupplyDelivery.identifier
Definition | Identifier assigned by the dispensing facility when the item(s) is dispensed.
Note | This is a business identifer, not a resource identifier (see discussion)
Control | 0..1
Type | Identifier
Comments | This is assigned by the dispenser, and used to refer to this order in other external standards.
SupplyDelivery.basedOn
Definition | A plan, proposal or order that is fulfilled in whole or in part by this event.
Control | 0..*
Type | Reference(SupplyRequest)
Requirements | Allows tracing of authorization for the event and tracking whether proposals/recommendations were acted upon.
Summary | true
SupplyDelivery.partOf
Definition | A larger event of which this particular event is a component or step.
Control | 0..*
Type | Reference(SupplyDelivery | Contract)
Requirements | E.g. Drug administration as part of a procedure, procedure as part of observation, etc.
Summary | true
Comments | Not to be used to link an event to an Encounter - use Event.context for that. [The allowed reference resources may be adjusted as appropriate for the event resource].
SupplyDelivery.status
Definition | A code specifying the state of the dispense event.
Control | 0..1
Terminology Binding | SupplyDeliveryStatus (Required)
Type | code
Is Modifier | true
Summary | true
Comments | This element is labeled as a modifier because the status contains codes that mark the resource as not currently valid.
SupplyDelivery.patient
Definition | A link to a resource representing the person whom the delivered item is for.
Control | 0..1
Type | Reference(Patient)
SupplyDelivery.type
Definition | Indicates the type of dispensing event that is performed. Examples include: Trial Fill, Completion of Trial, Partial Fill, Emergency Fill, Samples, etc.
Control | 0..1
Terminology Binding | Supply Item Type (Required)
Type | CodeableConcept
SupplyDelivery.suppliedItem
Definition | The item that is being delivered or has been supplied.
Control | 0..1
SupplyDelivery.suppliedItem.quantity
Definition | The amount of supply that has been dispensed. Includes unit of measure.
Control | 0..1
Type | SimpleQuantity
SupplyDelivery.suppliedItem.item[x]
Definition | Identifies the medication, substance or device being dispensed. This is either a link to a resource representing the details of the item or a code that identifies the item from a known list.
Control | 0..1
Terminology Binding | SNOMED CT Supply Item (Example)
Type | CodeableConcept|Reference(Medication | Substance | Device)
[x] Note | See Choice of Data Types for further information about how to use [x]
SupplyDelivery.occurrence[x]
Definition | The date or time(s) the activity occurred.
Control | 0..1
Type | dateTime|Period|Timing
[x] Note | See Choice of Data Types for further information about how to use [x]
Summary | true
Comments | [The list of types may be constrained as appropriate for the type of event].
SupplyDelivery.supplier
Definition | The individual responsible for dispensing the medication, supplier or device.
Control | 0..1
Type | Reference(Practitioner | Organization)
SupplyDelivery.destination
Definition | Identification of the facility/location where the Supply was shipped to, as part of the dispense event.
Control | 0..1
Type | Reference(Location)
SupplyDelivery.receiver

Definition | Identifies the person who picked up the Supply.
Control | 0..*
Type | Reference(Practitioner)

12.25.7 Resource SupplyDelivery - Mappings

URL: https://hl7.org/fhir/STU3/supplydelivery-mappings.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

12.25.7 Resource SupplyDelivery - Mappings

Orders and Observations Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Patient, Practitioner

Mappings:

RIM Mapping

HL7 v2 Mapping

Workflow Mapping

Mappings for the supplydelivery resource.

12.25.7.1 Mappings for RIM Mapping (http://hl7.org/v3)

SupplyDelivery | Supply[moodCode=RQO]
identifier | .id
basedOn | .outboundRelationship[typeCode=FLFS].target
partOf | .inboundRelationship[typeCode=COMP].source[moodCode=EVN]
status | .statusCode, also existence of fulfillment events
patient | .participation[typeCode=SBJ].role
type | .code
suppliedItem | .participation[typeCode=PRD].role
quantity | .quantity
item[x] | .participation[typeCode=PRD].role
occurrence[x] | .effectiveTime
supplier | .participation[typeCode=PFM].role
destination | .participation[typeCode=DST].role
receiver | .participation[typeCode=RCV].role

12.25.7.2 Mappings for HL7 v2 Mapping (http://hl7.org/v2)

SupplyDelivery
identifier
basedOn | ORC in proximity to EVN segment
partOf | Varies by domain
status
patient
type
suppliedItem
quantity
item[x]
occurrence[x] | Varies by domain
supplier
destination
receiver

12.25.7.3 Mappings for Workflow Mapping (http://hl7.org/fhir/workflow)

SupplyDelivery | Event
identifier | Event.identifier
basedOn | Event.basedOn
partOf | Event.partOf
status | Event.status
patient | Event.subject.reference(Patient)
type | Event.code
suppliedItem
quantity
item[x]
occurrence[x] | Event.occurrence[x]
supplier | Event.performer
destination
receiver

12.25.8 Resource SupplyDelivery - Design Notes

URL: https://hl7.org/fhir/STU3/supplydelivery-explanations.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

12.25.8 Resource SupplyDelivery - Design Notes

to do

12.25.9 Resource SupplyDelivery - Extensions & Profiles

URL: https://hl7.org/fhir/STU3/supplydelivery-profiles.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

12.25.9 Resource SupplyDelivery - Extensions & Profiles

Orders and Observations Work Group | Maturity Level: N/A | Ballot Status: Informative

This table lists profiles and extensions for the SupplyDelivery resource. For background information, see Profiling Resources and Extensibility. Additional profiles and resources may be found in published Implementation Guides , or in the Conformance resource registry .

12.25.9.1 Profiles

No Profiles defined for this resource

12.25.9.2 Extensions

No Extensions defined for this resource

Extensions that reference this resource:

No Extensions refer to this resource

12.25.9.3 Search Extensions

No Search Extensions defined for this resource

12.25.10 Resource SupplyDelivery - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/supplydelivery-version-maps.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

12.25.10 Resource SupplyDelivery - Detailed Descriptions

Orders and Observations Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Patient, Practitioner

R2 : R3 Converson maps for SupplyDelivery.

Functional status for this map: 1 test that all execute ok. All tests pass round-trip testing and all r3 resources are valid. (see documentation)

12.25.10.1 R2 to R3

```
map "http://hl7.org/fhir/StructureMap/SupplyDelivery2to3" = "R2 to R3 Conversions for SupplyDelivery"

uses "http://hl7.org/fhir/DSTU2/StructureDefinition/SupplyDelivery" alias SupplyDeliveryR2 as source
uses "http://hl7.org/fhir/StructureDefinition/SupplyDelivery" alias SupplyDelivery as target

imports "http://hl7.org/fhir/StructureMap/*2to3"

group for type+types  SupplyDelivery extends DomainResource
  input src : SupplyDeliveryR2 as source
  input tgt : SupplyDelivery as target

  "SupplyDelivery-identifier" : for src.identifier make tgt.identifier
  "SupplyDelivery-status" : for src.status make tgt.status
  "SupplyDelivery-patient" : for src.patient make tgt.patient
  "SupplyDelivery-type" : for src.type make tgt.type
  "SupplyDelivery-quantity" : for src.quantity as v make tgt.suppliedItem as si, si.quantity = v
  "SupplyDelivery-suppliedItem" : for src.suppliedItem as v make tgt.suppliedItem as si, si.item = v
  "SupplyDelivery-supplier" : for src.supplier make tgt.supplier
  "SupplyDelivery-whenPrepared" : for src.whenPrepared as v make tgt.extension as t, t.url =
    'http://hl7.org/fhir/SupplyDelivery-r2-whenPrepared', t.value = v
  "SupplyDelivery-time" : for src.time make tgt.occurrence
  "SupplyDelivery-destination" : for src.destination make tgt.destination
  "SupplyDelivery-receiver" : for src.receiver make tgt.receiver

endgroup
```

12.25.10.2 R3 to R2

```
map "http://hl7.org/fhir/StructureMap/SupplyDelivery3to2" = "R3 to R2 Conversion for SupplyDelivery"
```

```
uses "http://hl7.org/fhir/StructureDefinition/SupplyDelivery" alias SupplyDelivery as source
uses "http://hl7.org/fhir/DSTU2/StructureDefinition/SupplyDelivery" alias SupplyDeliveryR2 as target

imports "http://hl7.org/fhir/StructureMap/*3to2"

group for type+types  SupplyDelivery extends DomainResource
  input src : SupplyDelivery as source
  input tgt : SupplyDeliveryR2 as target

  "SupplyDelivery-identifier" : for src.identifier make tgt.identifier
  "SupplyDelivery-status" : for src.status make tgt.status
  "SupplyDelivery-patient" : for src.patient make tgt.patient
  "SupplyDelivery-type" : for src.type make tgt.type
  "SupplyDelivery-suppliedItem" : for src.suppliedItem as v then sdvItem(v, tgt)
  "SupplyDelivery-supplier" : for src.supplier make tgt.supplier
  "SupplyDelivery-whenPrepared" : for src.extension as e where e.url =
    'http://hl7.org/fhir/SupplyDelivery-r2-whenPrepared' make tgt.whenPrepared = (e.value)
  "SupplyDelivery-time" : for src.occurrence as v where v.is(dateTime) make tgt.time = v
  "SupplyDelivery-destination" : for src.destination make tgt.destination
  "SupplyDelivery-receiver" : for src.receiver make tgt.receiver

endgroup

group sdvItem extends BackboneElement
  input src as source
  input tgt as target

  "SupplyDelivery.item-quantity" : for src.quantity make tgt.quantity
  "SupplyDelivery.item-quantity" : for src.item make tgt.suppliedItem
endgroup
```

12.25.10.3 R3 Validation Errors